

炬芯 Actions

ATS3089x Datasheet

Actions® ATS3089x™ TFBGA129

Low Power Solution for
Dual mode Bluetooth smart watch

CPU + DSP Dual-core Single-chip
Bluetooth V5.3

Version: V1.3

2023-12-20

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Key Features

System

- 192MHz(Max) ARM MSTAR processor Core
- 192MHz(Max) CEVA TL420 DSP Core
- Memory Protect Unit(MPU)
- Internal 1168KB RAM for data and program
- Build in 32KB Execute-in-place(Xip) Cache to QSPI NOR flash and 32KB cache to HSPI/OSPI PSRAM
- Support QSPI STR/DTR serial Flash for custom defined software
- Support 32MHz OSC and 32.768KHz OSC with on-chip PLL
- Fully configurable PEQ, up to 20 segments
- Support RTC and Build-in timers
- Support Hi-res when A2DP
- Support echo cancellation
- Support ACTIONS single microphone noise reduction
- Support ACTIONS dual microphone noise reduction
- Support sensor hub

Secure Subsystem

- Support hardware AES-128/192/256 Encrypt & Decrypt in ECB/CTR/CBC/CBC-CTS mode
- Support hardware TRNG used to collects random bits from the ring oscillator, 64 random bits per time.
- Support SHA-1/SHA-224/SHA-256.
- Support hardware CRC-32/CRC-16
- Support 1024bits efuse
- Support 64bit CHIP unique ID
- Support RSA 2048bits signature
- Support secure boot
- Support firmware encryption
- Support On-the-fly decryption of external memories (QSPI NorFlash/Octo-SPI PSRAM), part of which can be encrypted by AES-128 in counter mode

Bluetooth

- Support Bluetooth V5.3
- Support LE Audio(LC3 Codec)
- Supports basic rate and enhanced data rate
- Supports Secure Simple Pairing
- Supports Low Power Mode
- Bluetooth Dual Mode support: Simultaneous LE and BR / EDR
- Supports multiple Low Energy states
- Fast AGC control to improve receiving dynamic range
- Supports AFH to dynamically detect channel quality to improve transmission quality
- Supports GFSK, π/4 DQPSK and 8DPSK modulation
- Supports Power / Enhanced Power Control
- LE Data Packet Length Extension
- LE 2M PHY
- Support long range BLE
- Support AOA/AOD
- Channel Selection Algorithm #2
- Support PTA (Packet Traffic Arbiter) when co-existing with WI-FI
- Support WCI2 when co-existing with 4G

Bluetooth Performance

- Max transmitting output power: BR 13dBm, BLE 14dBm
- Bluetooth receiving sensitivity: -96.5dBm@GFSK, -95dBm@π/4 DQPSK, -89.5dBm@8DPSK modulation
- Bluetooth receiving sensitivity (BLE mode): -99.5dBm@1Mbps, -96dBm@2Mbps

Audio

- Build in stereo 24-bit input sigma-delta ADC, SNR<=98dB, THD+N>=-88dB(A-weighting)
- ADC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48/88.2/96kHz
- Supports stereo single-ended input or differential analog microphone
- Supports stereo digital microphones.
- Build in single channel 24-bit input sigma-delta DAC, SNR<=113dB, THD+N>=-98dB, DR<=111dB, Noise>=2.7uVrms(A-weighting)
- DAC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48

k/88.2k/96kHz
- Support 1*I2S/PCM TX with master and slave mode, and another I2S TX with Master mode Only, both interfaces sampling rate from 8KHz to 192KHz
- Support 1*I2S/PCM RX with master and slave mode, sampling rate from 8KHz to 192KHz

Storage

- Support 2*QSPI interface for SPINOR or S PINAND, 1*HSPI for PSRAM.
- Support 2*SD/MMC card interface
- SIP 64Mbit QSPI NorFlash for for data and program.
- SIP 128Mbit HSPI PSRAM for gorgeous UI performance.

Graphics/Multimedia

- 192MHz 2.5D GPU
  Support any curves and any font rendering by HW tessellation, NO CPU overhead.
  2.5D perspective transformation.
  Any angle primitive rotation, flip and mirror.
  Anti-aliasing: 16x Anti-aliasing without extra performance drop and memory overhead.
  Solid, linear gradient paint and pattern paint.
  Texture mapping; Point sampling, linear and bilinear filtering.
  Scissoring and viewport clipping.
  Porter Duff Alpha Blending plus additional blend modes.
  8/16/24/32 bits source image and frame color formats.
  Index 1/2/4/8 format source image supported.
  Command buffer based DMA to minimize CPU overhead.
  Lossy compression to save system memory.

- 2D GPU
  RGB565/666/888/ARGB/ARGB8565/ARGB6666/Ax/ABMP/XRGB/CLUT source data format
  Support 256 Level Alpha
  Support three Image Layer
  Support any angle rotate with RGB565/ARGB/BGRA source
  The Layer1/2 Support Up/down scaling

- JPEG Baseline Decoder

Display

- Support RGB888/RGB666/RGB565 Source Data Format
- Support 8bit active (TFT) LCD panels with digital CPU interface
- Support (TFT) LCD panels with 3wire/4wire SPI option I/II interface
- Support 2lane SPI LCD interface
- Support SDR/DDR QSPI LCD interface
- Support TR-J/TR-S parallel HSYNC/VSYNC LCD interface

Power Management

- Operation voltage: I/O 1.8V or 3.3V, Core 0.6V~1.3V
- Supports Li-Ion battery and 5V power supply
- Supports 5V power supply plugged in reset
- Integrated linear battery charger with power path management up to 400mA charging current, which supports CC/CV mode
- Integrated 12bit SAR A/D converters for battery voltage monitor, temperature monitor and wire-controller
- Integrated 2*DC-DC buck converters, which can be switch to LDO mode
- Integrated 6*Load switch, which can be used as power supply for peripheral.
- Supports DC5V insertion detection and pull out detection
- Supports adaptive voltage scaling technology for power optimization
- Low Power Consumption:
  - Typical BR Sniff Current: <= 90μA@500ms, Txpower = 8dbm
  - Typical BLE Sniff Current: <= 65μA@500ms, Txpower = 8dbm
  - Typical Simultaneous BR and BLE Sniff Current: <= 110μA@500ms, Txpower = 8dbm
  - Standby mode: <= 20uA@Vbat = 3.8V
  - Shutdown mode: 2μA@Vbat = 3.8V
  - Run mode: <= 22 μA/MHz @BAT 3.8 V @CPU 192Mhz
  - Run mode: <= 12 μA/MHz @BAT 3.8 V @CPU 64Mhz

Flexible I/O

- Support 55 GPIOs at most
- Support 6 LRADC at most
- Support 4 PWM for lamp controller at most
- Support USB2.0 full speed
- Serial Interface:
  - 3*UART, Maximum Rate: 4Mbaud
  - 4*I2C, support for standard mode (100 kHz), Fast mode (400 kHz), and Fast mode+ (1 MHz)

- 2*SPIMT and 2*I2CMT for sensor device
- In addition to NOR, and QSPI_LCD, there are 3*SPI and 1*QSPI interfaces for users. The Maximum Rate of all the SPI/QPI interfaces is 96MHz, Maximum Rate of PSRAM interface is 140Mhz and Maximum Rate of NorFlash interface is 106Mhz.

Applications

- Dual mode Bluetooth smart watch

Package

- TFBGA129(5.5*5.5*0.93mm, Pitch 0.4mm)

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Revision History

Date	Revision	Description
2023-07-19	V1.0	Release version
2023-09-21	V1.1	Modify the Signal description of LCD SPI interface
2023-10-11	V1.2	Add the diagram of power on timing
2023-12-20	V1.3	1. Add the Device Marking and Label Information 2. Modify the feature description of PMU

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1 Introduction

1.1Overview

The ATS3089x is a highly integrated single-chip Bluetooth audio solution targeting Bluetooth smart watch markets with high performance, low power consumption, and low cost.

ATS3089x adopts Dual-core architecture. An ARM MSTAR core is used as application processor, and a companion CEVA TL420 DSP core is used to efficiently handle audio processing. The on-chip RAM and ROM are integrated for various Bluetooth audio processing applications.

ATS3089x integrates a Bluetooth controller that is fully compliant to V5.3 standard and is 100% backward compatible to V5.0/4.2/2.1 Bluetooth specifications. ATS3089x supports dual mode (BR/EDR + Low Energy Controllers) operations. Both links in BR/EDR and LE can be active simultaneously. Additionally, ATS3089x's embedded BLE Audio features bring new audio listening experiences and wonderful call experiences.

Well suited for wearable applications, ATS3089x integrates a 2.5D GPU, a 2D GPU and a built-in LCD controller providing excellent display effects while maintaining extremely low power consumptions. It supports LCD panels with 3-line SPI, 4-line SPI, Dual SPI, QSPI, DDR QSPI, TR-I/TR-S parallel HSYNC/VSYNC, or MCU interfaces. The SoC also integrates a sensor hub and a complete set of advanced power management circuits, which further enables ATS3089x to provide extremely low power consumption while maintaining high degrees of usability and achieving high marks in performance.

ATS3089x has a flexible memory configuration and a rich set of interfaces such as SD/SPI/USB2.0 FS/UART/I2C/PWM/I2S/PCM TX&RX. The ATS3089x architecture is fully programmable and can easily be adapted to work with application. It also has the minimum package & boasts the most compact BOM in industry, making it an ideal SoC candidate for all-in-one communication & wearable application devices.

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1.2Features table

Features	ATS3089P	ATS3089	ATS3089C
Package	TFBGA129		
Body Size	5.5x5.5x0.93		
Application	Smart Watch		
CPU	ARM MSTAR		
DSP	CEVA TL420		
2D GPU	√		
2.5D GPU	√		
RAM(KB)	1168		
Nor flash	Built-in 64Mbit, 4xIO, DTR	Built-in 64Mbit, 4xIO, DTR	Built-in 32Mbit, 4xIO, DTR
PSRAM	Built-in 256Mbit, 16xIO, DDR	Built-in 128Mbit, 16xIO, DDR	Built-in 64Mbit, 8xIO, DDR
Bluetooth	5.3		
Power	Li-ion		
Charger	400mA(Max)		
DCDC	2		
Shipping Mode	2uA		
Standby	20uA		
GPIO	55		
WIO	5(at most)		
SARADC channel	6		
LCD	QSPI/QSPI DDR/SPI/2lane SPI/CPU8080/TR-J/TR-S parallel HSYNC/VSYNC		
SDC	2		
Audio DAC	Differential/Single-End		
I2S/PCM	1*I2S IN+2*I2S OUT		
Microphones	2 channel Single-End/2 channel Differential/2DMIC		
USB 2.0	FS		
UART	3		
QSPI	3		
SPI(Include QSPI)	4		
SPIMT	2		
I2C	4		
I2CMT	2		
PWM	4		
LOSC	√		

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1.3Application Diagram

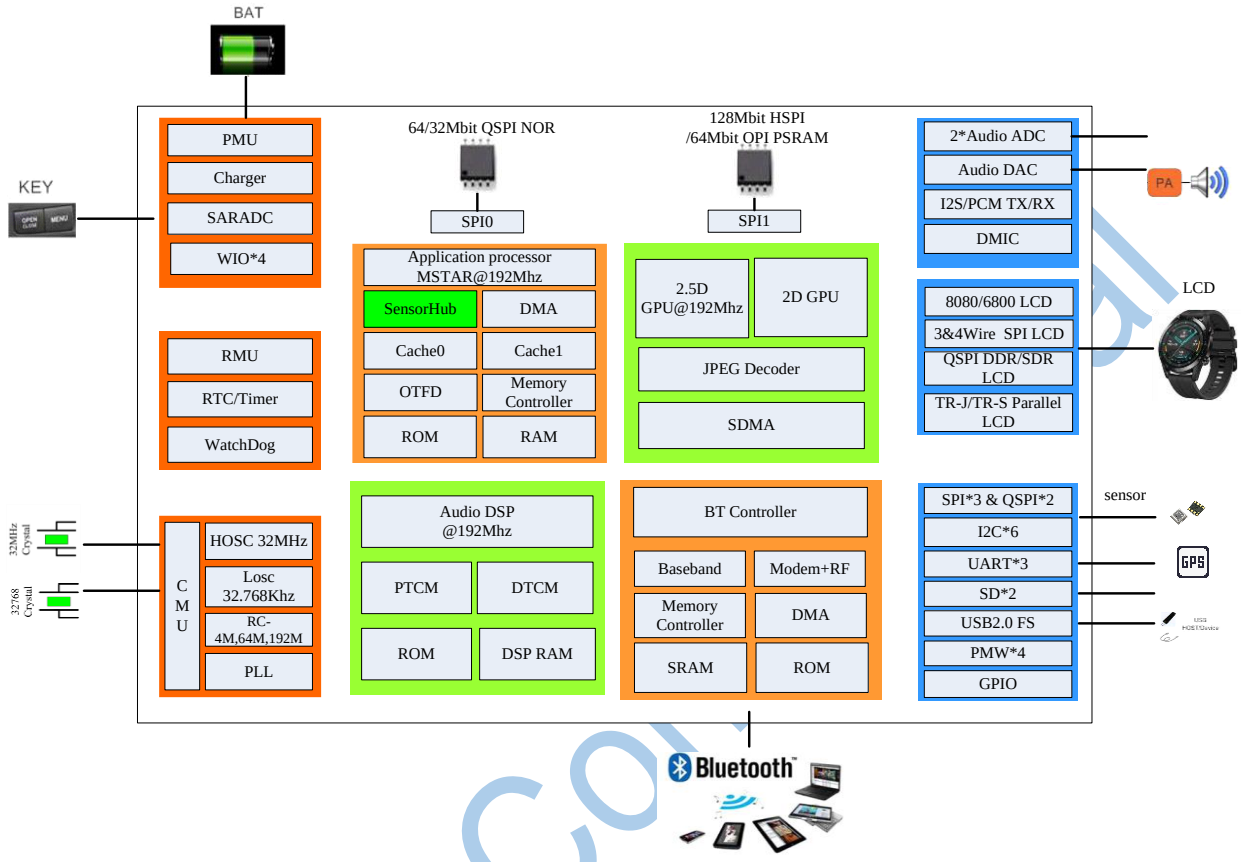


Figure 1-1 ATS3089x Application Diagram

1.4Pin Assignment and Descriptions

1.4.1 Pin Assignment

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	TRX	GND	HOSCI	HOSCO	IO35	IO32	IOVCC2	IO16	IO14	IO27	IO25	IO76/DM	IO75/DP
B	GND	GND	GND	IO77	IO34	IO30	IO17	IO15	IO33	IO26	IO24	IO28	IO29
C	VD12_RF	AGND_TXPA	AGND_TXPA	IO79	IO78	CMU_TST				IO23	IO22	IO74	IOVCC1
D	IO62	IO63	IO80								IO31	IO64	IO13
E	IO60	IO61			GND	GND	GND	GND	GND			IO12	IO11
F	IOVCC4	IO59		GND	GND	GND		GND	GND	GND		IO10	IO9
G	IO57	IO58		GND	GND				GND	GND		IO8	IOVCC5
H	IO55	IO56		GND	GND	GND		GND	GND	GND		NC	VCC2
J	IO53	IO54			AGND_AU	GND	VPSRAM	GND	GND			NC	VDD
K	IO51	IO52	IO42								NC	LOSCI	LOSCO
L	VREFO_ADC	IO49	IO50	IO47					NC	NC	NC	WIO4	WIO0
M	INPUT1N	INPUT0N	VMIC1	AOUTN	IO20	IO18	IO5	IO4	DC5V	VCC	BAT	WIO1	VD12
N	INPUT1P	INPUT0P	VMIC0	AOUTP	IO21	IO19	VC18	LXVDD18	DC5V	VCC	BAT	SVCC	LXVDD12

1.4.2 Pin Description

Table 1-1 ATS3089x Pin Description

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Pin No.	Pin Name	Pin Type	Pad drive level	GPIO Initial state	MFP Description (5bits)		Power domain
					AD_SELECT=0	AD_SELECT=1	
A1	TRX	RF			Bluetooth antenna IO		
A2	GND				The GND of the chip		
A3	HOSCI	AI			Input of HOSC(32MHz)		
A4	HOSCO	AO			Output of HOSC(32MHz)		
A5	GPIO35	I/O	Level 1/2/3/4	Z	Bit35 of General purpose I/O port GPIO SPI2_MISO I2C1_DAT PPI_TRIGGER4 I2C3_CLK DMIC01_DAT LCD_TE BT_UART_RX SPIMT0_SS1 I2CMT0_CLK I2CMT1_DAT LRADC4		IOVCC2
A6	GPIO32	I/O	Level 1/2/3/4	Z	Bit32 of General purpose I/O port GPIO SPI2_MISO I2C0_CLK PPI_TRIGGER1 I2C2_CLK LCD_RS_CD IISTX1_DOUT SPIMT0_MISO I2CMT0_CLK HOSCOUT LOSCOUT		IOVCC2
A7	IOVCC2	PWR			Power supply for GPIO Group2(1.8/3.1V)		IOVCC2
A8	GPIO16	I/O	Level 1/2/3/4	Z	Bit16 of General purpose I/O port GPIO SPI2_SS UART1_TX I2C1_DAT PPI_TRIGGER2 IISG0_MCLK IISG1_MCLK PWM2 LCD_D2 BT_REQ SPIMT1_SS1 I2CMT1_DAT		IOVCC2

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A9	GPIO14	I/O	Level 1/2/3/4	Z	Bit14 of General purpose I/O port GPIO UART0_TX I2C0_CLK PPI_TRIGGER0 IISTX0_DOUT PWM0 LCD_D0 BT_UART_TX SPIMT0_SS3 SPIMT1_MISO I2CMT0_CLK	IOVCC2
A10	GPIO27	I/O	Level 1/2/3/4	X	Bit27 of General purpose I/O port GPIO SPI3_IO3 UART2_RX I2C1_DAT SD0_DAT5 PPI_TRIGGER1 IISTX0_DOUT I2C3_CLK LCD_D7 SD1_CLK SPIMT1_MOSI I2CMT1_DAT LRADC6 VCC_OUT(3.3v)	VCC
A11	GPIO25	I/O	Level 1/2/3/4	Z	Bit25 of General purpose I/O port GPIO SPI3_IO2 UART1_RX I2C0_DAT SD0_CMD PPI_TRIGGER11 IISG0_BCLK IISG1_BCLK I2C2_CLK I2C3_DAT LCD_D5 SPIMT1_CLK I2CMT0_DAT LRADC5 HOSCOUT LOSCOUT WCI_TX	VCC
A12	GPIO76	I/O	Level 1/2/3/4	Firmware upgrade (L), Firmware boot(L)	Bit76 of General purpose I/O port GPIO SPI2_MOSI SPI3_MOSI UART0_RX UART2_TX IISG0_LRCLK IISG1_LRCLK I2C2_DAT	VCC

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					USBDM DMIC01_DAT PWM3 SD1_CLK BT_UART_RX SPIMT1_SS2 I2CMT0_DAT I2CMT1_DAT WCI_RX	
A13	GPIO75	I/O	Level 1/2/3/4	Firmware upgrade (H), Firmware boot(L)	Bit75 of General purpose I/O port GPIO SPI2_MISO SPI3_MISO UART0_TX UART2_RX IISG0_BCLK IISG1_BCLK I2C2_CLK USBDP DMIC01_CLK PWM2 SD1_DAT3 BT_UART_TX SPIMT1_SS3 I2CMT0_CLK I2CMT1_CLK WCI_TX	VCC
B1	GND				The GND of the chip	
B2	GND				The GND of the chip	
B3	GND				The GND of the chip	
B4	GPIO77	I/O	Level 1/2/3/4	Z	Bit77 of General purpose I/O port GPIO SPI2_MOSI SD0_CLK0 PPI_TRIGGER4 IISTX0_DOUT IISRX0_DIN I2C3_DAT DMIC01_DAT LCD_CS1 SD1_CLK IISTX1_DOUT SPIMT0_SS0 I2CMT0_DAT LRADC6 HOSCOUT LOSCOUT	IOVCC2
B5	GPIO34	I/O	Level 1/2/3/4	Z	Bit34 of General purpose I/O port GPIO SPI2_CLK I2C1_CLK	IOVCC2

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					PPI_TRIGGER3 DMIC01_CLK LCD_WR# BT_UART_TX SPIMT0_CLK I2CMT1_CLK	
B6	GPIO30	I/O	Level 1/2/3/4	Z	Bit30 of General purpose I/O port GPIO SPI2_SS PPI_TRIGGER0 IISG0_LRCLK IISG1_LRCLK I2C2_DAT I2C3_DAT DMIC01_CLK LCD_CS0 SPIMT1_CLK	IOVCC2
B7	GPIO17	I/O	Level 1/2/3/4	Z	Bit17 of General purpose I/O port GPIO SPI2_CLK UART1_RX I2C1_CLK PPI_TRIGGER3 IISG0_BCLK IISG1_BCLK I2C3_CLK PWM3 LCD_D3 BT_ACCESS SPIMT1_SS0 I2CMT1_CLK	IOVCC2
B8	GPIO15	I/O	Level 1/2/3/4	Z	Bit15 of General purpose I/O port GPIO UART0_RX I2C0_DAT PPI_TRIGGER1 IISRX0_DIN PWM1 LCD_D1 BT_UART_RX SPIMT0_SS2 SPIMT1_MOSI I2CMT0_DAT	IOVCC2
B9	GPIO33	I/O	Level 1/2/3/4	Z	Bit33 of General purpose I/O port GPIO SPI2_MOSI I2C0_DAT PPI_TRIGGER2 LCD_RD# SPIMT0_MOSI I2CMT0_DAT IOVCC2_OUT(1.8/3.3v)	IOVCC2
B10	GPIO26	I/O	Level 1/2/3/4	Z	Bit26 of General purpose I/O port GPIO	VCC

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					SPI3_SS UART2_TX I2C1_CLK SD0_DAT4 PPI_TRIGGER0 IISG0_LRCLK IISG1_LRCLK I2C2_DAT LCD_D6 SD1_CMD SPIMT1_MISO I2CMT1_CLK VCC_OUT(3.3v)	
B11	GPIO24	I/O	Level 1/2/3/4	Z	Bit24 of General purpose I/O port GPIO SPI3_CLK UART1_TX I2C0_CLK I2C1_DAT SD0_CLK0 PPI_TRIGGER10 IISG0_MCLK IISG1_MCLK I2C3_CLK LCD_D4 SPIMT0_SS1 SPIMT1_SS0 I2CMT0_CLK WCI_RX	VCC
B12	GPIO28	I/O	Level 1/2/3/4	X	Bit28 of General purpose I/O port GPIO SPI3_IO2 UART0_TX I2C0_CLK SD0_DAT6 PPI_TRIGGER2 IISTX0_DOUT I2C2_CLK SD1_DAT1 BT_UART_TX SPIMT1_SS0 I2CMT0_CLK	VCC
B13	GPIO29	I/O	Level 1/2/3/4	X	Bit29 of General purpose I/O port GPIO GPIO29 SPI2_SS SPI3_IO3 UART0_RX I2C0_DAT SD0_DAT7 PPI_TRIGGER3 IISRX0_DIN I2C2_DAT SD1_DAT0	VCC

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					IISTX1_DOUT BT_UART_RX SPIMT1_SS1 I2CMT0_DAT	
C1	VD12_RF	PWR			Output of regulator RF PA	VD12_RF
C2	AGND_TXPA				The GND of the RF TX PA	
C3	AGND_TXPA				The GND of the RF TX PA	
C4	GPIO79	I/O	Level 1/2/3/4	Z	Bit79 of General purpose I/O port GPIO SPI2_CLK SPI3_CLK UART1_RX UART2_RX I2C0_CLK I2C1_CLK SD0_DAT5 PPI_TRIGGER9 IISG0_LRCLK IISG1_LRCLK I2C3_CLK DMIC01_CLK PWM3 SD1_CMD SPIMT0_MOSI I2CMT0_DAT I2CMT1_CLK WCI_RX	IOVCC4
C5	GPIO78	I/O	Level 1/2/3/4	Z	Bit78 of General purpose I/O port GPIO SPI2_SS SPI3_SS UART1_TX I2C0_DAT SD0_DAT4 PPI_TRIGGER8 IISG0_BCLK IISG1_BCLK I2C2_CLK I2C3_CLK DMIC01_DAT PWM2 SD1_CLK SPIMT0_MISO I2CMT0_CLK I2CMT1_DAT WCI_TX	IOVCC4
C6	CMU_TST	I/O		Z	Test pad for CMU	VCC
C10	GPIO23	I/O	Level 1/2/3/4	Z	Bit23 of General purpose I/O port GPIO SPI3_MOSI	VCC

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					UART2_RTS I2C1_CLK SD0_DAT7 PPI_TRIGGER9 IISTX0_DOUT IISRX0_DIN I2C2_DAT SD1_DAT1 SPIMT0_SS2 SPIMT1_MOSI I2CMT0_DAT LRADC2 WCI_TX	
C11	GPIO22	I/O	Level 1/2/3/4	Z	Bit22 of General purpose I/O port GPIO SPI3_MISO UART2_CTS I2C0_DAT SD0_DAT6 PPI_TRIGGER8 I2C2_CLK PWM1 SD1_DAT0 IISTX1_DOUT SPIMT0_SS3 SPIMT1_MISO I2CMT0_CLK HOSCOUT LOSCOUT	VCC
C12	GPIO74	I/O	Level 1/2/3/4	Z	Bit74 of General purpose I/O port GPIO SPI2_MOSI SPI3_MOSI UART0_RX UART2_RX I2C0_CLK I2C1_DAT SD0_DAT6 PPI_TRIGGER4 IISG0_BCLK IISG1_BCLK I2C2_DAT I2C3_CLK DMIC01_CLK PWM1 SD1_DAT0 SPIMT0_CLK I2CMT1_CLK LRADC6 HOSCOUT LOSCOUT WCI_RX	IOVCC1
C13	IOVCC1	PWR			Power supply for GPIO Group1(1.8/3.1V)	IOVCC1

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D1	GPIO62	I/O	Level 1/2/3/4	Z	Bit62 of General purpose I/O port GPIO SPI3_IO3 UART0_RX UART1_RX UART2_RX SD0_DAT1 PPI_TRIGGER5 IISTX0_DOUT IISRX0_DIN I2C2_DAT SPIMT1_SS1 LRADC4	IOVCC4
D2	GPIO63	I/O	Level 1/2/3/4	Z	Bit63 of General purpose I/O port GPIO SPI2_MISO SPI3_MISO UART0_TX UART1_TX UART2_TX SD0_CMD PPI_TRIGGER6 IISG0_MCLK IISG1_MCLK I2C3_DAT IISTX1_DOUT SPIMT1_SS2 LRADC3 HOSCOUT LOSCOUT IOVCC4_OUT(1.8v/3.3v)	IOVCC4
D3	GPIO80	I/O	Level 1/2/3/4	Z	Bit80 of General purpose I/O port GPIO SPI2_MOSI SPI3_MOSI UART1_RX UART2_TX I2C1_DAT SD0_DAT7 PPI_TRIGGER8 IISTX0_DOUT IISRX0_DIN I2C2_DAT I2C3_DAT DMIC01_DAT SD1_DAT0 IISTX1_DOUT SPIMT1_SS0 HOSCOUT LOSCOUT	IOVCC4
D11	GPIO31	I/O	Level 1/2/3/4	X	Bit31 of General purpose I/O port GPIO SPI2_CLK SPI3_CLK	VCC

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					PPI_TRIGGER4 IISG0_MCLK IISG1_MCLK I2C3_DAT DMIC01_DAT SD1_DAT2 SPIMT1_CLK HOSCOUT LOSCOUT	
D12	GPIO64	I/O	Level 1/2/3/4	H	Bit64 of General purpose I/O port GPIO SPI2_MISO SPI3_MISO UART0_TX UART1_TX UART2_TX SD0_DAT7 PPI_TRIGGER11 IISG0_MCLK IISG1_MCLK I2C2_CLK I2C3_DAT PWM0 SD1_DAT1 IISTX1_DOUT SPIMT0_SS0 I2CMT1_DAT HOSCOUT LOSCOUT IOVCC1_OUT(1.8v/3.3v)	IOVCC1
D13	GPIO13	I/O	Level 1/2/3/4	Firmware upgrade (X), Firmware boot(Z)	Bit13 of General purpose I/O port GPIO SPI3_MOSI UART0_RTS UART1_RX I2C1_DAT SD0_DAT1 PPI_TRIGGER7 IISRX0_DIN I2C3_CLK DMIC01_DAT SD1_DAT0 SPI0_MOSI SPIMT0_SS1 WCI_TX	IOVCC1
E1	GPIO60	I/O	Level 1/2/3/4	Z	Bit60 of General purpose I/O port GPIO SPI2_MOSI I2C1_DAT SD0_CLK0 PPI_TRIGGER11 IISG0_BCLK IISG1_BCLK I2C3_CLK	IOVCC4

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					I2CMT1_DAT HOSCOUT LOSCOUT	
E2	GPIO61	I/O	Level 1/2/3/4	Z	Bit61 of General purpose I/O port GPIO SPI3_IO2 SD0_DAT0 PPI_TRIGGER4 IISG0_LRCLK IISG1_LRCLK I2C2_CLK SPIMT1_SS3 LRADC5 HOSCOUT LOSCOUT	IOVCC4
E5	GND				The GND of the chip	
E6	GND				The GND of the chip	
E7	GND				The GND of the chip	
E8	GND				The GND of the chip	
E9	GND				The GND of the chip	
E12	GPIO12	I/O	Level 1/2/3/4	Z	Bit12 of General purpose I/O port GPIO SPI3_MISO UART0_CTS UART1_TX I2C1_CLK SD0_CLK0 PPI_TRIGGER6 I2C2_CLK DMIC01_CLK PWM1 SD1_DAT3 IISTX1_DOUT SPI0_MISO SPIMT0_SS2 SPIMT1_SS3	IOVCC1
E13	GPIO11	I/O	Level 1/2/3/4	Firmware upgrade (X), Firmware boot(Z)	Bit11 of General purpose I/O port GPIO SPI3_CLK UART0_RX UART2_RTS I2C0_DAT SD0_DAT0 PPI_TRIGGER5 I2C2_DAT PWM3 SD1_CLK SPI0_CLK SPIMT0_SS3 SPIMT1_SS2	IOVCC1
F1	IOVCC4	PWR			Power supply for GPIO Group4(1.8/3.1V)	IOVCC1

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F2	GPIO59	I/O	Level 1/2/3/4	Z	Bit59 of General purpose I/O port GPIO SPI2_MISO I2C1_CLK SD0_CLK1 PPI_TRIGGER10 IISG0_MCLK IISG1_MCLK I2C3_DAT SPIMT0_SS1 I2CMT1_CLK WCI_RX	IOVCC4
F4	GND				The GND of the chip	
F5	GND				The GND of the chip	
F6	GND				The GND of the chip	
F8	GND				The GND of the chip	
F9	GND				The GND of the chip	
F10	GND				The GND of the chip	
F12	GPIO10	I/O	Level 1/2/3/4	Firmware upgrade (X), Firmware boot(Z)	Bit10 of General purpose I/O port GPIO SPI3_SS UART0_TX UART2_CTS I2C0_CLK SD0_CMD PWM2 SD1_DAT2 IISTX1_DOUT SPI0_SS SPIMT1_SS1	IOVCC1
F13	GPIO9	I/O	Level 1/2/3/4	Z	Bit9 of General purpose I/O port GPIO SPI2_MOSI SPI3_IO3 UART2_RX SD0_DAT3 PPI_TRIGGER3 IISG0_BCLK IISG1_BCLK I2C3_DAT PWM1 SD1_CLK SPIMT1_MOSI LRADC3	IOVCC1
G1	GPIO57	I/O	Level 1/2/3/4	Z	Bit57 of General purpose I/O port GPIO SPI2_SS I2C0_CLK SD0_DAT2 PPI_TRIGGER8 I2C2_DAT	IOVCC4

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					PWM1 IISTX1_DOUT SPIMT0_SS2 I2CMT0_CLK HOSCOUT LOSCOUT	
G2	GPIO58	I/O	Level 1/2/3/4	Z	Bit58 of General purpose I/O port GPIO SPI2_CLK I2C0_DAT SD0_DAT3 PPI_TRIGGER9 IISTX0_DOUT I2C3_CLK SPIMT0_SS3 I2CMT0_DAT WCI_TX	IOVCC4
G4	GND				The GND of the chip	
G5	GND				The GND of the chip	
G9	GND				The GND of the chip	
G10	GND				The GND of the chip	
G12	GPIO8	I/O	Level 1/2/3/4	Z	Bit8 of General purpose I/O port GPIO SPI2_MISO SPI3_IO2 UART2_TX SD0_DAT2 PPI_TRIGGER2 IISG0_MCLK IISG1_MCLK I2C2_DAT I2C3_CLK PWM0 SD1_CMD SPIMT1_MISO LRADC2	IOVCC1
G13	IOVCC5	PWR			Power supply for GPIO Group5(1.8V)	IOVCC5
H1	GPIO55	I/O	Level 1/2/3/4	Z	Bit55 of General purpose I/O port GPIO SPI2_MISO SPI3_MISO UART2_CTS I2C1_DAT SD0_DAT6 PPI_TRIGGER6 IISG1_LRCLK I2C3_CLK PWM2 SD1_CLK SPIMT1_MISO I2CMT1_CLK	IOVCC4

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H2	GPIO56	I/O	Level 1/2/3/4	Z	Bit56 of General purpose I/O port GPIO SPI2_MOSI SPI3_MOSI UART2_RTS SD0_DAT7 PPI_TRIGGER7 IISTX0_DOUT IISRX0_DIN I2C2_CLK I2C3_DAT PWM3 SPIMT1_MOSI I2CMT1_DAT	IOVCC4
H4	GND				The GND of the chip	
H5	GND				The GND of the chip	
H6	GND				The GND of the chip	
H8	GND				The GND of the chip	
H9	GND				The GND of the chip	
H10	GND				The GND of the chip	
H12	NC					
H13	VCC2	PWR			Output of regulator VCC2	VCC2
J1	GPIO53	I/O	Level 1/2/3/4	Z	Bit53 of General purpose I/O port GPIO SPI2_SS SPI3_SS UART2_TX I2C0_DAT SD0_DAT4 PPI_TRIGGER4 IISG1_MCLK I2C2_CLK PWM0 SD1_DAT2 SPIMT1_SS0 I2CMT0_CLK HOSCOUT LOSCOUT	IOVCC4
J2	GPIO54	I/O	Level 1/2/3/4	Z	Bit54 of General purpose I/O port GPIO SPI2_CLK SPI3_CLK UART2_RX I2C1_CLK SD0_DAT5 PPI_TRIGGER5 IISG1_BCLK I2C2_DAT PWM1	IOVCC4

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					SD1_DAT3 SPIMT1_CLK I2CMT0_DAT		
J5	AGND_AU				The GND of the Audio codec.		
J6	GND				The GND of the chip		
J7	VPSRAM				Power supply for PSRAM device. (1.8v)		
J8	GND				The GND of the chip		
J9	GND				The GND of the chip		
J12	NC						
J13	VDD	PWR			Output of regulator VDD(0.6~1.3V)		VDD
K1	GPIO51	I/O	Level 1/2/3/4	Z	Bit51 of General purpose I/O port GPIO SPI2_MISO UART1_CTS I2C1_CLK PPI_TRIGGER2 I2C3_CLK PWM2 SD1_DAT1 PTA_GRANT SPIMT0_MISO I2CMT1_CLK		IOVCC4
K2	GPIO52	I/O	Level 1/2/3/4	Z	Bit52 of General purpose I/O port GPIO SPI2_MOSI UART1_RTS I2C0_CLK I2C1_DAT PPI_TRIGGER3 IISTX0_DOUT I2C3_DAT PWM3 SD1_DAT0 IISTX1_DOUT SPIMT0_MOSI I2CMT1_DAT		IOVCC4
K3	GPIO42	I/O	Level 1/2/3/4	Z	Bit42 of General purpose I/O port GPIO SPI2_CLK UART2_RX I2C1_DAT PPI_TRIGGER6 IISG1_LRCLK PWM2 SD1_DAT3 SPIMT1_CLK		IOVCC3 (1.8v)
K11	NC						
K12	LOSCI	AI	Level 1/2/3/4	Z	Bit2 of wake up I/O I2C0_CLK	Input of LOSC(32768Hz)	SVCC
K13	LOSCO	AO	Level 1/2/3/4	Z	Bit3 of wake up I/O I2C0_DAT	Output of LOSC(32768Hz)	SVCC

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					LOSCOUT		
L1	VREF0_ADC	AO			Voltage reference of audio ADC		
L2	GPIO49	I/O	Level 1/2/3/4	Z	Bit49 of General purpose I/O port GPIO SPI2_SS UART1_TX I2C0_CLK PPI_TRIGGER0 I2C2_CLK PWM0 SD1_CMD BT_REQ SPIMT0_SS0 I2CMT0_CLK		IOVCC4
L3	GPIO50	I/O	Level 1/2/3/4	Z	Bit50 of General purpose I/O port GPIO SPI2_CLK UART1_RX I2C0_DAT PPI_TRIGGER1 I2C2_DAT PWM1 SD1_CLK BT_ACCESS SPIMT0_CLK I2CMT0_DAT		IOVCC4
L4	GPIO47	I/O	Level 1/2/3/4	Z	Bit47 of General purpose I/O port GPIO I2C3_CLK IISTX1_DOUT HOSCOUT LOSCOUT		IOVCC3 (1.8v)
L9	NC						
L10	NC						
L11	NC						
L12	WIO4	I/O	Level 1/2/3/4	Z	Bit4 of wake up I/O I2C1_CLK LOSCOUT		SVCC
L13	WIO0	I/O	Level 1/2/3/4	Z	Bit0 of wake up I/O ONOFF I2C1_CLK LRADC1		SVCC
M1	INPUT1N	I/O	Level 1/2/3/4	L	Bit93 of GPIO	INPUT1N/INPUT0	AVCC
M2	INPUT0N	I/O	Level 1/2/3/4	L	Bit91 of GPIO	INPUT0N/INPUT1	AVCC
M3	VMIC1	PWR			Power supply of MIC1		
M4	AOUTN	I/O	Level 1/2/3/4	L	Bit95 of GPIO	Audio DAC differential output(AOUTN);	AVCC

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M5	GPIO20	I/O	Level 1/2/3/4	Z	Bit20 of General purpose I/O port GPIO SPI2_SS SPI3_SS UART1_TX UART2_TX I2C1_CLK SD0_DAT2 PPI_TRIGGER6 IISG0_BCLK IISG1_BCLK PWM3 LCD_D6 SPIMT0_SS0 SPIMT1_SS0 I2CMT1_CLK LRADC2	VCC
M6	GPIO18	I/O	Level 1/2/3/4	Z	Bit18 of General purpose I/O port GPIO SPI2_MISO SPI3_MISO UART1_CTS I2C0_CLK SD0_DAT0 PPI_TRIGGER4 IISTX0_DOUT IISRX0_DIN PWM1 LCD_D4 SD1_DAT2 IISTX1_DOUT PTA_GRANT SPIMT0_MISO SPIMT1_MISO I2CMT0_CLK I2CMT1_CLK	VCC
M7	GPIO5	I/O	Level 1/2/3/4	Z	Bit5 of General purpose I/O port GPIO SPI2_CLK UART1_RX I2C0_DAT SD0_DAT5 PPI_TRIGGER1 IISTX0_DOUT IISRX0_DIN I2C2_CLK PWM1 SD1_DAT3 SPIMT1_CLK LRADC5	IOVCC1
M8	GPIO4	I/O	Level 1/2/3/4	Z	Bit4 of General purpose I/O port GPIO SPI2_SS UART1_TX	IOVCC1

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					I2C0_CLK SD0_DAT4 PPI_TRIGGER0 IISG0_LRCLK IISG1_LRCLK I2C2_DAT PWM0 SD1_DAT2 SPIMT1_SS0 LRADC4		
M9	DC5V	PWR			5.0V Voltage input		DC5V
M10	VCC	PWR			Output of regulator VCC		VCC
M11	VBAT	PWR			Battery Voltage input		VBAT
M12	WIO1	I/O	Level 1/2/3/4	Z	Bit1 of wake up I/O LRADC1 I2C1_DAT LOSCOUT		SVCC
M13	VD12	PWR			Output of buck VD12(0.6~1.4V)		VD12
N1	INPUT1P	I/O	Level 1/2/3/4	L	Bit92 of GPIO	INPUT1P/INPUT1	AVCC
N2	INPUT0P	I/O	Level 1/2/3/4	L	Bit90 of GPIO	INPUT0P/INPUT0	AVCC
N3	VMIC0	PWR			Power supply of MIC0		
N4	AOUTP	I/O	Level 1/2/3/4	L	Bit94 of GPIO	Audio DAC differential output(AOUTP); Audio DAC single end output(AOUT)	AVCC
N5	GPIO21	I/O	Level 1/2/3/4	Z	Bit21 of General purpose I/O port GPIO SPI2_CLK SPI3_CLK UART1_RX UART2_RX I2C0_CLK I2C1_DAT SD0_DAT3 PPI_TRIGGER7 IISG0_LRCLK IISG1_LRCLK PWM0 LCD_D7 SPIMT0_CLK SPIMT1_CLK I2CMT1_DAT LRADC3		VCC

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N6	GPIO19	I/O	Level 1/2/3/4	Z	Bit19 of General purpose I/O port GPIO SPI2_MOSI SPI3_MOSI UART1_RTS I2C0_DAT SD0_DAT1 PPI_TRIGGER5 IISG0_MCLK IISG1_MCLK PWM2 LCD_D5 SD1_DAT3 SPIMT0_MOSI SPIMT1_MOSI I2CMT0_DAT I2CMT1_DAT HOSCOUT LOSCOUT	VCC
N7	VC18	PWR			Output of buck VC18(1.8V)	VC18
N8	LXVC18	PWR			LX of buck VC18	VC18
N9	DC5V	PWR			5.0V Voltage input	DC5V
N10	VCC	PWR			Output of regulator VCC	VCC
N11	BAT	PWR			Battery Voltage input	VBAT
N12	SVCC				Standby power supply regulator	SVCC
N13	LXVD12	PWR			LX of buck VD12	VD12

Note: H: high level; L:low level; Z: high resistance; X: May be change in power on;
PWR: power supply PIN; I/O: GPIO; AI: analog input PIN; AO: analog output PIN;
GND: Ground pin;

1.4.3 Multi-Function of GPIO Pins

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Ball name	Ball num	Voltage	Power Domain	Analog Function	Pin Multi-Function																														
					0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
GPIO4	M8	3.3/1.8	IOVCC1		GPIO4			SPI2_SS			UART1_TX		I2C0_CLK		SD0_DAT4	PPI_Trigger0	IISG1_LRCLK	IISG1_LRCLK	I2C2_DAT				PWM0		SD1_DAT2					SPIMT1_SS0			LRADC4			
GPIO5	M7	3.3/1.8	IOVCC1		GPIO5			SPI2_CLK			UART1_RX		I2C0_DAT		SD0_DAT5	PPI_Trigger1	IISTX0_DOUT	IISRX0_DIN	I2C2_CLK				PWM1		SD1_DAT3					SPIMT1_CLK			LRADC5			
GPIO8	G12	3.3/1.8	IOVCC1		GPIO8			SPI2_MISO	SPI3_IO2			UART2_TX			SD0_DAT2	PPI_Trigger2	IISG0_MCLK	IISG1_MCLK	I2C2_DAT	I2C3_CLK			PWM0		SD1_CMD					SPIMT1_MISO			LRADC2			
GPIO9	F13	3.3/1.8	IOVCC1		GPIO9			SPI2_MOSI	SPI3_IO3			UART2_RX			SD0_DAT3	PPI_Trigger3	IISG0_BCLK	IISG1_BCLK		I2C3_DAT			PWM1		SD1_CLK					SPIMT1_MOSI			LRADC3			
GPIO10	F12	3.3/1.8	IOVCC1		GPIO10				SPI3_SS	UART0_TX		UART2_CTS	I2C0_CLK		SD0_CMD								PWM2		SD1_DAT2	IISTX1_DOUT	SPI0_SS			SPIMT1_SS1						
GPIO11	E13	3.3/1.8	IOVCC1		GPIO11				SPI3_CLK	UART0_RX		UART2_RTS	I2C0_DAT		SD0_DAT0	PPI_Trigger5			I2C2_DAT				PWM3		SD1_CLK		SPI0_CLK		SPIMT0_SS3	SPIMT1_SS2						
GPIO12	E12	3.3/1.8	IOVCC1		GPIO12				SPI3_MISO	UART0_CTS	UART1_TX			I2C1_CLK	SD0_CLK0	PPI_Trigger6			I2C2_CLK			DMIC01_CLK	PWM1		SD1_DAT3	IISTX1_DOUT	SPI0_MISO		SPIMT0_SS2	SPIMT1_SS3						
GPIO13	D13	3.3/1.8	IOVCC1		GPIO13				SPI3_MOSI	UART0_RTS	UART1_RX			I2C1_DAT	SD0_DAT1	PPI_Trigger7		IISRX0_DIN		I2C3_CLK		DMIC01_DAT			SD1_DAT0		SPI0_MOSI		SPIMT0_SS1							WCI_TX
GPIO14	A9	3.3/1.8	IOVCC2		GPIO14					UART0_TX			I2C0_CLK			PPI_Trigger0	IISTX0_DOUT						PWM0	LCD_D0				BT_UART_TX	SPIMT0_SS3	SPIMT1_MISO	I2CMT0_CLK				
GPIO15	B8	3.3/1.8	IOVCC2		GPIO15					UART0_RX			I2C0_DAT			PPI_Trigger1		IISRX0_DIN					PWM1	LCD_D1				BT_UART_RX	SPIMT0_SS2	SPIMT1_MOSI	I2CMT0_DAT				
GPIO16	A8	3.3/1.8	IOVCC2		GPIO16			SPI2_SS			UART1_TX			I2C1_DAT		PPI_Trigger2	IISG0_MCLK	IISG1_MCLK					PWM2	LCD_D2				BT_REQ		SPIMT1_SS1		I2CMT1_DAT			
GPIO17	B7	3.3/1.8	IOVCC2		GPIO17			SPI2_CLK			UART1_RX			I2C1_CLK		PPI_Trigger3	IISG0_BCLK	IISG1_BCLK		I2C3_CLK			PWM3	LCD_D3				BT_ACCESS		SPIMT1_SS0		I2CMT1_CLK			
GPIO18	M6	3.3	VCC		GPIO18			SPI2_MISO	SPI3_MISO		UART1_CTS		I2C0_CLK		SD0_DAT0	PPI_Trigger4	IISTX0_DOUT	IISRX0_DIN					PWM1	LCD_D4	SD1_DAT2	IISTX1_DOUT		PTA_GRANT	SPIMT0_MISO	SPIMT1_MISO	I2CMT0_CLK	I2CMT1_CLK				
GPIO19	N6	3.3	VCC		GPIO19			SPI2_MOSI	SPI3_MOSI		UART1_RTS		I2C0_DAT		SD0_DAT1	PPI_Trigger5	IISG0_MCLK	IISG1_MCLK					PWM2	LCD_D5	SD1_DAT3				SPIMT0_MOSI	SPIMT1_MOSI	I2CMT0_DAT	I2CMT1_DAT		HOSCOUT	LOSCOUT	
GPIO20	M5	3.3	VCC		GPIO20			SPI2_SS	SPI3_SS		UART1_TX	UART2_TX		I2C1_CLK	SD0_DAT2	PPI_Trigger6	IISG0_BCLK	IISG1_BCLK					PWM3	LCD_D6					SPIMT0_SS0	SPIMT1_SS0		I2CMT1_CLK	LRADC2			
GPIO21	N5	3.3	VCC		GPIO21			SPI2_CLK	SPI3_CLK		UART1_RX	UART2_RX	I2C0_CLK	I2C1_DAT	SD0_DAT3	PPI_Trigger7	IISG0_LRCLK	IISG1_LRCLK					PWM0	LCD_D7					SPIMT0_CLK	SPIMT1_CLK		I2CMT1_DAT	LRADC3			
GPIO22	C11	3.3	VCC		GPIO22				SPI3_MISO			UART2_CTS	I2C0_DAT		SD0_DAT6	PPI_Trigger8			I2C2_CLK				PWM1		SD1_DAT0	IISTX1_DOUT			SPIMT0_SS3	SPIMT1_MISO	I2CMT0_CLK			HOSCOUT	LOSCOUT	
GPIO23	C10	3.3	VCC		GPIO23				SPI3_MOSI			UART2_RTS		I2C1_CLK	SD0_DAT7	PPI_Trigger9	IISTX0_DOUT	IISRX0_DIN	I2C2_DAT						SD1_DAT1					SPIMT0_SS2	SPIMT1_MOSI	I2CMT0_DAT		LRADC2			WCI_TX
GPIO24	B11	3.3	VCC		GPIO24				SPI3_CLK		UART1_TX		I2C0_CLK	I2C1_DAT	SD0_CLK0	PPI_Trigger10	IISG0_MCLK	IISG1_MCLK		I2C3_CLK				LCD_D4					SPIMT0_SS1	SPIMT1_SS0	I2CMT0_CLK					WCI_RX
GPIO25	A11	3.3	VCC		GPIO25				SPI3_IO2		UART1_RX		I2C0_DAT		SD0_CMD	PPI_Trigger11	IISG0_BCLK	IISG1_BCLK	I2C2_CLK	I2C3_DAT				LCD_D5						SPIMT1_CLK	I2CMT0_DAT		LRADC5	HOSCOUT	LOSCOUT	WCI_TX
GPIO26	B10	3.3	VCC		GPIO26				SPI3_SS			UART2_TX		I2C1_CLK	SD0_DAT4	PPI_Trigger0	IISG0_LRCLK	IISG1_LRCLK	I2C2_DAT					LCD_D6	SD1_CMD						SPIMT1_MISO		I2CMT1_CLK				VCC_OUT
GPIO27	A10	3.3	VCC		GPIO27				SPI3_IO3			UART2_RX		I2C1_DAT	SD0_DAT5	PPI_Trigger1	IISTX0_DOUT			I2C3_CLK				LCD_D7	SD1_CLK						SPIMT1_MOSI		I2CMT1_DAT	LRADC6			VCC_OUT
GPIO28	B12	3.3	VCC		GPIO28				SPI3_IO2	UART0_TX			I2C0_CLK		SD0_DAT6	PPI_Trigger2	IISTX0_DOUT		I2C2_CLK						SD1_DAT1			BT_UART_TX		SPIMT1_SS0	I2CMT0_CLK					
GPIO29	B13	3.3	VCC		GPIO29			SPI2_SS	SPI3_IO3	UART0_RX			I2C0_DAT		SD0_DAT7	PPI_Trigger3		IISRX0_DIN	I2C2_DAT						SD1_DAT0	IISTX1_DOUT		BT_UART_RX		SPIMT1_SS1	I2CMT0_DAT					
GPIO30	B6	3.3/1.8	IOVCC2		GPIO30			SPI2_SS								PPI_Trigger0	IISG0_LRCLK	IISG1_LRCLK	I2C2_DAT	I2C3_DAT		DMIC01_CLK		LCD_CS0						SPIMT1_CLK						
GPIO31	D11	3.3	VCC		GPIO31			SPI2_CLK	SPI3_CLK							PPI_Trigger4	IISG0_MCLK	IISG1_MCLK		I2C3_DAT		DMIC01_DAT			SD1_DAT2					SPIMT1_CLK			HOSCOUT	LOSCOUT	
GPIO32	A6	3.3/1.8	IOVCC2		GPIO32			SPI2_MISO					I2C0_CLK			PPI_Trigger1			I2C2_CLK					LCD_RS_CD		IISTX1_DOUT			SPIMT0_MISO		I2CMT0_CLK			HOSCOUT	LOSCOUT	
GPIO33	B9	3.3/1.8	IOVCC2		GPIO33			SPI2_MOSI					I2C0_DAT			PPI_Trigger2								LCD_RD#					SPIMT0_MOSI		I2CMT0_DAT					IOVCC2_OUT
GPIO34	B5	3.3/1.8	IOVCC2		GPIO34			SPI2_CLK						I2C1_CLK		PPI_Trigger3						DMIC01_CLK		LCD_WR#				BT_UART_TX	SPIMT0_CLK			I2CMT1_CLK			
GPIO35	A5	3.3/1.8	IOVCC2		GPIO35			SPI2_MISO						I2C1_DAT		PPI_Trigger4				I2C3_CLK		DMIC01_DAT		LCD_TE				BT_UART_RX	SPIMT0_SS1		I2CMT0_CLK	I2CMT1_DAT	LRADC4		
GPIO42	K3	1.8	IOVCC3=VC18		GPIO42			SPI2_CLK				UART2_RX		I2C1_DAT		PPI_Trigger6		IISG1_LRCLK					PWM2		SD1_DAT3					SPIMT1_CLK						
GPIO47	L4	1.8	IOVCC3=VC18		GPIO47															I2C3_CLK						IISTX1_DOUT							HOSCOUT	LOSCOUT	
GPIO49	L2	3.3/1.8	IOVCC4		GPIO49			SPI2_SS			UART1_TX		I2C0_CLK			PPI_Trigger0			I2C2_CLK				PWM0		SD1_CMD			BT_REQ	SPIMT0_SS0		I2CMT0_CLK					
GPIO50	L3	3.3/1.8	IOVCC4		GPIO50			SPI2_CLK			UART1_RX		I2C0_DAT			PPI_Trigger1			I2C2_DAT				PWM1		SD1_CLK			BT_ACCESS	SPIMT0_CLK		I2CMT0_DAT					
GPIO51	K1	3.3/1.8	IOVCC4		GPIO51			SPI2_MISO			UART1_CTS			I2C1_CLK		PPI_Trigger2				I2C3_CLK			PWM2		SD1_DAT1			PTA_GRANT	SPIMT0_MISO			I2CMT1_CLK				
GPIO52	K2	3.3/1.8	IOVCC4		GPIO52			SPI2_MOSI			UART1_RTS		I2C0_CLK	I2C1_DAT		PPI_Trigger3	IISTX0_DOUT			I2C3_DAT			PWM3		SD1_DAT0	IISTX1_DOUT			SPIMT0_MOSI			I2CMT1_DAT				
GPIO53	J1	3.3/1.8	IOVCC4		GPIO53			SPI2_SS	SPI3_SS			UART2_TX	I2C0_DAT		SD0_DAT4	PPI_Trigger4		IISG1_MCLK	I2C2_CLK				PWM0		SD1_DAT2					SPIMT1_SS0	I2CMT0_CLK			HOSCOUT	LOSCOUT	
GPIO54	J2	3.3/1.8	IOVCC4		GPIO54			SPI2_CLK	SPI3_CLK			UART2_RX		I2C1_CLK	SD0_DAT5	PPI_Trigger5		IISG1_BCLK	I2C2_DAT				PWM1		SD1_DAT3					SPIMT1_CLK	I2CMT0_DAT						
GPIO55	H1	3.3/1.8	IOVCC4		GPIO55			SPI2_MISO	SPI3_MISO			UART2_CTS		I2C1_DAT	SD0_DAT6	PPI_Trigger6		IISG1_LRCLK		I2C3_CLK			PWM2		SD1_CLK					SPIMT1_MISO		I2CMT1_CLK				
GPIO56	H2	3.3/1.8	IOVCC4		GPIO56			SPI2_MOSI	SPI3_MOSI			UART2_RTS			SD0_DAT7	PPI_Trigger7	IISTX0_DOUT	IISRX0_DIN	I2C2_CLK	I2C3_DAT			PWM3							SPIMT1_MOSI		I2CMT1_DAT				
GPIO57	G1	3.3/1.8	IOVCC4		GPIO57			SPI2_SS					I2C0_CLK		SD0_DAT2	PPI_Trigger8			I2C2_DAT				PWM1			IISTX1_DOUT			SPIMT0_SS2		I2CMT0_CLK			HOSCOUT	LOSCOUT	
GPIO58	G2	3.3/1.8	IOVCC4		GPIO58			SPI2_CLK					I2C0_DAT		SD0_DAT3	PPI_Trigger9	IISTX0_DOUT			I2C3_CLK									SPIMT0_SS3		I2CMT0_DAT					WCI_TX
GPIO59	F2	3.3/1.8	IOVCC4		GPIO59			SPI2_MISO						I2C1_CLK	SD0_CLK1	PPI_Trigger10	IISG0_MCLK	IISG1_MCLK		I2C3_DAT									SPIMT0_SS1			I2CMT1_CLK				WCI_RX
GPIO60	E1	3.3/1.8	IOVCC4		GPIO60			SPI2_MOSI						I2C1_DAT	SD0_CLK0	PPI_Trigger11	IISG0_BCLK	IISG1_BCLK		I2C3_CLK												I2CMT1_DAT		HOSCOUT	LOSCOUT	
GPIO61	E2	3.3/1.8	IOVCC4		GPIO61				SPI3_IO2						SD0_DAT0	PPI_Trigger4	IISG0_LRCLK	IISG1_LRCLK	I2C2_CLK											SPIMT1_SS3			LRADC5	HOSCOUT	LOSCOUT	
GPIO62	D1	3.3/1.8	IOVCC4		GPIO62				SPI3_IO3	UART0_RX	UART1_RX	UART2_RX			SD0_DAT1	PPI_Trigger5	IISTX0_DOUT	IISRX0_DIN	I2C2_DAT											SPIMT1_SS1			LRADC4			
GPIO63	D2	3.3/1.8	IOVCC4		GPIO63			SPI2_MISO	SPI3_MISO	UART0_TX	UART1_TX	UART2_TX			SD0_CMD	PPI_Trigger6	IISG0_MCLK	IISG1_MCLK		I2C3_DAT						IISTX1_DOUT				SPIMT1_SS2			LRADC3	HOSCOUT	LOSCOUT	IOVCC4_OUT
GPIO64	D12	3.3/1.8	IOVCC1		GPIO64			SPI2_MISO	SPI3_MISO	UART0_TX	UART1_TX	UART2_TX			SD0_DAT7	PPI_Trigger11	IISG0_MCLK	IISG1_MCLK	I2C2_CLK	I2C3_DAT			PWM0		SD1_DAT1	IISTX1_DOUT			SPIMT0_SS0			I2CMT1_DAT		HOSCOUT	LOSCOUT	IOVCC1_OUT
GPIO74	C12	3.3/1.8	IOVCC1		GPIO74			SPI2_MOSI	SPI3_MOSI	UART0_RX		UART2_RX	I2C0_CLK	I2C1_DAT	SD0_DAT6	PPI_Trigger4	IISG0_BCLK	IISG1_BCLK	I2C2_DAT	I2C3_CLK		DMIC01_CLK	PWM1		SD1_DAT0				SPIMT0_CLK			I2CMT1_CLK	LRADC6	HOSCOUT	LOSCOUT	WCI_RX
GPIO75	A13	3.3	VCC		GPIO75			SPI2_MISO	SPI3_MISO	UART0_TX		UART2_RX					IISG0_BCLK	IISG1_BCLK	I2C2_CLK		USBDP	DMIC01_CLK	PWM2		SD1_DAT3			BT_UART_TX		SPIMT1_SS3	I2CMT0_CLK	I2CMT1_CLK				WCI_TX
GPIO76	A12	3.3	VCC		GPIO76			SPI2_MOSI	SPI3_MOSI	UART0_RX		UART2_TX					IISG0_LRCLK	IISG1_LRCLK	I2C2_DAT		USBDM	DMIC01_DAT	PWM3		SD1_CLK			BT_UART_RX		SPIMT1_SS2	I2CMT0_DAT	I2CMT1_DAT				WCI_RX
GPIO77	B4	3.3/1.8	IOVCC2		GPIO77			SPI2_MOSI							SD0_CLK0	PPI_Trigger4	IISTX0_DOUT	IISRX0_DIN		I2C3_DAT		DMIC01_DAT		LCD_CS1	SD1_CLK	IISTX1_DOUT			SPIMT0_SS0		I2CMT0_DAT		LRADC6	HOSCOUT	LOSCOUT	
GPIO78	C5	3.3/1.8	IOVCC4		GPIO78			SPI2_SS	SPI3_SS		UART1_TX		I2C0_DAT		SD0_DAT4	PPI_Trigger8	IISG0_BCLK	IISG1_BCLK	I2C2_CLK	I2C3_CLK		DMIC01_DAT	PWM2		SD1_CLK				SPIMT0_MISO		I2CMT0_CLK	I2CMT1_DAT				WCI_TX
GPIO79	C4	3.3/1.8	IOVCC4		GPIO79			SPI2_CLK	SPI3_CLK		UART1_RX	UART2_RX	I2C0_CLK	I2C1_CLK	SD0_DAT5	PPI_Trigger9	IISG0_LRCLK	IISG1_LRCLK		I2C3_CLK		DMIC01_CLK	PWM3		SD1_CMD				SPIMT0_MOSI		I2CMT0_DAT	I2CMT1_CLK				WCI_RX
GPIO80	D3	3.3/1.8	IOVCC4		GPIO80			SPI2_MOSI	SPI3_MOSI		UART1_RX	UART2_TX		I2C1_DAT	SD0_DAT7	PPI_Trigger8	IISTX0_DOUT	IISRX0_DIN	I2C2_DAT	I2C3_DAT		DMIC01_DAT			SD1_DAT0	IISTX1_DOUT				SPIMT1_SS0				HOSCOUT	LOSCOUT	
WIO0	L13	2.5	SVCC		WIO0	ONOFF	I2C1_CLK	LRADC1																												
WIO1	M12	2.5	SVCC		WIO1		I2C1_DAT	LRADC1			LOSCOUT																									
WIO2	K12	2.5	SVCC	LOSCI	WIO2		I2C0_CLK																													
WIO3	K13	2.5	SVCC	LOSCO	WIO3		I2C0_DAT				LOSCOUT																									
WIO4	L12	2.5	SVCC		WIO4		I2C1_CLK				LOSCOUT																									
AOUT_P	N4	1.7	AVCC	AOUT/AOUT_P	GPIO94																															
AOUT_N	M4	1.7	AVCC	AOUT_N	GPIO95																															
INPUT0N	M2	1.7	AVCC	INPUT0N	GPIO91																															

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INPUT0P	N2	1.7	AVCC	INPUT0P	GPIO90																													
INPUT1N	M1	1.7	AVCC	INPUT1N	GPIO93																													
INPUT1P	N1	1.7	AVCC	INPUT1P	GPIO92																													

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2 Bluetooth

2.1 Features

- Support Bluetooth V5.3
- Support LE Audio(LC3 Codec)
- Supports basic rate and enhanced data rate
- Supports Secure Simple Pairing
- Supports Low Power Mode
- Bluetooth Dual Mode support: Simultaneous LE and BR / EDR
- Supports multiple Low Energy states
- Fast AGC control to improve receiving dynamic range
- Supports AFH to dynamically detect channel quality to improve transmission quality
- Supports GFSK, π/4 DQPSK and 8DPSK modulation
- Supports Power / Enhanced Power Control
- LE Data Packet Length Extension
- LE 2M PHY
- Support long range BLE
- Support AOA/AOD
- Channel Selection Algorithm #2
- Support PTA (Packet Traffic Arbiter) when co-existing with WI-FI
- Support WCI2 when co-existing with 4G

2.2 Bluetooth Performance

- Max transmitting output power: BR 13dBm, BLE 14dBm
- Bluetooth receiving sensitivity: -96.5dBm@GFSK, -95dBm@π/4 DQPSK, -89.5dBm@8DPSK modulation
- Bluetooth receiving sensitivity (BLE mode): -99.5dBm@1Mbps, -96dBm@2Mbps

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3 Processor Core

3.1 Features

- 192MHz(Max) ARM MSTAR processor Core with FPU
- Support extended DSP instruction
- 32 bits data, 16/32 bits mixed coding command
- Ultra-low power consumption with integrated sleep modes.
- Platform robustness with optional integrated memory protection.
- 4kB trace RAM.

3.2 Nested Vectored Interrupt Controller

3.2.1 Features

This section describes the NVIC and the registers it uses. The NVIC supports:

- 1 to 64 interrupts.
- A programmable priority level of 0-255 for each interrupt. A higher level corresponds to a lower priority, so level 0 is the highest interrupt priority.
- Level and pulse detection of interrupt signals.
- Dynamic reprioritization of interrupts.
- Grouping of priority values into group priority and sub-priority fields.
- Interrupt tail-chaining.
- An external Non-maskable interrupt (NMI)
- Optional WIC, providing ultra-low power sleep mode support.

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4 DSP Core

- 192MHz(Max) CEVA TL420 DSP Core
- Audio Configuration Features set
  - High code compactness due to:
    -- Two instruction widths (16-bit or 32-bit)
    -- Powerful instruction set capabilities
  - 10 stage pipeline
  - All instructions can be conditional:
    -- Conditional execution
    -- Reduces cycle count and code size on control and overhead code
  - Computational units:
    -- One 32-bit x 32-bit Multiply-and-Accumulate (MAC) using 72-bit product
    -- One 32-bit x 16-bit MAC using 72-bit product
    -- One 32-bit x 32-bit MAC unit with automatic scaling
    -- One 32-bit x 16-bit MAC unit with automatic scaling
    -- One 36-bit arithmetic unit
    -- One 36-bit logical unit
    -- One 36-bit bit-manipulation unit, including a full barrel shifter and an exponent unit
    -- Four 36-bit accumulators
    -- Fully programmable product post-shifter for product scaling
  - 32-bit Scalar (SC) unit for integer operations
  - Two address generation units for two independent accesses to the data memories
    -- Up to two 32-bit load operations in parallel
    -- Single 32-bit store operation
  - Unaligned memory access for load and store operations
  - Multiple data addressing modes:
    -- Indirect addressing
    -- Modulo addressing
    -- Bit-reversed addressing
    -- Direct addressing
    -- Indexed addressing
    -- Stack addressing
  - Support for scaling data after multiplications or before storing to data memory
  - Three different saturation modes
  - Single-cycle exponent evaluation of 36-bit values
  - Four-cycle FFT double precision butterfly
  - Bits unpacking capabilities (for Huffman algorithms)
  - 2G byte program and data space
  - Word-addressable data space
  - Eight 32-bit memory addressing registers
  - Separate I/O space for peripherals
  - Fully registered memory interface
  - Direct memory access for upload and download to TCM
- Voice Configuration Feature Set
  - Two 16-bit x 16-bit MAC units
  - SIMD operations in word precisions
  - Advanced support for split operations in 16-bit precision
  - Two-cycle FFT single precision butterfly
  - Single-cycle division step support
- Legacy Compatibility Configuration Feature Set
  - Up to two 32-bit store operations in parallel
  - Viterbi instructions acceleration

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- Automatic context-switch operation on interrupts (task switching)
- Eight status and condition flags
- Legacy compatibility to previous TeakLite cores

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5 Memory Controller

5.1 Features

- Internal 1168KB RAM for data and program
- Build in 32KB Execute-in-place(Xip) Cache to QSPI NOR flash and 32KB cache to HSPI/OSPI PSRAM
- Memory Protect Unit (MPU)
- DMA/SDMA can access all the RAM blocks
- CPU can access all the RAM and ROM block through CPU bus
- DSP's data bus and program bus can access DSPRAM blocks
- 2.5D GPU and 2D GPU can access part of the RAM for graphics processing

6 Secure Subsystem

6.1 OTFD

6.1.1 Features

- Support On-the-fly decryption of external memories (QSPI NorFlash/HSPI PSRAM), part of which can be encrypted by AES-128 in counter mode

6.2 SecureEngine

6.2.1 Features

- Support hardware AES-128/192/256 Encrypt & Decrypt in ECB/CTR/CBC/CBC-CTS mode
- Support hardware TRNG used to collects random bits from the ring oscillator, 64 random bits per time.
- Support SHA-1/SHA-224/SHA-256
- Support hardware CRC-32/CRC-16
- Support 1024bits efuse
- Support 64bit CHIP unique ID
- Support RSA 2048bits signature

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7 DMA Controller

7.1Features

- DMA transmission is independent with the CPU and DSP
- Support for memory to memory, memory to peripheral, peripheral to memory transmission.
- 10-channel ordinary DMA, that supports for transmission in burst 8 mode or single mode. Only one of the ten DMA channels can transfer data at the same time.
- DMA0-9 transmission can be triggered on the occurrence of selected events
- Each channel can send two interrupts to the CPU on completion of certain operational events
- Transmission width includes 8-bit, 16-bit, and 32-bit, which is determined by DMA transmission type.

7.2Memory and Peripheral Access Description

7.2.1 Access memory

（1） Memory is accessed by DMA according to physical address.
（2） The following situations will lead to transfer error:
- Access rom
- Ram has been occupied by other modules
- The clock of ram is disabled
（3） Memory-to-memory transmission:
- The data in memory should be aligned by word (32-bit) when transferred.
（4） Memory-to-peripheral and peripheral-to-memory transmission:
- The data in memory should be aligned by word (32-bit) when the transfer width is 32-bit or 24-bit.
- The data in memory should be aligned by half-word (16-bit) when the transfer width is 16-bit.
- The data in memory should be aligned by byte (8-bit) when the transfer width is 8-bit.
（5） SRCTYPE/ DSTTYPE/SADDR/DADDR should be set when DMA access the peripherals.
（6） CPU cannot access the peripherals FIFO when the peripherals FIFO is accessed by DMA.

7.2.2 Access Peripheral Mode

Peripheral	Single	Burst8
SPI0/1/2/3	Y	Y
UART0/1/2	Y	Y
I2C0/1/2/3	Y	Y
USB	N	Y
SD0	N	Y
SD1	N	Y
PWM	Y	Y
DAC	N	Y
ADC	N	Y
IIS	N	Y

7.2.3 DMA channel priority

Each memory block can be accessed by only one master at the same time. The 10 channels of DMA share

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the same bus, so only one of these channels can use the bus to transfer data at the same moment.
While accessing one memory block, DMA submits an access request to the memory controller's arbiter. Meanwhile, CPU might send another request to access the same memory block. The arbiter grants the bus of this memory block according to the priority scheme. The priority scheme is known as round-robin algorithm. When DMA does not get the bus of this memory block, the memory controller will hold DMA. See memory controller specification for details.
Once DMA obtains the highest priority among the three masters, one of the 10 channels occupies the DMA bus according to the internal priority. The priority of DMA channel is DMA0 > DMA1 > DMA2 > DMA3 > DMA4 > DMA5 > DMA6 > DMA7>DMA8>DMA9.

7.2.4 DMA Access Channel

DMA supports memory-to-memory, memory-to-peripheral, and peripheral-to-memory transmission. Specific access paths are shown as follows:

DST / SRC	RAM	SPI01 23TX FIFO	UART 012TX	USB FIFO	SD1 FIFO	SD0 FIFO	DAC FIFO	IIS TX FIFO	I2C0/1 /2/3 FIFO	PWM
RAM	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
SPI0123 RX FIFO	Y	-	-	-		-				
UART012R X FIFO	Y	-	-	-		-				
USB FIFO	Y	-	-	-		Y				
SD1 FIFO	Y									
SD0 FIFO	Y	-	-	Y		-				
ADC FIFO	Y	-	-			-				
IIS RX FIFO	Y									
I2C0/1/2/3 FIFO	Y									

7.3DMA Register List

Table 7-1 DMA controller base address

Name	Physical Base Address	KSEG1 Base Address
DMAController	0x4001C000	0x4001C000

Table 7-2 DMA controller register list

Offset	Register Name	Description
0x00000000	DMAIP	DMA interrupt pending register
0x00000004	DMAIE	DMA interrupt enable register
0x00000100+0x100*x	DMAxCTL	DMAx control register (0<=x<=9)
0x00000104+0x100*x	DMAxSTART	DMAx start register (0<=x<=9)
0x00000108+0x100*x	DMAxSADDR0	DMAx source address register0 (0<=x<=9)
0x0000010c+0x100*x	DMAxSADDR1	DMAx source address register1 (0<=x<=9)
0x00000110+0x100*x	DMAxDADDR0	DMAx destination address register0 (0<=x<=9)
0x00000114+0x100*x	DMAxDADDR1	DMAx destination address register1 (0<=x<=9)

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0x00000118+0x100*x	DMAxBC	DMAx byte counter register (0<=x<=9)
0x0000011c+0x100*x	DMAxRC	DMAx remain counter register (0<=x<=9)

7.4 DMA Register Description

7.4.1 DMAIP

DMAIP (DMA Interrupt Pending Register, offset = 0x00000000)

Bit (s)	Name	Description	Access	Reset
31:26	Reserved	Reserved	R	x
25	DMA9HFIP	DMA9 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
24	DMA8HFIP	DMA8 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
23	DMA7HFIP	DMA7 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
22	DMA6HFIP	DMA6 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
21	DMA5HFIP	DMA5 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
20	DMA4HFIP	DMA4 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
19	DMA3HFIP	DMA3 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
18	DMA2HFIP	DMA2 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
17	DMA1HFIP	DMA1 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
16	DMA0HFIP	DMA0 Half Transmission IRQ Pending This bit can be written '1' to clear.	RW	0x0
15:10	Reserved	Reserved	R	x
9	DMA9TCIP	DMA9 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
8	DMA8TCIP	DMA8 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
7	DMA7TCIP	DMA7 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
6	DMA6TCIP	DMA6 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
5	DMA5TCIP	DMA5 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
4	DMA4TCIP	DMA4 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
3	DMA3TCIP	DMA3 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
2	DMA2TCIP	DMA2 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
1	DMA1TCIP	DMA1 Transmission Complete IRQ Pending This bit can be written '1' to clear.	RW	0x0
0	DMA0TCIP	DMA0 Transmission Complete IRQ Pending	RW	0x0

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		This bit can be written '1' to clear.		

7.4.2 DMAIE

DMAIE (DMA Interrupt Enable Register, offset = 0x00000004)

Bit (s)	Name	Description	Access	Reset
31:26	Reserved	Reserved	R	x
25	DMA9HFIE	DMA9 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
24	DMA8HFIE	DMA8 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
23	DMA7HFIE	DMA7 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
22	DMA6HFIE	DMA6 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
21	DMA5HFIE	DMA5 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
20	DMA4HFIE	DMA4 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
19	DMA3HFIE	DMA3 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
18	DMA2HFIE	DMA2 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
17	DMA1HFIE	DMA1 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
16	DMA0HFIE	DMA0 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	RW	0x0
15:10	Reserved	Reserved	R	x
9	DMA9TCIE	DMA9 Transmission Complete IRQ Enable: 0: disable DMA9 Transmission Complete interrupt 1: enable DMA9 Transmission Complete interrupt	RW	0x0
8	DMA8TCIE	DMA8 Transmission Complete IRQ Enable: 0: disable DMA8 Transmission Complete interrupt 1: enable DMA8 Transmission Complete interrupt	RW	0x0
7	DMA7TCIE	DMA7 Transmission Complete IRQ Enable: 0: disable DMA7 Transmission Complete interrupt 1: enable DMA7 Transmission Complete interrupt	RW	0x0
6	DMA6TCIE	DMA6 Transmission Complete IRQ Enable: 0: disable DMA6 Transmission Complete interrupt 1: enable DMA6 Transmission Complete interrupt	RW	0x0
5	DMA5TCIE	DMA5 Transmission Complete IRQ Enable: 0: disable DMA5 Transmission Complete interrupt 1: enable DMA5 Transmission Complete interrupt	RW	0x0

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4	DMA4TCIE	DMA4 Transmission Complete IRQ Enable: 0: disable DMA4 Transmission Complete interrupt 1: enable DMA4 Transmission Complete interrupt	RW	0x0
3	DMA3TCIE	DMA3 Transmission Complete IRQ Enable: 0: disable DMA3 Transmission Complete interrupt 1: enable DMA3 Transmission Complete interrupt	RW	0x0
2	DMA2TCIE	DMA2 Transmission Complete IRQ Enable: 0: disable DMA2 Transmission Complete interrupt 1: enable DMA2 Transmission Complete interrupt	RW	0x0
1	DMA1TCIE	DMA1 Transmission Complete IRQ Enable: 0: disable DMA1 Transmission Complete interrupt 1: enable DMA1 Transmission Complete interrupt	RW	0x0
0	DMA0TCIE	DMA0 Transmission Complete IRQ Enable: 0: disable DMA0 Transmission Complete interrupt 1: enable DMA0 Transmission Complete interrupt	RW	0x0

7.4.3 DMAxCTL

DMAxCTL (DMAx control Register, offset = 0x00000100+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:22	-	Reserved	-	0
21:20	TWS	Transmit data width select 0:32bit 1:16bit 2:8bit 3:reserved	RW	0
19	-	Reserved	-	0
18	reload	Reload the DMA controller registers and start DMA transmission after current DMA transmission is complete: 0: disable reload mode 1: enable reload mode	RW	0
17	TRM	0: Burst8 1: Single	RW	0
16	AUDIOTYPE	The method of audio data stored of DMA-ADC, DMA-DAC, DMA-I2S transmission: 0: interleaved stored in the memory 1: separated stored in the memory	RW	0x0
15	DAM	Destination address mode 0: increment 1: constant	RW	0
14:13	-	Reserved	-	0
12:8	DSTSL	Destination select: 5'b00000: memory 5'b00001: UART0 TX FIFO 5'b00010: UART1 TX FIFO 5'b00011: UART2 TX FIFO 5'b00100: USB FIFO 5'b00101: SD0 FIFO 5'b00110: SD1 FIFO 5'b00111: SPI0 TX FIFO 5'b01000: SPI1 TX FIFO 5'b01001: SPI2 TX FIFO 5'b01010: SPI3 TX FIFO 5'b01011: DAC FIFO0	RW	0

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		5'b01110: IIS TX0 FIFO 5'b10011: I2C0 TX FIFO 5'b10100: I2C1 TX FIFO 5'b10101: PWM FIFO 5'b11000: I2C2 TX FIFO 5'b11001: I2C3 TX FIFO Other: Reserved		
7	SAM	Source address mode 0: increment 1: constant	RW	0
6:5	-	Reserved	-	0
4:0	SRCSL	Source Select: 5'b00000: memory 5'b00001: UART0 RX FIFO 5'b00010: UART1 RX FIFO 5'b00011: UART2 RX FIFO 5'b00100: USB FIFO 5'b00101: SD0 FIFO 5'b00110: SD1 FIFO 5'b00111: SPI0 RX FIFO 5'b01000: SPI1 RX FIFO 5'b01001: SPI2 RX FIFO 5'b01010: SPI3 RX FIFO 5'b01011: ADC FIFO 5'b01110: IIS RX FIFO 5'b10011: I2C0 RX FIFO 5'b10100: I2C1 RX FIFO 5'b11000: I2C2 RX FIFO 5'b11001: I2C3 RX FIFO Other: Reserved	RW	0

7.4.4 DMAxSTART

DMAxSTART (offset = 0x00000104+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:1	Reserved	-	-	-
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.	RW	0x0

7.4.5 DMAxSADDR0

DMAxSADDR0 (DMAx Source Address Register0, offset = 0x00000108+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	RW	0x0

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7.4.6 DMAxSADDR1

DMAxSADDR1 (DMAx Source Address Register1, offset = 0x0000010c+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	RW	0x0

7.4.7 DMAxDADDR0

DMAxDADDR0 (DMAx Destination Address Register0, offset = 0x00000110+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	RW	0x0

7.4.8 DMAxDADDR1

DMAxDADDR1 (DMAx Destination Address Register1, offset = 0x00000114+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	RW	0x0

7.4.9 DMAxBC

DMAxBC (DMAx BYTE Counter Register, offset = 0x00000118+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:20	Reserved	Reserved	R	x
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	RW	0x0

7.4.10 DMAxRC

DMAxRC (DMAx Remain Counter Register, offset = 0x0000011c+0x100*x, 0 <= x <= 9)

Bit (s)	Name	Description	Access	Reset
31:20	Reserved	-	-	-
19:0	DMARemainCounterH	The Remain Counter of DMA transmission.	R	0x0

8 SDMA Controller

8.1Features

- 5-channel SDMA
- SDMA support stride mode
- SDMA support Y address decrease. This is designed for LCD Y-flip
- SDMA0 only support memory to LCD
- SDMA1~3 support memory to memory, memory to JPEG, JPEG to memory
- SDMA4 support memory to memory

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- Each channel can send two interrupts to the CPU on completion of certain operational events
- Transmission width is 32-bit. Max Burst length is configurable (8,16,32)

8.2Using mode Description

8.2.1 Stride Mode

Stride mode is mainly used for window cutting application of image display.

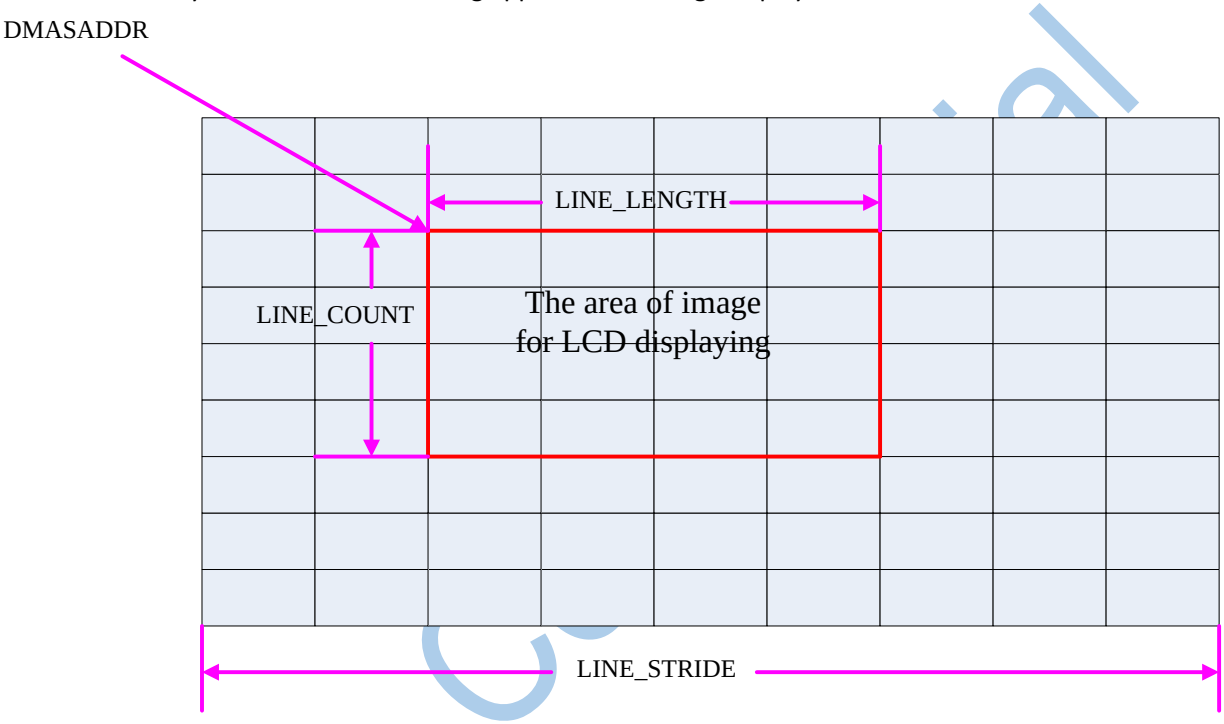


Figure 8-1 Data storage diagram of stride mode

When the image data is transmitted to LCD for displaying, As Figure 8-1 shown above, the red box is the part to be sent to LCD for displaying.
The DMASADDR register configures the start address of the image in the red box;
The LINE_ LENGTH register configures the length of the image for displaying;
The LINE_ COUNT register configures the height of the image for displaying;
The LINE_ STRIDE register configures the length of the original image.

When the image data is write to memory, the stride mode of the target address is the same as the above figure.

8.2.2 Couple Mode

The couple mode is used to improve the efficiency of JPEG decoding output to PSRAM. The basic principle is to set an intermediate buffer, and use two channels of SDMA to parallel the JPEG decoding and data writing to PSRAM through this intermediate buffer.
In couple mode, COUPLE_NUMBER is used to configure which two SDMA forms the couple mode. There are three optional groups, SDMA1 and SDMA4, SDMA2 and SDMA4, SDMA3 and SDMA4.
Couple mode requires a block of RAM as an intermediate buffer to speed up the write operation to PSRAM. The starting address and the size of this buffer are configured by register SDMA_COUPLE_BUF_ADDR and SDMA_COUPLE_BUF_SIZE.

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8.2.3 Color Fill Mode

Color Fill mode is mainly used to fill fixed colors in the image window quickly with SDMA.
Only SDMA1/2/3 can use Color Fill mode.

8.2.4 Priorities

The priority of SDMA controller is controlled by register SDMA_ PRIORITY, when the memory accessing commands of different master conflict. The SDMA controller select the higher priority command from SDMA1/SDMA2/SDMA3 as SDMA123 first, and then select the higher priority command from SDMA123, SDMA0 and SDMA4 to send to the memory device finally.

8.3 SDMA Register List

Table 8-1 SDMA controller base address

Name	Physical Base Address	KSEG1 Base Address
SDMAController	0x40078000	0x40078000

Table 8-2 SDMA controller register list

Offset	Register Name	Description
0x00000000	SDMAIP	SDMA interrupt pending register
0x00000004	SDMAIE	SDMA interrupt enable register
0x00000070	SDMA_PRIORITY	SDMA Priority Register
0x000000a0	SDMA_COUPLE_CONFIG	SDMA Couple Configure Register
0x000000a4	SDMA_COUPLE_BUF_ADDR	SDMA Couple Buffer Address Register
0x000000a8	SDMA_COUPLE_BUF_SIZE	SDMA Couple Buffer Size Register
0x000000ac	SDMA_COUPLE_START	SDMA Couple Start Register
0x000000b0	SDMA_COUPLE_WRITE_POINTER	SDMA Couple Write Pointer Register
0x000000b4	SDMA_COUPLE_READ_POINTER	SDMA Couple Read Pointer Register
0x00000100	SDMA0CTL	SDMA0 control register
0x00000104	SDMA0START	SDMA0 start register
0x00000108	SDMA0SADDR	SDMA0 source address0 register
0x00000118	SDMA0BC	SDMA0 byte counter register
0x0000011c	SDMA0RC	SDMA0 remain counter register
0x00000200	SDMA1CTL	SDMA1 control register
0x00000204	SDMA1START	SDMA1 start register
0x00000208	SDMA1SADDR	SDMA1 source address0 register
0x00000210	SDMA1DADDR	SDMA1 destination address0 register
0x00000218	SDMA1BC	SDMA1 byte counter register
0x0000021c	SDMA1RC	SDMA1 remain counter

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		register
0x00000300	SDMA2CTL	SDMA2 control register
0x00000304	SDMA2START	SDMA2 start register
0x00000308	SDMA2SADDR	SDMA2 source address0 register
0x00000310	SDMA2DADDR	SDMA2 destination address0 register
0x00000318	SDMA2BC	SDMA2 byte counter register
0x0000031c	SDMA2RC	SDMA2 remain counter register
0x00000400	SDMA3CTL	SDMA3 control register
0x00000404	SDMA3START	SDMA3 start register
0x00000408	SDMA3SADDR	SDMA3 source address0 register
0x00000410	SDMA3DADDR	SDMA3 destination address0 register
0x00000418	SDMA3BC	SDMA3 byte counter register
0x0000041c	SDMA3RC	SDMA3 remain counter register
0x00000500	SDMA4CTL	SDMA4 control register
0x00000504	SDMA4START	SDMA4 start register
0x00000508	SDMA4SADDR	SDMA4 source address0 register
0x00000510	SDMA4DADDR	SDMA4 destination address0 register
0x00000518	SDMA4BC	SDMA4 byte counter register
0x0000051c	SDMA4RC	SDMA4 remain counter register
0x00000c00	LINE_LENGTH0	Line Length Register
0x00000c04	LINE_COUNT0	Line Count Register
0x00000c08	LINE_SRC_STRIDE0	Line Stride Register
0x00000c10	LINE_REMAIN0	Line Remain Register
0x00000c14	BYTE_REMAIN_IN_LINE0	Byte Remain in Line Register
0x00000c20	LINE_LENGTH1	Line Length Register
0x00000c24	LINE_COUNT1	Line Count Register
0x00000c28	LINE_SRC_STRIDE1	Line Stride Register
0x00000c2c	LINE_DST_STRIDE1	Line Stride Register
0x00000c30	LINE_REMAIN1	Line Remain Register
0x00000c34	BYTE_REMAIN_IN_LINE1	Byte Remain in Line Register
0x00000c40	LINE_LENGTH2	Line Length Register
0x00000c44	LINE_COUNT2	Line Count Register
0x00000c48	LINE_SRC_STRIDE2	Line Stride Register
0x00000c4c	LINE_DST_STRIDE2	Line Stride Register
0x00000c50	LINE_REMAIN2	Line Remain Register
0x00000c54	BYTE_REMAIN_IN_LINE2	Byte Remain in Line Register
0x00000c60	LINE_LENGTH3	Line Length Register

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0x00000c64	LINE_COUNT3	Line Count Register
0x00000c68	LINE_SRC_STRIDE3	Line Stride Register
0x00000c6c	LINE_DST_STRIDE3	Line Stride Register
0x00000c70	LINE_REMAIN3	Line Remain Register
0x00000c74	BYTE_REMAIN_IN_LINE3	Byte Remain in Line Register
0x00000c80	LINE_LENGTH4	Line Length Register
0x00000c84	LINE_COUNT4	Line Count Register
0x00000c88	LINE_SRC_STRIDE4	Line Stride Register
0x00000c8c	LINE_DST_STRIDE4	Line Stride Register
0x00000c90	LINE_REMAIN4	Line Remain Register
0x00000c94	BYTE_REMAIN_IN_LINE4	Byte Remain in Line Register
0x00000d00	COLOR_FILL_DATA0	Color Fill Data0 Register
0x00000d04	COLOR_FILL_DATA1	Color Fill Data1 Register
0x00000d08	COLOR_FILL_DATA2	Color Fill Data2 Register

8.4 SDMA Register Description

8.4.1 SDMAIP

SDMAIP (DMA Interrupt Pending Register, offset = 0x00000000)

Bit(s)	Name	Description	R/W	Reset
31:21	-	Reserved	R	0x0
20	SDMA4HFIP	DMA4 Half Transmission IRQ Pending This bit can be written '1' to clear.	R/W	0x0
19	SDMA3HFIP	DMA3 Half Transmission IRQ Pending This bit can be written '1' to clear.	R/W	0x0
18	SDMA2HFIP	DMA2 Half Transmission IRQ Pending This bit can be written '1' to clear.	R/W	0x0
17	SDMA1HFIP	DMA1 Half Transmission IRQ Pending This bit can be written '1' to clear.	R/W	0x0
16	SDMA0HFIP	DMA0 Half Transmission IRQ Pending This bit can be written '1' to clear.	R/W	0x0
15:5	-	Reserved	R	0x0
4	SDMA4TCIP	DMA4 Transmission Complete IRQ Pending This bit can be written '1' to clear.	R/W	0x0
3	SDMA3TCIP	DMA3 Transmission Complete IRQ Pending This bit can be written '1' to clear.	R/W	0x0
2	SDMA2TCIP	DMA2 Transmission Complete IRQ Pending This bit can be written '1' to clear.	R/W	0x0
1	SDMA1TCIP	DMA1 Transmission Complete IRQ Pending This bit can be written '1' to clear.	R/W	0x0
0	SDMA0TCIP	DMA0 Transmission Complete IRQ Pending This bit can be written '1' to clear.	R/W	0x0

8.4.2 SDMAIE

SDMAIE (DMA Interrupt Enable Register, offset = 0x00000004)

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Bit(s)	Name	Description	R/W	Reset
31:21	-	Reserved	R	0x0
20	SDMA4HFIE	DMA4 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	R/W	0x0
19	SDMA3HFIE	DMA3 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	R/W	0x0
18	SDMA2HFIE	DMA2 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	R/W	0x0
17	SDMA1HFIE	DMA1 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	R/W	0x0
16	SDMA0HFIE	DMA0 Half Transmission Complete IRQ enable: 0: Disable Half Transmission Complete interrupt; 1: Enable Half Transmission Complete interrupt.	R/W	0x0
15:5	-	Reserved	R	0x0
4	SDMA4TCIE	DMA4 Transmission Complete IRQ Enable: 0: disable DMA4 Transmission Complete interrupt 1: enable DMA4 Transmission Complete interrupt	R/W	0x0
3	SDMA3TCIE	DMA3 Transmission Complete IRQ Enable: 0: disable DMA3 Transmission Complete interrupt 1: enable DMA3 Transmission Complete interrupt	R/W	0x0
2	SDMA2TCIE	DMA2 Transmission Complete IRQ Enable: 0: disable DMA2 Transmission Complete interrupt 1: enable DMA2 Transmission Complete interrupt	R/W	0x0
1	SDMA1TCIE	DMA1 Transmission Complete IRQ Enable: 0: disable DMA1 Transmission Complete interrupt 1: enable DMA1 Transmission Complete interrupt	R/W	0x0
0	SDMA0TCIE	DMA0 Transmission Complete IRQ Enable: 0: disable DMA0 Transmission Complete interrupt 1: enable DMA0 Transmission Complete interrupt	R/W	0x0

8.4.3 SDMA_PRIORITY

SDMA PRIORITY (DMA Priority Register, offset = 0x00000070)

Bit(s)	Name	Description	R/W	Reset
31:21	-	Reserved	R	0x0
20	SDMA123_PRIORITY	0: SDMA1>SDMA2>SDMA3 1: SDMA1/SDMA2/SDMA3 Round Robin	R/W	0x0
19:18	-	Reserved	R	0x0
17:16	SDMA1234_PRIORITY	0：SDMA1~3 FIFO and SDMA4 FIFO Round Robin; 1：SDMA1~3 FIFO > SDMA4 FIFO; 2：SDMA4 FIFO > SDMA1~3 FIFO; Others: Reserved	R/W	0x0
15:7	-	Reserved	R	0x0
6:0	FIFO_LEVEL	When the empty level of LCD FIFO is more than FIFO_LEVEL, the SDMA controller will set SDMA0 as highest priority. The	R/W	0xc

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		purpose is to ensure the real-time performance of LCD display.		

8.4.4 SDMA_COUPLE_CONFIG

SDMA_COUPLE_CONFIG (SDMA Couple Config Register, offset = 0x000000a0)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:2	BUF_THRESHOLD	The read and write threshold of the couple buffer. When SDMA1~3 writes more bytes to buffer than BUF_THRESHOLD, SDMA4 starts to read from the buffer.	R/W	0x1f
1:0	COUPLE_NUMBER	0: no couple mode 1: SDMA1 and SDMA4 2: SDMA2 and SDMA4 3: SDMA3 and SDMA4	R/W	0x0

8.4.5 SDMA_COUPLE_BUF_ADDR

SDMA_COUPLE_BUF_ADDR (SDMA Couple Buffer Address Register, offset = 0x000000a4)

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29:2	ADDR	The address of SDMA couple buffer, which is the destination address for SDMA1~3 and the source address for SDMA4.	R/W	0x0
1:0	-	Reserved	R	0x0

8.4.6 SDMA_COUPLE_BUF_SIZE

SDMA_COUPLE_BUF_SIZE (SDMA Couple Buffer Size Register, offset = 0x000000a8)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:2	SIZE	The size of SDMA couple buffer, which is 4 bytes aligned.	R/W	0x0
1:0	-	Reserved	R	0x0

8.4.7 SDMA_COUPLE_START

SDMA_COUPLE_START (SDMA Couple Start Register, offset = 0x000000ac)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	START	If the bit be set 1, the SDMA couple transmission will be started. SDMA1/2/3 works first. When SDMA1/2/3 writes more data to the couple buffer than BUF_THRESHOLD, SDMA4 will start to work and read from the couple buffer. After the SDMA couple transmission is completed, the bit clears to 0 automatically.	R/W	0x0

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8.4.8 SDMA_COUPLE_WRITE_POINTER

SDMA_COUPLE_WRITE_POINTER (SDMA Couple Write Pointer Register, offset = 0x000000b0)

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29:0	POINTER	Record the pointer of SDMA1/2/3 writing to the couple buffer.	R	0x0

8.4.9 SDMA_COUPLE_READ_POINTER

SDMA_COUPLE_READ_POINTER (SDMA Couple Read Pointer Register, offset = 0x000000b0)

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29:0	POINTER	Record the pointer of SDMA4 reading from the couple buffer.	R	0x0

8.4.10 SDMA0CTL

DMA0CTL (DMAx Control Register, offset = 0x00000100)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	STRME	Stride Mode Enable: 0: Disable 1: Enable	R/W	0x0
23:19	-	Reserved	R	0x0
18	RELOAD	Reload the DMA controller registers and start DMA transmission after current DMA transmission is complete: 0: disable reload mode 1: enable reload mode	R/W	0x0
17:14	-	Reserved	R	0x0
13:8	DSTSL	Destination select: 5'b00000: LCD	R	0x0
7	-	Reserved	R	0x0
6	YSAM	Y Source address mode 0: increment 1: decrement	R/W	0x0
5:0	SRCSL	Source Select: 5'b00000: memory	R	0x0

8.4.11 SDMA0START

DMA0START (DMA Start Register, offset = 0x00000104)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller.	R/W	0x0

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		This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.		

8.4.12 SDMA0SADDR

DMA0SADDR (DMA Source Address Register, offset = 0x00000108)

Bit(s)	Name	Description	R/W	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.13 SDMA0BC

DMA0BC (DMA Byte Counter Address Register, offset = 0x00000118)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	R/W	0x0

8.4.14 SDMA0RC

DMA0RC (DMA Remain Counter Address Register, offset = 0x0000011C)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMAREMAINCOUNTER	The Remain Counter of DMA transmission.	R	0x0

8.4.15 SDMA1CTL

DMA1CTL (DMAx Control Register, offset = 0x00000200)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	STRME	Stride Mode Enable: 0: Disable 1: Enable	R/W	0x0
23:21	-	Reserved	R	0x0
20	JPEG_OUT_FORMAT	0: RGB888 1: RGB565	R/W	0x0
19:15	-	Reserved	R	0x0
14	YDAM	Y Destination address mode 0: increment 1: decrement	R/W	0x0
13:9	-	Reserved	R	0x0

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8	DSTSL	Destination select: 0: memory 1: JPEG	R/W	0x0
7	-	Reserved	R	0x0
6	YSAM	Y Source address mode 0: increment 1: decrement	R/W	0x0
5:2	-	Reserved	R	0x0
1:0	SRCSL	Source Select: 0: memory 1: JPEG 2: color_fill 3: Reserved	R/W	0x0

8.4.16 SDMA1START

DMA1START (DMA Start Register, offset = 0x00000204)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.	R/W	0x0

8.4.17 SDMA1SADDR

DMA1SADDR (DMA Source Address Register, offset = 0x00000208)

Bit(s)	Name	Description	R/W	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.18 SDMA1DADDR

DMA1DADDR (DMA Destination Address Register, offset = 0x00000210)

Bit(s)	Name	Description	R/W	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

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8.4.19 SDMA1BC

DMA1BC (DMA Byte Counter Address Register, offset = 0x00000218)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	R/W	0x0

8.4.20 SDMA1RC

DMA1RC (DMA Remain Counter Address Register, offset = 0x0000021C)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMAREMAINCOUNTER	The Remain Counter of DMA transmission.	R	0x0

8.4.21 SDMA2CTL

DMA2CTL (DMAx Control Register, offset = 0x00000300)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	STRME	Stride Mode Enable: 0: Disable 1: Enable	R/W	0x0
23:21	-	Reserved	R	0x0
20	JPEG_OUT_FORMAT	0: RGB888 1: RGB565	R/W	0x0
19:15	-	Reserved	R	0x0
14	YDAM	Y Destination address mode 0: increment 1: decrement	R/W	0x0
13:9	-	Reserved	R	0x0
8	DSTSL	Destination select: 0: memory 1: JPEG	R/W	0x0
7	-	Reserved	R	0x0
6	YSAM	Y Source address mode 0: increment 1: decrement	R/W	0x0
5:2	-	Reserved	R	0x0
1:0	SRCSL	Source Select: 0: memory 1: JPEG 2: color_fill 3: Reserved	R/W	0x0

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8.4.22 SDMA2START

DMA2START (DMA Start Register, offset = 0x00000304)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.	R/W	0x0

8.4.23 SDMA2SADDR

DMA2SADDR (DMA Source Address Register, offset = 0x00000308)

Bit(s)	Name	Description	R/W	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.24 SDMA2DADDR

DMA2DADDR (DMA Destination Address Register, offset = 0x00000310)

Bit(s)	Name	Description	R/W	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.25 SDMA2BC

DMA2BC (DMA Byte Counter Address Register, offset = 0x00000318)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	R/W	0x0

8.4.26 SDMA2RC

DMA2RC (DMA Remain Counter Address Register, offset = 0x0000031C)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMAREMAINCOUNTER	The Remain Counter of DMA transmission.	R	0x0

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8.4.27 SDMA3CTL

DMA3CTL (DMAx Control Register, offset = 0x00000400)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	STRME	Stride Mode Enable: 0: Disable 1: Enable	R/W	0x0
23:21	-	Reserved	R	0x0
20	JPEG_OUT_FORMAT	0: RGB888 1: RGB565	R/W	0x0
19:15	-	Reserved	R	0x0
14	YDAM	Y Destination address mode 0: increment 1: decrement	R/W	0x0
13:9	-	Reserved	R	0x0
8	DSTSL	Destination select: 0: memory 1: JPEG	R/W	0x0
7	-	Reserved	R	0x0
6	YSAM	Y Source address mode 0: increment 1: decrement	R/W	0x0
5:2	-	Reserved	R	0x0
1:0	SRCSL	Source Select: 0: memory 1: JPEG 2: color_fill 3: Reserved	R/W	0x0

8.4.28 SDMA3START

DMA3START (DMA Start Register, offset = 0x00000404)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.	R/W	0x0

8.4.29 SDMA3SADDR

DMA3SADDR (DMA Source Address Register, offset = 0x00000408)

Bit(s)	Name	Description	R/W	Reset

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31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.30 SDMA3DADDR

DMA3DADDR (DMA Destination Address Register, offset = 0x00000410)

Bit(s)	Name	Description	R/W	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.31 SDMA3BC

DMA3BC (DMA Byte Counter Address Register, offset = 0x00000418)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	R/W	0x0

8.4.32 SDMA3RC

DMA3RC (DMA Remain Counter Address Register, offset = 0x0000041C)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMAREMAINCOUNTER	The Remain Counter of DMA transmission.	R	0x0

8.4.33 SDMA4CTL

DMA4CTL (DMAx Control Register, offset = 0x00000500)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	STRME	Stride Mode Enable: 0: Disable 1: Enable	R/W	0x0
23:15	-	Reserved	R	0x0
14	YDAM	Y Destination address mode 0: increment 1: decrement	R/W	0x0
13:9	-	Reserved	R	0x0
8	DSTSL	Destination select: 0: memory 1: Reserved	R	0x0
7	-	Reserved	R	0x0

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6	YSAM	Y Source address mode 0: increment 1: decrement	R/W	0x0
5:1	-	Reserved	R	0x0
0	SRCSL	Source Select: 0: memory	R	0x0

8.4.34 SDMA4START

DMA4START (DMA Start Register, offset = 0x00000504)

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	DMASTART	DMA start bit: A low-to-high conversion of this bit will lead to load source address, destination address, DRQ type to the DMA controller. This bit will be automatically cleared by the DMAx controller if the DMAx transmission is complete or DMAx transmission error occurs. This bit can be written '0' to abort DMAx transmission.	R/W	0x0

8.4.35 SDMA4SADDR

DMA4SADDR (DMA Source Address Register, offset = 0x00000508)

Bit(s)	Name	Description	R/W	Reset
31:0	DMASADDR	The source address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.36 SDMA4DADDR

DMA4DADDR (DMA Destination Address Register, offset = 0x00000510)

Bit(s)	Name	Description	R/W	Reset
31:0	DMADADDR	The destination address of DMA transmission. The bit[0] is no effect if data width is 16-bit. The bit[1..0] is no effect if data width is 32-bit.	R/W	0x0

8.4.37 SDMA4BC

DMA4BC (DMA Byte Counter Address Register, offset = 0x00000518)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMABYTECOUNTER	The frame length of DMA transmission.	R/W	0x0

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8.4.38 SDMA4RC

DMA4RC (DMA Remain Counter Address Register, offset = 0x0000051C)

Bit(s)	Name	Description	R/W	Reset
31:20	-	Reserved	R	0x0
19:0	DMAREMAINCOUNTER	The Remain Counter of DMA transmission.	R	0x0

8.4.39 LINE_LENGTH0

LINE_LENGTH0(Line Length Address Register, offset = 0x00000C00)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	LENGTH	Line Length.	R/W	0x0

8.4.40 LINE_COUNT0

LINE_COUNT0(Line Counter Address Register, offset = 0x00000C04)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNT	Line Count.	R/W	0x0

8.4.41 LINE_SRC_STRIDE0

LINE_STRIDE0(Line stride Address Register, offset = 0x00000C08)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.42 LINE_REMAIN0

LINE_REMAIN0(Line remain Address Register, offset = 0x00000C10)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Line remain counter in a transmission of stride mode	R	0x0

8.4.43 BYTE_REMAIN_IN_LINE0

BYTE_REMAIN_IN_LINE0(Line remain in line Address Register, offset = 0x00000C14)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0

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15:0	COUNTER	Byte remain counter in the line that transmitting.	R	0x0

8.4.44 LINE_LENGTH1

LINE_LENGTH1(Line Length Address Register, offset = 0x00000C20)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	LENGTH	Line Length.	R/W	0x0

8.4.45 LINE_COUNT1

LINE_COUNT1(Line Counter Address Register, offset = 0x00000C24)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNT	Line Count.	R/W	0x0

8.4.46 LINE_SRC_STRIDE1

LINE_STRIDE1(Line stride Address Register, offset = 0x00000C28)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.47 LINE_DST_STRIDE1

LINE_STRIDE1(Line stride Address Register, offset = 0x00000C2C)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.48 LINE_REMAIN1

LINE_REMAIN1(Line remain Address Register, offset = 0x00000C30)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Line remain counter in a transmission of stride mode	R	0x0

8.4.49 BYTE_REMAIN_IN_LINE1

BYTE_REMAIN_IN_LINE1(Line remain in line Address Register, offset = 0x00000C34)

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Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Byte remain counter in the line that transmitting.	R	0x0

8.4.50 LINE_LENGTH2

LINE_LENGTH2(Line Length Address Register, offset = 0x00000C40)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	LENGTH	Line Length.	R/W	0x0

8.4.51 LINE_COUNT2

LINE_COUNT2(Line Counter Address Register, offset = 0x00000C44)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNT	Line Count.	R/W	0x0

8.4.52 LINE_SRC_STRIDE2

LINE_STRIDE2(Line stride Address Register, offset = 0x00000C48)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.53 LINE_DST_STRIDE2

LINE_STRIDE2(Line stride Address Register, offset = 0x00000C4C)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.54 LINE_REMAIN2

LINE_REMAIN2(Line remain Address Register, offset = 0x00000C50)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Line remain counter in a transmission of stride mode	R	0x0

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8.4.55 BYTE_REMAIN_IN_LINE2

BYTE_REMAIN_IN_LINE2(Line remain in line Address Register, offset = 0x00000C54)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Byte remain counter in the line that transmitting.	R	0x0

8.4.56 LINE_LENGTH3

LINE_LENGTH3(Line Length Address Register, offset = 0x00000C60)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	LENGTH	Line Length.	R/W	0x0

8.4.57 LINE_COUNT3

LINE_COUNT3(Line Counter Address Register, offset = 0x00000C64)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNT	Line Count.	R/W	0x0

8.4.58 LINE_SRC_STRIDE3

LINE_STRIDE3(Line stride Address Register, offset = 0x00000C68)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.59 LINE_DST_STRIDE3

LINE_STRIDE3(Line stride Address Register, offset = 0x00000C6C)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.60 LINE_REMAIN3

LINE_REMAIN3(Line remain Address Register, offset = 0x00000C70)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Line remain counter in a transmission of stride mode	R	0x0

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8.4.61 BYTE_REMAIN_IN_LINE3

BYTE_REMAIN_IN_LINE3(Line remain in line Address Register, offset = 0x00000C74)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Byte remain counter in the line that transmitting.	R	0x0

8.4.62 LINE_LENGTH4

LINE_LENGTH4(Line Length Address Register, offset = 0x00000C80)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	LENGTH	Line Length.	R/W	0x0

8.4.63 LINE_COUNT4

LINE_COUNT43(Line Counter Address Register, offset = 0x00000C84)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNT	Line Count.	R/W	0x0

8.4.64 LINE_SRC_STRIDE4

LINE_STRIDE4(Line stride Address Register, offset = 0x00000C88)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.65 LINE_DST_STRIDE4

LINE_STRIDE4(Line stride Address Register, offset = 0x00000C8C)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	STRIDE	Line Stride.	R/W	0x0

8.4.66 LINE_REMAIN4

LINE_REMAIN4(Line remain Address Register, offset = 0x00000C90)

Bit(s)	Name	Description	R/W	Reset

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31:16	-	Reserved	R	0x0
15:0	COUNTER	Line remain counter in a transmission of stride mode	R	0x0

8.4.67 BYTE_REMAIN_IN_LINE4

BYTE_REMAIN_IN_LINE4(Line remain in line Address Register, offset = 0x00000C94)

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	COUNTER	Byte remain counter in the line that transmitting.	R	0x0

8.4.68 COLOR_FILL_DATA0

COLOR_FILL_DATA0(COLOR Fill Data0 Register, offset = 0x00000D00)

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	Color fill data	R/W	0x0

8.4.69 COLOR_FILL_DATA1

COLOR_FILL_DATA1(COLOR Fill Data1 Register, offset = 0x00000D04)

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	Color fill data	R/W	0x0

8.4.70 COLOR_FILL_DATA2

COLOR_FILL_DATA2(COLOR Fill Data2 Register, offset = 0x00000D08)

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	Color fill data	R/W	0x0

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9 PMU

9.1Features

The ATS3089x integrates a comprehensive power supply system, including the following features:

- Supports 4.2~4.5V Li-Ion battery and Integrate 400mA Linear battery charger with power path management.
- 2 buck converters output VC18, VD12 from BAT.
- Linear regulators output SVCC, VCC, VCC2, VDD.
- Integrated 6*Load switch (VPSRAM/2*VCC_OUT/IOVCC1_OUT/IOVCC2_OUT/IOVCC4_OUT), which can be used as power supply for peripheral.
- Internal bandgap reference.

9.2Module Description

9.2.1 Block Diagram

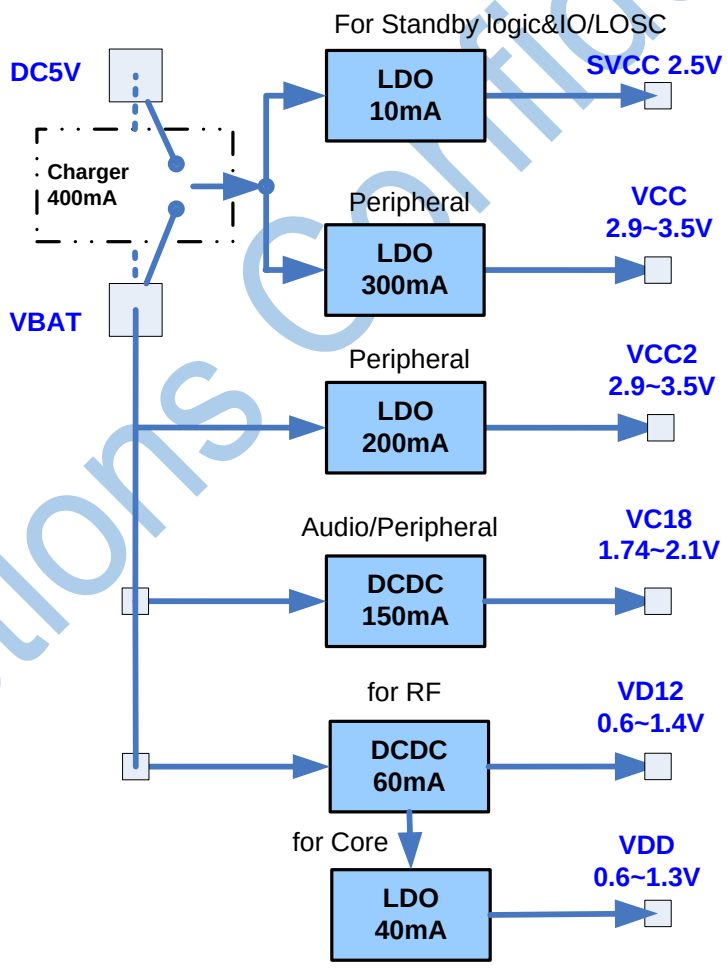


Figure 9-1 Power Supply Block Diagram

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9.2.2 Power on timing

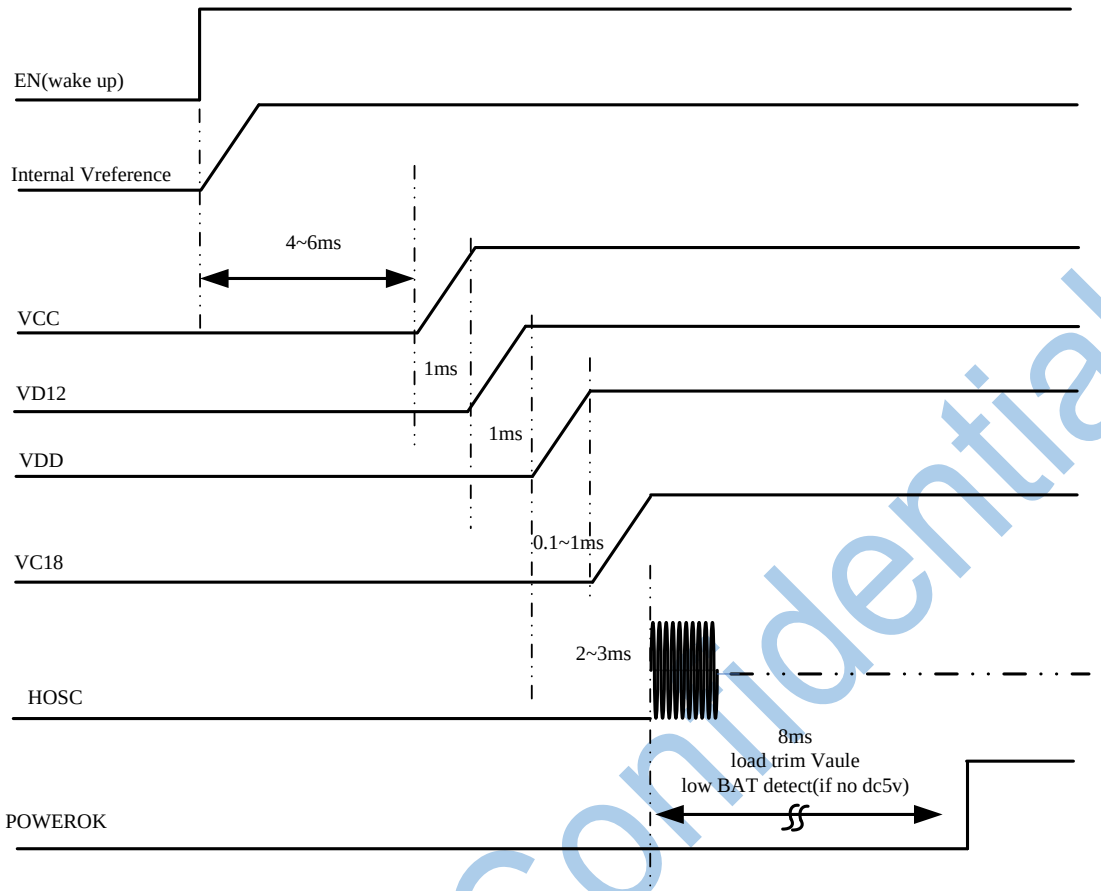


Figure 9-2 Diagram of power on timing

9.2.3 State machine of PMU

Description of PMU States:

- S1——active mode, Main CPU/BT is active
- S2——sensor transmit mode, all processors are in sleep, only sensor controller working
- S3——standby, all processors are in deep sleep, can be waked up by interrupt
- S4——shutdown, all regulators are power down except SVCC, can be waked up by DC5V/8HZ timer/WIO pins

Table 9-1 The states of Regulators

PMU State	VCC/VDD/VD12	VC18	VCC2	SVCC
S1/S2/S3	Enable	Programmable	Programmable	Enable
S4	Disable			Enable

9.2.4 DC-DC Converter

ATS3089x integrates 2 buck DC-DC converters which are named VD12/VC18. VD12/VC18 can also be configured as LDO mode. The DC-DC converter efficiently scales battery voltage to the required supply voltage. The DC-DC converters include several advanced features:

- Input power from BAT

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- Support 2.2uH power inductor (>300mA Isat).

Table 9-2 The feature of DC-DC

DCDC	Output Range(V)	Inductor	Capacitor	Inductor Peak Current	Max Load	Output Voltage	Efficiency (Typical)	Ripple	Freq
VD12	0.6~1.4	2.2uH	⩾4.7uF	200mA	60mA	0.6V	50% (20uA)	-	-
						1.2V	80% (10mA)	20mV	~2M
VC18	1.74~2.1	2.2uH	⩾4.7uF	300mA	150mA	1.86V	85% (10mA)	20mV	~2M

9.2.5 Linear Regulators

ATS3089x integrates 6 linear regulators respectively generate SVCC, VCC, VCC2, VD12, VC18, VDD.
All regulators have soft start, SVCC/VCC/VCC2/VC18/VDD have low voltage protection and VCC/VCC2 has over current protection.

9.2.5.1 Regulators Accurate and Maximum Output Current

The output voltages are precisely within ±2.5%, providing maximum output currents when voltages drop to 95%. Table9-3 shows data of maximum output current.

Table 9-3 Regulators Maximum Output Current

Block Name	Output Voltage	Maximum Output Current
SVCC	2.5V/2.0V(Shipping mode)	10 mA
VCC	2.9~3.5V	300 mA
VCC2	2.9~3.5V	200 mA
VC18	1.74~2.1	50 mA(LDO mode)
VD12	0.6~1.4V	60 mA(LDO mode)
VDD	0.6~1.3V	40 mA

9.2.6 Load switch

ATS3089x integrated 6*Load switch (VPSRAM/2*VCC_OUT/IOVCC1_OUT/IOVCC2_OUT/IOVCC4_OUT), which can be used as power supply for peripheral. VPSRAM is used for PSRAM power supply, while other power supplies can be sent out through specific GPIOs.

Table 9-4 Load switch description

Block Name	Output Voltage	Power source
VCC_OUT	2.9~3.5V	VCC
IOVCC1_OUT	2.9~3.5V	VCC
IOVCC2_OUT	2.9~3.5V	VCC/VC18
IOVCC4_OUT	1.74~2.1	VC18
VPSRAM	1.74~2.1	VC18

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9.2.7 A/D Converters

There are 12 bits multi-channel SARADC for system monitor.

Table 9-4 Input range of ADC channels

LRADC Channel	Input range
CHARGEI	0~512mA
BAT	0V~4.8V
DC5V	0V~6.0V
SENSOR	-40~120 ℃
SVCC	0~3.6V
LRADC1 (SVCC domain)	0~3.6V
VCCI	0~512mA
LRADC2/3/4/5/6 (VCC/VC18 domain)	0~3.6V/0~1.2V

Note: The input voltage of LRADC can not exceed its power supply voltage.

9.3PMUVDD Register List

Table 9-5 PMUVDD block base address

Name	Physical Base Address	KSEG1 Base Address
PMUVDD	0x40004000	0x40004000

Table 9-6 PMUVDD Block Configuration Registers List

Offset	Register Name	Description	Power domain
0x00	VOUT_CTL0	PMU control Register	VDD
0x04	VOUT_CTL1_S1	PMU control under S1	VDD
0x08	VOUT_CTL1_S2	PMU control under S2	VDD
0X0C	VOUT_CTL1_S3	PMU control under S3	VDD
0x10	PMU_DET	PMU status Register	VDD
0x14	VOUT_CTL1_S1M	PMU control under S1M Register	VDD
0X30	PWRGATE_DIG	Digital module Power gating control	VDD
0X34	PWRGATE_DIG_ACK	Digital module Power up acknowledgement	VDD
0x38	RAM_SHUTDOWN	RAM Power gating control register	VDD
0x40	PMU_INTMASK	PMU interrupt set register	VDD
0x44	RAM_DEEPSLEEP	RAM Retention control register	VDD
0x48	RAM_LIGHTSLEEP	RAM low power control register	VDD

9.4PMUVDD Register Description

9.4.1 VOUT_CTL0

Voltage set register (VDD), offset: 0x00

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Bit (s)	Name	Description	Access	Reset
31	VPSRAM_EN	VPSRAM output enable： 0：disable 1：enable	R/W	0
30:22	-	Reserved	R/W	X
21:20	VCC_SET_DC5V	VCC voltage level select when DC5V generate it. 0x0: 3.0V 0x1: 3.2V 0x2: 3.4V 0x3: 3.6V	R/W	0x2
19	-	Reserved	R/W	X
18	VC18_SW	The mode switch of VC18. 0：LDO 1：DCDC	R/W	0
17	VD12_SW	The mode switch of VD12. 0：LDO 1：DCDC	R/W	0
16	-	Reserved	R/W	0
15	VCC_OC_SET	VCC LDO Current limit: 0: 250mA 1: 450mA	R/W	0
14	VCC2_OC_SET	VCC2 LDO Current limit: 0: 250mA 1: 450mA	R/W	0
13:12	VCC_SET	VCC voltage level select when BAT generate it. 00 2.9V 01 3.1V *10 3.3V 11 3.5V	R/W	10
11:10	VC18_SET	VC18 voltage level select 00: 1.74V *01: 1.86V 10: 1.98V 11: 2.1V	R/W	01
9:8	VCC2_SET	VCC2 voltage level select 00 2.9V *01 3.1V 10 3.3V 11 3.5V	R/W	01
7:4	-	Reserved	R/W	0xc
3	VCC2_EN	VCC2 LDO enable. 0：disable 1：enable	R/W	0
2:0	-	Reserved	R/W	0x1

9.4.2 VOUT_CTL1_S1

Voltage set register (VDD), offset: 0x04

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0xc1
23:22	VDD_BYPASS_S1BT	VDD LDO enable control in S1BT state	R/W	11

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		00: VDD LDO disable, bypass to VDD12 01：auto bypass when VD12 <=VDD 10：auto bypass when VD12 < =VDD+50mV 11：enable LDO OP		
21	VC18_EN_S1	VC18 enable： 0: disable 1: enable		1
20:12	-	Reserved	R/W	0x1f8
11:8	VD12_SET_S1BT	VD12 voltage set by BT subsystem. 0000: 0.6V 0001: 0.65V 0010: 0.7V 0011: 0.75V 0100: 0.8V 0101: 0.85V 0110: 0.9V 0111: 0.95V 1000: 1.0V 1001: 1.05V 1010: 1.1V 1011: 1.15V 1100: 1.2V 1101: 1.25V *1110: 1.3V 1111: 1.4V	R	X
7:4	VDD_SET_DSP	VDD voltage set by DSP subsystem. 0000: Reserved 0001: 0.60V 0010: 0.65V 0011: 0.70V 0100: 0.75V 0101: 0.80V 0110: 0.85V 0111: 0.90V 1000: 0.95V 1001: 1.0V 1010: 1.05V 1011: 1.1V 1100: 1.15V 1101: 1.2V 1110: 1.25V 1111: 1.3V	R	X
3:0	VDD_SET_S1BT	VDD voltage set by BT subsystem. 0000: Reserved 0001: 0.60V 0010: 0.65V 0011: 0.70V 0100: 0.75V 0101: 0.80V 0110: 0.85V 0111: 0.9V 1000: 0.95V 1001: 1.0V 1010: 1.05V	R	X

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		1011: 1.1V 1100: 1.15V 1101: 1.2V 1110: 1.25V 1111: 1.3V		

9.4.3 VOUT_CTL1_S2

Voltage set register (VDD), offset: 0x08

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0x13
23:22	VDD_BYPASS_S2	VDD LDO enable control in S2 state： 00：bypass to VDD12, VDD LDO disable 01：auto bypass when VD12 =VDD 10：auto bypass when VD12 < =VDD+50mV 11：enable VDD LDO	R/W	11
21	VC18_EN_S2	VC18 enable 0: disable 1: enable	R/W	0x1
20:12	-	Reserved	R/W	0x0F8
11:8	VD12_SET_S2	VD12 voltage control of S2 state. 0000: 0.6V 0001: 0.65V 0010: 0.7V 0011: 0.75V 0100: 0.8V 0101: 0.85V *0110: 0.9V 0111: 0.95V 1000: 1.0V 1001: 1.05V 1010: 1.1V 1011: 1.15V 1100: 1.2V 1101: 1.25V 1110: 1.3V 1111: 1.4V	R/W	0110
7:3	-	Reserved	R/W	01010
2:0	VDD_SET_S2	VDD voltage control of S2 state. 000 Reserved 001 0.60V 010 0.65V 011 0.70V 100 0.75V *101 0.80V 110 0.85V 111 0.90V	R/W	101

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9.4.4 VOUT_CTL1_S3

Voltage set register (VDD), offset: 0x0C

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0x3
23:22	VDD_BYPASS_S3	VDD LDO enable control in S3 state： 00：bypass to VDD12, VDD LDO disable 01：auto bypass when VD12 =VDD 10：auto bypass when VD12 < =VDD+50mV 11：enable VDD LDO	R/W	11
21	VC18_EN_S3	VC18 enable 0: disable 1: enable	R/W	1
20:15	-	Reserved	R/W	0
14:12	VD12_SET_S3_HOSCEN	VD12 voltage control of S3 state when HOSC is enable. 000: 0.6V 001: 0.65V 010: 0.7V 011: 0.75V *100: 0.8V 101: 0.85V 110: 0.9V 111: 0.95V	R/W	100
11	-	Reserved	R	0
10:8	VD12_SET_S3	VD12 voltage control of S3 state when HOSC is disable. 000: 0.6V 001: 0.65V 010: 0.7V 011: 0.75V *100: 0.8V 101: 0.85V 110: 0.9V 111: 0.95V	R/W	100
7	-	Reserved	R	0
6:4	VDD_SET_S3_HOSCEN	VDD voltage control of S3 state when HOSC is enable. 000 Reserved 001 0.60V 010 0.65V 011 0.70V 100 0.75V *101 0.80V 110 0.85V 111 0.90V	R/W	101
3	-	Reserved	R	0
2:0	VDD_SET_S3	VDD voltage control of S3 state when HOSC is disable. 000 Reserved 001 0.60V 010 0.65V *011 0.70V 100 0.75V 101 0.80V 110 0.85V	R/W	011

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		111 0.90V		

9.4.5 PMU_DET

Charger detect register (VDD), Offset: 0x10

Bit(s)	Name	Description	R/W	Reset
31:13	-	Reserved	R	0
12	COUNTER8HZ_PD	Counter8hz reset pending occurred 0: Pending not occurred 1: Pending occurred Writing 1 to the bit will clear the bit	R/W	0
11	S1MAIN_REQ	PMU is in S1M state or not	R	X
10	S1BT_REQ	PMU is in S1BT state or not	R	X
9	-	reserved	R	0
8	S2_REQ	PMU is in S2 state or not	R	X
7:1	-	reserved	R	0
0	DC5VIN_DET	The indication of DC5V is plug in. 0: DC5V<BAT+0.033V 1: DC5V> BAT+0.080V	R	X

9.4.6 VOUT_CTL1_S1M

Voltage set register (VDD) offset: 0x14

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R	0x21
23:22	VDD_BYPASS_S1M	VDD LDO enable control in S1M state 00：bypass to VDD12, VDD LDO disable 01：auto bypass when VD12 =VDD 10：auto bypass when VD12 < =VDD+50mV 11：enable VDD LDO	R/W	0x3
21:12	-	Reserved	R/W	0x3F9
11:8	VD12_SET_S1M	0000: 0.6V 0001: 0.65V 0010: 0.7V 0011: 0.75V 0100: 0.8V 0101: 0.85V 0110: 0.9V 0111: 0.95V 1000: 1.0V 1001: 1.05V 1010: 1.1V 1011: 1.15V *1100: 1.2V 1101: 1.25V 1110: 1.3V 1111: 1.4V	R/W	0xC
7:4	VDD_SET_S1ML	VDD voltage coarse control : (S1M light sleep) 0000: Reserved	R/W	0x9

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		0001: 0.60V 0010: 0.65V 0011: 0.70V 0100: 0.75V 0101: 0.80V 0110: 0.85V 0111: 0.90V 1000: 0.95V *1001:1.0V 1010: 1.05V 1011: 1.1V 1100: 1.15V 1101: 1.2V 1110: 1.25V 1111: 1.3V		
3:0	VDD_SET_S1M	VDD voltage coarse control : (S1M) 0000: Reserved 0001: 0.60V 0010: 0.65V 0011: 0.70V 0100: 0.75V 0101: 0.80V 0110: 0.85V 0111: 0.90V 1000: 0.95V 1001: 1.0V 1010: 1.05V *1011: 1.1V 1100: 1.15V 1101: 1.2V 1110: 1.25V 1111: 1.3V	R/W	0xB

9.4.7 PWRGATE_DIG

Power gating Control Register (VDD), Offset: 0x30

Bit(s)	Name	Description	R/W	Reset
31	MAINCPU_PG	Main CPU sub system power gating bit: 0：disable 1：enable When disable gating, power will shut down	R/W	1
30:29	-	Reserved	-	0
28	BT_PG	BT sub system power gating bit: 0：disable 1：enable When disable gating, power will shut down	R/W	1
27	-	Reserved	-	0
26	AUDSP_PG	AUDSP sub system power gating bit: 0：disable 1：enable When disable gating, power will shut down	R/W	1
25	2.5DGPU_PG	2.5DGPU power gating bit:	R/W	1

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		0：disable 1：enable When disable gating, power will shut down		
24:22	-	reserved	R/W	X
21	DISPLAY_PG	Display system power gating bit: 0：disable 1：enable When disable gating, power will shut down	R/W	1
20:0	-	Reserved	R	0x 100000

9.4.8 PWRGATE_DIG_ACK

Power gating Control Register (VDD), Offset: 0x34

Bit(s)	Name	Description	R/W	Reset
31	MAINCPU_ACK	Main CPU sub system power OK Acknowledgement: 0：NO ACK 1：ACK	R	X
30:29	-	Reserved	R	0
28	BT_ACK	BT sub system power OK Acknowledgement: 0：NO ACK 1：ACK	R	X
27	-	Reserved	R	0
26	AUDSP_ACK	AUDSP sub system power OK Acknowledgement: 0：NO ACK 1：ACK	R	X
25	2.5DGPU_ACK	2.5DGPU power OK Acknowledgement: 0：NO ACK 1：ACK	R	X
24:22	-	reserved	R	X
21	DISPLAY_ACK	Display system power OK Acknowledgement: 0：NO ACK 1：ACK	R	X
20:0	-	RESERVED	R	0

9.4.9 RAM_SHUTDOWN

RAM Power gating Control Register (VDD), Offset: 0x38

Bit(s)	Name	Description	R/W	Reset
31:27	-	Reserved	R	0x0
26	RAM_USB_SD	USB RAM 0：NOT 1：shut down	R/W	0x0
25:19	-	Reserved	R	0x0
18	JPEGRAM_SD	JPEG_OUTRAM 0：NOT 1：Shut Down	R/W	0x0

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17	SHARERAM_SD	SHARERAM 32k 0：NOT 1：Shut Down	R/W	0x0
16	RAM16_SD	RAM16 48k 0：NOT 1：Shut Down	R/W	0x0
15	RAM15_SD	RAM15 32k 0：NOT 1：Shut Down	R/W	0x0
14	RAM14_SD	RAM14 64k 0：NOT 1：Shut Down	R/W	0x0
13	RAM13_SD	RAM13 16k 0：NOT 1：Shut Down	R/W	0x0
12	RAM12_SD	RAM12 64k 0：NOT 1：Shut Down	R/W	0x0
11	RAM11_SD	RAM11 64k 0：NOT 1：Shut Down	R/W	0x0
10	RAM10_SD	RAM10 64k 0：NOT 1：Shut Down	R/W	0x0
9	RAM9_SD	RAM9 64k 0：NOT 1：Shut Down	R/W	0x0
8	RAM8_SD	RAM8 64k 0：NOT 1：Shut Down	R/W	0x0
7	RAM7_SD	RAM7 64k 0：NOT 1：Shut Down	R/W	0x0
6	RAM6_SD	RAM6 64k 0：NOT 1：Shut Down	R/W	0x0
5	RAM5_SD	RAM5 64k 0：NOT 1：Shut Down	R/W	0x0
4	RAM4_SD	RAM4 64k 0：NOT 1：Shut Down	R/W	0x0
3	RAM3_SD	RAM3 64k 0：NOT 1：Shut Down	R/W	0x0
2	RAM2_SD	RAM2 64k 0：NOT 1：Shut Down	R/W	0x0
1	RAM1_SD	RAM1 64k 0：NOT 1：Shut Down	R/W	0x0

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0	RAM0_SD	RAM0 64k 0：NOT 1：Shut Down	R/W	0x0

9.4.10 PMU_INTMASK

Power gating Control Register (VDD), Offset: 0x40

Bit(s)	Name	Description	R/W	Reset
31:13	-	RESERVED	R	0
12	REMOTE_INTEN	REMOTE interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
11	BATLV_INTEN	BAT low voltage interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
10	DC5VLV_INTEN	DC5V low voltage interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
9:7	-	reserved	R	0
6	WIO0LV_INTEN	WIO0 low voltage interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
5	-	reserved	R	0
4	ALARM8HZ_INTEN	ALARM8HZ interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0x0
3	DC5VOUT_INTEN	DC5V plug out interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
2	DC5VIN_INTEN	DC5V plug in interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	1
1	ONOFF_S_INTEN	ONOFF short press interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	0
0	ONOFF_L_INTEN	ONOFF long press interrupt enable 0: Do not send interrupt signal to CPU 1: Send interrupt signal to CPU	R/W	1

9.4.11 RAM_DEEPSLEEP

RAM Retention Control Register (VDD), Offset: 0x44

Bit(s)	Name	Description	R/W	Reset
31:27	-	Reserved	R	0x0
26	RAM_USB_DS	USB RAM 0：NOT 1：Deep sleep	R/W	0x0
25:20	-	Reserved	R	0x0

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18	JPEGRAM_DS	JPEG_OUTRAM 0：NOT 1：Deep sleep	R/W	0x0
17	SHARERAM_DS	SHARERAM 32k 0：NOT 1：Deep sleep	R/W	0x0
16	RAM16_DS	RAM16 48k 0：NOT 1：Deep sleep	R/W	0x0
15	RAM15_DS	RAM15 32k 0：NOT 1：Deep sleep	R/W	0x0
14	RAM14_DS	RAM14 64k 0：NOT 1：Deep sleep	R/W	0x0
13	RAM13_DS	RAM13 16k 0：NOT 1：Deep sleep	R/W	0x0
12	RAM12_DS	RAM12 64k 0：NOT 1：Deep sleep	R/W	0x0
11	RAM11_DS	RAM11 64k 0：NOT 1：Deep sleep	R/W	0x0
10	RAM10_DS	RAM10 64k 0：NOT 1：Deep sleep	R/W	0x0
9	RAM9_DS	RAM9 64k 0：NOT 1：Deep sleep	R/W	0x0
8	RAM8_DS	RAM8 64k 0：NOT 1：Deep sleep	R/W	0x0
7	RAM7_DS	RAM7 64k 0：NOT 1：Deep sleep	R/W	0x0
6	RAM6_DS	RAM6 64k 0：NOT 1：Deep sleep	R/W	0x0
5	RAM5_DS	RAM5 64k 0：NOT 1：Deep sleep	R/W	0x0
4	RAM4_DS	RAM4 64k 0：NOT 1：Deep sleep	R/W	0x0
3	RAM3_DS	RAM3 64k 0：NOT 1：Deep sleep	R/W	0x0
2	RAM2_DS	RAM2 64k 0：NOT 1：Deep sleep	R/W	0x0

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1	RAM1_DS	RAM1 64k 0：NOT 1：Deep sleep	R/W	0x0
0	RAM0_DS	RAM0 64k 0：NOT 1：Deep sleep	R/W	0x0

9.4.12 RAM_LIGHTSLEEP

Ram Lightsleep Control Register (VDD), Offset: 0x48

Bit(s)	Name	Description	R/W	Reset
31:27	-	Reserved	R	0x0
26	RAM_USB_LS	USB RAM 0：NOT 1：Light sleep	R/W	0x0
25:19	-	Reserved	R/W	0x0
18	JPEGRAM_LS	JPEG_OUTRAM 0：NOT 1：Light sleep	R/W	0x0
17	SHARERAM_LS	SHARERAM 32k 0：NOT 1：Light sleep	R/W	0x0
16	RAM16_LS	RAM16 48k 0：NOT 1：Light sleep	R/W	0x0
15	RAM15_LS	RAM15 32k 0：NOT 1：Light sleep	R/W	0x0
14	RAM14_LS	RAM14 64k 0：NOT 1：Light sleep	R/W	0x0
13	RAM13_LS	RAM13 16k 0：NOT 1：Light sleep	R/W	0x0
12	RAM12_LS	RAM12 64k 0：NOT 1：Light sleep	R/W	0x0
11	RAM11_LS	RAM11 64k 0：NOT 1：Light sleep	R/W	0x0
10	RAM10_LS	RAM10 64k 0：NOT 1：Light sleep	R/W	0x0
9	RAM9_LS	RAM9 64k 0：NOT 1：Light sleep	R/W	0x0
8	RAM8_LS	RAM8 64k 0：NOT 1：Light sleep	R/W	0x0

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7	RAM7_LS	RAM7 64k 0：NOT 1：Light sleep	R/W	0x0
6	RAM6_LS	RAM6 64k 0：NOT 1：Light sleep	R/W	0x0
5	RAM5_LS	RAM5 64k 0：NOT 1：Light sleep	R/W	0x0
4	RAM4_LS	RAM4 64k 0：NOT 1：Light sleep	R/W	0x0
3	RAM3_LS	RAM3 64k 0：NOT 1：Light sleep	R/W	0x0
2	RAM2_LS	RAM2 64k 0：NOT 1：Light sleep	R/W	0x0
1	RAM1_LS	RAM1 64k 0：NOT 1：Light sleep	R/W	0x0
0	RAM0_LS	RAM0 64k 0：NOT 1：Light sleep	R/W	0x0

9.5PMUSVCC Register List

Table 9-7 PMUSVCC block base address

Name	Physical Base Address	KSEG1 Base Address
PMUSVCC	0x40004000	0x40004000

Table 9-8 PMUSVCC Block Configuration Registers List

Offset	Register Name	Description	Power domain
0X100	CHG_CTL_SVCC	CHARGING CONTROL	SVCC
0X108	SYSTEM_SET_SVCC	POWER SYSTEM & ONOFF SET	SVCC
0X10C	POWER_CTL_SVCC	POWER OFF CONTROL	SVCC
0X110	WAKE_CTL_SVCC	WAKE UP & INTERRUPT SOURCE CONTROL	SVCC
0X114	WAKE_PD_SVCC	WAKE UP PENDING	SVCC
0X118	COUNTER8HZ_SVCC	8-HZ COUNTER	SVCC
0X11C	ALARM8HZ_SVCC	COUNTER ALARM	SVCC

9.6PMUSVCC Register Description

9.6.1 CHG_CTL_SVCC

Charger control register (SVCC), Offset: 0x100

Bit(s)	Name	Description	R/W	Reset
31:19	-	reserved	R/W	X

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18	CV_IN100	Increase the voltage of BAT 100mv when charger is in CV mode. 0: Keep the output voltage of the charger at CV value. 1: Increase the output voltage of the charger at CV+100mv.	R/W	0
17:11	-	reserved	R/W	X
10	CHG_EN	CHARGER ENABLE BIT 0：DISABLE 1：ENABLE	R/W	1
9:5	CV_OFFSET	The voltage setting when charger in CV mode. 00000:4.1375V ……. 11111:4.525V Step=12.5mV	R/W	01001
4	CV_DE100	Reduce the voltage of BAT 100mv when charger is in CV mode. 0: Keep the output voltage of the charger at CV value. 1: Reduce the output voltage of the charger at CV-100mv.	R/W	0
3:0	CHG_CURRENT	The current setting when charger in CC mode. *0000: 20mA 0001: 30mA 0010: 40mA 0011: 50mA 0100: 60mA 0101: 70mA 0110: 80mA 0111: 90mA 1000: 100mA 1001: 150mA 1010: 200mA 1011: 250mA 1100: 300mA 1101: 350mA 1110: 400mA 1111: 450mA	R/W	0x0

9.6.2 SYSTEM_SET_SVCC

System set Register (SVCC), Offset: 0x108

Bit(s)	Name	Description	R/W	Reset
31:28	-	reserved	R	0
27	DC5VRST_EN	DC5V reset enable 0:disable 1:enable	R/W	0
26:24	-	reserved	R/W	010
23:21	ONOFF_PRESS_TIME	ONOFF press time setting 000: 50ms < t < 0.125s, regarded as short press; t >=0.125s, regarded as long press; 001: 50ms < t < 0.25s, regarded as short press; t >=0.25s, regarded as long press; 010: 50ms < t < 0.5s, regarded as short press;	R/W	001

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		t >=0.5s, regarded as long press; 011: 50ms < t < 1s, regarded as short press; t >=1s, regarded as long press; 100: 50ms < t < 1.5s, regarded as short press; t >=1.5s, regarded as long press; 101: 50ms < t < 2s, regarded as short press; t >=2s, regarded as long press; 110: 50ms < t < 3s, regarded as short press; t >=3s, regarded as long press; 111: 50ms < t < 4s, regarded as short press; t >=4s, regarded as long press;		
20:19	ONOFF_RST_EN	ONOFF long press function selection. 00：no action 01：reset the chip 10：restart the chip	R/W	01
18	ONOFF_RST_TIME	ONOFF long press reset/restart time configuration. 0:8s 1:12s	R/W	0
17	ONOFF_PRES	ONOFF press state indication. 0: ONOFF is not press down 1: ONOFF is press down	R	0
16	PULLR_SEL	ONOFF press down voltage select. 0：Regarded as ONOFF pin press down When ONOFF detect high 1：Regarded as ONOFF pin press down When ONOFF detect low	R/W	0
15:12	-	Reserved	R/W	0
11	LB_EN	Entry shutdown mode when low battery. 0：disable 1：enable	R/W	1
10:9	LB_VOL	LB (Low battery) voltage setting 00: 2.8V 01: 2.9V *10: 3.0V 11: 3.3V	R/W	10
8	OC_EN	VCC over current protection enable bit. 0：disable 1：enable	R/W	1
7:0	-	Reserved	R/W	X

9.6.3 POWER_CTL_SVCC

Power Control Register (SVCC), Offset: 0x10C

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0
0	ENTER_S4	0: enter S4	R/W	1

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		1: default		

9.6.4 WAKE_CTL_SVCC

Wake up source enable Register (SVCC), Offset: 0x110

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R/W	0
29:27	BATLV_VOL	The threshold of BAT low voltage wakeup. 000：3.4V 001：3.5V 010：3.6V 011：4.1V 100：4.2V 101：4.25V 110：4.3V 111：4.35V	R/W	000
26:24	DC5VLV_VOL	The threshold of DC5V low voltage wakeup. 000：0.2V 001：0.5V 010：1.0V 011：1.5V 100：2.0V 101：3V 110：4V 111：4.5V	R/W	000
23:22	-	Reserved	R/W	00
21:20	WIO0LV_VOL	WIO0 low voltage selection. 00：<0.45V; 01：<1V; 10：<1.25V; 11：<1.5V;	R/W	01
19:13	-	Reserved	R/W	X
12	REMOTE_WKEN	Remote wakeup enable 0：disable 1：enable	R/W	0
11	BATLV_WKEN	BAT low voltage wakeup enable 0：disable 1：enable	R/W	0
10	DC5VLV_WKEN	DC5V low voltage wakeup enable 0：disable 1：enable	R/W	0
9	-	Reserved	R/W	0
8	WIO0LV_DETEN	WIO0 low voltage detect enable 0：disable 1：enable	R/W	1
7	-	Reserved	R/W	0
6	WIO0LV_WKEN	WIO0 low voltage wakeup enable 0：disable 1：enable	R/W	0
5	WIO_WKEN	WIO wakeup enable	R/W	0

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		0：disable 1：enable		
4	ALARM8HZ_WKEN	ALARM8hz wakeup enable 0：disable 1：enable	R/W	0
3	DC5VOUT_WKEN	DC5V plug out wakeup enable 0：disable 1：enable	R/W	0
2	DC5VIN_WKEN	DC5V plug in wakeup enable 0：disable 1：enable	R/W	1
1	SHORT_WKEN	ONOFF short press wakeup enable 0: disable 1: enable	R/W	0
0	LONG_WKEN	ONOFF long press wakeup enable 0: disable 1: enable	R/W	1

9.6.5 WAKE_PD_SVCC

WAKE up source pending Register (SVCC), Offset: 0x114

Bit(s)	Name	Description	R/W	Reset
31:23	-	Reserved	R/W	0
22	ONOFF_8S_RESET	ONOFF_8S reset pending occurred 0: Pending not occurred 1: Pending occurred Writing 1 to the bit will clear the bit	R/W	0
21	WD_RST_PD	WD reset pending occurred 0: Pending not occurred 1: Pending occurred Writing 1 to the bit will clear the bit	R/W	0
20	DC5VIN_RST_PD	DC5VIN reset pending： 0: No DC5VIN reset occurred 1: DC5VIN reset occurred	R/W	0
19	-	Reserved	R/W	0
18	LB_PD	Low battery protection pending. 0: No Low battery occurred 1: Low battery occurred	R/W	0
17	OC_PD	Over current pending 0: No Over current occurred 1: Over current occurred	R/W	0
16	LVPRO_PD	Low voltage protect pending. 0: Low voltage protect is not occurred 1: Low voltage protect is occurred	R/W	0
15:13	-	Reserved	R	0
12	REMOTE_PD	Remote pending. 0: Remote pending is not occurred 1: Remote pending is occurred	R/W	0
11	BATLV_PD	BAT low voltage pending. 0: BAT low voltage is not occurred 1: BAT low voltage is occurred	R/W	0

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10	DC5VLV_PD	DC5V low voltage pending. 0: DC5V low voltage is not occurred 1: DC5V low voltage is occurred	R/W	0
9:7	-	Reserved	R	0
6	WIO0LV_PD	WIO0 low voltage pending. 0: WIO0 low voltage is not occurred 1: WIO0 low voltage is occurred	R/W	0
5	WIO_PD	WIO wakeup pending. 0: WIO wakeup is not occurred 1: WIO wakeup is occurred	R/W	0
4	ALARM8HZ_PD	ALARM8HZ pending. 0: ALARM8HZ is not occurred 1: ALARM8HZ is occurred	R/W	0
3	DC5VOUT_PD	DC5V plug out pending. 0: DC5V plug out is not occurred 1: DC5V plug out is occurred	R/W	0
2	DC5VIN_PD	DC5V plug in pending. 0: DC5V plug in is not occurred 1: DC5V plug in is occurred	R/W	0
1	ONOFF_S_PD	ONOFF short press pending. 0: ONOFF short press is not occurred 1: ONOFF short press is occurred	R/W	0
0	ONOFF_L_PD	ONOFF long press pending. 0: ONOFF long press is not occurred 1: ONOFF long press is occurred	R/W	0

9.6.6 COUNTER8HZ_SVCC

SVCC 8hz counter Register (SVCC), offset: 0x118

Bit(s)	Name	Description	R/W	Reset
31:28	-	Reserved	R	0x0
27	CLK_SEL	Counter8hz clock source select. 0：RC32k 1：LOSC	RW	0x0
26	COUNTER_EN	Counter enable 0: disable 1: enable	RW	0x0
25:0	COUNTER_VAL	8hz counter value.	R	0x0

9.6.7 ALARM8HZ_SVCC

SVCC 8hz Alarm Register (SVCC), offset: 0x11c

Bit(s)	Name	Description	R/W	Reset
31:27	-	Reserved	R	0x0
26	ALARM8HZ_PD	ALARM8HZ pending	R	0
25:0	ALARM_VAL	8hz Alarm value setting	RW	0x0

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9.7PMUADC Register List

Table 9-9 PMUADC block base address

Name	Physical Base Address	KSEG1 Base Address
PMUADC	0x40004000	0x40004000

Table 9-10 PMUADC Block Configuration Registers List

Offset	Register Name	Description	Power domain
0x200	PMUADC_CTL	SARADC control	VDD
0X204	PMUADC_INTMASK	SARADC sample interrupt set	VDD
0X208	PMUADC_PD	SARADC sample pending	VDD
0X20C	PMUADCDIG_CTL	SARADC digital control	VDD
0x210	CHARGI_DATA	Measure Charging current	VDD
0x214	BATADC_DATA	Measure battery voltage	VDD
0x218	DC5VADC_DATA	Measure DC5V voltage	VDD
0x21C	SENSADC_DATA	IC Temperature sensor	VDD
0x220	SVCCADC_DATA	Measure SVCC voltage	VDD
0x224	LRADC1_DATA	Measure IO voltage	VDD
0x228	VCCI_DATA	Measure BAT to VCC current voltage	VDD
0x22C	LRADC2_DATA	Measure IO voltage	VDD
0x230	LRADC3_DATA	Measure IO voltage	VDD
0x234	LRADC4_DATA	Measure IO voltage	VDD
0x238	LRADC5_DATA	Measure IO voltage	VDD
0x23C	LRADC6_DATA	Measure IO voltage	VDD

9.8PMUADC Register Description

9.8.1 PMUADC_CTL

PMUADC Control Register (VDD), Offset = 0x200

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	-	X
23	PMUADC_EN	ADC enable： 0：disable 1：enable	R/W	0
22:18	-	Reserved	-	0x0a
17	LRADC6_SCAL	The voltage range of ADC6 input. 0: 0~VCC, enable the voltage dividing resistor; 1: 0~1.2V, bypass the voltage dividing resistor;	R/W	0
16	LRADC5_SCAL	The voltage range of ADC5 input. 0: 0~VCC, enable the voltage dividing resistor; 1: 0~1.2V, bypass the voltage dividing resistor;	R/W	0
15	LRADC4_SCAL	The voltage range of ADC4 input. 0: 0~VCC, enable the voltage dividing resistor; 1: 0~1.2V, bypass the voltage dividing resistor;	R/W	0
14	LRADC3_SCAL	The voltage range of ADC3 input. 0: 0~VCC, enable the voltage dividing resistor; 1: 0~1.2V, bypass the voltage dividing resistor;	R/W	0

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13	LRADC2_SCAL	The voltage range of ADC2 input. 0: 0~VCC, enable the voltage dividing resistor; 1: 0~1.2V, bypass the voltage dividing resistor;	R/W	0
12	-	Reserved	-	00
11	LRADC6_CHEN	LRADC6 A/D channel enable. 0: Disable 1: Enable	R/W	0
10	LRADC5_CHEN	LRADC5 A/D channel enable. 0: Disable 1: Enable	R/W	0
9	LRADC4_CHEN	LRADC4 A/D channel enable. 0: Disable 1: Enable	R/W	0
8	LRADC3_CHEN	LRADC3 A/D channel enable. 0: Disable 1: Enable	R/W	0
7	LRADC2_CHEN	LRADC2 A/D channel enable. 0: Disable 1: Enable	R/W	0
6	VCCI_CHEN	BAT-VCC current A/D channel enable. 0: Disable 1: Enable	R/W	0
5	LRADC1_CHEN	LRADC1 A/D channel enable. 0: Disable 1: Enable	R/W	1
4	SVCC_CHEN	SVCC A/D channel enable 0: Disable 1: Enable	R/W	0
3	SENSOR_CHEN	Sensor A/D channel enable 0: Disable, sensor circuit and output disable 1: Enable, sensor circuit and output enable	R/W	0
2	DC5V_CHEN	DC5V A/D channel enable 0: Disable 1: Enable	R/W	0
1	BATV_CHEN	Battery A/D channel enable 0: Disable 1: Enable	R/W	1
0	CHARGI_CHEN	CHARGI A/D channel enable 0: Disable 1: Enable	R/W	0

9.8.2 PMUADC_INTMASK

PMUADC Control Register (VDD), Offset = 0x204

Bit(s)	Name	Description	R/W	Reset
31:12	-	Reserved	R	0x0
11	LRADC6_INTEN	LRADC6 A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
10	LRADC5_INTEN	LRADC5 A/D interrupt enable. 0: Disable	R/W	0x0

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		1: Enable		
9	LRADC4_INTEN	LRADC4 A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
8	LRADC3_INTEN	LRADC3 A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
7	LRADC2_INTEN	LRADC2 A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
6	VCCI_INTEN	VCCI A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
5	LRADC1_INTEN	LRADC1 A/D interrupt enable. 0: Disable 1: Enable	R/W	0x0
4	SVCC_INTEN	SVCC A/D interrupt enable 0: Disable 1: Enable	R/W	0x0
3	SENSOR_INTEN	Sensor A/D interrupt enable 0: Disable 1: Enable	R/W	0x0
2	DC5V_INTEN	DC5V A/D interrupt enable 0: Disable 1: Enable	R/W	0x0
1	BATV_INTEN	Battery A/D interrupt enable 0: Disable 1: Enable	R/W	0x0
0	CHARGI_INTEN	CHARGI A/D interrupt enable 0: Disable 1: Enable	R/W	0x0

9.8.3 PMUADC_PD

PMUADC Control Register (VDD), Offset = 0x208

Bit(s)	Name	Description	R/W	Reset
31:21	-	Reserved	R	0
11	LRADC6_PD	LRADC6 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
10	LRADC5_PD	LRADC5 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
9	LRADC4_PD	LRADC4 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
8	LRADC3_PD	LRADC3 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
7	LRADC2_PD	LRADC2 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
6	VCCI_PD	VCCI A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
5	LRADC1_PD	LRADC1 A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0

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4	SVCC_PD	SVCC A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
3	SENSOR_PD	Sensor A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
2	DC5V_PD	DC5V A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
1	BATV_PD	Battery A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0
0	CHARGI_PD	CHARGI A/D data ready pending Writing 1 to the bit will clear the bit	R/W	0

9.8.4 PMUADCDIG_CTL

PMUADC Control Register (VDD), Offset = 0x20C

Bit(s)	Name	Description	R/W	Reset
31:30	LRADC6_AVG	LRADC6 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
29:28	LRADC5_AVG	LRADC5 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
27:26	LRADC4_AVG	LRADC4 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
25:24	LRADC3_AVG	LRADC3 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
23:22	LRADC2_AVG	LRADC2 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
21:20	VCCI_AVG	VCCI over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
19:18	LRADC1_AVG	LRADC1 over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0

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17:16	SVCCADC_AVG	SVCCADC over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
15:14	SENSADC_AVG	SENSADC over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
13:12	DC5VADC_AVG	DC5VADC over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
11:10	BAT_AVG	BATADC over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
9:8	CHARGI_AVG	CHARGI over sample setting. 00：do not oversampling 01：average after 4 oversampling 10：average after 8 oversampling 11：average after 16 oversampling	R/W	0x0
7:6	-	Reserved	R/W	0x0
5	LRADC2_R_SEL	LRADC2 VCC pull up resistance enable. The precision of this resistance is 100k±1%(TYP). 0: disable 1: enable	R/W	0x0
4:0	-	Reserved	R/W	X

9.8.5 CHARGI_DATA

Charger current ADC DATA Register (VDD), Offset = 0x210

Bit(s)	Name	Description	R/W	Reset
31:12	-	RESERVED	R	0
12:0	CHARGI	Charger current ADC, used to detect Battery charging current.	R	xx

9.8.6 BATADC_DATA

BAT Voltage ADC DATA Register (VDD), Offset = 0x214

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	BATV	BAT Voltage ADC, used to detect Battery voltage.	R	xx

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		The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling. Input voltage range is: Li-ion: 0-4.8V		

9.8.7 DC5VADC_DATA

DC5V ADC DATA Register (VDD), Offset = 0x218

Bit(s)	Name	Description	R/W	Reset
31:12	-	RESERVED	R	0
11:0	DC5V	DC5V Voltage ADC, used to detect DC5V voltage. Input voltage range is: 0-6V using half range	R	xx

9.8.8 SENSADC_DATA

Sensor ADC DATA Register (VDD), Offset = 0x21C

Bit(s)	Name	Description	R/W	Reset
31:12	-	RESERVED	R	0
11:0	SENSOR	TEMPSENSOR ADC, used to detect TEMPSENSOR voltage.	R	xx

9.8.9 SVCCADC_DATA

SVCCADC DATA Register (VDD), Offset = 0X220

Bit(s)	Name	Description	R/W	Reset
31:12	-	RESERVED	R	0
11:0	SVCC	SVCC voltage output SVCCADC input voltage range is from 0 to 3.6V	R	xx

9.8.10 LRADC1_DATA

LRADC1 DATA Register (VDD), Offset = 0x224

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC1	LRADC1 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

9.8.11 VCCI_DATA

VCCI DATA Register (VDD), Offset = 0x228

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
12:0	VCCI	VCCI data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

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9.8.12 LRADC2_DATA

LRADC2 DATA Register (VDD), Offset = 0x22C

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC2	LRADC2 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

9.8.13 LRADC3_DATA

LRADC3 DATA Register (VDD), Offset = 0x230

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC3	LRADC3 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

9.8.14 LRADC4_data

LRADC4 DATA Register (VDD), Offset = 0x234

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC4	LRADC4 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

9.8.15 LRADC5_DATA

LRADC5 DATA Register (VDD), Offset = 0x238

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC5	LRADC5 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

9.8.16 LRADC6_DATA

LRADC6 DATA Register (VDD), Offset = 0x23C

Bit(s)	Name	Description	R/W	Reset
31:14	-	RESERVED	R	0
13:0	LRADC6	LRADC6 data output. The ADC sampling values are aligned with high bits, and the extra 2 low bits are used when oversampling.	R	xx

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10 System Control

10.1 RMU

10.1.1 Features

The RMU Controller of ATS3089x has following features:

- The RMU (Reset Management Unit) can reset all the peripherals.
- The MCU can enter power-saving mode by setting the registers of RMU.

10.1.2 RMU Register List

Table 10-1 RMU base address

Name	Physical Base Address	KSEG1 Base Address
RMU	0x40000000	0x40000000

Table 10-2 RMU register list

Offset	Register Name	Description
0x00000000	MRCR0	Module Reset Control Register0
0x00000004	MRCR1	Module Reset Control Register1
0x00000008	RST_CTL	watch dog Reset Control Register

10.1.3 RMU Register Description

10.1.3.1 MRCR0

MRCR0 (Module Reset Control Register0, offset = 0x00000000)

Bit(s)	Name	Description	R/W	Reset
31:30	-	-	R	0x0
29	I2C2RESET	I2C2 controller reset bit: 0: reset 1: normal	R/W	0x0
28	I2C1RESET	I2C1 controller reset bit: 0: reset 1: normal	R/W	0x0
27	I2C0RESET	I2C0 controller reset bit: 0: reset 1: normal	R/W	0x0
26	UART2RESET	UART2 controller reset bit: 0: reset 1: normal	R/W	0x0
25	UART1RESET	UART1 controller reset bit: 0: reset 1: normal	R/W	0x0
24	UART0RESET	UART0 controller reset bit: 0: reset 1: normal	R/W	0x0
23	I2C3RESET	I2C3 controller reset bit:	R/W	0x0

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		0: reset 1: normal		
22	-	Reserved	R	0x0
21	SDMA_RESET	Special DMA controller RESET: 0: reset 1: normal	R/W	0x0
20	-	Reserved	R	0x0
19	LRADCRESET	LRADC Controller reset bit: 0: reset 1: normal	R/W	0x1
18	-	Reserved	R	0x0
17	PWMRESET	PWM reset bit: 0: reset 1: normal	R/W	0x0
16	-	Reserved	R	0x0
15	2.5GPU_RESET	2.5DGPU RESET: 0: reset 1: normal	R/W	0x0
14	LCDRESET	LCD controller reset bit: 0: reset 1: normal	R/W	0x0
13	JPEG_RESET	JPEG controller RESET: 0: reset 1: normal	R/W	0x0
12	2DGPURESET	2DGPU Controller reset bit: 0: reset 1: normal	R/W	0x0
11	USBRESET2	USB Controller reset2 bit: 0: reset 1: normal	R/W	0x0
10	USBRESET	USB Controller reset bit: 0: reset 1: normal	R/W	0x0
9	SPI1DCACHERESET	SPI1DCACHE Controller reset bit: 0: reset 1: normal	R/W	0x0
8	SPI0CACHERESET	SPICACHE Controller reset bit: 0: reset 1: normal	R/W	0x0
7	SPI3RESET	SPI3 controller reset bit: 0: reset 1: normal	R/W	0x0
6	SPI2RESET	SPI2 controller reset bit: 0: reset 1: normal	R/W	0x0
5	SPI1RESET	SPI1 controller reset bit: 0: reset 1: normal	R/W	0x0
4	SPI0RESET	SPI0 controller reset bit: 0: reset 1: normal	R/W	0x0
3	-	Reserved	R	0x0
2	SD1RESET	SD1 card controller reset bit:	R/W	0x0

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		0: reset 1: normal		
1	SD0RESET	SD0 card controller reset bit: 0: reset 1: normal	R/W	0x0
0	DMARESET	DMA reset bit: 0: reset 1: normal	R/W	0x0

10.1.3.2 MRCR1

MRCR1 (Module Reset Control Register1, offset = 0x00000004)

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	BTHUBRESET	BluetoothHub reset bit: 0: reset 1: normal	R/W	0x0
23:20	-	Reserved	R	0x0
19	I2CMT1RESET	I2CMT1 controller reset bit: 0: reset 1: normal	R/W	
18	I2CMT0RESET	I2CMT0 controller reset bit: 0: reset 1: normal	R/W	
17	SPIMT1RESET	SPIMT1 controller reset bit: 0: reset 1: normal	R/W	
16	SPIMT0RESET	SPIMT0 controller reset bit: 0: reset 1: normal	R/W	0x0
15	TIMER5_RESET	TIMER5 controller RESET: 0: reset 1: normal	R/W	0x0
14	TIMER4_RESET	TIMER4 controller RESET: 0: reset 1: normal	R/W	0x0
13	TIMER3_RESET	TIMER3 controller RESET: 0: reset 1: normal	R/W	0x0
12	TIMER2_RESET	TIMER2 controller RESET: 0: reset 1: normal	R/W	0x0
11	TIMER1_RESET	TIMER1 controller RESET: 0: reset 1: normal	R/W	0x0
10	TIMER0_RESET	TIMER0 controller RESET: 0: reset 1: normal	R/W	0x0
9:7	-	Reserved	R/W	0x0
6	I2STX1RESET	I2STX1 Controller RESET: 0: reset	R/W	0x0

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		1: normal		
5	I2SRXRESET	I2SRX reset bit: 0: reset 1: normal	R/W	0x0
4	I2STX0RESET	I2STX0 reset bit: 0: reset 1: normal	R/W	0x0
3	ADCRESET	ADC reset bit 0: reset 1: normal	R/W	0x0
2	DACRESET	DAC reset bit: 0: reset 1: normal	R/W	0x0
1	-	Reserved	R	0x1
0	AUDIODSP_ALLRESET	All DSP reset 0: reset all dsp 1: normal	R/W	0x0

10.1.3.3 RST_CTL

WDRST_CTL (watch dog reset Control Register, offset = 0x00000008)

Bit(s)	Name	Description	R/W	Reset
31:4		-	R	0
3	SYSRESETREQ	0: DISABLE 1: ENABLE	RW	1
2	-	-	RW	0
1	WD_RST_BTCPU	0: DISABLE 1: ENABLE	RW	0
0	-	-	RW	0

10.2 UID

10.2.1 Features

- Support 64bit CHIP unique ID.

10.2.2 UID Register List

Table 10-3 **UID Registers Address**

Name	Physical Base Address	KSEG1 Base Address
UID	0x400000A0	0x400000A0

Table 10-4 UID Registers **List**

Offset	Register Name	Description
0x0000	UID0	Unique Device ID Register0
0x0004	UID1	Unique Device ID Register1

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10.2.3 UID Register Description

10.2.3.1 UID0

Unique Device ID Register0
Offset = 0x0000

Bit(s)	Name	Description	R/W	Reset
31:0	UID0	Unique Device ID Information	R	0x0

10.2.3.2 UID1

Unique Device ID Register1
Offset = 0x0004

Bit(s)	Name	Description	R/W	Reset
31:0	UID1	Unique Device ID Information	R	0x0

10.3 CMU Analog

10.3.1 Features

- Support only one oscillator inputs: 32M.
- Supply 1 HOSC circuit.
- Supply 1 LOSC as an option.
- Supply 3 RC circuits: RC4M, RC64M, and RC192M.
- Supply 3 PLLs and special clocks of all modules. The 3 PLLs are CORE PLL, AUDIOPLL and SPLL. Special clocks may be from the LOSC, RC32K, RC4M or RC64M and RC192M.
- COREPLL supports spread spectrum.
- The source of all PLL are HOSC.

10.3.2 CMU Analog Register List

Table 10-5 CMU Analog Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
CMU_ANALOG_REGISTER	0x40000100	0x40000100

Table 10-6 CMU Analog Controller Registers

Offset	Register Name	Description
0x00	HOSC_CTL	HOSC Control Register
0x04	HOSCLDO_CTL	HOSC LDO Control Register
0x08	RC64M_CTL	RC64M Control Register
0x10	LOSC_CTL	LOSC Control Register
0x14	RC4M_CTL	RC4M Control Register
0x18	RC192M_CTL	RC128M Control Register
0x1C	AVDDLDO_CTL	AVDD LDO Control Register
0x20	COREPLL_CTL	COREPLL Control Register
0x24	SPLL_CTL	SPLL Control Reigster

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0x28	AUDIOPLL_CTL	AUDIOPLL Control Register
0x64	RC32K_CTL	RC32K Control Register
0x68	RC32K_CAL	RC32K Calibration Register
0x6C	RC32K_COUNT	RC32K Calibration Count Register
0x74	HGLFCLK_CTL	HOSC Generate Low Frequency Clock Control Register

10.3.3 CMU Analog Register Description

10.3.3.1 HOSC_CTL

HOSC control register.
Offset = 0x00(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:29	-	Reserved	R	0x0
28	HOSC_READY	HOSC READY Signal	R	0x0
27:25	-	Reserved	R/W	0x2
24:17	HOSC_CAP	HOSCO PAD base cap select, range from 0pF to 20pF: Trim cap = 0.1pF * HOSCI_CAP	R/W	0x64
16:15	-	Reserved	R/W	0x0
14:13	HOSC_AMPLMT	HOSC amplitude Adjust 00：450mV 01：500mV 10：550mV 11：600mV	R/W	0x3
12	HOSC_AMPLMT_EN	HOSC amplitude Adjust Enable: 0: disabled 1: enable	R/W	0x0
11:9	HGMC	High Frequency crystal Oscillator GMMIN select bits	R/W	0x7
8:0	-	Reserved	R/W	0x75

10.3.3.2 HOSCLDO_CTL

HOSC LDO Control Register
Offset = 0x04(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:6	--	reserved	R	X
5	OSCVDD0_PD_BIASSET	OSCVDD LDO0 ipd set: 0: disable 1: enable, save 1/2 bias current	R/W	0x0
4	OSCVDD0_PD	OSCVDD0 pd set: 0: disable 1: enable	R/W	0x1
3:2	OSCVDD0_VSET	OSCVDD0 output voltage set: 00: 1.0V 01: 1.1V**** 10: 1.2V 11: 1.3V	R/W	0x1
1	--	reserved	R/W	0x0
0	OSCVDD0_EN	OSCVDD0 LDO0 enable:	R/W	0x0

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		0: disable 1: enable		

Note: OSCVDDLDO should be enabled or disabled by hardware according to OSCVDDx_EN and HOSCEN in different PMU State.

10.3.3.3 RC64M_CTL

RC64M Clock Control Register
Offset = 0x08(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:25	FRQCAL	RC64M Calibration Frequency The frequency calibration result, read only.	R	0x40
24	CALDONE	RC64M FREQ Calibration done 0: uncalibrated or calibrating 1: calibration done Cleared when CALEN/RC64M_EN is cleared, or when the FREQ of RC64M exceeds INRANGE	R	0x0
23:14	FRQASET	RC64M Hardware Auto Calibration frequency setting When CALMODE is 1, hardware may adjust the calibration reference freq of RC64M through this field, one step is 500KHz, FOSC=500KHz*N	R/W	0x80
13:11	-	Reserved.	R/W	010
10:4	FRQMSET	RC64M Manual Calibration frequency setting When CALMODE is 0, software may adjust the freq of RC64M through this field.	R/W	0x40
3	CALMODE	RC64M Calibration Mode 0: manual calibration by FRQMSET 1: auto calibration by FRQASET	R/W	0x0
2	CALEN	RC64M Calibration Enable: 0: disable 1: enable	R/W	0x0
1:0	-	Reserved, be read as zero.	R/W	0x0

10.3.3.4 LOSC_CTL

LOSC Control Register.
Offset = 0x10(SVCC domain)

Bit(s)	Name	Description	R/W	Reset
31:21	-	Reserved, be read as zero.	R	0x0
20:16	LOSCI_CAP	LOSCI PAD Capacitance select: 00000: 0pf 00001: 0.8pf 00010: 1.6pf 00011: 2.4pf ... 10000: 12.8pf** ... 11111: 24.8pf	R/W	0x19
15	LOSC_RDY	LOSC Status:	R	0

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		0: LOSC not Ready 1: LOSC ready		
14:13	-	Reserved, be read as zero.	R/W	0
12:8	LOSCO_CAP	LOSCO PAD Capacitance select: 00000: 0pf 00001: 0.8pf 00010: 1.6pf 00011: 2.4pf … 10000: 12.8pf** … 11111: 24.8pf	R/W	0x19
7	-	Reserved	R/W	0x0
6:4	LGMC	Low Frequency crystal Oscillator GMMIN select bits	R/W	0x7
3:1	-	Reserved	R/W	0x0
0	LOSC_EN	LOSC enable: 0: disable 1: enable	R/W	0x0

10.3.3.5 RC4M_CTL

RC4M Clock Control Register
Offset = 0x14(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:7	-	Reserved	R/W	0x0
6:1	RC4M_FRQSET	OSC4M frequency set bits:	R/W	0x2C
0	-	Reserved, be read as zero.	R/W	0x0

Note:

10.3.3.6 RC192M_CTL

RC192M Clock Control Register
Offset = 0x18(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:25	FRQCAL	RC192M Calibration Frequency The frequency calibration result, read only.	R	0x40
24	CALDONE	RC192M FREQ Calibration done 0: uncalibrated or calibrating 1: calibration done Cleared when CALEN/RC192M _EN is cleared, or when the FREQ of RC192M exceeds INRANGE	R	0x0
23:14	FRQASET	RC192M Hardware Auto Calibration frequency setting 000: 128M 001: 160M 010: 192M ***** 011: 224M 100: 256M Others: reserved	R/W	0x2
13:11	-	Reserved	R/W	010

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10:4	FRQMSET	RC192M Manual Calibration frequency setting When CALMODE is 0, software may adjust the freq of RC192M through this field.	R/W	0x48
3	CALMODE	RC192M Calibration Mode 0: manual calibration by FRQMSET 1: auto calibration by FRQASET	R/W	0x0
2	CALEN	RC192M Calibration Enable: 0: disable 1: enable	R/W	0x0
1	-	Reserved, be read as zero.	R/W	0x0
0	RC192MEN	RC192M Enable bit: 0: Disable 1: Enable	R/W	0x0

Note:

10.3.3.7 AVDDLDO_CTL

AVDD LDO Control Register
Offset = 0x1C(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:6	--	reserved	R	0x0
5	AVDD_PD_BIASSET	AVDD LDO ipd set: 0: disable 1: enable, save 1/2 bias current	R/W	0x0
4	AVDD_PD_EN	AVDD pd set: 0: disable 1: enable	R/W	0x1
3:2	AVDD_VSET	AVDD output voltage set: 00: 1.0V 01: 1.1V**** 10: 1.2V 11: 1.3V	R/W	0x1
1	--	reserved	R/W	0x0
0	AVDD_EN	AVDD LDO enable: 0: disable 1: enable	R/W	0x0

10.3.3.8 COREPLL_CTL

COREPLL Control Register
Offset = 0x20(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:9	-	Reserved, be read as zero.	R	0x0
8	COREPLLPMD	COREPLL phase match detect: 0: CORE PLL phase not match 1: CORE PLL phase match	R	X
7	COREPLLEN	COREPLL Enable： 0: Disable 1: Enable	R/W	0x0
6	-	Reserved	R/W	0x0

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5:0	COREPLLSCORE	COREPLL Frequency Select: Formula: 8M* SCORE Range:40 ~ 320M Value must be bigger than 4 0-4: reserved 5: 5*8M=40M …… 40: 40*8M=320M Others reserved.	R/W	0x05

10.3.3.9 SPLL_CTL

SPLL Control Register
Offset = 0x24(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:9	-	Reserved	R	X
8	SPLL_LOCK	SPLL phase match detect: 0: SPLL phase not match 1: SPLL phase match This bit is only used for debug. Software must not detect PLL state according to this bit.	R	X
7:1	-	Reserved	R/W	0x0
0	SPLL_EN	SPLL Enable: 0: disable 1: enable	R/W	0x0

10.3.3.10 AUDIOPLL_CTL

AUDIOPLL Control Register
Offset = 0x28(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:9	-	Reserved	R/W	0x0
8	AUDPLL_LOCK	AUDIO PLL phase match detect: 0: AUDIO PLL phase not match 1: AUDIO PLL phase match This bit is only used for debug. Software must not detect PLL state according to this bit. This bit is updated by master CPU. This bit could only be read by master DSP.	R	X
7	AUDPLLEN	Audio PLL Enable 0：Disable 1：Enable	R/W	0x0
6	AUDPLL_MODE	AUDIO PLL mode select: 0: integer-N 1: fractional-N	R/W	0x0
5:0	AUDPLL_APS	AUDIO PLL Clock Select Refer to the following table to see the configuration.	R/W	0x0

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10.3.3.11 RC32K_CTL

RC32K Clock Control Register
Offset = 0x64(SVCC domain)

Bit(s)	Name	Description	R/W	Reset
31:7	-	Reserved, be read as zero.	R	0x0
6:1	RC32K_FRQSET	RC32K frequency set bits	R/W	0x2C
0	-	Reserved, be read as zero.	R	0x0

10.3.3.12 RC32K_CAL

RC32K Clock Calibration Register
Offset = 0x68(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved, be read as zero.	R	0x0
23:8	CALCYCLE	RC32K Calibration Cycles The calibration internal = CALCYCLE*Frequency of RC32K	R/W	0x0
7:5	-	Reserved, be read as zero.	R	0x0
4	CALDONE	RC32K calibration Done 0: not calibration 1: calibration done	R	0x0
3:2	-	Reserved, be read as zero.	R	0x0
1	CALSRC	RC32K calibration source: 0: RC32K 1: LOSC	R/W	0x0
0	CALEN	RC32K Calibration Start Enable: 0: disable 1: enable	R/W	0x0

10.3.3.13 RC32K_COUNT

RC32K Clock Calibration Count Register
Offset = 0x6C(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:0	CALCOUNT	RC32K Calibration Count	R	0x0

10.3.3.14 HGLFCLK_CTL

HOSC Generate Low Frequency Clock Control Register
Offset = 0x74(VDD domain)

Bit(s)	Name	Description	R/W	Reset
31:2	-	Reserved, be read as zero.	R	0x0
1	CK32K768SRC	CK32K768 Source: 0: LOSC 1: HGLFCLK	R/W	0x0
0	HGLFCLKEN	HOSC Generate Low Frequency Clock Enable:	R/W	0x0

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		0: disable 1: enable		

10.4 CMU Digital

10.4.1 Features

The CMU Controller of ATS3089x has following features:

- The CMU (Clock Management Unit) can select RC4M, HOSC, CORE_PLL, RC64M and RC192M as the clock of each peripheral.
- The clock sources of the memory blocks can be selected by CMU.
- 4 CMU states: S1, S1BT, S2 and S3.

10.4.2 CMU Digital Register List

Table 10-7 CMU Digital Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
CMU_DIGITAL_REGISTER	0x40001000	0x40001000

Table 10-8 CMU Digital Controller Registers

Offset	Register Name	Description
0x0000	CMU_SYSCLK	SYSCLK Control Register
0x0004	CMU_DEVCLKEN0	DEVCLKEN Control Register0
0x0008	CMU_DEVCLKEN1	DEVCLKEN Control Register1
0x0010	CMU_SD0CLK	SD0 CLK Control Register
0x0014	CMU_SD1CLK	SD1 CLK Control Register
0x001C	CMU_JPEGCLK	JPEG Clock Control Register
0x0020	CMU_SPI0CLK	SPI0 CLK Control Register
0x0024	CMU_SPI1CLK	SPI1 CLK Control Register
0x0028	CMU_SPI2CLK	SPI2 CLK Control Register
0x002C	CMU_SPI3CLK	SPI3 CLK Control Register
0x0030	CMU_2DGPUCLK	2DGPUCLK Control Register
0x0034	CMU_LCDCLK	LCDCLK Control Register
0x0038	CMU_25DGPUCLK	2.5DGPU Clock Control Register
0x0040	CMU_TIMER0CLK	TIMER0 Clock Control Register
0x0044	CMU_TIMER1CLK	TIMER1 Clock Control Register
0x0048	CMU_TIMER2CLK	TIMER2 Clock Control Register
0x004c	CMU_TIMER3CLK	TIMER3 Clock Control Register
0x0050	CMU_TIMER4CLK	TIMER4 Clock Control Register
0x0054	CMU_TIMER5CLK	TIMER5 Clock Control Register
0x0068	CMU_PWM0CLK	PWMCLK0 Control Register
0x006C	CMU_PWM1CLK	PWMCLK1 Control Register
0x0070	CMU_PWM2CLK	PWMCLK2 Control Register
0x0074	CMU_PWM3CLK	PWMCLK3 Control Register
0x0078	CMU_LRADCCLK	LRADC CLK Control Register
0x0080	CMU_I2C0CLK	I2C0CLK Control Register
0x0084	CMU_I2C1CLK	I2C1CLK Control Register
0x0088	CMU_I2C2CLK	I2C2CLK Control Register
0x008C	CMU_I2C3CLK	I2C3CLK Control Register

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0x0090	CMU_AUDDSPCLK	Audio DSPCLK Control Register
0x0098	CMU_DACCLK	DACCLK Control Register
0x009c	CMU_ADCCLK	ADCCLK Control Register
0x00A0	CMU_I2STXCLK	I2STXCLK Control Register
0x00A4	CMU_I2SRXCLK	I2SRXCLK Control Register
0x00B0	CMU_MEMCLKEN0	MEMCLKEN Control Register0
0x00B4	CMU_MEMCLKEN1	MEMCLKEN Control Register1
0x00C0	CMU_MEMCLKSRC0	MEMCLKSRC Control Register0
0x00C4	CMU_MEMCLKSRC1	MEMCLKSRC Control Register1
0x00D0	CMU_S1CLKCTL	S1 Clock Control Register
0x00D4	CMU_S1BTCLKCTL	S1BT Clock Control Register
0x00DC	CMU_S2CLKCTL	S2 Clock Control Register
0x00E0	CMU_S3CLKCTL	S3 Clock Control Register
0x00E4	CMU_WDCLK	CMU WatchDog Clock Control Register
0x00E8	CMU_GPIOCLKCTL	CMU GPIO interrupt Clock Control register
0x00EC	CMU_PMUWKUPCLKCTL	PMU Wakeup Clock Control Register
0x0100	CMU_SPIMT0CLK	CMU SPIMT0 Clock Control Register
0x0104	CMU_SPIMT1CLK	CMU SPIMT1 Clock Control Register
0x0110	CMU_I2CMT0CLK	CMU I2CMT0 Clock Control Register
0x0114	CMU_I2CMT1CLK	CMU I2CMT1 Clock Control Register

10.4.3 CMU Digital Register Description

10.4.3.1 CMU_SYSCLK

CMU_SYSCLK Control register
Offset = 0x00

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	CPUTIMERCLKSRC	CPU Timer Clock Source: 0: HOSC/32 1: RC4M/4	R/W	0x0
23:10	-	Reserved	R	0x0
9:8	AHBCLKDIV	HCLK divisor 00: /2 01: /4 10/11: /1	R/W	0x0
7:4	CPUCLKDIV	CPU_CLK divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0
3	-	Reserved	R	0x0
2:0	CORECLKSRC	CORE_CLK Source: 000: RC4M 001: HOSC	R/W	0x0

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		010: COREPLL 011: RC64M 100: reserved 101: RC192M 110: RC32K		

10.4.3.2 CMU_DEVCLKEN0

CMU_DEVCLKEN Control register0
Offset = 0x04

Bit(s)	Name	Description	R/W	Reset
31	-	Reserved	R/W	0x0
30	EXINTCLKEN	External Interrupt clock enable bit: 0: disable 1: enable The source clock of external interrupt is RC3M	R/W	0x1
29	I2C2CLKEN	I2C2 controller clock enable bit: 0: disable 1: enable	R/W	0x0
28	I2C1CLKEN	I2C1 controller clock enable bit: 0: disable 1: enable	R/W	0x0
27	I2C0CLKEN	I2C0 controller clock enable bit: 0: disable 1: enable	R/W	0x0
26	UART2CLKEN	UART2 controller clock enable bit: 0: disable 1: enable	R/W	0x0
25	UART1CLKEN	UART1 controller clock enable bit: 0: disable 1: enable	R/W	0x0
24	UART0CLKEN	UART0 controller clock enable bit: 0: disable 1: enable	R/W	0x0
23	I2C3CLKEN	I2C3 controller clock enable bit: 0: disable 1: enable	R/W	0x0
22	-	Reserved	R/W	0x0
21	SDMACLKEN	SPECIAL DMA Clock Enable bit : 0: disable 1: enable	R/W	0x0
20	CPUTIMERCLKEN	CPU Timer Clock Enable bit: 0: disable 1: enable	R/W	0x0
19	LRADCCLKEN	LRADC Controller clock enable bit: 0: disable 1: enable	R/W	0x1
18	-	Reserved	R/W	0x0
17	PWM0CLKEN	PWM0 clock enable bit: 0: disable 1: enable This bit controls all the clock gating of PWM0	R/W	0x0

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16	-	Reserved	R/W	0x0
15	25DGPUCLKEN	25DGPU clock enable bit: 0: disable 1: enable	R/W	0x0
14	LCDCLKEN	LCD controller clock enable bit: 0: disable 1: enable	R/W	0x0
13	JPEGCLKEN	JPEG Controller clock enable bit: 0: disable 1: enable	R/W	0x0
12	2DGPUCLKEN	2DGPU Controller clock enable bit: 0: disable 1: enable	R/W	0x0
11	USBHCLKEN	USB Controller HCLK enable bit: 0: disable 1: enable This bit controls the HCLK and RC32K to the USB controller.	R/W	0x0
10	USBCLKEN	USB Controller clock enable bit: 0: disable 1: enable	R/W	0x0
9	SPI1DCACHECLKEN	SPI1DCACHE Controller clock enable bit: 0: disable 1: enable	R/W	0x0
8	SPI0CACHECLKEN	SPICACHE Controller clock enable bit: 0: disable 1: enable	R/W	0x0
7	SPI3CLKEN	SPI3 controller clock enable bit: 0: disable 1: enable	R/W	0x0
6	SPI2CLKEN	SPI2 controller clock enable bit: 0: disable 1: enable	R/W	0x0
5	SPI1CLKEN	SPI1 controller clock enable bit: 0: disable 1: enable	R/W	0x0
4	SPI0CLKEN	SPI0 controller clock enable bit: 0: disable 1: enable	R/W	0x0
3	-	Reserved	R/W	0x0
2	SD1CLKEN	SD1 card controller clock enable bit: 0: disable 1: enable	R/W	0x0
1	SD0CLKEN	SD0 card controller clock enable bit: 0: disable 1: enable	R/W	0x0
0	DMACLKEN	DMA clock enable bit: 0: disable 1: enable	R/W	0x0

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10.4.3.3 CMU_DEVCLKEN1

CMU_DEVCLKEN Control register1
Offset = 0x08

Bit(s)	Name	Description	R/W	Reset
31	32KCLKOUTEN	32K Clock Output Enable 0: disable 1: enable	R/W	0x0
30	32MCLKOUTEN	32M Clock Output Enable 0: disable 1: enable	R/W	0x0
29:23	-	Reserved	R/W	0x0
22	PWM3CLKEN	PWM3 clock enable bit: 0: disable 1: enable	R/W	0x0
21	PWM2CLKEN	PWM2 clock enable bit: 0: disable 1: enable	R/W	0x0
20	PWM1CLKEN	PWM1 clock enable bit: 0: disable 1: enable	R/W	0x0
19	I2CMT1CLKEN	I2CMT1 controller clock enable bit: 0: disable 1: enable	R/W	0x0
18	I2CMT0CLKEN	I2CMT0 controller clock enable bit: 0: disable 1: enable	R/W	0x0
17	SPIMT1CLKEN	SPIMT1 controller clock enable bit: 0: disable 1: enable	R/W	0x0
16	SPIMT0CLKEN	SPIMT0 controller clock enable bit: 0: disable 1: enable	R/W	0x0
15	TIMER5CLKEN	Timer5 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER5_CLK	R/W	0x0
14	TIMER4CLKEN	Timer4 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER4_CLK	R/W	0x0
13	TIMER3CLKEN	Timer3 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER3_CLK	R/W	0x0
12	TIMER2CLKEN	Timer2 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER2_CLK	R/W	0x0
11	TIMER1CLKEN	Timer1 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER1_CLK	R/W	0x0

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10	TIMER0CLKEN	Timer0 controller clock: 0: disable 1: enable This bit controls the clock gating of TIMER0_CLK	R/W	0x0
9	AUDDSPDMACLKEN	Audio DSP DMA Clock Enable bit: 0: disable 1: enable	R/W	0x0
8	DACANACLKEN	DAC Analog Clock Enable bit: 0: disable 1: enable	R/W	0x0
7	I2SHCLKEN	I2STX/RX AHB clock enable bit: 0: disable 1: enable	R/W	0x0
6	I2SSRDCLKEN	I2SSRD Mclock enable bit: 0: disable 1: enable	R/W	0x0
5	I2SG1CLKEN	I2SG1 clock enable bit: 0: disable 1: enable	R/W	0x0
4	I2SG0CLKEN	I2SG0 clock enable bit: 0: disable 1: enable	R/W	0x0
3	ADCCLKEN	ADC controller clock enable bit: 0: disable 1: enable	R/W	0x0
2	DACCLKEN	DAC controller clock enable bit: 0: disable 1: enable	R/W	0x0
1	AUDDSPTIMERCLKEN	Audio DSP Timer clock enable bit: 0: disable 1: enable	R/W	0x0
0	AUDDSPCLKEN	Audio DSP clock enable bit: 0: disable 1: enable	R/W	0x0

10.4.3.4 CMU_SD0CLK

SD0 Clock Control Register
Offset = 0x0010

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SD0CLKSRC	SD0 Card Controller Clock Source Select: 00: HOSC 01: CORE_PLL 10: RC192M 11: Reserved	R/W	0x0
7	-	Reserved	R	0x0
6	SD0CLKPOSTDIV	SD0 Card Controller Clock Post-Divisor	R/W	0x0

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		0: /1 1: /128		
5:4	-	Reserved	R	0x0
3:0	SD0CLKDIV	SD0 Card Controller Clock Divisor 0000: /1 0001: /2 0010: /3 0011: /4 0100: /5 0101: /6 0110: /7 0111: /8 1000: /9 1001: /10 1010: /11 1011: /12 1100: /13 1101: /14 1110: /15 1111: /16	R/W	0x0

10.4.3.5 CMU_SD1CLK

SD1 Clock Control Register
Offset = 0x0014

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SD1CLKSRC	SD1 Card Controller Clock Source Select: 00: HOSC 01: CORE_PLL 10: RC192M 11: Reserved	R/W	0x0
7	-	Reserved	R	0x0
6	SD1CLKPOSTDIV	SD1 Card Controller Clock Post-Divisor 0: /1 1: /128	R/W	0x0
5:4	-	Reserved	R	0x0
3:0	SD1CLKDIV	SD1 Card Controller Clock Divisor 0000: /1 0001: /2 0010: /3 0011: /4 0100: /5 0101: /6 0110: /7 0111: /8 1000: /9 1001: /10 1010: /11 1011: /12	R/W	0x0

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		1100: /13 1101: /14 1110: /15 1111: /16		

10.4.3.6 CMU_JPEGCLK

JPEG Clock Control Register
Offset = 0x001C

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	JPEGCLKSRC	JPEG Controller Clock Source: 00: Reserved 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x1
7:4	-	Reserved	R	0x0
3:0	JPEGCLKDIV	JPEG Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0

10.4.3.7 CMU_SPI0CLK

SPI0 Clock Control Register
Offset = 0x0020

Bit(s)	Name	Description	R/W	Reset
31:13	-	Reserved	R	0x0
12	SPI0MODE	SPI0 Controller Mode Select: 0: Normal Mode 1: DDR Mode	R/W	0x0
11:10	-	Reserved	R	0x0
9:8	SPI0CLKSRC	SPI0 Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPI0CLKDIV	SPI0 Clock Divisor:	R/W	0x0

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		0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5		

10.4.3.8 CMU_SPI1CLK

SPI1 Clock Control Register
Offset = 0x0024

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SPI1CLKSRC	SPI1 Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPI1CLKDIV	SPI1 Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0

10.4.3.9 CMU_SPI2CLK

SPI2 Clock Control Register
Offset = 0x0028

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SPI2CLKSRC	SPI2 Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPI2CLKDIV	SPI2 Clock Divisor:	R/W	0x0

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		0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5		

10.4.3.10 CMU_SPI3CLK

SPI3 Clock Control Register
Offset = 0x002C

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SPI3CLKSRC	SPI3 Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPI3CLKDIV	SPI3 Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0

10.4.3.11 CMU_2DGPUCLK

2DGPU Clock Control Register
Offset = 0x0030

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	2DGPUCLKSRC	2DGPU Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	2DGPUCLKDIV	2DGPU Controller Clock Divisor: 0: /1 1: /2 2: /3 3: /4	R/W	0x0

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		4: /5 … 13: /14 14: /1.5 15: /2.5		

10.4.3.12 CMU_LCDCLK

LCD Clock Control Register
Offset = 0x0034

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	LCDCLKSRC	LCD Controller Clock Source: 00: HOSC 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x0
7:5	-	Reserved	R	0x0
4	LCDCLKPREDIV	LCD Controller Clock PRE-divisor: 0: /1 1: /6	R/W	0x0
3:0	LCDCLKDIV	LCD Controller Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 11: /12 Others: Reserved	R/W	0x0

10.4.3.13 CMU_25DGPUCLK

2.5DGPU Clock Control Register
Offset = 0x0038 default value: 0x00001100

Bit(s)	Name	Description	R/W	Reset
31:13	-	Reserved	R	0x0
12	25DGPURAMCLKGATING	25DGPU RAM Clock Gating Disable: 0: enable 1: disable	R/W	0x1
11:10	-	Reserved	R	0x0
9:8	25DGPUCLKSRC	25DGPU Controller Clock Source: 00: Reserved 01: CORE_PLL 10: RC192M 11: RC64M	R/W	0x1
7:4	-	Reserved	R	0x0
3:0	25DGPUCLKDIV	25DGPU Clock Divisor: 0: /1	R/W	0x0

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		1: /2 2: /3 3: /4 4: /5 ... 13: /14 14: /1.5 15: /2.5		

10.4.3.14 CMU_TIMER0CLK

TIMER0 Clock Control Register
Offset = 0x0040

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER0CLKSRC	Timer0 clock Source: 000: HOSC 001: HOSC/32 010: CK32K768 011: Reserved 100: RC32K	R/W	0x0

10.4.3.15 CMU_TIMER1CLK

TIMER1 Clock Control Register
Offset = 0x0044

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER1CLKSRC	Timer1 clock Source: 000: HOSC 001: HOSC/32 010: CK32K768 011: Reserved 100: RC32K	R/W	0x0

10.4.3.16 CMU_TIMER2CLK

TIMER2 Clock Control Register
Offset = 0x0048

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER2CLKSRC	Timer2 clock Source: 000: HOSC 001: HOSC/32 010: CK32K768 011: HOSC/32000 100: RC32K	R/W	0x0

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10.4.3.17 CMU_TIMER3CLK

TIMER3 Clock Control Register
Offset = 0x004C

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER3CLKSRC	Timer3 clock Source: 000: HOSC 001: HOSC/32 010: CK32K768 011: HOSC/32000 100: RC32K	R/W	0x0

10.4.3.18 CMU_TIMER4CLK

TIMER4 Clock Control Register
Offset = 0x0050

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER4CLKSRC	Timer4 clock Source: 000: HOSC 001: HOSC/32 010: RC4M 011: HOSC/32000 100: RC32K	R/W	0x0

10.4.3.19 CMU_TIMER5CLK

TIMER5 Clock Control Register
Offset = 0x0054

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2:0	TIMER5CLKSRC	Timer5 clock Source: 000: HOSC 001: HOSC/32 010: CK32K768 011: HOSC/32000 100: RC32K	R/W	0x0

10.4.3.20 CMU_PWM0CLK

CMU_PWM0CLK Control register
Offset = 0x0068

Bit(s)	Name	Description	R/W	Reset
31:11	-	Reserved	R	0x0
10:9	PWM0CLKSRC	PWM0 Controller Clock Source 00: RC32K 01: HOSC	R/W	0x0

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		10/11: RC64M		
8:0	PWM0CLKDIV	PWM0 Controller Clock Divisor: 0: /1 1: /2 ... 255: /256 256: /512 257: /1024 258: /2048 259: /4096 260: /8192 261~511: reserved	R/W	0x0

10.4.3.21 CMU_PWM1CLK

CMU_PWM1CLK Control register
Offset = 0x006C

Bit(s)	Name	Description	R/W	Reset
31:11	-	Reserved	R	0x0
10:9	PWM1CLKSRC	PWM1 Controller Clock Source 00: RC32K 01: HOSC 10/11: RC64M	R/W	0x0
8:0	PWM1CLKDIV	PWM1 Controller Clock Divisor: 0: /1 1: /2 ... 255: /256 256: /512 257: /1024 258: /2048 259: /4096 260: /8192 261~511: reserved	R/W	0x0

10.4.3.22 CMU_PWM2CLK

CMU_PWM2CLK Control register
Offset = 0x0070

Bit(s)	Name	Description	R/W	Reset
31:11	-	Reserved	R	0x0
10:9	PWM2CLKSRC	PWM2 Controller Clock Source 00: RC32K 01: HOSC 10/11: RC64M	R/W	0x0
8:0	PWM2CLKDIV	PWM2 Controller Clock Divisor: 0: /1 1: /2 ... 255: /256 256: /512 257: /1024	R/W	0x0

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		258: /2048 259: /4096 260: /8192 261~511: reserved		

10.4.3.23 CMU_PWM3CLK

CMU_PWM3CLK Control register
Offset = 0x0074

Bit(s)	Name	Description	R/W	Reset
31:11	-	Reserved	R	0x0
10:9	PWM3CLKSRC	PWM3 Controller Clock Source 00: RC32K 01: HOSC 10/11: RC64M	R/W	0x0
8:0	PWM3CLKDIV	PWM3 Controller Clock Divisor: 0: /1 1: /2 … 255: /256 256: /512 257: /1024 258: /2048 259: /4096 260: /8192 261~511: reserved	R/W	0x0

10.4.3.24 CMU_LRADCCLK

CMU_LRADCCLK Control register
Offset = 0x0078

Bit(s)	Name	Description	R/W	Reset
31:7	-	Reserved	R	0x0
6:4	LRADCCLKSRC	LRADC Clock Source 00: RC32K 01: CK32K768 10: RC4M /16 11: HOSC/128	R/W	0x0
3:2	-	Reserved	R	0x0
1:0	LRADCCLKDIV	LRADC Clock Divisor 00: /1 01: /2 10: /4 11: /8	R/W	0x0

10.4.3.25 CMU_I2C0CLK

CMU I2C0CLK Control register
Offset = 0x0080

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Bit(s)	Name	Description	R/W	Reset
31:5	-	Reserved	R	0x0
4	I2C0CLKSRC	I2C0 Clock Source 0: HOSC 1: RC192M/6	R/W	0x0
1:0	I2C0CLKDIV	I2C0 Controller Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0

10.4.3.26 CMU_I2C1CLK

CMU I2C1CLK Control register
Offset = 0x0084

Bit(s)	Name	Description	R/W	Reset
31:5	-	Reserved	R	0x0
4	I2C1CLKSRC	I2C1 Clock Source 0: HOSC 1: RC192M/6	R/W	0x0
1:0	I2C1CLKDIV	I2C1 Controller Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0

10.4.3.27 CMU_I2C2CLK

CMU I2C2CLK Control register
Offset = 0x0088

Bit(s)	Name	Description	R/W	Reset
31:5	-	Reserved	R	0x0
4	I2C2CLKSRC	I2C2 Clock Source 0: HOSC 1: RC192M/6	R/W	0x0
1:0	I2C2CLKDIV	I2C2 Controller Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0

10.4.3.28 CMU_I2C3CLK

CMU I2C3CLK Control register
Offset = 0x008C

Bit(s)	Name	Description	R/W	Reset
31:5	-	Reserved	R	0x0
4	I2C3CLKSRC	I2C3 Clock Source	R/W	0x0

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		0: HOSC 1: RC192M/6		
1:0	I2C3CLKDIV	I2C3 Controller Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0

10.4.3.29 CMU_AUDDSPCLK

CMU_AUDDSPCLK Control register
Offset = 0x90

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	DSPAPBCLKDIV	DSP_APB_CLK divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0
7:4	DSPCLKDIV	DSP Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0
3	-	Reserved	R	0x0
2:0	DSPCLKSRC	DSP CLK Source: 000: Reserved 001: HOSC 010: COREPLL 011: RC64M 100: Reserved 101: RC192M 110: Reserved	R/W	0x1

10.4.3.30 CMU_DACCLK

DAC Clock Control Register
Offset = 0x0098

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24	DACFIFOCLKEN	DAC FIFO Access Clock Enable: 0: disable 1: enable	R/W	0x0
23:22	-	Reserved	R	0x0

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21	DACCICFIRCLKEN	DAC CIC FIR Clock Enable: 0: disable 1: enable	R/W	0x0
20	DACSDMCLKEN	DAC SDM Clock Enable: 0: disable 1: enable	R/W	0x0
19:18	-	Reserved	R	0x0
17:16	DACFIR2XCLKDIV	DAC 2X FIR Clock Divisor 00: /1 01: /2 10: /3 11: reserved	R/W	0x0
15	-	Reserved	R	0x0
14	DACFIRCLKDIV	DAC FIR Clock Divisor 0: /1 1: /2	R/W	0x0
13	DACHUMDIV	DAC Hnum Divisor 0: /1 1: /2	R/W	0x0
12	DACCICCLKDIV	DAC CICCLK and SDMLCK Clock Divisor 0: /1 1: /2	R/W	0x0
11:9	-	Reserved	R	0x0
8	DACCLKSRC	DAC Clock Source Selection: 0: Audiopll/4 1: HOSC	R/W	0x0
7:2	-	Reserved	R	0x0
1:0	DACCLKDIV	DAC Clock Divisor. 00: /1 01: /2 10: /3 11: /4	R/W	0x0

10.4.3.31 CMU_ADCCLK

ADC Clock Control Register
Offset = 0x009C

Bit(s)	Name	Description	R/W	Reset
31	-	Reserved	R/W	0x0
30	ADCSWITCHRC4M	ADC Clock Switch to RC4M: 0: ADC Clock from Audiopll 1: ADC Clock from RC4M	R/W	0x0
29:25	-	Reserved	R	0x0
24	ADCFIFOCLKEN	ADC FIFO Access Clock Enable: 0: disable 1: enable	R/W	0x0
23	ADCFIREN	ADC FIR Filter CLK enable： 0： disable 1： enable	RW	0x0
22	ADCCICEN	ADC CIC Filter enable： 0： disable	RW	0x0

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		1： enable		
21	ADCANAEN	ADC Analog CLK enable： 0： disable 1： enable	RW	0x0
20	ADCDMICEN	DMIC CLK enable： 0： disable 1： enable	RW	0x0
19	ADCFIRCLKRVS	ADC FIR Clock Reverse control 0： Normal 1： Reverse	RW	0x0
18	ADCCICCLKRVS	ADC CIC Clock Reverse control 0： Normal 1： Reverse	RW	0x0
17	ADCANACLKRVS	ADC Analog Clock Reverse control 0： Normal 1： Reverse	RW	0x0
16	ADCDMICCLKRVS	ADC DMIC Clock Reverse control 0： Normal 1： Reverse	RW	0x0
15	ADCFIRCLKDIV	ADC FIR Clock Divisor: 0: /1 1: /3	RW	0x0
14	-	Reserved	R	0x0
13:12	ADCOVFSCLKDIV	ADC OVFS Clock Divisor: 00: /1 01: /2 10: /4 11: reserved	RW	0x0
11:9	-	Reserved	R	0x0
8	ADCCLKSRC	ADC Clock Source 0: AudioPLL0 1: AudioPLL1	R/W	0x0
7:5	-	Reserved	R	0x0
4	ADCCLKPREDIV	ADC Clock Pre-Divisor 0: /2 1: /4	R/W	0x0
3	-	Reserved	R	0x0
2:0	ADCCLKDIV	ADC Clock Divisor 000: /1 001: /2 010: /3 011: /4 100: /6 101: /8 110: /12 111: ---	R/W	0x0

10.4.3.32 CMU_I2STXCLK

I2STX Clock Control register
Offset = 0x00A0

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Bit(s)	Name	Description	R/W	Reset
31	I2SG0LRCLKOEN	I2SG0 LRCLK Output to PAD enable 0: disable 1: enable This bit enables the LRCLK sending to the I2SG0 clock output PAD.	R/W	0x0
30	I2SG0BCLKOEN	I2SG0 BCLK Output to PAD enable 0: disable 1: enable This bit enables the BCLK sending to the I2SG0 clock output PAD.	R/W	0x0
29	I2SG0MCLKOEN	I2SG0 MCLK Output to PAD enable 0: disable 1: enable This bit enables the MCLK sending to the I2SG0 clock output PAD.	R/W	0x0
28	I2STX0CLKSRC	I2STX0 Clock Source: 0: I2SG0 1: I2SG1	R/W	0x0
27:25	-	Reserved	R	0x0
24:23	I2SG0LRCLKPROC	I2SG0 LRCLK Process: 00: 50% DUTY for I2S 01: 1 BCLK for I2S 10: 50% DUTY for PCM 11: 1 BCLK for PCM This bit is valid when I2SG0BLRCLKSRC is set to 0.	R/W	0x0
22:20	I2SG0LRCLKDIV	I2SG0 LRCLK Divisor： 000: /32 001: /64 010: /128 011: /256 100: /16 Others: reserved This bit is valid when I2SG0BLRCLKSRC is set to 0.	R/W	0x0
19:18	I2SG0BCLKDIV	I2SG0 BCLK Divisor： 00: /4 01: /6 10: /8 11: /12 This bit is valid when I2SG0BLRCLKSRC is set to 0.	R/W	0x0
17	-	Reserved	R	
16	I2SG0BLRCLKSRC	I2SG0 BCLK/LRCLK Source： 0: internal from I2SG0_MCLK 1: external from PAD	R/W	0x0
15	-	Reserved	R	0x0
14	I2SG0MCLKEXTREV	I2SG0_MCLK_EXT Reverse： 0：Normal 1：Reversed	R/W	0x1
13:12	I2SG0MCLKSRC	I2SG0_MCLK source： 00：I2SG0CLK 01：DAC_256fs_CLK 10：DAC_128fs_CLK 11：I2SG0_MCLK_EXT	R/W	0x0

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11	I2STX0FIFOCLKEN	I2STX0 FIFO Clock Enable 0: disable 1: enable	R/W	0x0
10:4	-	Reserved	R	0x0
3:0	I2SG0CLKDIV	I2SG0 Clock Divisor. 0000: /1 0001: /2 0010: /3 0011: /4 0100: /6 0101: /8 0110: /12 0111: /16 1000: /24 1001: /32 1010: /48 1011: /96 Others Reserved.	R/W	0x0

10.4.3.33 CMU_I2SRXCLK

I2STX Clock Control register
Offset = 0x00A4

Bit(s)	Name	Description	R/W	Reset
31	I2SG1LRCLKOEN	I2SG1 LRCLK Output to PAD enable 0: disable 1: enable This bit enables the LRCLK sending to the I2SG1 clock output PAD.	R/W	0x0
30	I2SG1BCLKOEN	I2SG1 BCLK Output to PAD enable 0: disable 1: enable This bit enables the BCLK sending to the I2SG1 clock output PAD.	R/W	0x0
29	I2SG1MCLKOEN	I2SG1 MCLK Output to PAD enable 0: disable 1: enable This bit enables the MCLK sending to the I2SG1 clock output PAD.	R/W	0x0
28	I2SRX0CLKSRC	I2SRX Clock Source: 0: I2SG0 1: I2SG1	R/W	0x0
27:25	-	Reserved	R	0x0
24:23	I2SG1LRCLKPROC	I2SG1 LRCLK Process: 00: 50% DUTY for I2S 01: 1 BCLK for I2S 10: 50% DUTY for PCM 11: 1 BCLK for PCM This bit is valid when I2SG1BLRCLKSRC is set to 0.	R/W	0x0
22:20	I2SG1LRCLKDIV	I2SG1 LRCLK Divisor： 000: /32 001: /64	R/W	0x0

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		010: /128 011: /256 100: /16 Others: reserved This bit is valid when I2SG1BLRCLKSRC is set to 0.		
19:18	I2SG1BCLKDIV	I2SG1 BCLK Divisor： 00: /4 01: /6 10: /8 11: /12 This bit is valid when I2SG1BLRCLKSRC is set to 0.	R/W	0x0
17	-	Reserved	R	0x0
16	I2SG1BLRCLKSRC	I2SG1 BCLK/LRCLK Source： 0: internal from I2SG1_MCLK 1: external from PAD	R/W	0x0
15	-	Reserved	R	0x0
14	I2SG1MCLKEXTREV	I2SG1_MCLK_EXT Reverse： 0：Normal 1：Reversed	R/W	0x1
13:12	I2SG1MCLKSRC	I2SG1_MCLK source： 00：I2SG1CLK 01：reserved 10：reserved 11：I2SG1_MCLK_EXT	R/W	0x0
11	I2SRX0FIFOCLKEN	I2SRX0 FIFO Clock Enable 0: disable 1: enable	R/W	0x0
10:4	-	Reserved	R	0x0
3:0	I2SG1CLKDIV	I2SG1 Clock Divisor. 0000: /1 0001: /2 0010: /3 0011: /4 0100: /6 0101: /8 0110: /12 0111: /16 1000: /24 1001: /32 1010: /48 1011: /96 Others Reserved.	R/W	0x0

10.4.3.34 CMU_MEMCLKEN0

CMU_MEMCLKEN Control register0
Offset = 0x00B0

Bit(s)	Name	Description	R/W	Reset
31	SPI1CACHERAMCLKEN	SPI1CACHE RAM clock enable bit: 0: disable 1: enable	R/W	0x1

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30	SPI0CACHERAMCLKEN	SPI0CACHE RAM clock enable bit: 0: disable 1: enable	R/W	0x1
29:26	-	Reserved	R/W	0x0
25	SRAMCLKEN	Share RAM clock enable bit : 0: disable 1: enable	R/W	0x1
24:23	-	Reserved	R/W	0x0
22	RAM16CLKEN	RAM16 clock enable bit : 0: disable 1: enable	R/W	0x1
21	-	Reserved	R/W	0x0
20	RAM15CLKEN	RAM15 clock enable bit : 0: disable 1: enable	R/W	0x1
19	-	Reserved	R/W	0x0
18	RAM14CLKEN	RAM14 clock enable bit : 0: disable 1: enable	R/W	0x1
17	-	Reserved	R/W	0x0
16	RAM13CLKEN	RAM13 clock enable bit : 0: disable 1: enable	R/W	0x1
15	-	Reserved	R/W	0x0
14	RAM12CLKEN	RAM12 clock enable bit : 0: disable 1: enable	R/W	0x1
13	-	Reserved	R/W	0x0
12	RAM11CLKEN	RAM11 clock enable bit : 0: disable 1: enable	R/W	0x1
11	RAM10CLKEN	RAM10 clock enable bit : 0: disable 1: enable	R/W	0x1
10	RAM9CLKEN	RAM9 clock enable bit : 0: disable 1: enable	R/W	0x1
9	RAM8CLKEN	RAM8 clock enable bit : 0: disable 1: enable	R/W	0x1
8	RAM7CLKEN	RAM7 clock enable bit : 0: disable 1: enable	R/W	0x1
7	RAM6CLKEN	RAM6 clock enable bit : 0: disable 1: enable	R/W	0x1
6	RAM5CLKEN	RAM5 clock enable bit : 0: disable 1: enable	R/W	0x1
5	RAM4CLKEN	RAM4 clock enable bit : 0: disable 1: enable	R/W	0x1
4	RAM3CLKEN	RAM3 clock enable bit :	R/W	0x1

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		0: disable 1: enable		
3	RAM2CLKEN	RAM2 clock enable bit : 0: disable 1: enable	R/W	0x1
2	RAM1CLKEN	RAM1 clock enable bit : 0: disable 1: enable	R/W	0x1
1	RAM0CLKEN	RAM0 clock enable bit : 0: disable 1: enable	R/W	0x1
0	ROMCLKEN	ROM clock enable bit : 0: disable 1: enable	R/W	0x1

10.4.3.35 CMU_MEMCLKEN1

CMU_MEMCLKEN Control register1
Offset = 0x00B4

Bit(s)	Name	Description	R/W	Reset
31:29	-	Reserved	R/W	0x0
28	JPEGOUTRAM3CLKEN	JPEG OUTRAM3 Clock Enable: 0: disable 1: enable	R/W	0x0
27	JPEGOUTRAM2CLKEN	JPEG OUTRAM2 Clock Enable: 0: disable 1: enable	R/W	0x0
26	JPEGOUTRAM1CLKEN	JPEG OUTRAM1 Clock Enable: 0: disable 1: enable	R/W	0x0
25	JPEGOUTRAM0CLKEN	JPEG OUTRAM0 Clock Enable: 0: disable 1: enable	R/W	0x0
24	JPEGRAMCLKEN	JPEG RAM Clock Enable: 0: disable 1: enable	R/W	0x0
23:9	-	Reserved	R/W	X
8	USBRAMCLKEN	URAM0,1 Clock Enable: 0: disable 1: enable	R/W	0x1
7:1	-	Reserved	R/W	0x0
0	AUDDSPRAMCLKEN	AUDDSPDTCRAM, PTCRAM Clock Enable: 0: disable 1: enable	R/W	0x1

10.4.3.36 CMU_MEMCLKSRC0

CMU_MEMCLK Source Control register0
Offset = 0x00C0

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R/W	0x0

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29:28	SRAMCLKDIV	Share RAM Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0
27:25	SRAMCLKSRC	Share RAM Clock Source: 000: RC4M 001: HOSC 010: COREPLL 011: RC192M 100: RC64M	R/W	0x0
24	-	Reserved	R/W	0x0
23:22	RAM16CLKSRC	RAM16 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
21:20	RAM15CLKSRC	RAM15 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
19:18	RAM14CLKSRC	RAM14 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
17:16	RAM13CLKSRC	RAM13 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
15:14	RAM12CLKSRC	RAM12 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
13:12	RAM11CLKSRC	RAM11 Clock Source: 00: CPUCLK 01: AUDIODSPCLK Others Reserved.	R/W	0x0
11	RAM10CLKSRC	RAM10 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
10	RAM9CLKSRC	RAM9 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
9	RAM8CLKSRC	RAM8 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
8	RAM7CLKSRC	RAM7 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
7	RAM6CLKSRC	RAM6 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
6	RAM5CLKSRC	RAM5 Clock Source: 0: CPUCLK	R/W	0x0

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		1: AUDIODSPCLK		
5	RAM4CLKSRC	RAM4 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
4	RAM3CLKSRC	RAM3 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
3	RAM2CLKSRC	RAM2 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
2	RAM1CLKSRC	RAM1 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
1	RAM0CLKSRC	RAM0 Clock Source: 0: CPUCLK 1: AUDIODSPCLK	R/W	0x0
0	-	Reserved	R/W	0x0

10.4.3.37 CMU_MEMCLKSRC1

CMU_MEMCLK Source Control register1
Offset = 0x00C4

Bit(s)	Name	Description	R/W	Reset
31:29	-	Reserved	R/W	0x0
28	JPEG_OUTRAM3CLKSRC	JPEG OUTRAM3 Clock Source: 0: CPUCLK 1: JPEGCLK	R/W	0x0
27	JPEG_OUTRAM2CLKSRC	JPEG OUTRAM2 Clock Source: 0: CPUCLK 1: JPEGCLK	R/W	0x0
26	JPEG_OUTRAM1CLKSRC	JPEG OUTRAM1 Clock Source: 0: CPUCLK 1: JPEGCLK	R/W	0x0
25	JPEG_OUTRAM0CLKSRC	JPEG OUTRAM0 Clock Source: 0: CPUCLK 1: JPEGCLK	R/W	0x0
24	JPEGRAMCLKSRC	JPEG internal RAM Clock Source: 0: CPUCLK 1: JPEGCLK	R/W	0x0
23:14	-	Reserved	R/W	0x0
13:12	USBRAMPRECLKDIV	USBRAM0, RAM1 Pre-Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0
11:9	USBRAMPRECLKSRC	USBRAM0, RAM1 Pre-Clock Source: 000: RC4M 001: HOSC 010: COREPLL 011: RC192M 100: RC64M	R/W	0x0

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		Others: Reserved		
8	USBRAMCLKSRC	USBRAM0, RAM1 Clock Source: 0: USBRAMPRECLK 1: USBctl_URAM_CLK	R/W	0x0
7:6	-	Reserved	R/W	0x0
5:4	AUDDSPTCRAMPRECLKDIV	AUDDSPDTCRAM, PTCRAM Pre-Clock Divisor: 00: /1 01: /2 10: /4 11: /8	R/W	0x0
3:1	AUDDSPTCRAMPRECLKSRC	AUDDSPDTCRAM, PTCRAM Pre-Clock Source: 000: RC4M 001: HOSC 010: COREPLL 011: RC192M 100: RC64M	R/W	0x0
0	AUDDSPTCRAMCLKSRC	AUDDSPDTCRAM, PTCRAM Clock Source: 0: AUDDSPTCRAMPRECLK 1: AUDDSPCLK	R/W	0x0

10.4.3.38 CMU_S1CLKCTL

CMU S1CLK Control register
Offset = 0x00D0

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2	RC64M_S1EN	RC64M S1 Entering Control: 0: disable 1: enable	R/W	0x0
1	HOSC_S1EN	HOSC S1 Entering Control: 0: disable 1: enable	R/W	0x1
0	RC4M_S1EN	RC4M S1 Entering Control: 0: disable 1: enable	R/W	0x1

10.4.3.39 CMU_S1BTCLKCTL

CMU S1BTCLK Control register
Offset = 0x00D4

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2	RC64M_S1BTEN	RC64M S1BT Entering Control: 0: disable 1: enable	R/W	0x0
1	HOSC_S1BTEN	HOSC S1BT Entering Control: 0: disable 1: enable If this bit is set, HOSC will be enabled all the S1BT time.	R/W	0x1
0	RC4M_S1BTEN	RC4M S1BT Entering Control: 0: disable	R/W	0x1

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		1: enable		

10.4.3.40 CMU_S2CLKCTL

CMU S2SCLK Control register
Offset = 0x00DC

Bit(s)	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2	RC64M_S2EN	RC64M S2 Entering Control: 0: disable 1: enable	R/W	0x0
1	HOSC_S2EN	HOSC S2 Entering Control: 0: disable 1: enable If this bit is set, HOSC will be enabled all the S2S time.	R/W	0x0
0	RC4M_S2EN	RC4M S2 Entering Control: 0: disable 1: enable	R/W	0x0

10.4.3.41 CMU_S3CLKCTL

CMU S3CLK Control register
Offset = 0x00E0

Bit(s)	Name	Description	R/W	Reset
31:4	-	Reserved	R	0x0
3	S3RAM4CLKON	S3 ShareRAMCLK, SPIMTxCLK and I2CMTxCLK on: 0: disable 1: enable	R/W	0x0
2	-	Reserved	R	0x0
1	HOSC_S3EN	HOSC S3 Entering Control: 0: disable 1: enable If this bit is set, HOSC will be enabled all the S3 time.	R/W	0x0
0	RC4M_S3EN	RC4M S3 Entering Control: 0: disable 1: enable	R/W	0x0

10.4.3.42 CMU_WDCLK

CMU Watchdog CLK Control register
Offset = 0x00E4

Bit(s)	Name	Description	R/W	Reset
31:2	-	Reserved	R	0x0
1	CPUDS_WDCLKEN	CPU Deepsleep WatchDog Clock Enable 0: disable 1: enable	R/W	0x0
0	WDCLKSRC	WatchDog Clock Source 0: RC32K	R/W	0x0

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		1: CK32K768		

10.4.3.43 CMU_GPIOCLKCTL

CMU GPIO interrupt Clock Control register
Offset = 0x00E8

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	GPIOINTCLKSRC	GPIO interrupt Clock Source: 0: RC32K 1: RC4M	R/W	0x1

10.4.3.44 CMU_PMUWKUPCLKCTL

CMU PMU STATE And WAKEUP Clock Control register
Offset = 0x00EC

Bit(s)	Name	Description	R/W	Reset
31:1	-	Reserved	R	0x0
0	PMUWKUPCLKSRC	PMU Wakeup Clock Source: 0: RC32K 1: RC4M/4	R/W	0x1

10.4.3.45 CMU_SPIMT0CLK

SPIMT0 Clock Control Register
Offset = 0x0100

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SPIMT0CLKSRC	SPIMT0 Controller Clock Source: 00: RC4M 01: RC64M/2 10: HOSC Others: Reserved	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPIMT0CLKDIV	SPIMT0 Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0

10.4.3.46 CMU_SPIMT1CLK

SPIMT1 Clock Control Register
Offset = 0x0104

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Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	SPIMT1CLKSRC	SPIMT1 Controller Clock Source: 00: RC4M 01: RC64M/2 10: HOSC Others: Reserved	R/W	0x0
7:4	-	Reserved	R	0x0
3:0	SPIMT1CLKDIV	SPIMT1Clock Divisor: 0: /1 1: /2 2: /3 3: /4 4: /5 … 13: /14 14: /1.5 15: /2.5	R/W	0x0

10.4.3.47 CMU_I2CMT0CLK

I2CMT0 Clock Control Register
Offset = 0x0110

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	I2CMT0CLKSRC	I2CMT0 Controller Clock Source: 00: RC4M 01: RC64M/2 10: HOSC Others: Reserved	R/W	0x0
7:2	-	Reserved	R	0x0
1:0	I2CMT0CLKDIV	I2CMT0 Clock Divisor: 0: /1 1: /2 2: /4 3: /8	R/W	0x0

10.4.3.48 CMU_I2CMT1CLK

I2CMT1 Clock Control Register
Offset = 0x0114

Bit(s)	Name	Description	R/W	Reset
31:10	-	Reserved	R	0x0
9:8	I2CMT1CLKSRC	I2CMT1 Controller Clock Source: 00: RC4M 01: RC64M/2 10: HOSC Others: Reserved	R/W	0x0
7:2	-	Reserved	R	0x0
1:0	I2CMT1CLKDIV	I2CMT1Clock Divisor: 0: /1 1: /2	R/W	0x0

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		2: /4 3: /8		

10.5 RTC

This part have 3 individual modules: Calendar, Watch Dog (WD) and Timer0/1/2/3/4/5.

10.5.1 Features

- An individual power supply: VDD
- Internal oscillator, LOSC optional
- Calendar with an alarm IRQ which can wake up the PMU
- Interval Timer can wake up CPU periodically
- Six Timers with IRQS, all of them are low power timers
- A watch dog which can be configured optional as IRQ or Reset

10.5.2 RTC Register List

Table 10-9 RTC block base address

Name	Physical Base Address	KSEG1 Base Address
RTC	0x4000C000	0x4000C000

Table 10-10 RTC Controller Registers

Offset	Register Name	Description
0x00	RTC_CTL	RTC Control Register
0x08	RTC_MSALM	RTC MilliSecond Alarm Register
0x0C	RTC_HMSALM	RTC Day Hour Minute and Second Alarm Register
0x10	RTC_YMDALM	RTC Year Month Date Alarm Register
0x14	RTC_MS	RTC MilliSecond Register
0x18	RTC_HMS	RTC Day Hour Minute and Second Register
0x1C	RTC_YMD	RTC Year Month Date Register
0x20	WD_CTL	Watch Dog Control register
0x24	RTC_IT_VAL	RTC Interval Timer Value Register

10.5.3 RTC Register Description

10.5.3.1 RTC_CTL

Calendar Control Register
Offset=0x0000

Bits	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29	IT_IRQ_PD	Interval Timer IRQ Pending 0:normal 1: Pending Write 1 auto clean	R/W	0x0
28	IT_IRQ_EN	Interval Timer IRQ Enable 0:disable 1:enable	R/W	0x0
27:25	-	Reserved	R	0x0

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24	IT_EN	Interval Timer Enable 0:disable 1:enable	R/W	0x0
23:18	-	Reserved	R	0x0
17	CAL_CLK_SEL	CAL clock Select 0:4HZ Select 1:100HZ Select	R/W	0x0
16	RTC32768HZ_SEL	RTC32768HZ Clock select 0 : LO32768Hz 1 : EXT_32768HZ	R/W	0x0
15:8	-	Reserved	R	0x0
7	LEAP	RTC Leap Year bit 1: leap year 0: not leap year	R	0x1
6:5	-	Reserved	R	0x0
4	CAL_EN	Calendar Enable 1: Enable 0: Disable	R/W	0x0
3:2	4HZ_CLK_SEL	01: 32768_division 10: HOSC_division 00: HCL_division Others: reserved	R/W	0x0
1	ALIE	Alarm IRQ Enable 1: Enable 0: Disable	R/W	0x0
0	ALIP	Alarm IRQ Pending bit, writing 1 to this bit will clear it	R/W	0x0

NOTE: The CAL_EN bit must be disabled when The RTC_HMS/RTC_YMD/RTC_MS register being written Or error will occur.

10.5.3.2 RTC_MSALM

Offset=0x0008

Bits	Name	Description	R/W	Reset
31:7	-	Reserved	R	0x0
6:0	MS	MilliSecond setting 00H – 1FH	R/W	0x0

Note: MilliSecond =MS*10, ignore when RTC_4HZ Select.

10.5.3.3 RTC_HMSALM

Offset=0x000C

Bits	Name	Description	Access	Reset
31:21	-	Reserved	R	0
20:16	HOURALM	Alarm hour setting 00H – 17H	R/W	0
15:14	-	Reserved	R	0
13:8	MINALM	Alarm minute setting 00H – 3BH	R/W	0
7:6	-	Reserved	R	0

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5:0	SECALM	Alarm second setting 00H – 3BH	R/W	0

10.5.3.4 RTC_YMDALM

Offset=0x0010

Bits	Name	Description	R/W	Reset
31:23	-	Reserved	R	0x0
22:16	YEARALM	Alarm year setting 00H – 63H	R/W	0x0
15:12	-	Reserved	R	0x0
11:8	MONALM	Alarm month setting 01H – 0CH	R/W	0x1
7:5	-	Reserved	R	0x0
4:0	DATEALM	Alarm day setting 01H – 1FH	R/W	0x1

Note: When this register is set to zero, the RTC alarm alarms daily. Otherwise when it is set to nonzero, the RTC alarm alarms on the specific date and time.

10.5.3.5 RTC_MS

Offset=0x0014

Bits	Name	Description	R/W	Reset
31:7	-	Reserved	R	0x0
6:0	MS	MilliSecond setting 00H –63H	R/W	0x0

Note: MilliSecond=MS*10, ignore when RTC_4HZ Select.

10.5.3.6 RTC_HMS

Offset=0x0018

Bits	Name	Description	R/W	Reset
31:21	-	Reserved	R	0x0
20:16	HOUR	Time hour setting 00H – 17H	R/W	0x0
15:14	-	Reserved	R	0x0
13:8	MIN	Time minute setting 00H – 3BH	R/W	0x0
7:6	-	Reserved	R	0x0
5:0	SEC	Time second setting 00H – 3BH	R/W	0x0

10.5.3.7 RTC_YMD

Offset=0x001C

Bits	Name	Description	R/W	Reset
31:23	-	Reserved	R	0x0

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22:16	YEAR	Time year setting 00H – 63H	R/W	0x0
15:12	-	Reserved	R	0x0
11:8	MON	Time month setting 01H – 0CH	R/W	0x1
7:5	-	Reserved	R	0x0
4:0	DATE	Time day setting 01H – 1FH	R/W	0x1

10.5.3.8 WD_CTL

Offset=0x0020（VDD）

Bits	Name	Description	R/W	Reset
31..7	-	reserved	R	0x0
6	IRQP	Watch dog IRQ pending bit, writing 1 to this bit will clear it	RW	0x0
5	SIGS	Watchdog Signal (IRQ or Reset-) Select.0: Reset,1: irq-. 0:Send Reset signal when watchdog overflow. 1:Send IRQ signal when watchdog overflow.	RW	0x0
4	WDEN	Watch Dog timer enable, when WD timer is enabled and the WD timer overflows, an internal reset (WDRST-) is generated to force the system into reset status and then reboot. 1: Enable 0: Disable	RW	0x0
3..1	CLKSEL	Watch Dog timer Clock Select, WDCKS Clock Selected Watch Dog Length The watch dog's overflow value is 180. 000 1khz 176 ms 001 512hz 352 ms 010 256hz 703ms 011 128hz 1.4 s 100 64hz 2.8s 101 32hz 5.6 s 110 16hz 11.2s 111 10ms	RW	0x0
0	CLR	Clear bit, write 1 to clear WD timer, cleared automatically	RW	0x0

10.5.3.9 RTC_IT_VAL

RTC Interval Timer Value Register (VDD)
Offset=0x24

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	VAL	Read or write Interval Timer/Counter value register Note: If set IT_VAL=n, IRQ will happen after n+1 RTC_CLK.	R/W	0x0

10.5.4 TIMER Register List

Table 10-11 TIMER block base address

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Name	Physical Base Address	KSEG1 Base Address
TIMER_REGISTER	0x4000C100	0x4000C100

Table 10-12 TIMER Controller Registers

Offset	Register Name	Description
0x00	T0_CTL	Timer0 Control register
0x04	T0_VAL	Timer0 Value
0x08	T0_CNT	Timer0 current counter register
0x20	T1_CTL	Timer1 Control register
0x24	T1_VAL	Timer1 Value
0x28	T1_CNT	Timer1 current counter register
0x40	T2_CTL	Timer2 Control register
0x44	T2_VAL	Timer2 Value
0x48	T2_CNT	Timer2 current counter register
0x60	T3_CTL	Timer3 Control register
0x64	T3_VAL	Timer3 Value
0x68	T3_CNT	Timer3 current counter register
0x80	T4_CTL	Timer4 Control register
0x84	T4_VAL	Timer4 Value
0x88	T4_CNT	Timer4 current counter register
0xA0	T5_CTL	Timer5 Control register
0xA4	T5_VAL	Timer5 Value register
0xA8	T5_CNT	Timer5 current counter register

10.5.5 TIMER Register Description

10.5.5.1 T0_CTL

Timer0 control register
Offset=0x0000(VDD)

Bit(s)	Name	Description	R/W	Reset
31:12	-	reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	Reserved	R	0x0
5	EN	Timer 0 Enable 0: Disable 1: Enable	R/W	0x0
4	-	Reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0

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2	RELO	Timer 0 Reload. 0: Not reload 1: Reload	R/W	0x0
1	ZIEN	Timer0 IRQ Enable When this bit is enabled, Timer0_Zero_IRQ sent out the irq signal until the pending bit was cleared.	R/W	0x0
0	ZIPD	Timer0 IRQ Pending, Writing 1 to clear this bit.	R/W	0x0

10.5.5.2 T0_VAL

Timer0 value register
Offset=0x0004(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	VAL	Read or write Timer/Counter value register Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK.	R/W	0x0

10.5.5.3 T0_CNT

Timer0 current counter register
Offset=0x0008(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	CNT	Read or write current Timer0 value	R	0x0

10.5.5.4 T1_CTL

Timer1 control register
Offset=0x0020(VDD)

Bit(s)	Name	Description	R/W	Reset
31:12	-	reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	Reserved	R	0x0
5	EN	Timer Enable 0: Disable 1: Enable	R/W	0x0
4	-	Reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0

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2	RELO	Timer Reload. 0: Not reload 1: Reload	R/W	0x0
1	ZIEN	Timer IRQ Enable When this bit is enabled, TimerX_Zero_IRQ sent out the irq signal until the pending bit was cleared.	R/W	0x0
0	ZIPD	Timer IRQ Pending, Writing 1 to clear this bit.	R/W	0x0

10.5.5.5 T1_VAL

Timer1 value register
Offset=0x0024(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	VAL	Read or write current Timer1 value Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK.	R/W	0x0

10.5.5.6 T1_CNT

Timer1 current counter register
Offset=0x0028(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	CNT	Read or write current Timer1 value	R	0x0

10.5.5.7 T2_CTL

Timer2 control register
Offset=0x0040(VDD)

Bits	Name	Description	R/W	Reset
31:12	-	Reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	reserved	R	0x0
5	EN	Timer Enable 0: Disable 1: Enable	R/W	0x0
4	-	reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0
2	RELO	Timer Reload enable 0: Not reload 1: Reload	R/W	0x0
1	ZIEN	Timer IRQ Enable When this bit is enabled, TimerX_IRQ sent out the irq signal until the pending bit was cleared.	R/W	0x0

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		If DIR='0',T2_CNT compare with ZERO **If DIR='1',T2_CNT compare with T2_VAL.** **For more detail reference to timer2/3 block diagram**		
0	ZIPD	Timer2 IRQ Pending, Writing 1 to clear this bit. **If occurred a timer overflow or zero, this pending would be set to '1'**	R/W	0x0

10.5.5.8 T2_VAL

Timer2 value register
Offset=0x0044(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	VAL	Set timer counter value. If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero. If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL. When in counter mode,Tx_VAL[7:0] was used. Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK.	R/W	0x0

10.5.5.9 T2_CNT

Timer2 current counter register
Offset=0x0048(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	CNT	Timer current value registers.	R	0x0

10.5.5.10 T3_CTL

Timer3 control register
Offset=0x0060(VDD)

Bits	Name	Description	R/W	Reset
31:12	-	Reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	reserved	R	0x0
5	EN	Timer Enable	R/W	0x0

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		0: Disable 1: Enable		
4	-	reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0
2	RELO	Timer Reload enable 0: Not reload 1: Reload	R/W	0x0
1	ZIEN	Timer IRQ Enable When this bit is enabled, Timer3_IRQ sent out the irq signal until the pending bit was cleared. If DIR='0', T3_CNT compare with ZERO If DIR='1', T3_CNT compare with T3_VAL.	R/W	0x0
0	ZIPD	Timer3 IRQ Pending, Writing 1 to clear this bit. If occurred a timer overflow or zero, this pending would be set to '1'	R/W	0x0

10.5.5.11 T3_VAL

Timer3 value register
Offset=0x0064(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	VAL	Set timer counter value. If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero. If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL. When in counter mode,Tx_VAL[7:0] was used. Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK.	R/W	0x0

10.5.5.12 T3_CNT

Timer3 current counter register
Offset=0x0068(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	CNT	Timer current value registers.	R	0x0

10.5.5.13 T4_CTL

Timer4 control register
Offset=0x0080(VDD)

Bit(s)	Name	Description	R/W	Reset

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31:12	-	reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	Reserved	R	0x0
5	EN	Timer 0 Enable 0: Disable 1: Enable	R/W	0x0
4	-	Reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0
2	RELO	Timer Reload. 0: Not reload 1: Reload	R/W	0x0
1	ZIEN	Timer IRQ Enable When this bit is enabled, TimerX_Zero_IRQ sent out the irq signal until the pending bit was cleared.	R/W	0x0
0	ZIPD	Timer IRQ Pending, Writing 1 to clear this bit.	R/W	0x0

10.5.5.14 T4_VAL

Timer4 value register
Offset=0x0084(VDD)

Bit(s)	Name	Description	R/W	Reset
31:0	VAL	Set timer counter value. If timer setting in count up mode, time would count up from zero to the value set in Tx_VAL. when the current timer counter equal to Tx_VAL, it would cause an irq. If timer reload mode was set, Tx_VAL would be reload auto and timer current value was reset to zero. If timer setting in countdown mode, timer would countdown form Tx_VAL to zero. When the current timer counter equal to zero. it would cause an irq. If timer reload mode was set, zero would be reload auto and timer current value was reset to Tx_VAL. When in counter mode,Tx_VAL[7:0] was used. Note: If set Tx_VAL=n, irq would cause after n+1 Tx_CLK.	R/W	0x0

10.5.5.15 T4_CNT

Timer4 current counter register
Offset=0x0088(VDD)

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Bit(s)	Name	Description	R/W	Reset
31:0	CNT	Timer current value registers.	R	0x0

10.5.5.16 T5_CTL

Timer5 control register
Offset=0xA0

Bits	Name	Description	R/W	Reset
31:12	-	Reserved	R	0x0
11	DIR	Timer Counting direction set: 0: down 1: up	R/W	0x0
10:9	MODE_SEL	Timer mode select: 00: normal timer 11: continue mode Others : reserved	R/W	0x0
8:6	-	Reserved	R	0x0
5	EN	Timer Enable 0:Disable 1:Enable	R/W	0x0
4	-	reserved	R	0x0
3	PAUSE	PAUSE Counter 0: normal 1: pause	R/W	0x0
2	RELO	Timer Reload enable 0:Not reload 1:Reload	R/W	0x0
1	ZIEN	Timer IRQ Enable When this bit is enabled, Timer5_Zero_IRQ sent out the irq signal until the pending bit was cleared. **If DIR='0', T5_CNT compare with ZERO** **If DIR='1', T5_CNT compare with T5_VAL.** **For more detail reference to block diagram**	R/W	0x0
0	ZIPD	Timer IRQ Pending, Writing 1 to clear this bit.	R/W	0x0

10.5.5.17 T5_VAL

Timer5 value register
Offset=0xA4

Bits	Name	Description	R/W	Reset
31:0	VAL	Set timer counter value. If timer setting in count up mode, time would count up from zero to the value set in T5_VAL. When the current timer counter equal to T5_VAL, it would cause an irq. If timer reload mode was set, T5_VAL would be reload auto and timer current value was reset to zero. If timer setting in countdown mode, timer would countdown form T5_VAL to zero. When the current timer counter equal to zero. It would cause an irq. If timer reload mode was set, zero	R/W	0x0

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		would be reload auto and timer current value was reset to T5_VAL. When in counter mode,T5_VAL[7:0] was used. Note: If set T5_VAL=n, irq would cause after n+1 T5_CLK.		

10.5.5.18 T5_CNT

Timer5 current counter register
Offset=0xA8

Bits	Name	Description	R/W	Reset
31:0	CNT	current Timer5 value	R	0x0

10.6 PPI & Wake up Controller

10.6.1 Features

The Programmable peripheral interconnect (PPI)

- 34 triggers to interact autonomously with each tasks independent of the CPU
- 16 tasks can be used

Wake up Controller

- Eight interrupts as wake up source
- Four power status：S4, S3, S2, S1
- Eight interrupts can be configured as wake up source for each power status
- Wake up Controller can change the power status by the interrupt

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11 Storage

- ATS3089x integrates two SD/MMC controller: SD0 and SD1
- SIP 128Mbit HSPI PSRAM for gorgeous UI performance through the SPI1 interface.

11.1 SD/MMC

SD/MMC Card Controller Features

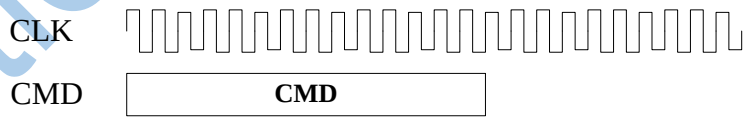
- Fully compliant with MMC Specification 4.3
- Support SDcard up to 2.0
- Support SDIO up to 2.0
- Integrated Clock Delay Chain Technique to Regulate Card Interface Timing: Latching Delay Chain for input Signal, Output Delay Chain for output signal.
- Integrated Watchdog timeout Counter to report Exception happening.
- Integrated Pull up resistance (value 50Komh) for Data and CMD Line.
- Integrated CRC calculate and check circuit.
- Send continuous clock to support SDIO card.
- Support 3.1/1.8V CLK PAD voltage.
- Support 3.1/1.8V CMD PAD voltage.
- Support 3.1/1.8V DAT PAD voltage.
- Band Width: 50MByte/S
- Maximal SD interface Clock: 50MHz

11.1.1 Function Description

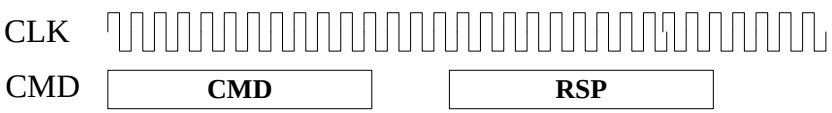
11.1.1.1 Transfer Mode Description

We summarized all SD card operation and ranged them into 10 transfer modes, which the design of SD card controller is based on. Use the sequence set of propriety transfer modes, it can manage all SD bus operations. These transfer modes are described as following:

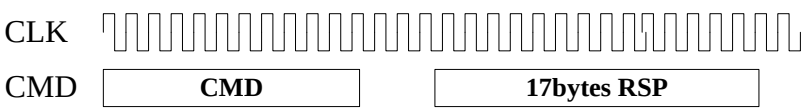
TM_0: Transfer only CMD



TM_1: Send CMD and receive 6 byte RSP:



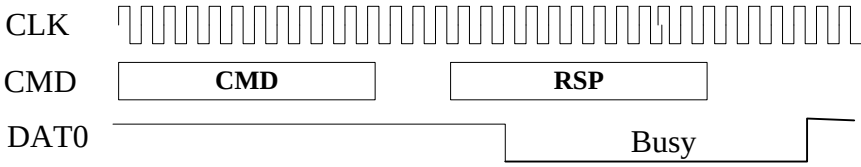
TM_2: Send CMD and receive 17 byte RSP:



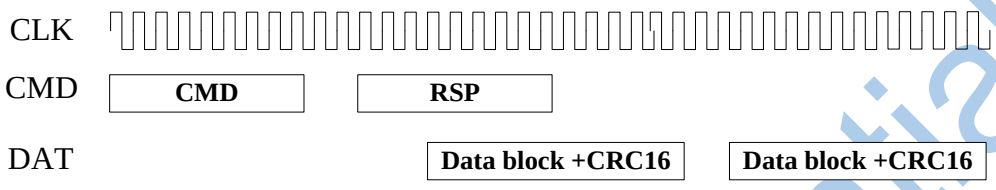
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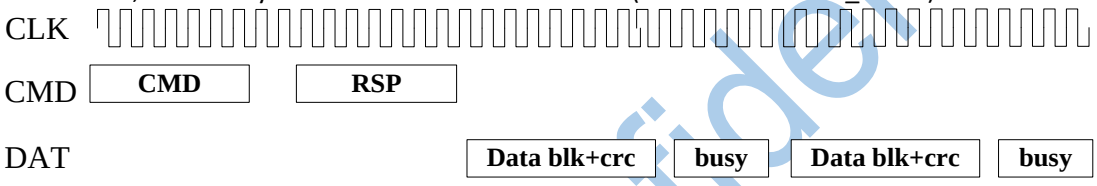
TM_3: Send CMD, receive 6byte RSP and wait busy signal on data0:



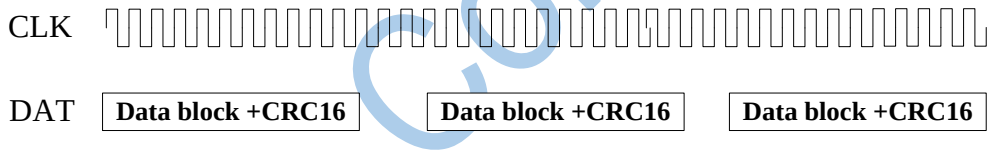
TM_4: Send CMD, receive 6byte RSP and receive data in (in sight of SD controller) mode:



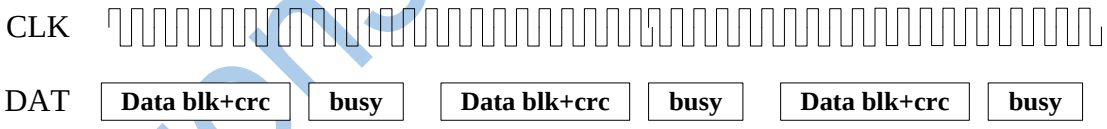
TM_5: Send CMD, receive6byte RSP and send data out mode (include check RCR_status).



TM_6: Transfer data in without command and response:



TM_7: Transfer data out without command and response:



TM_8: Transfer only Clocks (with a cycle number determined by TCN in SD_CTL bit 11~8):

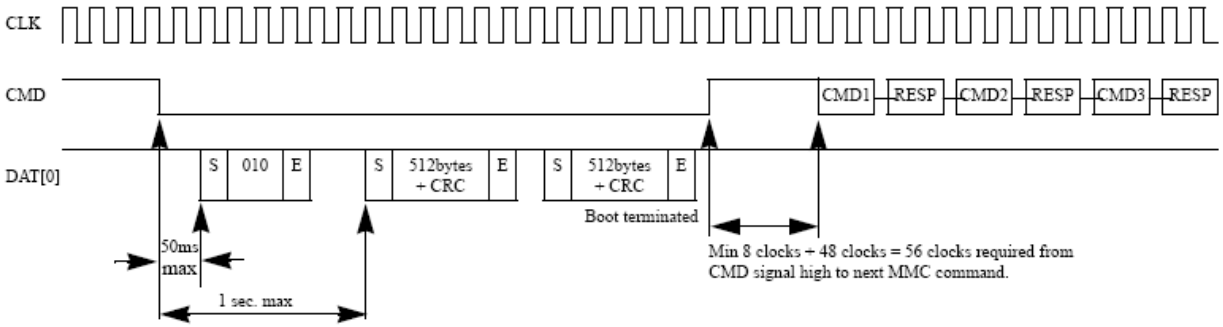


Note: the clock number will be a multiplicative of 16

TM_9&TM_10: BOOT mode transfer data in (TM_Mode9). (Hardware check data crc16, do not check boot mode acknowledge.):

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BOOT mode Transfer data in mode (TM_mode10). (Hardware check data crc16, check boot mode acknowledge).TM_mode10, the same as mode9 except the controller need to check boot acknowledge.

We manage the SD card based on it's state machine, so before every operation the driver developer must know the current state of the card, then select a certain CMD to send to change the state of card or read/write it. Every CMD has an ARG, a RSP type, and may has data in or out operation.
Frist, select a correct transfer mode, and configure the ARG register. When there are data need to be transmitted, configure the data transfer master (DMA or CPU) as needed.
Then, Start the transfer. When the data to transfer is ready, start data transfer master, and then wait both of them to be finished.
Finally, check every state that related to the transfer: CRC7_ERROR, CMD_NO_RSP, CRC16_ERROR, TIMEOUT.

11.1.1.2 Hardware watchdog timeout counter

Since there are lots of waiting situation of transfer mode, like wait until a card quit the busy state, wait until a card to send back data for a read CMD. To simplification the Software timeout configuration, we build a watchdog timeout counter in SDC module. When the transfer is no data, or just has one sector to transfer, this timeout counter behave normally. When transfer multiple sectors (TM_4 or TM_5) the value of the counter is reset at every beginning of each sector, it called the "watchdog" timeout counter.
If Transfer Watchdog timeout function enabled (SD_CTL bit31 is set), then Hardware Watchdog timer starts counting after transfer start. The value of the counter is keep counting until the transfer is finished, auto cleared at the end of every transfer. When the counter's value equals to the value set in TOUTCNT (SD_CTL, bits 30~24), then a timeout error will be set in SD_STATE bit15, and the current transfer will stop at the same time.
The time out counter running clock is described as following:

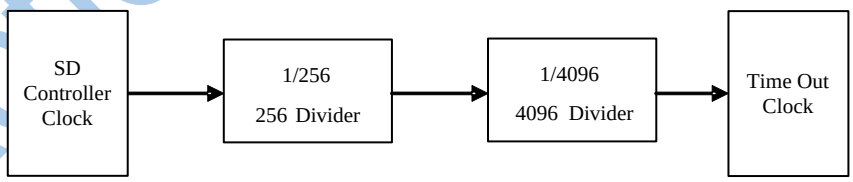


Figure 11-1 timeout clock generating diagram.

Hardware Timeout was only used in data in/data out mode/busy checked mode/boot mode, such as mode3, 4, 5, 6, 7, 9, 10.
If Hardware Timeout Counter's value equals '0', the timeout time is 128*(Timeout Clock Period), is the Max Timeout time.
If Hardware Timeout Counter equal '1', the timeout time is 1*(Timeout Clock Period),The Minimum Timeout time.
As Example, If SD Card Clock frequency is 10MHz, SD Card Clock period is 100ns, Hardware Timeout clock

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frequency is 10/256/4096MHz, Timeout clock period is 100*256*4096ns =104.8576ms. If Hardware Timeout Counter's value equals 0, the timeout time is 128*104.8576ms; If Hardware Timeout Counter's value equals 1, the timeout time is 1*104.8576ms.
If the SD clock is another frequency, the timeout period can be analogy in a simple way like table below:

Table11-1 The relationship of Timeout & Clock

Card clk	Min timeout time (1)	Max timeout time (128)
10Mhz	104.8576ms	13.4218s
20Mhz	104.8576ms/2	13.4218s /2
30Mhz	104.8576ms/3	13.4218s /3
40Mhz	104.8576ms/4	13.4218s /4
50Mhz	104.8576ms/5	13.4218s /5

11.1.2 Operation Manual

11.1.2.1 Typical software operation sequence

1, SDC send CMD and received RSP operation sequence
1) Configure SD_CMD with CMD_index.
2) Configure SD_ARG with Appropriate argument.
3) Select one of the 10 Transfer mode (Configure SD_CTL Bits0～3)
4) Set SD_CTL bit7, start transfer.
5) Wait until SD_CTL bit7 is clear.
6) Check NO_RSP/CRC7/CRC16 in SD_STATE register.

2, SDC send CMD received RSP and transfer data operation sequence
1) Configure SD_CMD with CMD_index.
2) Configure SD_ARG with appropriate argument (card data LBA).
3) Configure SD_BLK_SZ/SD_BLK_NUM with appropriate transfer block size and block number.
4) Select one of the 4 data trans Transfer modes (Configure SD_CTL BIT0～3), one of TM4,5,6,7.
5) Configure DMA Source/destination/transfer mode/transfer number eg.
6) Start DMA transfer
7) Start SDC transfer (set SD_CTL bit7)
8) Wait until DMA transfer finished.
9) Wait until SDC transfer finished. (SD_CTL bit7 is clear)
10) Check NO_RSP/CRC7/CRC16/TIMEOUT status of SDC_STATE register.

3, eMMC BOOT mode operation sequence
1) Configure SD_BLK_SZ/SD_BLK_NUM with appropriate transfer block size and block number.
2) Select one of the 2 BOOT modes (Configure SD_CTL BIT0～3), as one of TM9、10.
3) Configure DMA Source/destination/transfer mode/transfer number eg.
4) Pull CMD line LOW by set CMD_LOW_EN.
5) Start SDC transfer. (Set SD_CTL bit7)
6) Wait DMA transfer finish
7) Wait SDC transfer finished. (SD_CTL bit7 is clear)
8) Check BOOT_ACK/CRC16/TIME_OUT status of SD_STATE

11.1.3 SDC Register List

Table 11-2 SD Controller Registers Address

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Name	Physical Base Address	KSEG1 Base Address
SD0_CONTROLLER_REGISTERS	0x40054000	0x40054000
SD1_CONTROLLER_REGISTERS	0x40058000	0x40058000

Table 11-3 SD Controller Registers

Offset	Register Name	Description
0x00	SDx_EN	SD MMC card enable register
0x04	SDx_CTL	SD MMC card control register
0x08	SDx_STATE	SD MMC card STATU register
0x0C	SDx_CMD	SD MMC card send command register
0x10	SDx_ARG	SD MMC argument register
0x14	SDx_RSPBUF0	SD MMC RSP Buffer0 (Bit31~0)register
0x18	SDx_RSPBUF1	SD MMC RSP Buffer1 (Bit63~32)register
0x1C	SDx_RSPBUF2	SD MMC RSP Buffer2 (Bit95~64)register
0x20	SDx_RSPBUF3	SD MMC RSP Buffer3 (Bit127~96)register
0x24	SDx_RSPBUF4	SD MMC RSP Buffer4 (Bit135~128)register
0x28	SDx_DAT	SD MMC DATA register
0x2C	SDx_BLK_SIZE	SD Block size
0x30	SDx_BLK_NUM	SD Block number

11.1.4 SD0 Controller Register Description

11.1.4.1 SD0_EN

SD/MMC controller enable register
Offset = 0x0000

Bit	Name	Description	RW	Reset
31:10	-	Reserved	R	x
9	CSS	Clock signal status select in idle mode 0:drive to low 1:drive to high	RW	0
8	CLKS	SD MMC clock output pad select 0: CLK to pad SD_CLK0 1: CLK to pad SD_CLK1	RW	0
7	EN	SD module Enable 1: Enable 0: Disable	RW	0
6	AHB_DMA_SEL	SDC CHANEL(AHB OR DMA BUS)SELECTION 1: DMA BUS 0: AHB BUS	RW	0
5	FIFO_WIDTH	FIFO width selection 1: 8bit 0: 32bit	RW	0
4	-	Reserved	R	0
3	SDIOEN	SDIO function enable 1:enable 0:disable	RW	0
2	-	Reserved	R	0
1:0	DATAWID	SD Interface Data bus Width selection 00b :1 bit	RW	00

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		01b: 4 bit 10b: 8 bit 11b: Reserved		

11.1.4.2 SD0_CTL

SD/MMC controller control register
Offset = 0x0004

Bit	Name	Description	RW	Reset
31	TOUTEN	Enable /disable hardware r/w time out function. 0: Disable 1: Enable The timeout period is configured by Data Timeout Counter	RW	0
30:24	TOUTCNT	Hardware Time out counter value: This counter determine the timeout time of SD card data output. (used in TM_mode3~7 and boot mode), for details in section 9.1.1.2	RW	0
23:20	RDELAY	DATA input (Read) delay time selection: When host latch data delay the inside latching data clock to compensate signal transmission delay.	RW	0x1
19:16	WDELAY	DATA output(write) delay Time selection: When sending data, controller will delay the output signals relative to clock ,this will guarantee a suitable hold time for the signal	RW	0xF
15:14	-	Reserved	R	0
13	CMDLEN	CMD LOW Enable Enable CMD line drive low level by Software 1: Drive CMD line to low. 0: Don't drives CMD line to low. NOTE: Can Used in boot mode.	RW	0
12	SCC	Sending continuous clock. 1: Enable 0: Disable. Write 0 to this bit, stop continuance sending clock (should clear it after transfer start bit is set)	RW	0
11:8	TCN	Transfer clock number. The clock number is 16 times of this field value and a '0' means 256 cycles. (used with TM_MODE8)	RW	0
7	TS	Transfer Start: 1: When set 1 by software, SDC transfer starts according to The Transfer Mode, access mode, and other control situations .It will automatically cleared after transfer complete or any Err occurred. 0 : When cleared by software, The controller needs a few CLOCKs to stop transfer and reset the state machine fully so the software must check the result after writing "0" to	RW	0

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		it. Note: Software can not start a new transfer until it comes to “0” actually.		
6	LBE	Data in mode the last block send 8 more clocks enable bit 0: Disable 1: Enable When it’s set, at the end of the read last block, the SDC will send 8 more clocks when normally transfer is finished. It’s for the card to complete the operation.	RW	0
5	C7EN	Command CRC Check 1: Disable CRC7 checked 0: Enable CRC7 checked When it’s set, it’s indicates the SDC don’t check CRC7, and don’t generate the CRC7_ERROR.	RW	0
4	-	Reserved	R	0
3:0	TM	Transfer Mode[3:0] Specifies the transfer mode when transfer start bit is set 0000: Transfer command without response 0001: Transfer command with 6 bytes response (not including Data transfer) 0010: Transfer command with 17 bytes response (not including Data transfer) 0011: Transfer command with 6 bytes response, and with busy (not including Data transfer) 0100: Transfer data in mode with command and response (include crc16 checked) 0101: Transfer data out mode with command and response(include CRC and busy checked) 0110: Transfer data in mode without command and response 0111: Transfer data out mode without command and response (include CRC and busy checked) 1000: Transfer only clock (without any command, response, and data) 1001: BOOT mode ,Transfer data in mode with Command (Hardware check data CRC16, do not check boot mode acknowledge.) 1010: BOOT mode ,Transfer data in mode with Command (Hardware check data CRC16, check boot mode acknowledge). 1011~1111: Reserved NOTE: After each transfer the value of these bits will be reset to “0xf” by hardware.	RW	1111

11.1.4.3 SD0_STATE

SD/MMC controller STATE register
Offset = 0x0008

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Bits	Name	Description	R/W	Reset
31:16	-	Reserved	R	0
15	TOUTE	Time out error If set means a timeout error has happened. Next transfer started will clear it.	R	0
14	BAEP	Boot mode acknowledge error When in TM_MODE9, (boot mode with hardware check acknowledge), if set means a acknowledge error or timeout has happened. Next transfer started will clear it.	R	0
13	-	Reserved	R	0
12	MEMRDY	Memory ready 1: memory ready for read or write 0: memory is not ready for read and write	R	0
11	CMDS	CMD Status: This bit reflects the level of the CMD Signal of SD/MMC card	R	1
10	DAT1S	DAT1 Status: This bit reflects the level of the DAT1 Signal of SD/MMC card	R	1
9	SDIOP	SDIO IRQ pending bit 1: One SDIO interrupt has happened 0: No SDIO interrupt happened. Write1 to this bit will clear it.	RW	0
8	SDIOEN	SDIO IRQ enable 1: SDIO IRQ enable (the interrupt trigger form DAT1 PIN) 0: SDIO interrupt disable	RW	0
7	DAT0S	DAT0 Status: This bit reflects the level of the DAT0 Signal of SD/MMC card	R	1
6	TEIE	Transfer end IRQ enable: When set, enable interrupt request.	RW	0
5	TEI	Transfer end IRQ pending. Write1 to this bit will clear it.	R	0
4	CLNR	Command Line No response (only for command with response) This bit is auto cleared when Transfer Start is set.	R	0
3	CLC	Command Line transfer Complete: This bit is auto cleared when Transfer Start is set, and is set when command line transfer is complete.	R	0
2	WC16ER	CRC Write data Error: When set, this indicated a CRC write error detected over the data line. This bit is auto cleared when Transfer Start is set.	R	0
1	RC16ER	CRC Read data Error: When set, this indicated a CRC16 error detected over the received data. This bit is auto cleared when Transfer Start is set.	R	0
0	CRC7ER	CRC command response Error: When set, this indicated CRC7 error detected over the response. This bit is auto cleared when Transfer Start is set.	R	0

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11.1.4.4 SD0_CMD

SD/MMC controller send command register
Offset=0x000c

Bits	Name	Description	RW	Reset
31:8	-	Reserved		0
7:0	CMD	CMD-index to be transmitted Command register. Fixed the bit7 is '0' The bit6 is '1'	RW	01xxxxxxb

Note: The value in this register will be kept until next be written

11.1.4.5 SD0_ARG

SD/MMC controller CMD argument register
Offset=0x0010

Bits	Name	Description	RW	Reset
31:0	ARG	CMD's arguments: Be written before SD_CMD	RW	0

Note: The value in this register will be kept until next be written

11.1.4.6 SD0_RSPBUF0

SD/MMC controller RSP Buffer0 (Bit31~0) register
Offset=0x0014

Bits	Name	Description	R/W	Reset
31:0	RSP0	CMD's response Bit31~0 including the CRC and end bit	R	0

11.1.4.7 SD0_RSPBUF1

SD/MMC controller RSP Buffer1 (Bit63~32) register
Offset=0x0018

Bits	Name	Description	RW	Reset
31:0	RSP1	Bit63~32	R	0

11.1.4.8 SD0_RSPBUF2

SD/MMC controller RSP Buffer2 (Bit95~64) register
Offset=0x001c

Bits	Name	Description	RW	Reset
31:0	RSP2	Bit95~64	R	0

11.1.4.9 SD0_RSPBUF3

SD/MMC controller RSP Buffer3 (Bit127~96) register
Offset=0x0020

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Bits	Name	Description	RW	Reset
31:0	RSP3	Bit127~96	R	0

11.1.4.10 SD0_RSPBUF4

SD/MMC controller RSP Buffer4 (Bit135~128) register
Offset=0x0024

Bits	Name	Description	RW	Reset
31:8	-	Reserved	R	0
7:0	RSP4	Bit135~128	R	0

11.1.4.11 SD0_DAT

SD/MMC controller DATA register
Offset=0x0028

Bits	Name	Description	RW	Reset
31:0	DATA	Data register	R	xx

Note1: This register map to RAM access address. Can't be accessed except data transfer
Note2: When SD_EN[5] set to 0, SDC RAM width fix to 32bit.
For example:
If CPU or DMA need to sequential access 6(Not a multiple of 4) byte data: B1, B2, B3, B4, B5, B6.
Transfer data out: please write SD_DAT twice: B4B3B2B1 and XXB6B5. ('X' means don't care)
Transfer data in : CPU or DMA will read out B4B3B2B1 and XXB6B5 in twice from RAM.
Both master and slave should ignore these invalid bytes XX in the last 32-bit data.
Note3: When SD_EN[5] set to 1, only low 8bit of SD_DAT is valid. When SD in DMA mode, the developer should configure the width of DMA to be 8bit too.

11.1.4.12 SD0_BLK_SIZE

SD/MMC controller Block size Register
Offset=0x002c

Bits	Name	Description	RW	Reset
31:12	-	Reserved	R	0
11:0	BC	Block Count[9:0]. This field determines a block size, which is how many bytes founded a block. In SDR50/SDR100 mode, only 512bytes/block can be set.	RW	0

11.1.4.13 SD0_BLK_NUM

SD/MMC controller Block number Register
Offset=0x0030

Bits	Name	Description	RW	Reset
31:16	-	Reserved	R	0
15:0	BN	This field determines block number in one read or	RW	01B

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		write operations. Default value is 1. '0' means no block will be transferred.		

Note: The host hardware do not support multiple blocks transfer when the Byte counter is not the multiple of 4 bytes.

11.1.5 SD1 Controller Register Description

11.1.5.1 SD1_EN

SD/MMC controller enable register
Offset = 0x0000

Bits	Name	Description	RW	Reset
31:10	-	Reserved	R	x
9	CSS	Clock signal status select in idle mode 0:drive to low 1:drive to high	RW	0
8	-	Reserved	R	x
7	EN	SD module Enable 1: Enable 0: Disable	RW	0
6	AHB_DMA_SEL	SDC CHANEL（AHB OR DMA BUS) SELECTION 1:DMA BUS 0:AHB BUS	RW	0
5:4	-	Reserved	R	x
3	SDIOEN	SDIO function enable 1:enable 0:disable	RW	0
2	-	Reserved	R	0
1:0	DATAWID	SD Interface Data bus Width selection 00b :1 bit 01b: 4 bit Others: Reserved	RW	00

11.1.5.2 SD1_CTL

SD/MMC controller control register
Offset = 0x0004

Bit	Name	Description	RW	Reset
31:24	-	Reserved	R	x
23:20	RDELAY	DATA input (Read) delay time selection: when host latch data delay the inside latching data clock to compensate signal transmission delay.	RW	0x8
19:16	WDELAY	DATA output(write) delay Time selection: When sending data, controller will delay the output signals relative to clock , this will guarantee a suitable hold time for the signal.	RW	0xF

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15:13	-	Reserved	R	x
12	SCC	Sending continuous clock. 1: Enable 0: Disable. Write 0 to this bit, stop continuance sending clock （should clear it after transfer start bit is set）	RW	0
11:8	TCN	Transfer clock number. The clock number is 16 times of this field value and a '0' means 256 cycles. (used with TM_MODE8)	RW	0
7	TS	Transfer Start: 1: When set 1 by software, SDC transfer starts according to The Transfer Mode, access mode, and other control situations .It will automatically cleared after transfer complete or any Err occurred. 0 : When cleared by software, The controller needs a few CLOCKs to stop transfer and reset the state machine fully so the software must check the result after writing "0" to it. Note: Software can not start a new transfer until it comes to "0" actually..	RW	0
6	LBE	Data in mode the last block send 8 more clocks enable bit 0: Disable 1: Enable When set , at the end of the read last block, the SDC will send 8 more clocks when normally transfer is finished. It's for the card to complete the operation.	RW	0
5:4	-	Reserved	R	x
3:0	TM	Transfer Mode[3:0] Specifies the transfer mode when transfer start bit is set 0000: Transfer command without response 0001: Transfer command with 6 bytes response (not including Data transfer) 0010: Transfer command with 17 bytes response (not including Data transfer) 0011: Transfer command with 6 bytes response, and with busy (not including Data transfer) 0100: Transfer data in mode with command and response (include crc16 checked) 0101: Transfer data out mode with command and response(include CRC and busy checked) 0110: Transfer data in mode without command and response 0111: Transfer data out mode without command and response (include CRC and busy checked)	RW	1111

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		1000: Transfer only clock (without any command, response, and data) NOTE: After each transfer the value of these bits will be reset to “0xf” by hardware.		

11.1.5.3 SD1_STATE

SD/MMC controller STATE register
Offset = 0x0008

Bits	Name	Description	R/W	Reset
31:13	-	Reserved	R	x
12	MEMRDY	Memory ready 1: memory ready for read or write 0: memory is not ready for read and write	R	0
11	CMDS	CMD Status: This bit reflects the level of the CMD Signal of SD/MMC card	R	1
10	DAT1S	DAT1 Status: This bit reflects the level of the DAT1 Signal of SD/MMC card	R	1
9	SDIOP	SDIO IRQ pending bit 1: One SDIO interrupt has happened 0: No SDIO interrupt happened. Write1 to this bit will clear it.	RW	0
8	SDIOEN	SDIO IRQ enable 1: SDIO IRQ enable (the interrupt trigger form DAT1 PIN) 0: SDIO interrupt disable	RW	0
7	DAT0S	DAT0 Status: This bit reflects the level of the DAT0 Signal of SD/MMC card	R	1
6	TEIE	Transfer end IRQ enable: When set, enable interrupt request.	RW	0
5	TEI	Transfer end IRQ pending. Write1 to this bit will clear it.	R	0
4	CLNR	Command Line No response (only for command with response) This bit is auto cleared when Transfer Start is set.	R	0
3	CLC	Command Line transfer Complete: This bit is auto cleared when Transfer Start is set, and is set when command line transfer is complete.	R	0
2	WC16ER	CRC Write data Error: When set, this indicated a CRC write error detected over the data line. This bit is auto cleared when Transfer Start is set.	R	0
1	RC16ER	CRC Read data Error: When set, this indicated a CRC16 error detected over the received data. This bit is auto cleared when Transfer Start is set.	R	0
0	CRC7ER	CRC command response Error: When set, this indicated CRC7 error detected over the response. This bit is auto cleared when Transfer Start	R	0

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		is set.		

11.1.5.4 SD1_CMD

SD/MMC controller send command register
Offset=0x000c

Bits	Name	Description	RW	Reset
31:8	-	Reserved		x
7:0	CMD	CMD-index to be transmitted Command register. Fixed the bit7 is '0' The bit6 is '1'	RW	01000000b

Note: The value in this register will be kept until next be written

11.1.5.5 SD1_ARG

SD/MMC controller CMD argument register
Offset=0x0010

Bits	Name	Description	RW	Reset
31:0	ARG	CMD's arguments: Be written before SD_CMD	RW	0

Note: The value in this register will be kept until next be written

11.1.5.6 SD1_RSPBUF0

SD/MMC controller RSP Buffer0 (Bit31~0) register
Offset=0x0014

Bits	Name	Description	R/W	Reset
31:0	RSP0	CMD's response Bit31~0 including the CRC and end bit	R	0

11.1.5.7 SD1_RSPBUF1

SD/MMC controller RSP Buffer1 (Bit63~32) register
Offset=0x0018

Bits	Name	Description	RW	Reset
31:0	RSP1	Bit63~32	R	0

11.1.5.8 SD1_RSPBUF2

SD/MMC controller RSP Buffer2 (Bit95~64) register
Offset=0x001c

Bits	Name	Description	RW	Reset
31:0	RSP2	Bit95~64	R	0

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11.1.5.9 SD1_RSPBUF3

SD/MMC controller RSP Buffer3 (Bit127~96) register
Offset=0x0020

Bits	Name	Description	RW	Reset
31:0	RSP3	Bit127~96	R	0

11.1.5.10 SD1_RSPBUF4

SD/MMC controller RSP Buffer4 (Bit135~128) register
Offset=0x0024

Bits	Name	Description	RW	Reset
31:8	-	Reserved	R	0
7:0	RSP4	Bit135~128	R	0

11.1.5.11 SD1_DAT

SD/MMC controller DATA register
Offset=0x0028

Bits	Name	Description	RW	Reset
31:8	-	Reserved	R	xx
7:0	DATA	Data register	R	xx

Note1: This register map to RAM access address. Can't be accessed except data transfer
Note2: SDC1 RAM width fix to 8bit.

11.1.5.12 SD1_BLK_SIZE

SD/MMC controller Block size Register
Offset=0x002c

Bits	Name	Description	RW	Reset
31:10	-	Reserved	R	x
9:0	BC	Block Count[9:0]. This field determines a block size, which is how many bytes founded a block.	RW	0

11.1.5.13 SD1_BLK_NUM

SD/MMC controller Block number Register
Offset=0x0030

Bits	Name	Description	RW	Reset
31:16	-	Reserved	R	x
15:0	BN	This field determines block number in one read or write operations. Default value is 1. "0" means no block will be transferred.	RW	01B

Note: The host hardware do not support multiple blocks transfer when the Byte counter is not the multiple of 4 bytes.

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12 Transfer and Communication

12.1 USB

12.1.1 Features

- Complies with the USB2.0 full speed Specification
- Supports point-to-point communication with one full-speed device in Host mode(no HUB support).
- Supports full-speed in peripheral mode.
- Supports 6 IN endpoints and 6 OUT endpoint except endpoint0.
- Supports bulk Isochronous and Interrupt transfer.
- Partially configurable endpoint type and single, double triple or quad buffering.
- Integrated synchronous RAM as endpoint FIFOs.
- Supports suspend, resume and power managements function.
- Support remote wakeup for device mode.
- Support DMA transfer for endpoint in1 and out2.

12.1.2 USB Register List

Table 12-1 USB Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
USB_CONTROLLER_REGISTERS	0x40050000	0x40050000

Table 12-2 USB Controller Registers

Offset	Register Name	Description
0x421	DPDMCTRL	DPDM control register
0x422	LINESTATUS	Line status register

12.1.3 USB Register Description

12.1.3.1 DPDMCTRL

DP DM control register
Offset = 0x421

Bit (s)	Name	Description	Access	Reset
7	-	Reserved	R	0B
6	PLUGIN	This bit Indicated the usb connection status when Linedeten is enable. 1: connect 0: disconnect	R	x
5	-	Reserved	R	0B
4	LSDETEN	Line status detect enable 1：enable 0：disable	R/W	1B
3	DMPUEN	500Kohm DM pull up resistor enable. 1：enable 0：disable	R/W	0B
2	DPPUEN	500Kohm DP pull up resistor enable. 1：enable 0：disable	R/W	0B
1	DMPDDIS	DM pull down disable. 1：disable 0：enable	R/W	0B

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0	DPPDDIS	DP pull down disable. 1：disable 0：enable	R/W	0B

12.1.3.2 LINESTATUS

Line status register
Offset = 0x422

Bit (s)	Name	Description	Access	Reset
7	CEASESUSPEND	When in suspend state and USB clock has been suspended (CLKGATE.GOSUSPEND is set to 1), Once the controller is ready to resume (for Host) or remotely wakeup (for Peripheral) actively, software should firstly set this bit to resume USB clock(UTMISUSPENDM is driven to high). Then the other relative registers can be configured to generate Resume Signal on USB bus. This bit can be set by hardware or by software when UTMISUSPENDM needs driving to high. This bit will be cleared by hardware, when UTMISUSPENDM is driven to low. When read, this bit reflects the UTMISUSPENDM signal state.	R/W	0
6:5	-	Reserved	R	0H
4:3	LINESTATE	USB linestate[1:0] Linestate0:DP Linestase1:DM	R	0B
2:1	-	Reserved	R	0H
0	OTGRESET	USB OTG reset. If AOTG is in reset state, this bit will be set, otherwise it will be 0.	R/W	0B

12.2 I2C

12.2.1 Features

ATS3089x contains four normal I2C interfaces:

- Both master and slave functions supported
- Support for standard mode (100 kHz), Fast mode (400 kHz), and Fast mode+ (1 MHz)
- Both master and slave supports DMA mode
- Only 7-bit address mode support
- 8 bit x 8 TX FIFO and 8bit x 8 RX FIFO
- Supports general call
- Pull-up resistors are required on both of the I2C signal lines as the I2C drivers are open drain. Typically external 2.2k-Ohm resisters are used to pull the signals up to VCC if not select internal Pull-Up resistor.

12.2.2 Function Description

An Inter-IC bus, used to communicate across circuit-board distances. At the low end of the spectrum of communication options for "inside the box" communication is I2C.
I2C provides support for communication with various slow, on-board peripheral devices that are accessed intermittently, while being extremely modest in its hardware resource needs. It is a simple,

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low-bandwidth, short-distance protocol. Most available I2C devices operate at speeds up to 400Kbit/s, with some venturing up into the low megahertz range. I2C is easy to use to link multiple devices together since it has a built-in addressing scheme.

Note:

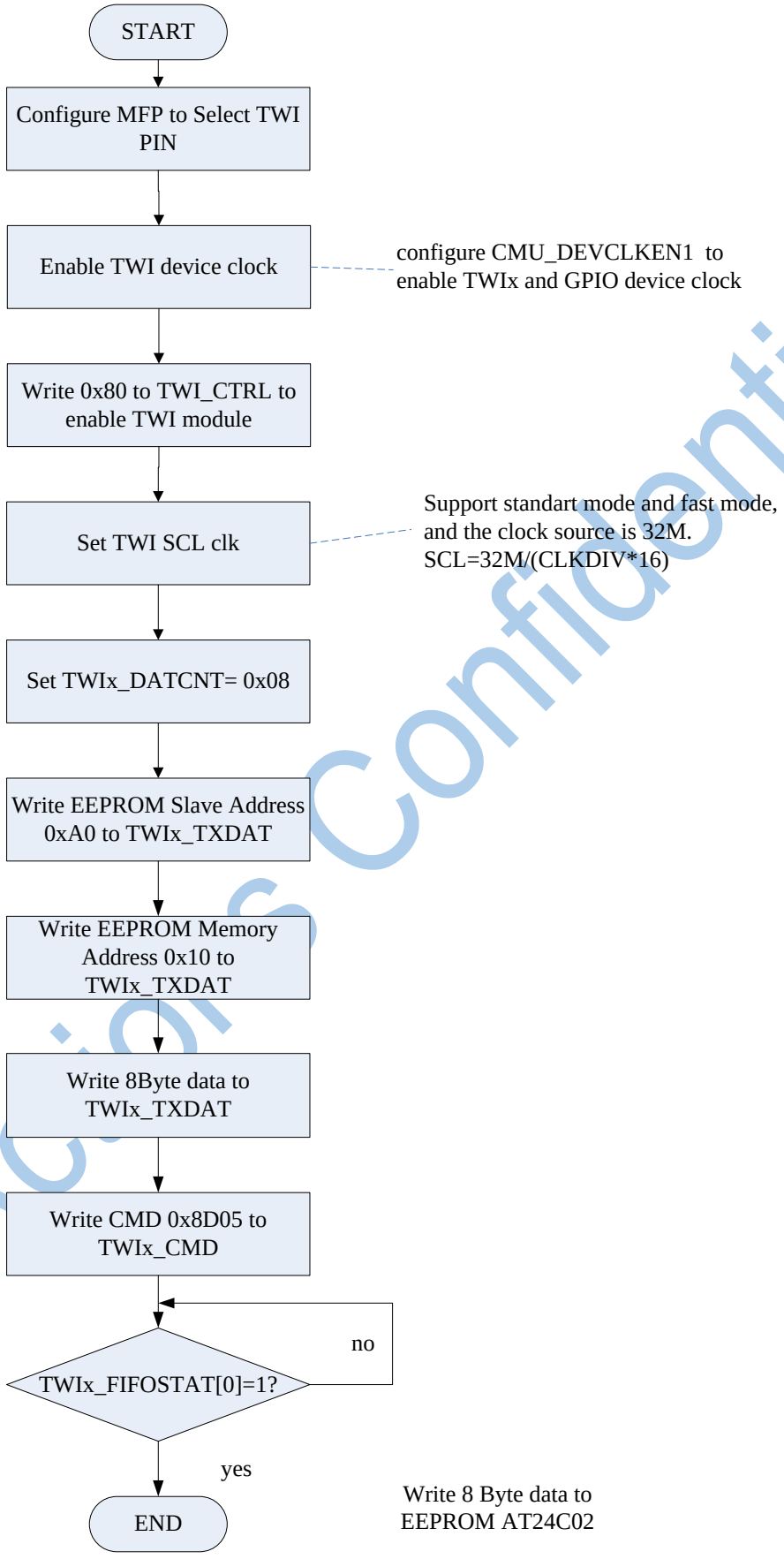
1. The I2C module is Slave mode when in IDLE status.
2. When write start command, the I2C module change from slave to master mode; and after a stop command change to slave mode.
3. Generate the IRQ while the bus statuses change.
   - transfer complete
   - detect normal stop bit (no bus error)

   If the address received don't match the content of I2C address, don't generate the IRQ and reset to IDLE when detect the start bit followed the slave address in slave mode.
4. Release the bus by software after receive data or address.
5. In multi-master application, when miss the control of the bus, ignore the bus command before a stop command detected. After a stop command, the device start an arbitration procedure. The arbitration will end with one of the following:
   - The I2C module enters master mode (it won the arbitration)
   - The I2C module enters slave mode (it lost the arbitration or the other master addresses the I2C module)
   - The I2C module enter master-in-waiting mode, when the I2C module is waiting for the other master (that won control) to complete its transaction so that it can once again try to grain the arbitration of the bus.

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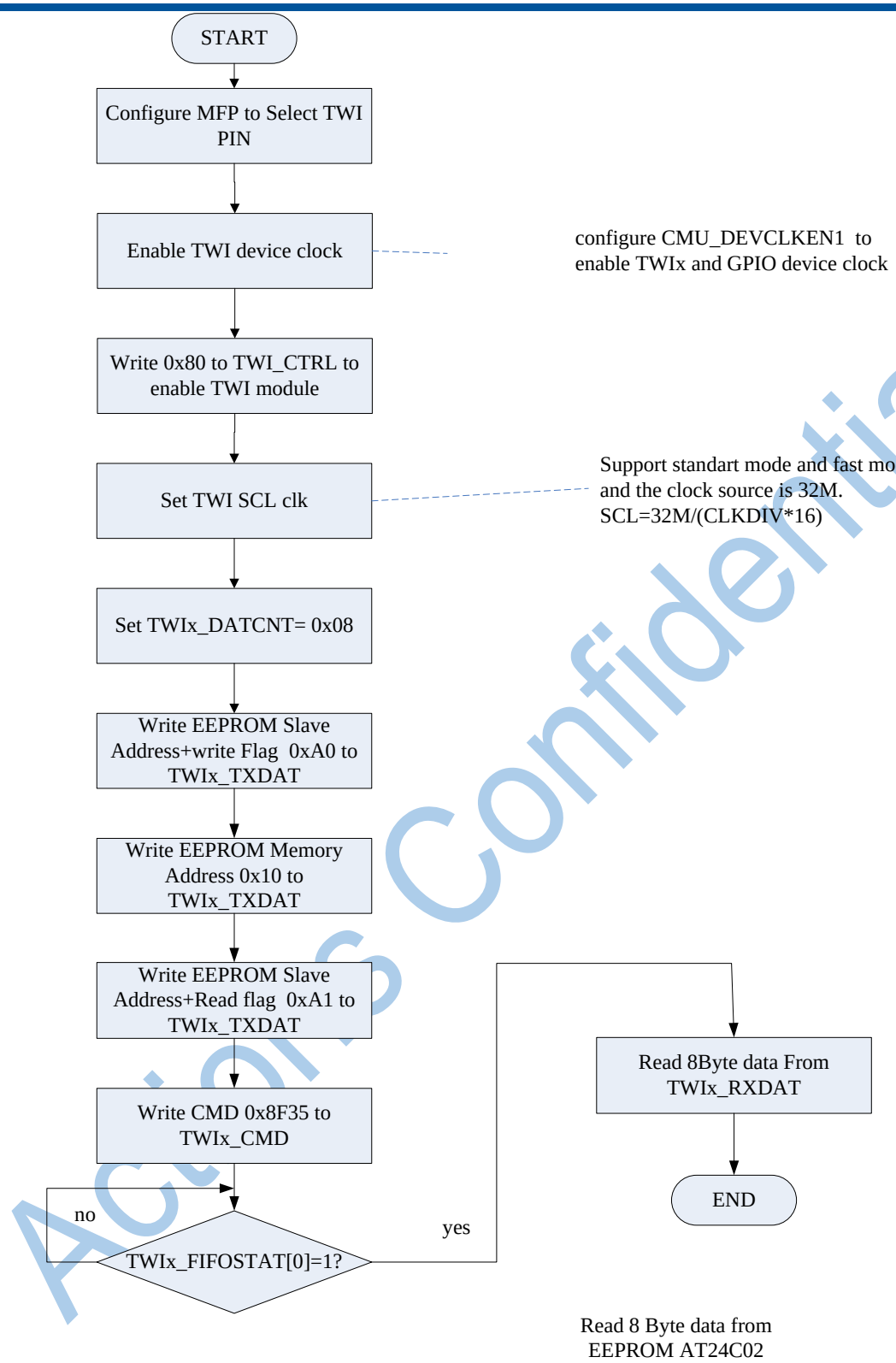
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12.2.3 Operation Manual



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12.2.4 I2C Register List

Table 12-3 I2C Register Block Base Address

Name	Physical Base Address	KSEG1 Base Address
I2C0	0x40044000	0x40044000

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I2C1	0x40048000	0x40048000
I2C2	0x4004C000	0x4004C000
I2C3	0x40074000	0x40074000

Table 12-4 I2C Registers Offset Address

Offset	Register Name	Description
0x0000	I2C_CTL	I2C Control Register
0x0004	I2C_CLKDIV	I2C Clock Divider Register
0x0008	I2C_STAT	I2C Status Register
0x000C	I2C_ADDR	I2C Address Register
0x0010	I2C_TXDAT	I2C TX Data Register
0x0014	I2C_RXDAT	I2C RX Data Register
0x0018	I2C_CMD	I2C Command Register
0x001C	I2C_FIFOCTL	I2C FIFO Control Register
0x0020	I2C_FIFOSTAT	I2C FIFO Status Register
0x0024	I2C_DATCNT	I2C Data Transmit Counter Register
0x0028	I2C_RCNT	I2C Data Transmit Remain Counter Register

12.2.5 I2C Register Description

12.2.5.1 I2Cx_CTL

I2C Control Register
Offset=0x00

Bits	Name	Description	R/W	Reset
31:21	-	Reserved	R	0x0
20	TCP_FIFO_IRQ_EN	Transfer Complete in fifo mode IRQ Enable 0:disable 1:enable	RW	0x0
19	STD_IRQ_EN	Stop Detect IRQ Enable 0:disable 1:enable	RW	0x0
18	-	Reserved	R	0x0
17	NACK_IRQ_EN	NACK Detect IRQ Enable 0:disable 1:enable	RW	0x0
16	ADM_IRQ_EN	Address match as a slave IRQ Enable 0:disable 1:enable	RW	0x0
15:11	-	Reserved	R	0x0
10	IRQC	I2C IRQ Control 0: set when RX FIFO received 1 bytes data in IRQ mode 1: set when RX FIFO received 4 bytes data in IRQ mode	RW	0x0
9	DRQTE	TX DRQ enable 0：disable 1：enable	RW	0x0
8	DRQRE	RX DRQ enable 0：disable 1：enable	RW	0x0

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7	BUSSEL	TX/RX FIFO Bus select 0: AHB. 1: DMA	RW	0x0
6	IRQE	IRQ Enable. 0: Disable 1: Enable	RW	0x0
5	EN	Enable. When enable, reset the status machine to IDLE 0: Disable 1: Enable	RW	0x0
4	-	Reserved	RW	0x0
3:2	GBCC	Generating Bus Control Condition (only for master mode). 00: No effect 01: Generating START condition 10: Generating STOP condition 11: Generating Repeated START condition Write the slave address to the I2C_DAT register, select start or restart, and then the start or restart command follow by the slave address will occur on the bus.	RW	0x0
1	RB	Release Bus. Write 1 to this bit will release the bus.	RW	0x0
0	GRAS	Generate ACK or NACK Signal. When receive data 0: generate the ACK signal at 9th clock of SCL 1: generate the NACK signal at 9th clock of SCL	RW	0x0

12.2.5.2 I2Cx_CLKDIV

I2C Clock Divide Control Register
Offset=0x04

Bits	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7:0	CLKDIV	Clock Divider Factor (only for master mode). I2C clock (SCL) can select standard (100kbps) mode and fast (400kbps) mode. Calculating SCL is as following: SCL=I2Cx_CLK/(CLKDIV*16)	RW	0x0

12.2.5.3 I2Cx_STAT

I2C Status Register
Offset=0x08

Bits	Name	Description	R/W	Reset
31:11	-	Reserved	R	0x0
10	SRGC	Slave receive general call 0: not receive a general call 1: receive a general call	R	0x0
9	SAMB	Slave address match bit 0: slave address not match 1: slave address match	R	0x0

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8	LBST	Last Byte Status Bit. 0: Indicate the last byte received or transmitted is address 1: Indicate the last byte received or transmitted is data	R	0x0
7	TCB	Transfer complete bit 0: not finish transfer 1: In normal mode: A byte transfer finish, include transfer the ACK or NACK bit Writing 1 to this bit will clear it.	RW	0x0
6	BBB	Bus busy bit 0: Not busy 1: Busy This bit will set to 1 while the start command detected, and set to 0 after the stop command	R	0x0
5	STAD	Start detect bit, include restart. The bit is clear when the I2C module is disable or when the STOP condition is detected. Writing 1 to the bit will clear it. 0: Start bit is not detected 1: Start bit is detected	RW	0x0
4	STPD	Stop detect bit The bit is clear when the I2C module is disable or when the START condition is detected. Writing 1 to the bit will clear it. 0: Stop bit is not detected 1: Stop bit is detected	RW	0x0
3	-	Reserved	RW	0x0
2	IRQP	IRQ Pending Bit. 1: IRQ 0: No IRQ Set condition: 1. transfer complete in fifo mode 2. detect normal stop bit (no bus error) 3. transfer complete in cmd mode 4. Receive NACK when not overlook the NACK 5. Address match as a slave Clear condition: Writing 1 to this bit will clear it.	RW	0x0
1	BEB	Bus error bit 0: No error occur 1: Bus error occur Write “1” to clear this bit The below conditions occur generate error bit: Detect stop bit right after detect start/restart bit. Detect stop, start bit when sending or receiving data.	RW	0x0
0	RACK	Receive ACK or NACK when transmit data or address 0: NACK	R	0x0

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		1: ACK The bit will be updated when the 9th of next byte clock arrived		

12.2.5.4 I2Cx_ADDR

I2C Address Register
Offset=0x0c

Bits	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7:1	SDAD	Own Slave Device Address. Only use in slave mode. I2C_Addr contains the own address of the module when the device is use in slave mode. Content of the register is irrelevant when the I2C module is functioning as a master.	RW	0x0
0	-	Reserved.	R	0x0

12.2.5.5 I2Cx_TXDAT

I2C Data Register
Offset=0x10

Bits	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7:0	DA	The registers of Data or address to be transfer, or received to. I2CDAT contains the byte to be transmitted on the I2C-bus or a byte that has been received from the I2C-bus. In master mode, along with the data byte to be transmitted, it also includes the slave address. The seven MSB's are the slave I2C device address while the LSB is the Read/Write bit. 8 level FIFO, 8 x 8bit	W	0x0

12.2.5.6 I2Cx_RXDAT

I2C receive Data Register
Offset=0x14

Bits	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7:0	DA	The Receive data Register 8 level FIFO, 8 x 8bit	R	0x0

12.2.5.7 I2Cx_CMD

I2C Command Register
Offset=0x18

Bits	Name	Description	R/W	Reset

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31:16	-	Reserved	R	0x0
15	SECL	Start to execute the command list 0: not execute 1: execute command	RW	0x0
14:13	-	Reserved	R	0x0
12	WRS	Write or Read select 0: write 1: read This bit only used in Slave mode.	RW	0x0
11	MSS	Master or slave mode select 0: slave mode 1: Master mode	RW	0x0
10	SE	Stop enable 0: disable 1: enable	RW	0x0
9	NS	NACK select 0: not select 1: select generate the NACK signal at 9th clock of SCL of the last byte when read data	RW	0x0
8	DE	Data enable 0: disable 1: enable The counts of data transmitted depend on the I2Cx_CNT register.	RW	0x0
7:5	SAS	Second address select 000: no address 001: 1 byte address 010: 2 byte address 011: 3 byte address 100: 4 byte address 101: 5 byte address 110: 6 byte address 111: 7 byte address	RW	0x0
4	RBE	Restart bit enable 0: not send restart bit 1: send restart bit	RW	0x0
3:1	AS	Address select 000: no address 001: 1 byte address 010: 2 byte address 011: 3 byte address 100: 4 byte address 101: 5 byte address 110: 6 byte address 111: 7 byte address The address include slave address and slave internal memory address.	RW	0x0
0	SBE	Start bit enable 0: not send start bit 1: send start bit	RW	0x0

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12.2.5.8 I2Cx_FIFOCTL

I2C FIFO control Register
Offset=0x1c

Bits	Name	Description	R/W	Reset
31:3	-	Reserved	R	0x0
2	TFR	TX FIFO reset bit Write 1 to reset TX FIFO, auto clear to 0 when Tx FIFO reset complete.	RW	0x0
1	RFR	RX FIFO reset bit Write 1 to reset RX FIFO, auto clear to 0 when Rx FIFO reset complete.	RW	0x0
0	NIB	NACK Ignore Bit 0: not ignore, when receive NACK when write, generate Error, do not continue the command list execute, generate IRQ 1: ignore NACK, when receive NACK, don't generate error, and will continue the command list execute	RW	0x0

12.2.5.9 I2Cx_FIFOSTAT

I2C FIFO status Register
Offset=0x20

Bits	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:12	TFD	Tx FIFO level display This field indicate the current Tx FIFO level	R	0x0
11:8	RFD	Rx FIFO level display This field indicate the current Rx FIFO level	R	0x0
7	-	Reserved	R	0x0
6	WRS	Write or read status bit when acts as slave, used only in FIFO mode 0: master write to slave 1: master read from slave	R	0x0
5	TFF	TX FIFO full bit 0: not full 1: full	R	0x0
4	TFE	TX FIFO empty bit 1: empty 0: not empty	R	0x1
3	RFF	RX FIFO full bit 0: not full 1: full	R	0x0
2	RFE	RX FIFO empty bit 1: empty 0: not empty	R	0x1
1	RNB	Receive NACK Error bit 0: not receive NACK 1: receive NACK when write data Write 1 to clear this bit	RW	0x0

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0	CECB	Command Execute Complete bit 0: not complete 1: complete	R	0x1

12.2.5.10 I2Cx_DATCNT

I2C data Counter Register
Offset=0x24

Bits	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	TC	Data Transmit counter	RW	0x0

12.2.5.11 I2Cx_RCNT

I2C remain Counter Register
Offset=0x28

Bits	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15:0	TC	Remain counter	R	0x0

12.3 I2CMT

12.3.1 Features

ATS3089x contains two I2CMT interfaces:

- Normal mode and Auto mode.
- I2C master compatible at Normal mode.
- I2C master read/write one/multi-byte operation executed automatically at auto mode.
- Standard mode 100 kbps, up to fast mode 400 kbps.
- Support for clock stretching.
- Dedicated AHB-like DMA Bus for data access.
- Internal Pull-Up Resistor (50K) optional.

12.3.2 Function Description

I2CMT is a two-wire half-duplex master which can communicate with multiple slave devices connected to the same bus. The I2CMT master can be started by one of the four tasks. The I2CMT master implements dedicated AHB-like Bus for reading and writing to and from the RAM.

I2CMT master define four tasks, which triggered by events from other peripherals, such as Timer, interrupt. And each task can start a transmission (write or read one/multiple bytes).

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12.4 UART

12.4.1 Features

ATS3089x contains three UART interfaces: UART0/1/2. UART0/1/2has the following features:

- 5-8 Data Bits and LSB first in Transmit and Received
- 1-2 Stop Bits
- Even, Odd, or No Parity
- 16 Byte Transmit and Receive FIFOs
- Capable of speeds up to 4Mbps to enable connections with Bluetooth and other peripherals
- Support IRQ and DMA mode to transmit data
- Support RTS/CTS Automatic Hardware Flow Control to reduce interrupts to host system
- UART RX Support DMA single mode
- Add UART RX DMA counter for valid data in RAM.
- RX Baud Rate tolerance≤±2%
- Support Rx timeout IRQ pending

12.4.2 UART Register List

Table 12-5 UART Registers Block Base Address

Name	Physical Base Address	KSEG1 Base Address
UART0	0x40038000	0x40038000
UART1	0x4003C000	0x4003C000
UART2	0x40040000	0x40040000

Table 12-6 UART Registers Offset Address

Offset	Register Name	Description
0x0000	UARTx_CTL	UARTx Control Register
0x0004	UARTx_RXDAT	UARTx Receive FIFO Data Register
0x0008	UARTx_TXDAT	UARTx Transmit FIFO Data Register
0x000C	UARTx_STA	UARTx Status Register
0x0010	UARTx_BR	UARTx BAUDRATE divider Register
0x0014	UARTx_CTL2	UART Control2 Register

12.4.3 UARTX Register Description

12.4.3.1 UARTx_CTL

UART Control Register
Offset=0x0000

Bits	Name	Description	R/W	Reset
31	RXENABLE	UART RX disable 1:normal 0:disable	R/W	0x0
30	TXENABLE	UART TX disable 1:normal 0:disable	RW	0x0
29	TX_FIFO_EN	UART TX FIFO enable: 0: Disable	R/W	0x0

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		1: Enable		
28	RX_FIFO_EN	UART RX FIFO enable: 0: Disable 1: Enable	R/W	0x0
27:26	TX_FIFO_SEL	UART TX FIFO Input Select. 0: From CPU 1: From DMA 2:Reserved 3:From DSP Others: Reserved	RW	0x0
25	-	Reserved	RW	0x0
24	RX_FIFO_SEL	UART RX FIFO Output Select. 0: To CPU 1: To DMA	RW	0x0
23:22	AFL	Level for trigger RTS to high in autoflow when the autoflow bit is set 00:14byte 01:12byte 10:8byte 11:4byte	RW	0x0
21:20	-	Reserved	RW	0x0
19	TXIE	UART TX IRQ Enable. 0: Disable 1: Enable	RW	0x0
18	RXIE	UART RX IRQ Enable. 0: Disable 1: Enable	RW	0x0
17	TXDE	UART TX DRQ Enable. 0: Disable 1: Enable	RW	0x0
16	RXDE	UART RX DRQ Enable. 0: Disable 1: Enable	RW	0x0
15	EN	UART Enable. When this bit is clear, the UART clock source is inhibited. This can be used to place the module in a low power standby state. 0:disable 1: enable	RW	0x0
14	-	Reserved	RW	0x0
13	RTSE	RTS Enable. When this bit is set, request to send data. Note: This bit has no effect if Autoflow enable bit is set. 0:no request 1: request to send data	RW	0x0
12	AFE	Autoflow mode Enable Setting this bit enables automatic hardware flow control. Enabling this mode overrides software control of the signals. 0: Autoflow mode disable (normal mode) 1: Autoflow mode enable	RW	0x0
11:10	RDIC	UART RX DRQ/IRQ Control 00: set when RX FIFO received at least one byte data in	RW	0x0

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		IRQ/DRQ mode. 01: set when RX FIFO received 4 bytes data in IRQ mode 10: set when RX FIFO received 8 bytes data in IRQ/DRQ mode 11: set when RX FIFO received 12 bytes data in IRQ/DRQ mode In DMA burst mode (normal DMA), DO not set 00,01 because at least 8 bytes necessary. In DMA single mode (special DMA), DO set 00 for 1 Bytes transfer for each DRQ.		
9:8	TDIC	UART TX DRQ/IRQ Control 00: set when TX FIFO is at least 1 byte empty in IRQ/DRQ mode. 01: set when TX FIFO is 4 bytes empty in IRQ mode. 10: set when TX FIFO is 8 bytes empty in IRQ/DRQ mode. 11: set when TX FIFO is 12 bytes empty in IRQ/DRQ mode. In DMA burst mode (normal DMA), DO not set 00,01 because at least 8 bytes necessary. In DMA single mode (special DMA), DO set 00 for 1 Bytes transfer for each DRQ.	RW	0x0
7	CTSE	CTS Enable. If this bit is 1,the transmitter check CTS before send the next byte, Note: no effect if autoflow enable bit is set 0:don't check CTS before send 1:check CTS before send	RW	0x0
6:4	PRS	Parity Select. Bit 6: PEN, Parity enable Bit 5: STKP, Stick parity Bit 4: EPS, Even parity PEN STKP EPS Selected Parity 0 x x None 1 0 0 Odd 1 0 1 logic 1 1 1 0 Even 1 1 1 logic 0	RW	0x0
3	-	Reserved	R	0
2	STPS	STOP Select. If this bit is 0, 1 stop bit is generated in transmission. If this bit is 1, 2 stop bits are generated. 0:1 stop bit 1:2 stop bit	RW	0x0
1:0	DWLS	Data Width Length Select. 00: 5 bits 01: 6 bits 10: 7 bits 11: 8 bits	RW	0x0

12.4.3.2 UARTx_RXDAT

UART Receive FIFO Data Register

Offset=0x0004

Bits	Name	Description	R/W	Reset

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31:8	Reserved		R	0x0
7:0	RXDAT	Received Data. The depth of FIFO is 8bit×16levels.	R	x

12.4.3.3 UARTx_TXDAT

UART Transmit FIFO Data Register

Offset=0x0008

Bits	Name	Description	R/W	Reset
31:8	Reserved		R	0
7:0	TXDAT	Transmitted Data. The depth of FIFO is 8bit×16 levels	W	0

12.4.3.4 UARTx_STA

UART Status Register

Offset=0x000c

Bits	Name	Description	R/W	Reset
31:25	Reserved		R	0
24	RXTO_PD	Rx Timeout IRQ Pending bit 0:NO IRQ 1:IRQ Write 1 auto clean	R/W	0x0
23	PAER	Parity Status. 0: Parity OK 1: Parity error. Writing 1 to the bit will clear the bit. When parity error.	R/W	0x0
22	STER	Stop Status. 0: Stop OK 1: Stop error. Writing 1 to the bit will clear the bit. When stop bit detect error.	R/W	0x0
21	UTBB	UART TX busy bit 0:not busy, TX FIFO is empty and all data be shift out 1:busy	R	0
20:16	TXFL	TX FIFO Level. The field indicates the current TX FIFO empty level.	R	10000
15:11	RXFL	RX FIFO Level. The field indicates the current RX FIFO level of valid data.	R	0
10	TFES	TX FIFO empty Status 0: not empty 1: empty	R	1
9	RFFS	RX FIFO full Status 0: not full 1: full	R	0
8	RTSS	RTS Status. The bit reflects the status of the external RTS- pin.	R	0
7	CTSS	CTS Status. The bit reflects the status of the external CTS- pin.	R	x

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6	TFFU	TX FIFO Full. 1: Full 0: No Full	R	0
5	RFEM	RX FIFO Empty. 1: Empty 0: No Empty	R	1
4	RXST	Receive Status. 0: receive OK 1: receive error. Writing 1 to the bit will clear the bit. When stop bit detect error， or parity error， or clock error	RW	0
3	TFER	TX FIFO Error. 0: No Error 1: Error Writing 1 to the bit will clear the bit and reset the TX FIFO.	RW	0
2	RXER	RX FIFO Error. 0: No Error 1: Error Writing 1 to the bit will clear the bit and reset the RX FIFO.	RW	0
1	TIP	TX IRQ Pending Bit. 0: No IRQ 1: IRQ Writing 1 to the bit to clear the bit.	RW	1
0	RIP	RX IRQ Pending Bit. 0: No IRQ 1: IRQ Writing 1 to the bit to clear it.	RW	0

12.4.3.5 UARTx_BR

UART BAUDRATE divider register

Offset=0x0010

Bits	Name	Description	R/W	Reset
31:16	TXBRDIV	UART TX BAUDRATE divider BaudRate = Colck_source/BaudRate divider Clock_source=HOSC	R/W	0x028
15:0	RXBRDIV	UART RX BAUDRATE divider BaudRate = Colck_source/BaudRate divider Clock_source=HOSC	R/W	0x028

12.4.3.6 UARTx_CTL2

UART Control register2

Offset=0x0014

Bit(s)	Name	Description	R/W	Reset
31	RXTO_IRQ_EN	Rx Timeout IRQ Pending Enable 0：disable 1：enable The IRQ will happen when the RXTO time pass and the UARTx Rx in IDLE.	R/W	0x0
30:8	-	Reserved	R	0x0

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7:0	RXTO	UARTx Rx IDLE Timeout(in byte time)	R/W	0x10

12.5 SPI0

12.5.1 Features

SPI0 is a combination module which includes conventional SPI and SPI_Cache interface.

- Only support master mode
- Support mode0, mode1, mode2 and mode3
- Support AHB and Cache access SPI and DMA interface
- Only Support 8bit transfer mode
- Support 32 bit seed 8bit randomize
- Support 3wire
- Support 64 level delay chain, 0.25ns/step
- Support 1x/2x/dual/4x/quad/ DTR 4x access SPI NOR
- Support 3-byte/4-byte address modes access SPINOR
- Support 16bit CRC only when Cache access fifo, not support CRC when AHB access fifo
- Support QPI 4x/quad/DTR 4x for Cache access SPI NOR
- Support DTR(Double Transfer Rate)
- Support cache abort
- Support 106MHz spi_clk as highest speed
- Support 3.1V/1.8V IO voltage

12.6 SPI1

12.6.1 Features

- Support SPI normal mode: mode 0/1/2/3
- Support SPI/QPI PSRAM and Double-Data-Rate Octal SPI PSRAM(X8/X16)
- Support 1x/2x/dual/4x/quad access SPI NOR
- Support IRQ and DMA mode to transmit data
- Support 64 level delay chain, 0.15ns/step
- Support slave mode
- Support 160MHz spi_clk as highest speed

12.7 SPI2

12.7.1 Features

- Support SPI normal mode: mode 0/1/2/3
- Support normal 4 wire mode/3 wire mode
- Support IRQ and DMA mode to transmit data
- Support 32 level delay chain, 0.5ns/step
- Support slave mode
- Support 96MHz spi_clk as highest speed

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12.7.2 SPI2 Register List

Table 12-7 SPI2 Registers Block Base Address

Name	Physical Base Address	KSEG1 Base Address
SPI2	0x40030000	0x40030000

Table 12-8 SPI2 Registers Offset Address

Offset	Register Name	Description
0x0000	SPI2_CTL	SPI2 Control Register
0x0004	SPI2_STA	SPI2 Status Register
0x0008	SPI2_TXDAT	SPI2 Transmit FIFO Data Register
0x000C	SPI2_RXDAT	SPI2 Receive FIFO Data Register
0x0010	SPI2_BC	SPI2 Byte Counter Register

12.7.3 SPI2 Register Description

12.7.3.1 SPI2_CTL

SPI2 Control Register
Offset=0x0000

Bits	Name	Description	R/W	Reset
31	CLKSEL	FIFO write or read clock select 0: use CPU clock 1: use DMA clock	R/W	0x0
30	FWS	FIFO width select 0: 8bit 1: 32bit	R/W	0x0
29:28	SPI_MODE_SELECT	SPI Mode Select 00 : Mode0 01 : Mode1 10 : Mode2 11 : Mode3	R/W	0x3
27	-	Reserved	R/W	0x0
26	RX_WRITE_SEL	SPI Rx Write Select, Select suitable cycle To sample the right rx data 0: delay 2 spi_clk cycle (used when SPI_DELAY <= 4'b1000) 1: delay 3 spi_clk cycle (used when SPI_DELAY <= 4'b1111)	R/W	0x0
25	DMS	DMA transmit mode select 0: burst8 mode 1: single mode	R/W	0x0
24	TXCEB	TX Convert Endian bit, only used in 32Bit mode: 0: not convert Endian 0x76543210 ->0x76543210 1: convert Endian 32bit mode: 0x76543210 ->0x10325476 When in 8 bit mode, this bit have no effect	R/W	0x0
23	RXCEB	RX Convert Endian bit, only used in 32Bit mode: 0: not convert Endian	R/W	0x0

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		0x76543210 ->0x76543210 1: convert Endian 32bit mode: 0x76543210 ->0x10325476 When in 8 bit mode, this bit have no effect		
22	MSS	Master or Slave mode select 0: Master mode 1: Slave mode	R/W	0x0
21	MSB	SPI LSB/MSB First Select 0: SPI transmit and receive MSB first 1: SPI transmit and receive LSB first	R/W	0x0
20	RILS	RX IRQ Level select 0: RX FIFO not empty, generate IRQ 1: RX FIFO at least 8 level data, generate IRQ Note: this bit have no effect when SPI_RIRQ_EN is disable, SPI2_CTL[8].	R/W	0x0
19:15	SPI_DELAY	**SPI2 Master read clock delay time (valid when SPI_WR select write/read and read mode)** 00000: no delay 00001: delay 0.5 ns 00010: delay 1 ns 00011: delay 1.5 ns 00100: delay 2 ns 00101: delay 2.5 ns 00110: delay 3 ns 00111: delay 3.5 ns 01000: delay 4 ns 01001: delay 4.5 ns 01010: delay 5 ns 01011: delay 5.5 ns 01100: delay 6 ns 01101: delay 6.5 ns 01110: delay 7 ns 01111: delay 7.5 ns 10000: delay 8 ns 10001: delay 8.5 ns 10010: delay 9 ns 10011: delay 9.5 ns 10100: delay 10 ns 10101: delay 10.5 ns 10110: delay 11 ns 10111: delay 11.5 ns 11000: delay 12 ns 11001: delay 12.5 ns 11010: delay 13 ns 11011: delay 13.5 ns 11100: delay 14 ns 11101: delay 14.5 ns 11110: delay 15 ns 11111: delay 15.5 ns	R/W	0x0
14:11	-	Reserved	R/W	0x0
10	SPI_3WIRE	SPI2 Master 3-wire mode enable 0:disable 1:enable use 3-wire mode only in master read only or write only mode.	R/W	0x0

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9	SPI_TIRQ_EN	SPI TX IRQ Enable, trigger SPI TX IRQ when SPI TX FIFO at least 8 level empty 0: disable 1: enable	R/W	0x0
8	SPI_RIRQ_EN	SPI RX IRQ Enable, this trigger of SPI RX IRQ controlled by SPI2_CTL[20]. 0: disable 1: enable	R/W	0x0
7	SPI_TDRQ_EN	SPI TX DRQ Enable, trigger DRQ when SPI TX FIFO at least 8 level empty; When DMA remain counter < 8, trigger DRQ until all data transfer completely; 0: disable 1: enable	R/W	0x0
6	SPI_RDRQ_EN	SPI RX DRQ Enable, trigger DRQ when SPI RX FIFO at least 8 level full; When DMA remain counter < 8, trigger DRQ until all data received completely; 0: disable 1: enable	R/W	0x0
5	SPI_TX_FIFO_EN	SPI Tx FIFO Enable 0: Disable 1: Enable	R/W	0x0
4	SPI_RX_FIFO_EN	SPI Rx FIFO Enable 0: Disable 1: Enable	R/W	0x0
3	SPI_SS	SPI NSS pin control output 0: output low 1: output high	R/W	0x1
2	-	Reserved	R/W	0x0
1:0	SPI_WR	SPI Read/Write Mode (AHB Interface Only) 00: disable 01: Read only 10: Write only 11: Read and Write	R/W	0x0

12.7.3.2 SPI2_STA

SPI2 Status Register
Offset=0x0004

Bits	Name	Description	Access	Reset
31:14	-	Reserved	R	0x0
13	SPI_RXHF	SPI RX FIFO Half Full 0: not half full 1: half full	R	0x0
12	SPI_TXHF	SPI TX FIFO Half Full 0: not half full 1: half full	R	0x0
11	TFWO	TX FIFO error pending if Write FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
10	-	Reserved	R	0x0

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9	RFWO	RX FIFO error pending if Write FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
8	TFRO	RX FIFO error pending if Read FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
7	SPI_RXFU	SPI2 RX FIFO Full 0: not full 1: full	R	0x0
6	SPI_RXEM	SPI2 RX FIFO Empty 0: not empty 1: empty	R	0x1
5	SPI_TXFU	SPI2 TX FIFO Full 0: not full 1: full	R	0x0
4	SPI_TXEM	SPI2 TX FIFO Empty 0: not empty 1: empty	R	0x1
3	SPI_TIRQ_PD	SPI2 TX IRQ Pending, Writing '1' to this bit will clear it. 0: No TX IRQ Pending 1: TX IRQ Pending.	R/W	0x0
2	SPI_RIRQ_PD	SPI2 RX IRQ Pending, Writing '1' to this bit will clear it. 0: No RX IRQ Pending 1: RX IRQ Pending.	R/W	0x0
1	-	Reserved	R	0x0
0	SPI_BUSY	SPI2 master busy status bit. 0: SPI idle status 1: SPI busy status (Clock is transmiting or Rx Remain Counter doesn't reduced to zero)	R	0x0

12.7.3.3 SPI2_TXDAT

SPI2 Transmit FIFO Data Register
Offset=0x0008

Bits	Name	Description	Access	Reset
31:0	SPI2_TXDAT	SPI2 TX FIFO, 32bitx16 levels When SPI2_CTL[30] select 8bit width, bit[7:0] is valid. 8bitx64levels When SPI2_CTL[30] select 32bit width, bit[31:0] is valid. Be read as zero.	W	0x0

12.7.3.4 SPI2_RXDAT

SPI2 Receive FIFO Data Register
Offset=0x000C

Bits	Name	Description	Access	Reset
31:0	SPI2_RXDAT	SPI RX FIFO, When SPI2_CTL[30] select 8bit width, bit[7:0] is valid. 8bitx64levels When SPI2_CTL[30] select 32bit width, bit[31:0] is valid.	R	0x0

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		32bitx16 levels		

12.7.3.5 SPI2_BC

SPI2 Bytes Count Register, this register is used for setting SPI2 bytes counter bits in the SPI read mode only.

Offset=0x0010

Bits	Name	Description	Access	Reset
31:16	REMAIN_CNT	Indicate how many bytes need to be received, only use in master mode	R	0x0
15:0	SPI2_BC	Bytes Counter [15: 0]	R/W	0x0

12.8 SPI3

12.8.1 Features

- Support SPI normal mode: mode 0/1/2/3
- support normal 4 wire mode
- Support 1x/2x/dual/4x/quad access SPI NAND
- Support IRQ and DMA mode to transmit data
- Support 32 level delay chain, 0.5ns/step
- Support slave mode
- Support 96MHz spi_clk as highest speed
- Support auto check RB

12.8.2 SPI3 Register List

Table 12-9 SPI3 Registers Block Base Address

Name	Physical Base Address	KSEG1 Base Address
SPI3	0x40034000	0x40034000

Table 12-10 SPI3 Registers Offset Address

Offset	Register Name	Description
0x0000	SPI3_CTL	SPI3 Control Register
0x0004	SPI3_STA	SPI3 Status Register
0x0008	SPI3_TXDAT	SPI3 Transmit FIFO Data Register
0x000C	SPI3_RXDAT	SPI3 Receive FIFO Data Register
0x0010	SPI3_BC	SPI3 Byte Counter Register
0x0018	SPI3_CHECKRB	SPI3 CHECKRB configure
0x001c	SPI3_CHECKRB_COUNTER	SPI3 auto check RB counter register

12.8.3 SPI3 Register Description

12.8.3.1 SPI3_CTL

SPI3 Control Register

Offset=0x0000

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Bits	Name	Description	R/W	Reset
31	CLKSEL	FIFO write or read clock select 0: use CPU clock 1: use DMA clock	R/W	0x0
30	FWS	FIFO width select 0: 8bit 1: 32bit	R/W	0x0
29:28	SPI_MODE_SELECT	SPI Mode Select 00 : Mode0 01 : Mode1 10 : Mode2 11 : Mode3	R/W	0x3
27	-	Reserved	R/W	0x0
26	RX_WRITE_SEL	SPI Rx Write Select, Select suitable cycle To sample the right rx data 0: delay 2 spi_clk cycle (used when SPI_DELAY <= 4'b1000) 1: delay 3 spi_clk cycle (used when SPI_DELAY <= 4'b1111)	R/W	0x0
25	DMS	DMA transmit mode select 0: burst8 mode 1: single mode	R/W	0x0
24	TXCEB	TX Convert Endian bit, only used in 32Bit mode: 0: not convert Endian 0x76543210 ->0x76543210 1: convert Endian 32bit mode: 0x76543210 ->0x10325476 When in 8 bit mode, this bit have no effect	R/W	0x0
23	RXCEB	RX Convert Endian bit, only used in 32Bit mode: 0: not convert Endian 0x76543210 ->0x76543210 1: convert Endian 32bit mode: 0x76543210 ->0x10325476 When in 8 bit mode, this bit have no effect	R/W	0x0
22	MSS	Master or Slave mode select 0: Master mode 1: Slave mode	R/W	0x0
21	MSB	SPI LSB/MSB First Select 0: SPI transmit and receive MSB first 1: SPI transmit and receive LSB first	R/W	0x0
20	RILS	RX IRQ Level select 0: RX FIFO not empty, generate IRQ 1: RX FIFO at least 8 level data, generate IRQ Note: this bit have no effect when SPI_RIRQ_EN is disable, SPI3_CTL[8].	R/W	0x0
19:15	SPI_DELAY	**SPI3 Master read clock delay time (valid when SPI_WR select write/read and read mode)** 00000: no delay 00001: delay 0.5 ns 00010: delay 1 ns 00011: delay 1.5 ns 00100: delay 2 ns	R/W	0x0

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		00101: delay 2.5 ns 00110: delay 3 ns 00111: delay 3.5 ns 01000: delay 4 ns 01001: delay 4.5 ns 01010: delay 5 ns 01011: delay 5.5 ns 01100: delay 6 ns 01101: delay 6.5 ns 01110: delay 7 ns 01111: delay 7.5 ns 10000: delay 8 ns 10001: delay 8.5 ns 10010: delay 9 ns 10011: delay 9.5 ns 10100: delay 10 ns 10101: delay 10.5 ns 10110: delay 11 ns 10111: delay 11.5 ns 11000: delay 12 ns 11001: delay 12.5 ns 11010: delay 13 ns 11011: delay 13.5 ns 11100: delay 14 ns 11101: delay 14.5 ns 11110: delay 15 ns 11111: delay 15.5 ns		
14:13	-	Reserved	R/W	0x0
12	SPI_3WIRE	SPI3 Master 3-wire mode enable 0:disable 1:enable use 3-wire mode only in master read only or write only mode.	R/W	0x0
11:10	SPI_IO_MODE	SPI data I/O mode select 00: 1x I/O mode select 01: 1x I/O mode select 10: 2x I/O mode select 11: 4x I/O mode select	R/W	0x0
9	SPI_TIRQ_EN	SPI TX IRQ Enable, trigger SPI TX IRQ when SPI TX FIFO at least 8 level empty 0: disable 1: enable	R/W	0x0
8	SPI_RIRQ_EN	SPI RX IRQ Enable, this trigger of SPI RX IRQ controlled by SPI3_CTL[20]. 0: disable 1: enable	R/W	0x0
7	SPI_TDRQ_EN	SPI TX DRQ Enable, trigger DRQ when SPI TX FIFO at least 8 level empty; When DMA remain counter < 8, trigger DRQ until all data transfer completely; 0: disable 1: enable	R/W	0x0
6	SPI_RDRQ_EN	SPI RX DRQ Enable, trigger DRQ when SPI RX FIFO at least 8 level full.; When DMA remain counter < 8, trigger DRQ until all data received completely; 0: disable	R/W	0x0

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		1: enable		
5	SPI_TX_FIFO_EN	SPI Tx FIFO Enable 0: Disable 1: Enable	R/W	0x0
4	SPI_RX_FIFO_EN	SPI Rx FIFO Enable 0: Disable 1: Enable	R/W	0x0
3	SPI_SS	SPI NSS pin control output 0: output low 1: output high	R/W	0x1
2	-	Reserved	R/W	0x0
1:0	SPI_WR	SPI Read/Write Mode (AHB Interface Only) 00: disable 01: Read only 10: Write only 11: Read and Write	R/W	0x0

12.8.3.2 SPI3_STA

SPI3 Status Register
Offset=0x0004

Bits	Name	Description	Access	Reset
31:14	-	Reserved	R	0x0
13	SPI_RXHF	SPI RX FIFO Half Full 0: not half full 1: half full	R	0x0
12	SPI_TXHF	SPI TX FIFO Half Full 0: not half full 1: half full	R	0x0
11	TFWO	TX FIFO error pending if Write FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
10	-	Reserved	R	0x0
9	RFWO	RX FIFO error pending if Write FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
8	TFRO	RX FIFO error pending if Read FIFO overflow occur 0: no error 1: error occur Writing '1' to clear this bit	R/W	0x0
7	SPI_RXFU	SPI3 RX FIFO Full 0: not full 1: full	R	0x0
6	SPI_RXEM	SPI3 RX FIFO Empty 0: not empty 1: empty	R	0x1
5	SPI_TXFU	SPI3 TX FIFO Full 0: not full 1: full	R	0x0
4	SPI_TXEM	SPI3 TX FIFO Empty	R	0x1

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		0: not empty 1: empty		
3	SPI_TIRQ_PD	SPI3 TX IRQ Pending, Writing '1' to this bit will clear it. 0: No TX IRQ Pending 1: TX IRQ Pending.	R/W	0x0
2	SPI_RIRQ_PD	SPI3 RX IRQ Pending, Writing '1' to this bit will clear it. 0: No RX IRQ Pending 1: RX IRQ Pending.	R/W	0x0
1	-	Reserved	R	x
0	SPI_BUSY	SPI3 master busy status bit. 0: SPI idle status 1: SPI busy status (Clock is transmiting or Rx Remain Counter doesn't reduced to zero)	R	0x0

12.8.3.3 SPI3_TXDAT

SPI3 Transmit FIFO Data Register
Offset=0x0008

Bits	Name	Description	Access	Reset
31:0	SPI3_TXDAT	SPI3 TX FIFO, 32bitx16 levels When SPI3_CTL[30] select 8bit width, bit[7:0] is valid. 8bitx64levels When SPI3_CTL[30] select 32bit width, bit[31:0] is valid. Be read as zero.	W	0x0

12.8.3.4 SPI3_RXDAT

SPI3 Receive FIFO Data Register
Offset=0x000C

Bits	Name	Description	Access	Reset
31:0	SPI3_RXDAT	SPI RX FIFO, When SPI3_CTL[30] select 8bit width, bit[7:0] is valid. 8bitx64levels When SPI3_CTL[30] select 32bit width, bit[31:0] is valid. 32bitx16 levels	R	0x0

12.8.3.5 SPI3_BC

SPI3 Bytes Count Register, this register is used for setting SPI3 bytes counter bits in the SPI read mode only.
Offset=0x0010

Bits	Name	Description	Access	Reset
31:16	REMAIN_CNT	Indicate how many bytes need to be received, only use in master mode	R	0x0
15:0	SPI3_BC	Bytes Counter [15:0]	R/W	0x0

12.8.3.6 SPI3_CHECKRB

SPI3 Check RB Register
Offset=0x0018

Bit(s)	Name	Description	R/W	Reset

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31:9	-	Reserved	R	0x0
8	RB	SNAND RB status bit 0: not busy 1: busy	R	0x0
7	CHECKRB_PD	SNAND auto check RB finish pending bit. After check RB finish, this bit will keep 1. Write 1 to clear this bit. 0: no check RB finish pending 1: check RB finish pending	R/W	0x0
6:2	-	Reserved	R	0x0
1	CHECKRB_IRQ_EN	SNAND check RB IRQ enable bit SPI will set check RB finish pending when SNAND auto check RB finish. 0: Disable 1: Enable	R/W	0x0
0	CHECKRB	SNAND auto check RB start bit 0: disable auto check RB 1: start auto check RB	R/W	0x0

12.8.3.7 SPI3_CHECKRB_COUNTER

SPI3 auto check RB counter Register
Offset=0x001C

Bit(s)	Name	Description	R/W	Reset
31:12	-	Reserved	R	0x0
11:0	RBCOUNTER	Auto check RB counter, count by SPICLK	R/W	0xFFF

12.9 SPIMT

12.9.1 Features

ATS3089x contains two SPIMT interfaces:

- Normal mode and Auto mode.
- SPI master compatible at Normal mode.
- SPI master read/write one/multi-byte operation executed automatically at auto mode.
- 4 chip select signal.
- Clock may be 4MHz or 32MHz.
- Dedicated AHB-like DMA Bus for data access.

12.10 PWM

12.10.1 Features

- Independent of 4 PWM
- The frequency of PWM comes from the division of 32KHz/HOSC/CK64M
- Support normal mode , breath mode and programmable mode
  - Normal mode: can output 65536 kinds of duty
  - Breath mode: supports breathing lights with various flicker frequencies

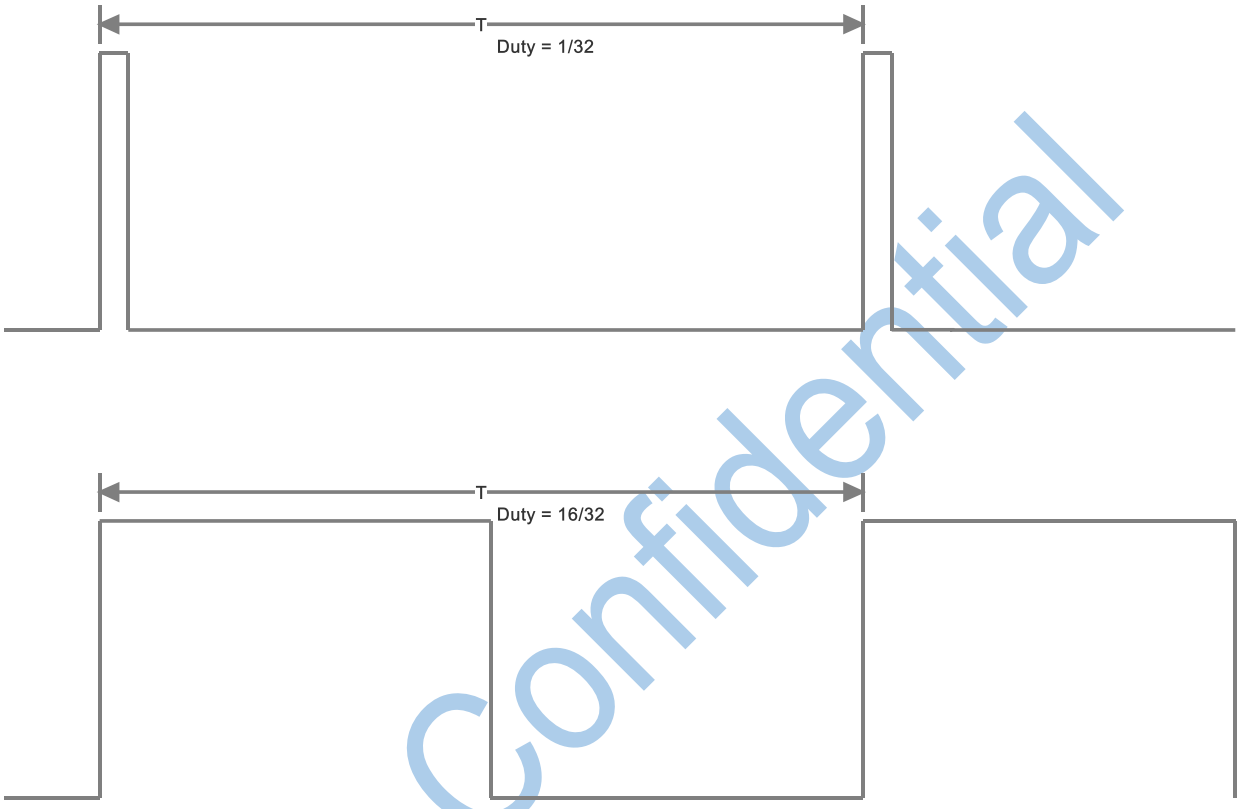
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- Programmable mode: The duty of output signal can be changed according the ram data.

12.10.2 Module Description

12.10.2.1 Normal Mode Timing

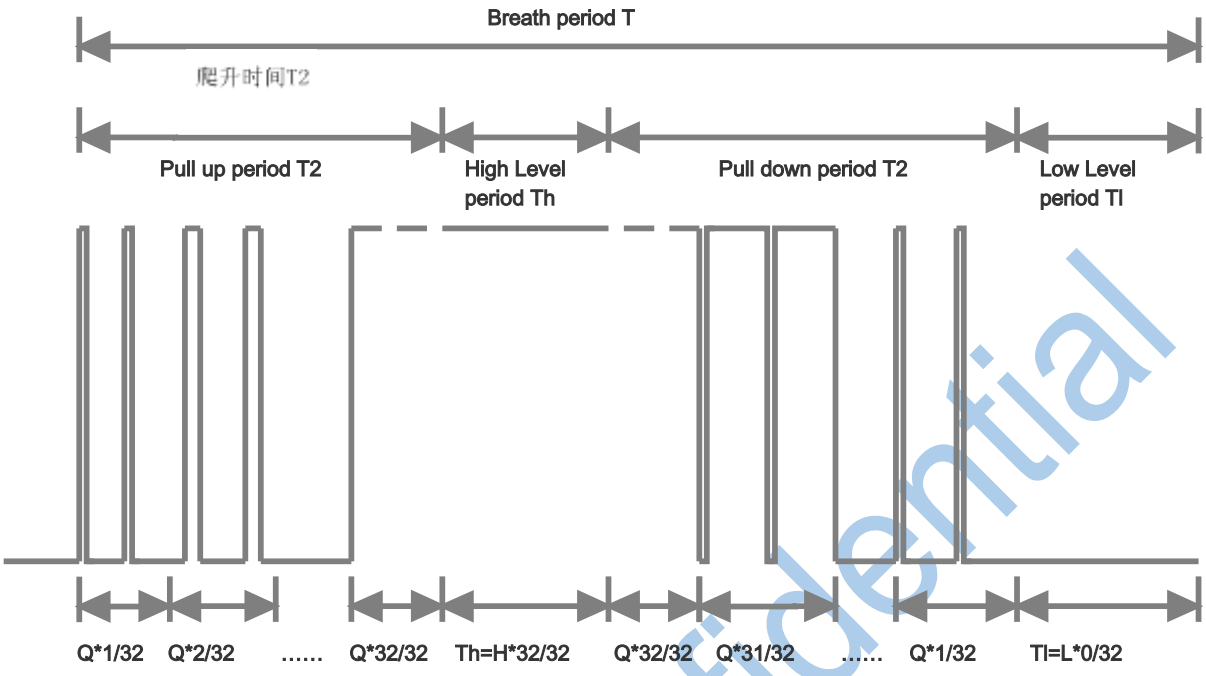


PWM can be used as backlights enable signal of LCD. The brightness of the backlights is decided by the duty cycle of PWM.

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12.10.2.2 Breath Mode Timing



The breath mode of PWM can be used for driver of breathing light. For example, if PWM CLK from CMU is f=1/t and we need the time of pull up or pull down to be T2=Q*32*32t=0.5s, and the time of high level Th=H*32T=0.5s and the time of low level Tl=L*32t=2s, when we set Q=2, the f should be 4096 and H is 64, L is 256.

12.10.2.3 Programmable Mode Timing

Example used DMA transmission:

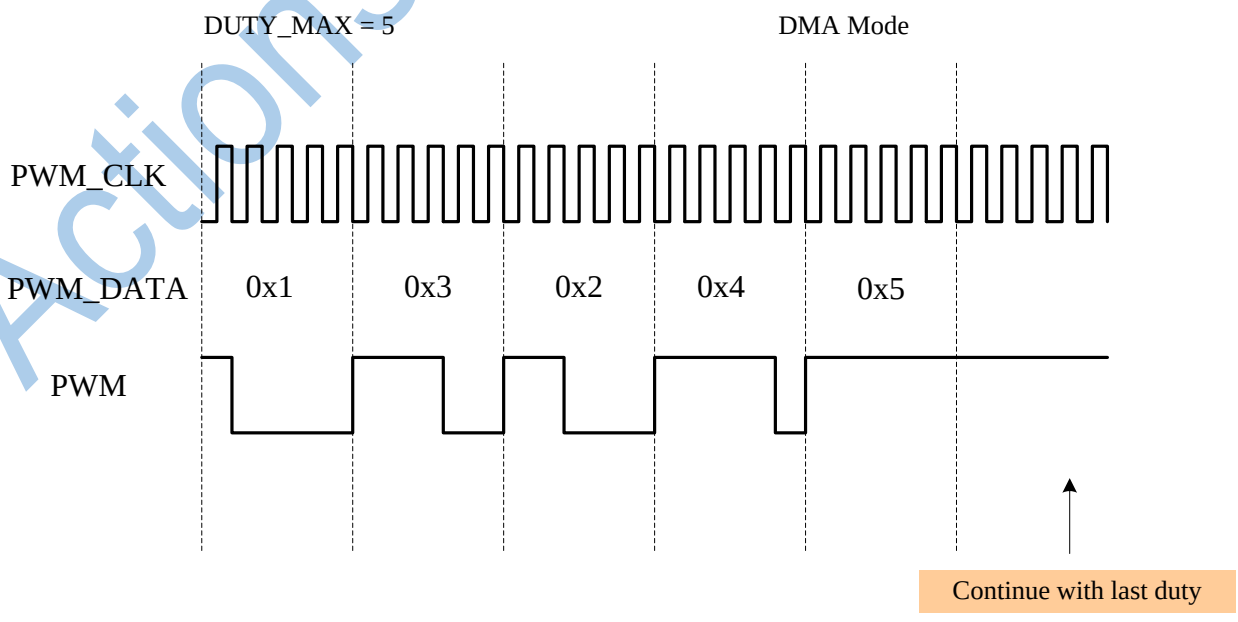


Figure12-1 Programmable mode example used DMA transmission

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12.10.3 Operation Manual

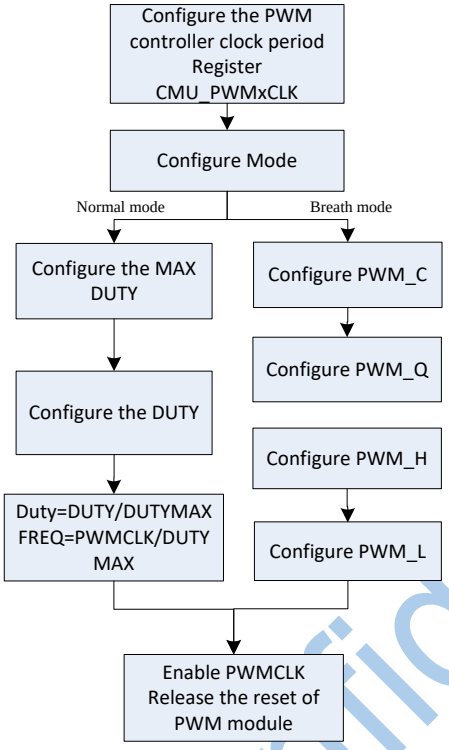


Figure 12-2 PWM Configuration

12.10.4 PWM Register List

Table 12-11 PWM Registers Block Base Address

Name	Physical Base Address	KSEG1 Base Address
PWM	0x40060000	0x40060000

Table 12-12 PWM Registers Offset Address

Offset	Register Name	Description
0x000+0x100*x	PWMx_CTL	PWM Output Control register, 0<=x<=3
0x004+0x100*x	PWMx_C	PWM breath mode C register, 0<=x<=3
0x008+0x100*x	PWMx_Q	PWM breath mode Q register, 0<=x<=3
0x00c+0x100*x	PWMx_H	PWM breath mode H register, 0<=x<=3
0x010+0x100*x	PWMx_L	PWM breath mode L register, 0<=x<=3
0x014+0x100*x	PWMx_DUTYMAX	PWM Duty Max register , 0<=x<=3
0x018+0x100*x	PWMx_DUTY	PWM Duty register , 0<=x<=3
0xf00	PWM_DMACTL	PWM Enable Control register
0xf04	PWM_FIFODAT	PWM FIFO Data register
0xf08	PWM_FIFOSTA	PWM FIFO Status register

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12.10.5 PWM Register Description

12.10.5.1 PWMx_CTL

PWMx Output Control Register, 0<=x<=3
Offset = 0x000 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:5	Reserved	Reserved	R	0x0
4	CH_START	PWMx Channel start 0: stop after this PWM wave period 1: PWM start	RW	0x0
3	CH_EN	PWMx Channel Enable 0: disable 1: enable	RW	0x0
2	POL_SEL	Polarity select: 0:PWMx low voltage level active 1:PWMx high voltage level active Active in fixed Mode and Breath Mode	RW	0x0
1:0	MODE_SEL	PWMx Mode Select: 0: Fixed Mode 1: Breath Mode 2: Programmable Mode 3: Reserved	RW	0x0

12.10.5.2 PWMx_C

PWMx Breath mode C configuration Register, 0<=x<=3
Offset = 0x004 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:24	Reserved	Reserved	R	0x0
7:0	C	Counter Top of PWM Duty For Example: In Breath Mode if C = 32, The PWM Duty is 1/32, 2/32, .. , 32/32	RW	0x20

12.10.5.3 PWMx_Q

PWMx Breath mode Q configuration Register, 0<=x<=3
Offset = 0x008 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:24	Reserved	Reserved	R	0x0
23:16	QD	Time of Every Duty =C/C …1/C: Falling time T2=QD*C*C*t t is the period of CMU_PWM C*t is the period of PWM Wave Only Active in Breath Mode	RW	0x0
15:8	Reserved	Reserved	R	0x0
7:0	QU	Time of Every Duty =1/C …C/C: Rising time T1=QU*C*C*t t is the period of CMU_PWM C*t is the period of PWM Wave	RW	0x0

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		Only Active in Breath Mode		

12.10.5.4 PWMx_H

PWMx Breath mode H/L configuration Register, 0<=x<=3
Offset = 0x00c + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:16	Reserved	Reserved	R	0x0
15:0	H	Time of Duty =C/C : High Level Time = H*C*t t is the period of CMU_PWM\ C*t is the period of PWM Wave Only Active in Breath Mode	RW	0x0

12.10.5.5 PWMx_L

PWMx Breath mode H/L configuration Register, 0<=x<=3
Offset = 0x010 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:16	Reserved	Reserved	R	0x0
15:0	L	Time of Duty =0/C: Low Level Time = L*C*t t is the period of CMU_PWM C*t is the period of PWM Wave Only Active in Breath Mode	RW	0x0

12.10.5.6 PWMx_DUTYMAX

PWMx Duty Max Register, 0<=x<=3
Offset = 0x014 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:16	Reserved	Reserved	R	0x0
15:0	DUTYMAX	Configurable maximum duty. PWM period = PWM_CLK period * DUTY_MAX Note: DUTYMAX >= 0x1	RW	0x1

12.10.5.7 PWMx_DUTY

PWMx Duty Register, 0<=x<=3
Offset = 0x018 + 0x100*x

Bit(s)	Name	Description	R/W	Reset
31:16	Reserved	Reserved	R	0x0
15:0	DUTY	PWM DUDY = DUTY / DUTY_MAX Only Active in Normal Mode	RW	0x0

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12.10.5.8 PWM_DMACTL

PWM Module Enable Register
Offset = 0xf00

Bit(s)	Name	Description	R/W	Reset
31:1	Reserved	Reserved	R	0x0
7:4	PWM_FIFO_CLK_SEL	0000:PWM0 0001:PWM1 0010:PWM2 0011:PWM3 Other: Reserved	RW	0x0
3:1	-	-	RW	0x0
0	START	PWM DMA START 0: disable 1: enable	RW	0x0

12.10.5.9 PWM_FIFODAT

PWM FIFO Data Register
Offset = 0xf04

Bit(s)	Name	Description	R/W	Reset
31:16	Reserved	Reserved	R	0x0
15:0	DATA	DUTY DATA PWM DUDY = DUTY / DUTY_MAX Only Active in programmable Mode	R	0x0

Note: DUTY must be configured ≤ (DUTY_MUX), failing to do so may result in unpredictable behavior.

12.10.5.10 PWM_FIFOSTA

PWM FIFO Status Register
Offset = 0xf08

Bit(s)	Name	Description	R/W	Reset
31:5	Reserved	Reserved	R	0x0
4:3	LEVEL	PWM FIFO level This field indicates pwm fifo empty level	R	0x2
2	EMPTY	PWM FIFO empty status: 0: not empty 1: empty	R	0x1
1	FULL	PWM FIFO full status: 0: not full 1: full	R	0x0
0	ERROR	PWM FIFO error status: 0: no error 1: error Writing 1 to the bit is clear	RW	0x0

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13 Audio Interface

13.1 DAC

13.1.1 Features

- Build in single channel 24-bit input sigma-delta DAC, SNR>=108dB, THD+N<=-90dB, DR>=107dBA
- DAC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/88.2k/96kHz
- Support digital volume of 256 steps with zero cross detection
- Build in single channel 20mW PA (Power Amplifier). PA output supports traditional mode, and differential mode. The noise floor of Audio PA on A-WT is 3.5uVrms (Typical) when in differential mode.
- An anti-pop circuit for suppressing noise of PA when enable and disable

13.1.2 Module Description

13.1.2.1 Data Block Diagram

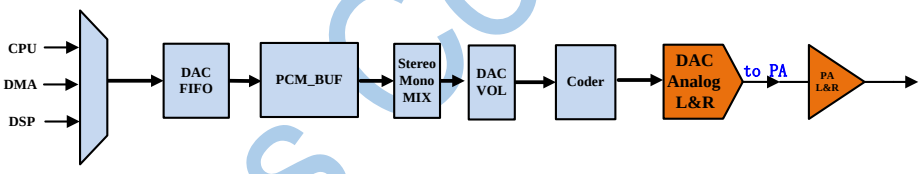


Figure 13-1 DAC Main Block

13.1.3 Signal List

Table 13-1 DAC Signals

SIGNAL Name	I/O Type	Short Description
AOUT/AOUTP	AO	output of PA / Positive output of PA
AOUTN	AO	Negative output of PA

13.1.4 Register List

Table 13-2 DAC Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
DAC_Control_Register	0x4005C000	0x4005C000

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Table 13-3 DAC Controller Registers

Offset	Register Name	Description
0x00	DAC_DIGCTL	DAC Digital Control Register
0x08	DAC_STAT	DAC State Register
0x10	PCM_BUF_CTL	PCM Buffer Control Register
0x14	PCM_BUF_STAT	PCM Buffer State Register
0x18	PCM_BUF_THRES_HE	Threshold to generate half empty signal
0x1C	PCM_BUF_THRES_HF	Threshold to generate half full signal
0x28	VOL_LCH	Volume Left Channel Control Register
0x2C	VOL_RCH	Volume Right Channel Control Register

13.1.5 Register Description

13.1.5.1 DAC_DIGCTL

DAC Digital Control Register
Offset = 0x00

Bit(s)	Name	Description	R/W	Reset
31:1	-	R/W forbidden	-	-
0	DDEN	DAC Digital Enable. 0x0: Disable 0x1: Enable	RW	0x0

13.1.5.2 DAC_STAT

DAC State Register
Offset = 0x08

Bit(s)	Name	Description	R/W	Reset
31:8	-	R/W forbidden	-	-
7	DAF0EIP	DAC FIFO0 half-Empty IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
6	DAF0F	DAC FIFO0 Full Flag. 0x0: Not Full 0x1: Full	R	0x0
5	-	Reserved	R	0x0
4:0	DAF0S	DAC FIFO0 Status These 5 bits shows how many sample data can be filled to the FIFO. For example, when read as 11, means CPU/DMA can write 11 samples to FIFO. Display one by one	R	x

13.1.5.3 PCM_BUF_CTL

PCM BUF Control Register
Offset = 0x10

Bit(s)	Name	Description	R/W	Reset
31:8	-	R/W forbidden	-	-

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7	PCMBEPIE	PCM_BUF **Empty** IRQ Enable. 0x0: Disable 0x1: Enable	RW	0x0
6	PCMBFUIE	PCM_BUF **Full** IRQ Enable. 0x0: Disable 0x1: Enable	RW	0x0
5	PCMBHEIE	PCM_BUF **Half Empty** IRQ Enable. 0x0: Disable 0x1: Enable	RW	0x0
4	PCMBHFIE	PCM_BUF **Half Full** IRQ Enable. 0x0: Disable 0x1: Enable	RW	0x0
3:0	-	R/W forbidden	-	-

13.1.5.4 PCM_BUF_STAT

PCM BUF State Register
Offset = 0x14

Bit(s)	Name	Description	R/W	Reset
31:20	-	R/W forbidden	-	-
19	PCMBEIP	PCM BUF **Empty** IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
18	PCMBFIP	PCM BUF **Full** IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
17	PCMBHEIP	PCM BUF **Half Empty** IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
16	PCMBHFIP	PCM BUF **Half Full** IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
15:11	-	R/W forbidden	-	-
10:0	PCMBS	PCM_BUF Status These 11 bits shows how many samples FIFO are not filled. For example, when read as 512, means CPU/DMA can write 512 samples to FIFO. Display one sample	R	0x400

13.1.5.5 PCM_BUF_THRES_HE

PCM BUF Threshold of Half Empty Register
Offset = 0x18

Bit(s)	Name	Description	R/W	Reset
31:12	-	R/W forbidden	-	-
11:0	THRESHOLD	The threshold to generate **half empty** signal: When the number of samples equal to the half	RW	0x100

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		empty threshold, pcm_buf generates half-empty IRQ. (**The Threshold value should be set even number, and not zero**).		

13.1.5.6 PCM_BUF_THRES_HF

PCM BUF Threshold of Half Full Register
Offset = 0x1C

Bit(s)	Name	Description	R/W	Reset
31:12	-	R/W forbidden	-	-
11:0	THRESHOLD	The threshold to generate **half full** signal: When the number of samples equal to the half full threshold, pcm_buf generates half-full IRQ. (**The Threshold value should be set even number, and not zero**).	RW	0x300

13.1.5.7 VOL_LCH

Volume Left Channel Control Register
Offset = 0x28

Bit(s)	Name	Description	R/W	Reset
31:12	-	R/W forbidden	-	-
11	DONE_STA	DAC volume left channel soft stepping gain reach target set status: 0 : marching 1 : done	R	0x1
10	SOFT_STEP_EN	DAC volume left channel soft stepping gain control enable 0 : disable 1 : enable	RW	0x0
9:8	-	R/W forbidden	-	-
7:0	VOLL	Volume Left Channel Control (step:0.375dB): 0xff: +24.375 dB 0xfe: +24dB 0xfd: +23.625dB … 0xbf: +0.375dB 0xbe: 0dB … 0x00: mute	RW	0xbf

Note: Only VOL_RCH is valid when DAC is working. VOL_LCH and VOL_RCH are valid both when IIS TX is working.

13.1.5.8 VOL_RCH

Volume Right Channel Control Register
Offset = 0x2C

Bit(s)	Name	Description	R/W	Reset
31:12	-	R/W forbidden	-	-
11	DONE_STA	DAC volume right channel soft stepping gain reach target set status: 0 : marching	R	0x1

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		1 : done		
10	SOFT_STEP_EN	DAC volume right channel soft stepping gain control enable 0 : disable 1 : enable	RW	0x0
9:8	-	R/W forbidden	-	-
7:0	VOLR	Volume right Channel Control (step:0.375dB): 0xff: +24.375 dB 0xfe: +24dB 0xfd: +23.625dB … 0xbf: +0.375dB 0xbe: 0dB … 0x00: mute	RW	0xbf

Note: Only VOL_RCH is valid when DAC is working. VOL_LCH and VOL_RCH are valid both when IIS TX is working.

13.2 ADC

13.2.1 Features

- Build in stereo 24 bit input sigma-delta ADCs, SNR>=97dB(A-weighting), THD+N<=-90db
- ADC supports sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/88.2k/96kHz
- A digital high-pass filter can be used to remove dc offsets when ADC use
- Every ADC has independent Automatic Amplitude Limitation
- Supports single-ended input analog microphones and full differential input microphone
- Supports 2 Digital microphones

13.2.2 Module Description

13.2.2.1 Block Diagram

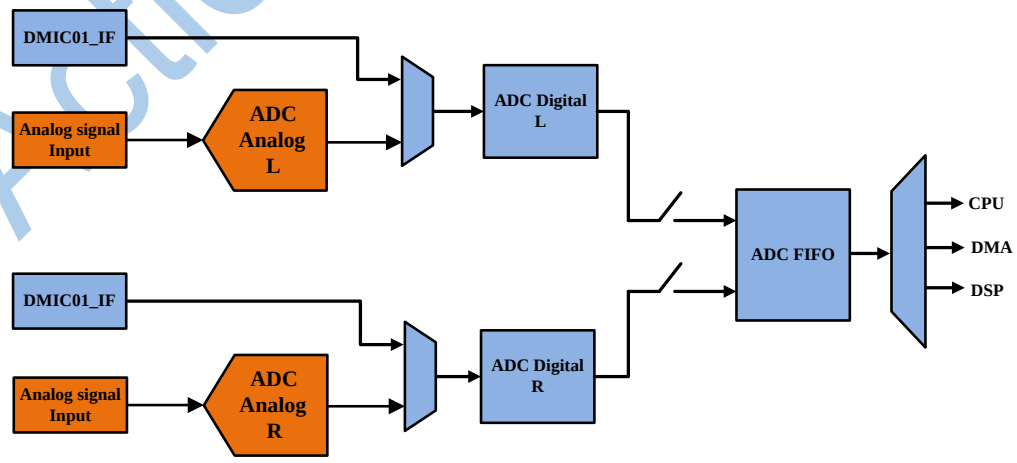


Figure 13-2 ADC Main Block

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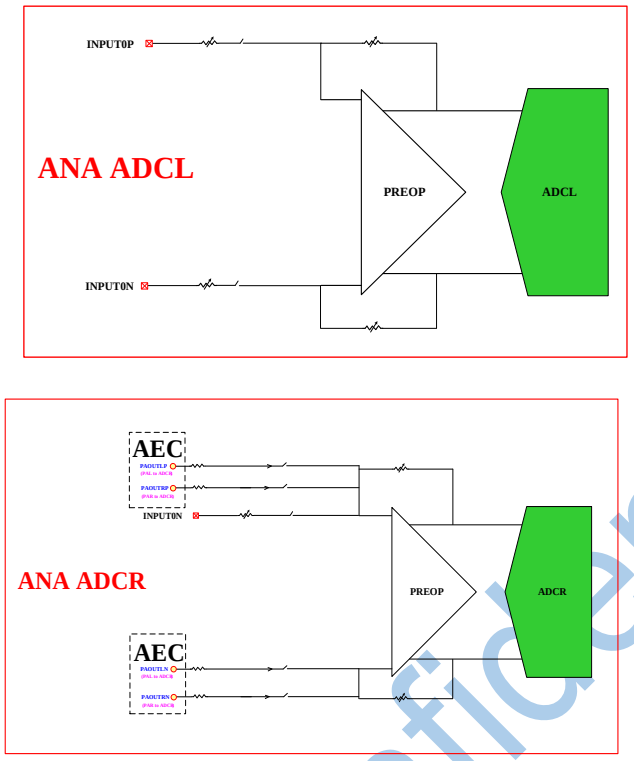


Figure 13-3 ADC analog block

13.2.2.2 ADC frequency response curves

ATS3089x provides 3 frequency response curves for use.

Table 13-4 The parameters of 3 frequency response curves

Mode Select	Pass Band Ripple (db)	Stopband Attenuation (db)	-1db Cutoff Frequency(Hz)
mode A	-0.04~0.06	-71	-0.463
mode B	-0.09~0.00	-82	-0.449
mode C	-0.04~0.08	-71	-0.235

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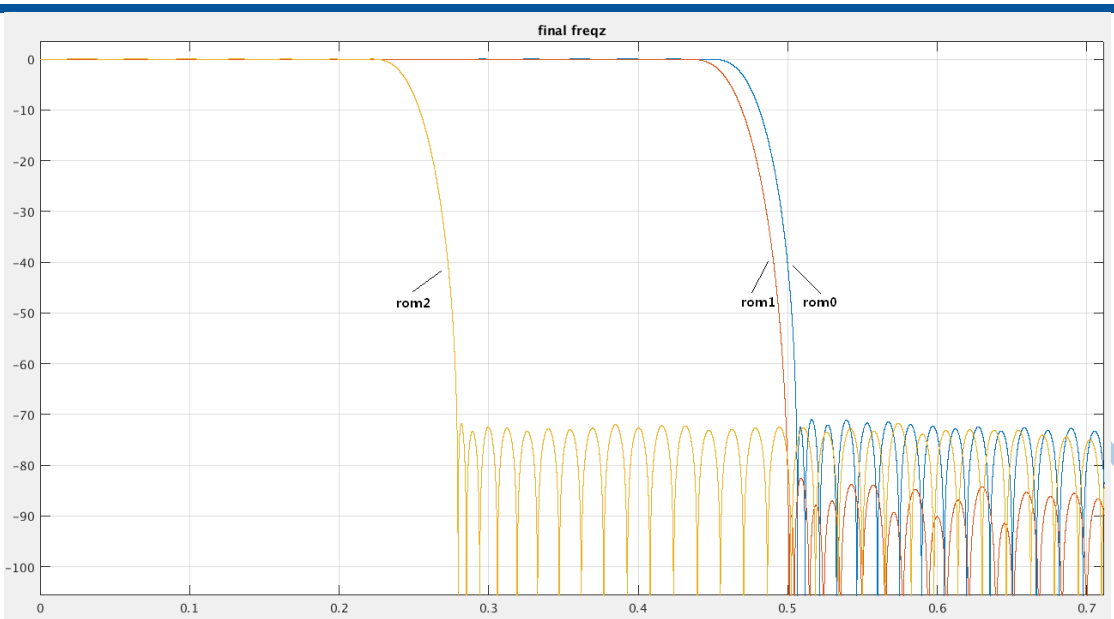


Figure 13-4 The frequency response curves

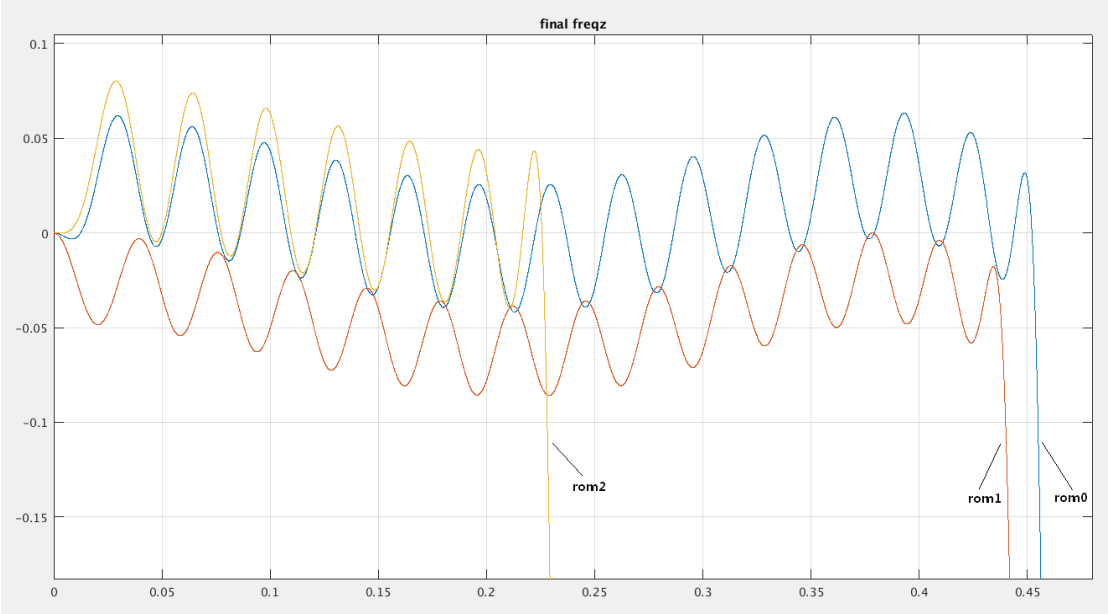


Figure 13-5 In-band ripple

13.2.2.3 Signal List

Table 13-5 ADC Signals

SIGNAL Name	I/O Type	Short Description
INPUT0P	AI	Analog single-end input to ADC0, or can form differential input（ADC0） with INPUT0N
INPUT0N	AI	Analog single-end input to ADC1, or can form differential input（ADC0） with INPUT0P
DMIC_CLK	DO	DMIC clock output
DMIC_DATA	DI	DMIC data input

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13.2.2.4 DMIC

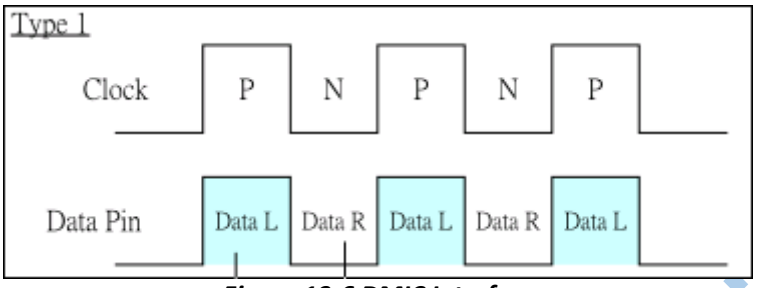


Figure 13-6 DMIC Interface

ATS3089x has 4 DMIC which can form 2 stereo voice inputs.

13.2.3 Register List

Table 13-6 ADC Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
ADC_Control_Register	0x4005C100	0x4005C100

Table 13-7 ADC Controller Registers

Offset	Register Name	Description
0x00	ADC_DIGCTL	ADC Digital Control Register
0x04	CH0_DIGCTL	CH0 Digital Control Register
0x08	CH1_DIGCTL	CH1 Digital Control Register
0x10	ADC_STAT	ADC State Register
0x20	ADC0_CTL	ADC0 analog control Register
0x24	ADC1_CTL	ADC1 analog control Register
0x2C	ADC_VMIC_CTL	ADC VMIC control Register

13.2.4 Register Description

13.2.4.1 ADC_DIGCTL

ADC Digital Control Register
Offset = 0x00

Bit(s)	Name	Description	R/W	Reset
31:14	-	R/W forbidden	-	-
13	ADC1_DIG_EN	ADC1 digital enable: 0:disable 1:enable	RW	0x0
12	ADC0_DIG_EN	ADC0 digital enable: 0:disable 1:enable	RW	0x0
11	-	R/W forbidden	-	-
10:8	DMIC_PRE_GAIN	Pre-Gain control for DMIC	RW	0x0

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		000 : 1x 001 : 2x 010 : 4x 011 : 8x 100 : 16x 101 : 32x 110 : 63x 111 : 63x for all DMIC interface If CHx data select DMIC and CHx_DIG_EN was enable channel x DMIC interface was enable.		
7	-	Reserved	R/W	0x0
6	DMIC01_CHS	DMIC01 channel select： 0：L/R 1：R/L When set to 0, DMIC will latch L channel data in DMIC_CLK rising edge. When set to 1,DMIC will latch R channel data in DMIC_CLK rising edge.	RW	0x0
5:4	ADC_MD_SEL	mode select 00 : FIR mode A 01 : FIR mode B 10 : FIR mode C 11 : FIR mode D	RW	0x0
3:0	-	R/W forbidden	-	-

13.2.4.2 CH0_DIGCTL

CH0 Digital Control Register
Offset = 0x04

Bit(s)	Name	Description	R/W	Reset
31:4	-	R/W forbidden	-	-
3:0	ADCGC	ADC0 Digital Gain Control 0x0: 0dB 0x1: 3.5dB 0x2: 7dB 0x3: 10.5dB 0x4: 14dB 0x5: 17.5dB 0x6: 21dB 0x7: 24.5dB 0x8: 28dB 0x9: 31.5dB 0xa: 35dB 0xb: 38.5dB 0xc: 42dB 0xd: 45.5dB 0xe: 49dB 0xf: 52.5dB	RW	0x0

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13.2.4.3 CH1_DIGCTL

CH1 Digital Control Register
Offset = 0x08

Bit(s)	Name	Description	R/W	Reset
31:4	-	R/W forbidden	-	-
3:0	ADCGC	ADC1 Digital Gain Control 0x0: 0dB 0x1: 3.5dB 0x2: 7dB 0x3: 10.5dB 0x4: 14dB 0x5: 17.5dB 0x6: 21dB 0x7: 24.5dB 0x8: 28dB 0x9: 31.5dB 0xa: 35dB 0xb: 38.5dB 0xc: 42dB 0xd: 45.5dB 0xe: 49dB 0xf: 52.5dB	RW	0x0

13.2.4.4 ADC_STAT

ADC State Register
Offset = 0x10

Bit(s)	Name	Description	R/W	Reset
31:10	-	R/W forbidden	-	-
9	FIFO0_ER	FIFO0 error 0 : no error 1 : error When fifo0 was full, system bus write ADCFIFO0 would cause error Writing 1 to the bit will clear it	RW	0
8	ADF0EF	ADC FIFO0 Empty Flag. 0x0: Not Empty 0x1: Empty	R	0x1
7	ADF0IP	ADC FIFO half-Full IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it.	RW	0x0
6:0	ADF0S	ADC FIFO Status. These 6 bits shows how many samples fifo0 filled. For example, when read as 9, means CPU can read 9 samples from fifo0.	R	0x0

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13.2.4.5 ADC0_CTL

ADC0 Control Register (all input can be used as mic/aux)
Offset = 0x20

<table>
<tr><th>Bit(s)</th><th>Name</th><th>Description</th><th>R/W</th><th>Reset</th></tr>
<tr><td>31:19</td><td>-</td><td>R/W forbidden except bit[20]</td><td>-</td><td>-</td></tr>
<tr><td>18:16</td><td>FDBUF0_IRS</td><td>FDBUF0 input resistor select
0xx: not connect, disable
100:40k
101:20k
110:10k
For se mode, this res should be connected. If only one channel input ,then set the res the same as AUX input res, the gain is calculated as PREAM_PG.</td><td>RW</td><td>0x0</td></tr>
<tr><td>15:12</td><td>PREAM0_PG</td><td>PREAMP0 OP feedback res select to form different gain with different input res:
For use, first select input mode and input res, then select preamp feedback res, then gain is fixed.
For diff mode (se mode -6db)
<table>
<tr><td></td><td>40k</td><td>20k</td><td>10k</td></tr>
<tr><td>0000</td><td>-12db</td><td>-6db</td><td>0db</td></tr>
<tr><td>0001</td><td>-9db</td><td>-3db</td><td>3db</td></tr>
<tr><td>0010</td><td>-6db</td><td>0db</td><td>6db</td></tr>
<tr><td>0011</td><td>-3db</td><td>3db</td><td>9db</td></tr>
<tr><td>0100</td><td>0db</td><td>6db</td><td>12db</td></tr>
<tr><td>0101</td><td>3db</td><td>9db</td><td>15db</td></tr>
<tr><td>0110</td><td>6db</td><td>12db</td><td>18db</td></tr>
<tr><td>0111</td><td>7.5db</td><td>13.5db</td><td>19.5db</td></tr>
<tr><td>1000</td><td>9db</td><td>15db</td><td>21db</td></tr>
<tr><td>1001</td><td>10.5db</td><td>16.5db</td><td>22.5db</td></tr>
<tr><td>1010</td><td>12db</td><td>18db</td><td>24db</td></tr>
<tr><td>1011</td><td>13.5db</td><td>19.5db</td><td>25.5db</td></tr>
<tr><td>1100</td><td>15db</td><td>21db</td><td>27db</td></tr>
<tr><td>1101</td><td>16.5db</td><td>22.5db</td><td>28.5db</td></tr>
<tr><td>1110</td><td>18db</td><td>24db</td><td>30db</td></tr>
<tr><td>1111</td><td>19.5db</td><td>25.5db</td><td>31.5db</td></tr>
</table></td><td>RW</td><td>0x0</td></tr>
<tr><td>11</td><td>-</td><td>R/W forbidden</td><td>-</td><td>-</td></tr>
<tr><td>10</td><td>FDBUF0_EN</td><td>FD BUF OP 0 enable for SE mode
0:disable
1:enable</td><td>RW</td><td>0x0</td></tr>
<tr><td>9</td><td>PREOP0_EN</td><td>PREOP 0 enable
0:disable
1:enable</td><td>RW</td><td>0x0</td></tr>
<tr><td>8</td><td>ADC0_EN</td><td>ADC0 channel SDM enable
0:disable
1:enable
When enable this bit, must enable bit[20]</td><td>RW</td><td>0x0</td></tr>
<tr><td>7:6</td><td>INPUT0N_EN</td><td>INPUT0N pad to ADC0/1 channel input enable
00~01: disable
11: enable channel</td><td>RW</td><td>0x0</td></tr>
<tr><td>5:4</td><td>INPUT0P_EN</td><td>INPUT0P pad to ADC0 channel input enable
00~01: disable
11: enable channel</td><td>RW</td><td>0x0</td></tr>
</table>

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3	-	R/W forbidden	-	-
2	INPUT0_IN_MODE	INPUT0 input mode select 1: single end input mode: INPUT0P to ADCL channel, INPUT0N to ADCR channel; 0: differential input mode: INPUT0P and INPUT0N form the differential input to ADCL channel INPUT0P is the positive end ; INPUT0N is the negative end.	RW	0x0
1:0	INPUT0_IRS	INPUT0 input resistor select 00: 40k 01:20k 10:10k	RW	0x0

13.2.4.6 ADC1_CTL

ADC1 Control Register (all input can be used as mic/aux)
Offset = 0x24

Bit(s)	Name	Description	R/W	Reset
31:29	-	R/W forbidden	-	-
28	MIX2AD1FDSE	Mixout to ADC1 input mode select 0:fd mode 1:se mode	RW	0x0
27	PAR2AD1_EN	PA OUTR mix to ADC1 channel 0:disable 1:enable Input res is 40k.	RW	0x0
26	PAL2AD1_EN	PA OUTL mix to ADC1 channel 0:disable 1:enable Input res is 40k.	RW	0x0
25	PAR2AD1_PD_EN	PA OUTR mix to ADC1 channel pull down 0:disable 1:enable	RW	0x0
24	PAL2AD1_PD_EN	PA OUTL mix to ADC1 channel pull down 0:disable 1:enable	RW	0x0
23:19	-	R/W forbidden except bit[20]	-	-
18:16	FDBUF1_IRS	FDBUF1 input resistor select 0xx: not connect, disable 100:40k 101:20k 110:10k *For se mode, this res should be connected. If only one channel input ,then set the res the same as AUX input res, the gain is calculated as PREAM_PG.*	RW	0x0
15:12	PREAM1_PG	PREAMP1 OP feedback res select to form different gain with different input res: *For use, first select input mode and input res, then select preamp feedback res, then gain is fixed.* For diff mode (se mode -6db) 40k / 20k / 10k 0000: -12db / -6db / 0db 0001: -9db / -3db / 3db	RW	0x0

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		0010	-6db	0db	6db		
		0011	-3db	3db	9db		
		0100	0db	6db	12db		
		0101	3db	9db	15db		
		0110	6db	12db	18db		
		0111	7.5db	13.5db	19.5db		
		1000	9db	15db	21db		
		1001	10.5db	16.5db	22.5db		
		1010	12db	18db	24db		
		1011	13.5db	19.5db	25.5db		
		1100	15db	21db	27db		
		1101	16.5db	22.5db	28.5db		
		1110	18db	24db	30db		
		1111	19.5db	25.5db	31.5db		
11	-	R/W forbidden				-	-
10	FDBUF1_EN	FD BUF OP 1 enable for SE mode 0:disable 1:enable				RW	0x0
9	PREOP1_EN	PREOP 1 enable 0:disable 1:enable				RW	0x0
8	ADC1_EN	ADC1 channel SDM enable 0:disable 1:enable When enable this bit, must enable bit[20]				RW	0x0
7:0	-	R/W forbidden				-	-

13.2.4.7 ADC_VMIC_CTL

ADC VMIC Register
Offset = 0x2C

Bit(s)	Name	Description	R/W	Reset
31:4	-	R/W forbidden	-	-
3:2	VMIC0_VOL	VMIC0 output voltage control 00: 1.57V 01: 1.61V 10: 1.65V 11: 1.7V	R/W	0x2
1:0	VMIC0_EN	VMIC0 enable 0x: disable VMIC0 10: output voltage fix in VC18; 11: enable VMIC0 and voltage set by bit[3:2]; **For mems mic, try to use 10 bypass mode to save power.**	R/W	0x0

13.3 I2S/PCM

13.3.1 Features

I2S :

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- TX0/RX0 Support Master / Slave mode; TX1 Support Master mode Only.
- TX0/RX0 Support Slave Mode Sample Rate Detect.
- 5 wire mode：TX0+RX0 Support Master / Slave; TX1+TX0+RX0 Support Master Only.
- TX0/RX0 Formats include I2S /L-J /R-J /TDM4 /TDM8; TX1 NOT Support TDM4/TDM8.
- TX0/RX0/TX1 Support data width 16/20/24 bit.
- TX0/RX0 Support Sample Rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/96k/192k for Dual channel; TX1 NOT Support 192k.
- TX1 Support Volume adjustment.

PCM:

- TX0/RX0 Support Master and Slave mode; TX1 Support Master mode Only.
- TX0/RX0 Support Slave Mode Sample Rate Detect.
- TX0/RX0/TX1 Support 13~16/24 bits linear, MSB first.
- TX0/RX0/TX1 Support Short Frame Sync and Long Frame Sync.
- TX0/RX0 Support Sample rate 8k/12k/11.025k/16k/22.05k/24k/32k/44.1k/48k/96k/192k for Dual channel; TX1 NOT Support 192k.
- TX0/RX0 Support 1 /2 /4 /8 channel; TX1 Only Support 2 channel.
- TX0/RX0/TX1 Support 16/20/24 bit data width.
- TX0/RX0/TX1 Per channel can set 16 or 32 pcm_clk cycles.

I2S&PCM:

- TX0 FIFO with 24bit*32 levels, accessed by CPU, DMA, DSP, DSPDMA.
- RX0 FIFO with 24bit*32 levels, accessed by CPU, DMA, DSP, DSPDMA.
- TX1 share 24bit*16 levels DACFIFO, with 3Kbyte PCM RAM.
- I2S_MCLK, I2S_BCLK/PCM_CLK, I2S_LRCLK/PCM_SYNC can independently shut down.
- The max setting frequency of Bclk/PCM_clk/TX0/RX0 is 12.288 MHz, TX1 is 6.144MHz.

13.3.1 Module Description

The timing format I2S support are as follows.

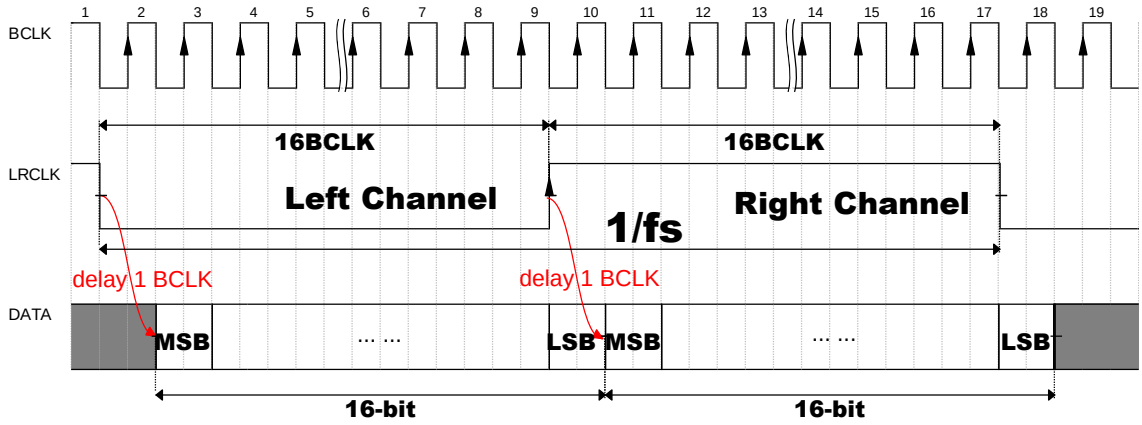


Figure 13-7 I2S- format -16bit

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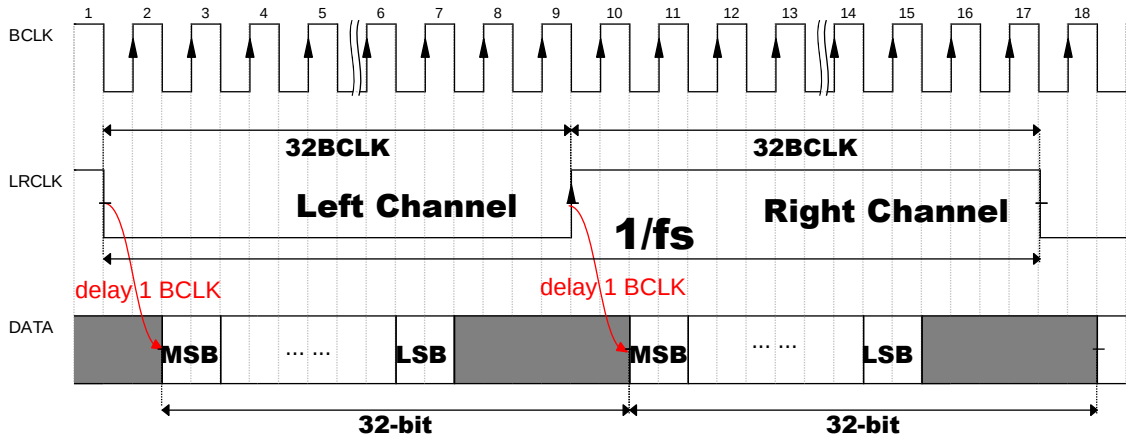


Figure 13-8 I2S- format -32bit

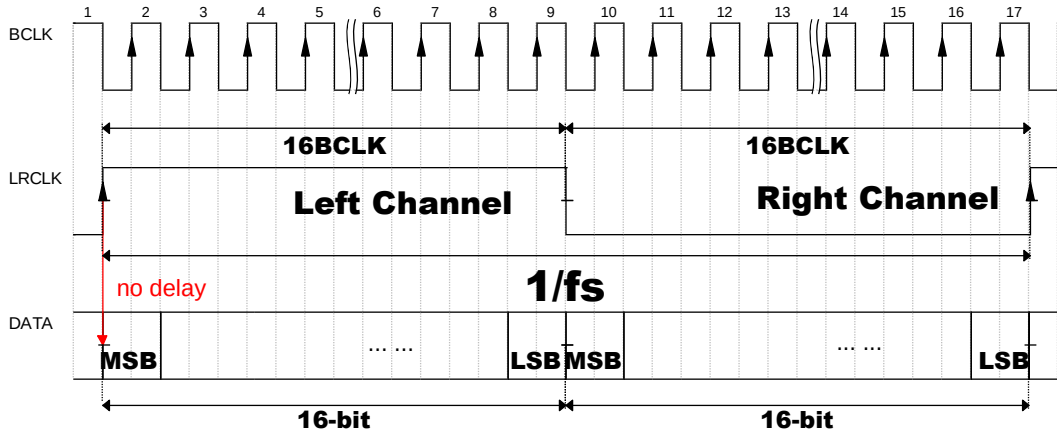


Figure 13-9 Left justified - format - 16bit

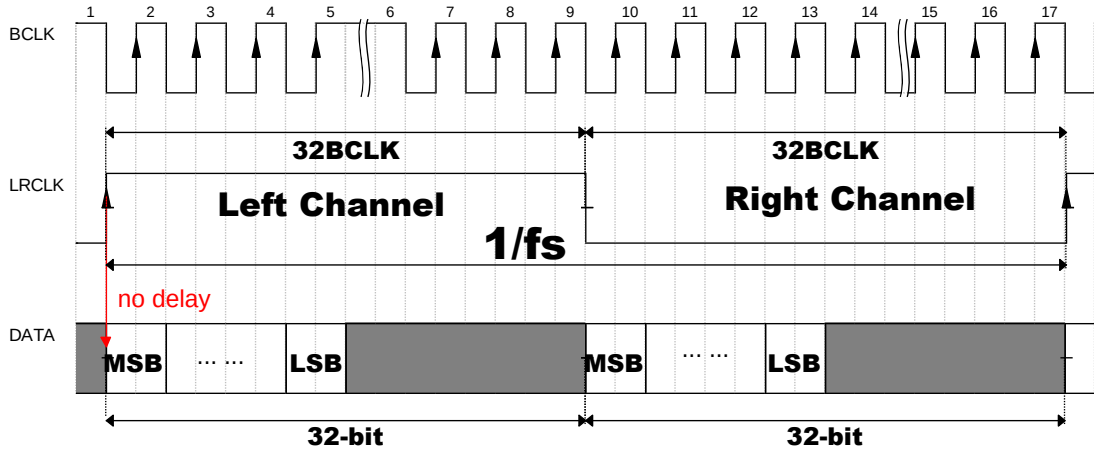


Figure 13-10 Left justified - format - 32bit

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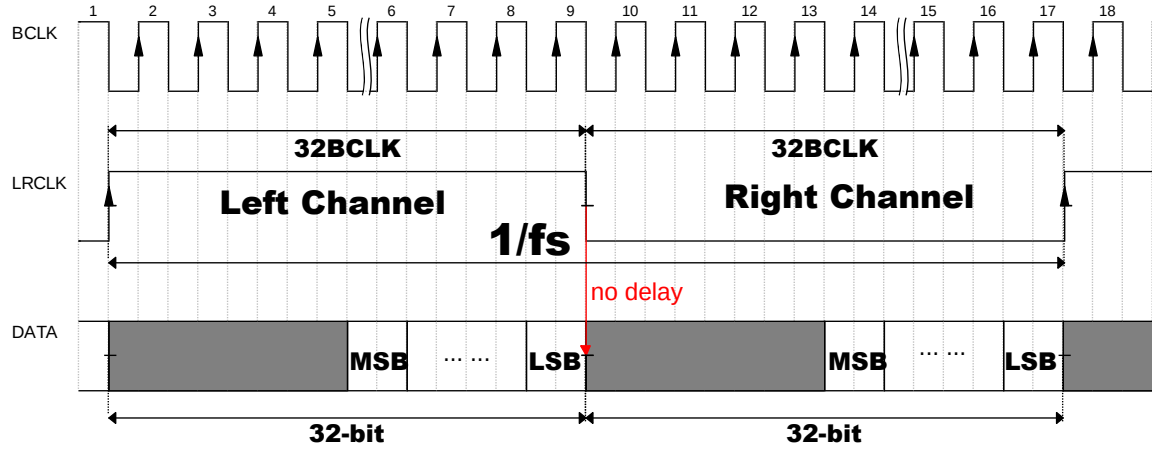


Figure 13-11 Right justified - format - 32bit

13.3.2 Timing Parameter

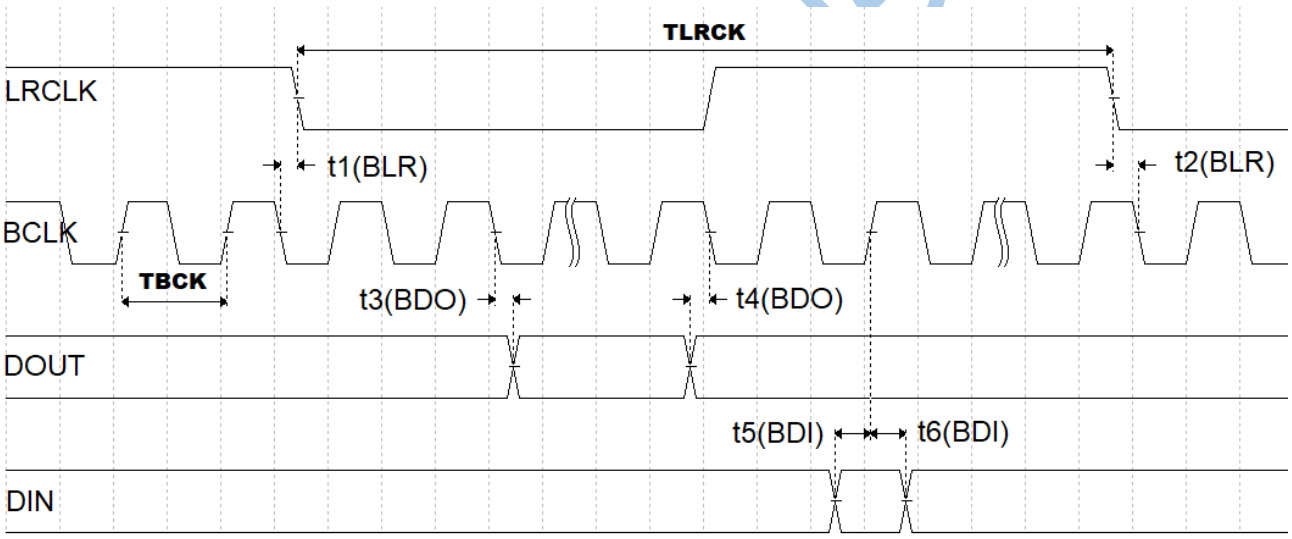


Figure 13-12 I2S Timing

Table 13-8 I2S Timing parameter in Host Mode

SYMBOL	PARAMETER	MIN	TYP	MAX	UINT
TLRCK	LRCLK Period	1/fs*0.98	1/fs	1/fs*1.02	ns
TBCK	BCLK Period	1/(64fs)*0.98	1/(64fs)	1/(64fs)*1.02	ns
TBCKH	BCLK High Pulse Duration		50% TBCK		ns
TBCKL	BCLK Low Pulse Duration		50% TBCK		ns
t1(BLR)	BCLK falling edge ahead LRCLK falling edge time			6	ns
t2(BLR)	BCLK falling edge delay LRCLK falling edge time			6	ns
t3(BDO)	BCLK falling edge ahead data vaild edge time			6	ns
t4(BDO)	BCLK falling edge delay data vaild edge time			6	ns
t5(BDI)	DIN setup time	16			
t6(BDI)	DIN hold time	12			

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Note: TA = 25° C, vdd=1.1V

Table 13-9 I2S Timing parameter in **Slave Mode**

SYMBOL	PARAMETER	MIN	TYP	MAX	UINT
TLRCK	LRCLK Period	1/fs*0.98		1/fs*1.02	ns
TBCK	BCLK Period	1/(64fs)*0.98		1/(64fs)*1.02	ns
TBCKH	BCLK High Pulse Duration		50% TBCK		ns
TBCKL	BCLK Low Pulse Duration		50% TBCK		ns
t1(BLR)	BCLK falling edge ahead LRCLK falling edge time			15	ns
t2(BLR)	BCLK falling edge delay LRCLK falling edge time			15	ns
t3(BDO)	BCLK falling edge ahead data vaild edge time	12			ns
t5(BDI)	DIN setup time	12			
t6(BDI)	DIN hold time	12			

Note: TA = 25° C, vdd=1.1V

13.3.3 I2STX Register List

Table 13-10 I2STX Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
I2STX_Register	0x4005C400	0x4005C400

Table 13-11 I2STX Controller Registers

Offset	Register Name	Description
0x0000	I2STX0_CTL	I2STX0 Control Register
0x0004	I2STX0_FIFOCTL	I2STX0 FIFO control register
0x0008	I2STX0_FIFOSTAT	I2STX0 FIFO status register
0x000C	I2STX0_DAT	I2STX0 FIFO data register
0x0010	I2STX0_SRDCTL	I2STX0 Slave Mode TX Sample Rate Detect Register
0x0014	I2STX0_SRDSTA	I2STX0 Slave Mode TX Sample Rate Detect State Register
0x0018	I2STX0_FIFO_CNT	I2STX0 Out FIFO Counter Register
0x0020	I2STX1_CTL	I2STX1 Control Register

13.3.4 I2STX Register Description

13.3.4.1 I2STX0_CTL

I2STX0 Control Register
Offset = 0x00

Bit(s)	Name	Description	R/W	Reset
31	TX_ER_PD	The TX0 Channel alignment error pending: 0: no irq pending 1: irq pending Write '1' to clean this bit.	R/W	0x0

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30	TX_ER_IE	Transmitting Error interrupt enable: 0：disable 1：enable	R/W	0x0
29:27	-	Reserved	R	0x0
26	PCM_SYNC	PCM Frame Mode Select 0: Short Frame Sync Mode 1: Long Frame Sync Mode	R/W	0x0
25:24	PCM_CHAN	How many Channels be transmitted per frame 0x0: l Slot. 0x1: 2 Slot. 0x2: 4 Slot. 0x3: 8 Slot.	R/W	0x0
23	-	Reserved for [22:21]TXD_DELAY	R	0x0
22:21	TXD_DELAY	**tx data output at:** 0x0: 0.5 bclk cycles ahead of the bclk active edge. 0x1: 1.0 bclk cycles ahead of the bclk active edge. 0x2: 1.5 bclk cycles ahead of the bclk active edge. 0x3: 2.0 bclk cycles ahead of the bclk active edge.	R/W	0x0
20	TDMTX_CHAN	0:4 channel 1:8 channel	R/W	0x0
19	TDMTX_MODE	0:TDM-A (I2S data format) 1:TDM-B (left-justified data format)	R/W	0x0
18:17	TDMTX_SYNC	**TDM Frame Start at:** 00: the rising edge of LRCLK with a pulse (duration of one BCLK); 01: the rising edge of LRCLK with a 50% duty cycle; 10: the falling edge of LRCLK with a 50% duty cycle; 11: reserved;	R/W	0x0
16	-	Reserved for MULT_DEVICE.	R/W	0x0
15:9	-	Reserved	R	0x0
8	LPEN0	I2STX0 and I2SRX0 data loopback enable 0：disable 1：enable When enable,TX0 send clock and data to RX0.	R/W	0x0
7	-	Reserved	R/W	0x0
6	PCMTX0_EN	PCM TX Enable: 0: Disable 1: Enable	R/W	0x0
5:4	TXWIDTH	Effective width 00: data are 16 bit effective 01: data are 20 bit effective 10: data are 24 bit effective ***** 11: reserved	R/W	0x2
3	TXBCLKSET	0:32bit 1:16bit	R/W	0x0
2:1	TXMODELSEL	I2S transfer format select: 00: S format 01: left-justified format 10: right-justified format 11: TDM	R/W	0x0
0	TXEN	I2S TX Enable. 0: Disable	R/W	0x0

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		1: Enable		

13.3.4.2 I2STX0_FIFOCTL

I2STX0 FIFO Control Register
Offset = 0x04

Bit(s)	Name	Description	R/W	Reset
31:8	-	Reserved	R/W	0x3
7	TXFIFO_DMAWIDTH	I2STXFIFO DMA/DSPDMA transfer width configured 0 : 32bit 1 : 16bit Match with DMA/DSPDMA width which configured at register DMAx_CTL /AUDDSP_DMAx_CTL. I2STXFIFO need not to support 8bit and 64bit DMA data width.	R/W	0x0
6:4	FIFO_IN_SEL	I2STX0_FIFO Input Select: 0x00: CPU 0x01: DMA 0x02: Reserved 0x03: DSP 0x04: DSPDMA Others: Reserved	R/W	0x0
3	FIFO_DSPDMA_DEN	I2STX0_FIFO DSPDMA DRQ Enable. 0x0: Disable 0x1: Enable	R/W	0x0
2	FIFO_IEN	I2STX0_FIFO Half-Empty IRQ Enable. 0x0: Disable 0x1: Enable	R/W	0x0
1	FIFO_DEN	I2STX0_FIFO Half-Empty DRQ Enable. 0x0: Disable 0x1: Enable	R/W	0x0
0	FIFO_RST	I2STX0_FIFO Reset. 0x0: Reset FIFO 0x1: Enable FIFO	R/W	0x0

13.3.4.3 I2STX0_FIFOSTA

I2STX0 FIFO State Register
Offset = 0x08

Bit(s)	Name	Description	R/W	Reset
31:9	-	Reserved	R	0x0
8	TFEP	TX0 FIFO Error Pending Bit. 0: No Error 1: Error Writing 1 to the bit will clear it	RW	0x0
7	IP	I2STX0_FIFO Half-Empty IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ	RW	0x0

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		Writing 1 to the bit is clear it.		
6	TFFU	I2STX0_FIFO Full Flag. 0x0: Not Full 0x1: Full	R	0x0
5:0	STA	I2STX0_FIFO Status These 6 bits shows how many fifo level not filled (can be written into fifo). If no I2STX0_CLK register would display zero. When read as 4, mean can write 4 samples to fifo.	R	0x0

13.3.4.4 I2STX0_DAT

I2STX0 FIFO Data Register
Offset = 0x0c

Bit(s)	Name	Description	R/W	Reset
31:8	DAT	I2STX_FIFO Data FIFO is 24bit x 32 levels.	W	x
7:0	-	Reserved	R	0x0

13.3.4.5 I2STX0_SRDCTL

I2STX0 Slave Mode Sample Rate Detect Control Register
Offset = 0x10

Bit(s)	Name	Description	R/W	Reset
31:13	-	Reserved	R	0x0
12	MUTE_EN	If detect sample rate or channel width changing, mute the TX output as 0. 0:not mute 1:mute **Mute would continue until SRC_PD was clear.**	R/W	0x0
11	TO_IE	sample rate detection **timeout** interrupt enable 0:disable 1:enable	R/W	0x0
10	SRC_IE	sample rate change detection interrupt enable 0:disable 1:enable	R/W	0x0
9	-	Reserved	R/W	0x0
8	CHW_IE	**channel width change** interrupt enable 0:disable 1:enable	R/W	0x0
7:6	-	reserved	R/W	0x0
5:4	CNT_TIM	Slave mode sample rate detect counter period select 0:2 LRCLK cycle 1:4 LRCLK cycle Other: reserved	R/W	0x0
3:1	SRD_TH	The threshold of sample rate detection. 0: 8	R/W	0x0

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		1: 16 … 6: 56 7: 64 Value=(SRD_TH+1)*8		
0	SRD_EN	Slave mode sample rate detect enable: 0:disable 1:enable	R/W	0x0

13.3.4.6 I2STX0_SRDSTA

I2STX0 Slave Mode Sample Rate Detect State Register
Offset = 0x14

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24:12	CNT	CNT of LRCLK which sampling by SRDCLK. CNT= SRDCLK / LRCLK. It would clear when module was enabling at first time.	R	0x0
11	TO_PD	sample rate change detection **timeout** interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit. If CNT overflow (0x9c00), if would cause TIMEOUT irq.	RW	0x0
10	SRC_PD	**sample rate change** detection interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit.	RW	0x0
9	-	Reserved for future use	R	0x0
8	CHW_PD	**channel width change** interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit.	RW	0x0
7:3	-	Reserved	R	0x0
2:0	WL	Channel width length (the rate of BCLK to LRCLK)： 000： 32 rate(I2S: 16*2bit) 001： 64 rate(I2S: 32*2bit; TDM4: 16*4bit) 010：128 rate(TDM8: 16*8bit; TDM4: 32*4bit) 011：256 rate(TDM8: 32*8bit) 1xx：Reserved	R	0x0

13.3.4.7 I2STX0_FIFO_CNT

I2STX0 FIFO Out Counter Register
Offset = 0x20

Bit(s)	Name	Description	R/W	Reset
31:19	-	Reserved		
18	IP	I2STX FIFO counter overflow irq pending 0: no pending 1: pending	RW	0x0

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		Write '1' to clear this bit		
17	IE	I2STX FIFO counter overflow irq enable: 0:disable 1:enable If CNT overflow, it would cause an interrupt.	RW	0x0
16	EN	I2STX FIFO counter enable 0:disable 1:enable Disable this function could reset the whole counter.	RW	0x0
15:0	CNT	I2STX FIFO counter If overflow count would be clear to zero and cause an interrupt This counter count the valid data output by I2STX FIFO, it means that if FIFO is empty, this counter would not add till FIFO had been written data in again.	R	0x0

13.3.4.8 I2STX1_CTL

I2STx1 Control Register
Offset = 0x20

Bit(s)	Name	Description	R/W	Reset
31:27	-	Reserved	R/W	0x0
26	PCM_SYNC	PCM Frame Mode Select 0: Short Frame Sync Mode 1: Long Frame Sync Mode	R/W	0x0
25:24	PCM_CHAN	How many Channels be transmitted per frame 0x0: l Slot. 0x1: 2 Slot. 0x2: 4 Slot. 0x3: 8 Slot.	R/W	0x1
23:21	-	Reserved for TXD_DELAY	R/W	0x0
20	-	Reserved for TDMTX_CHAN	R/W	0x0
19	-	Reserved for TDMTX_MODE	R/W	0x0
18:17	-	Reserved for TDMTX_SYNC	R/W	0x0
16	-	Reserved for MULT_DEVICE.	R/W	0x0
15:10	-	Reserved	R	0x0
9	TX_MULT	I2STX1 and I2STX0 Multi-Link 0x0: disable 0x1: enable	R/W	0x0
8	LPEN1	I2STX1 and I2SRX0 data loopback enable 0：disable 1：enable When enable,TX1 send clock and data to RX0.	R/W	0x0
7	-	Reserved	R/W	0x0
6	PCMTX_EN	PCM TX Enable: 0: Disable 1: Enable	R/W	0x0
5:4	TXWIDTH	Effective width 00: data are 16 bit effective	R/W	0x2

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		01: data are 20 bit effective 10: data are 24 bit effective ***** 11: reserved		
3	TXBCLKSET	0:32bit 1:16bit	R/W	0x0
2:1	TXMODELSEL	I2S transfer format select: 00: I2S format 01: left-justified format 10: right-justified format 11: TDM	R/W	0x0
0	TXEN	I2S TX Enable. 0: Disable 1: Enable	R/W	0x0

13.3.5 I2SRX Register List

Table 13-12 I2SRX Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
I2SRX_Register	0x4005C500	0x4005C500

Table 13-13 I2SRX Controller Registers

Offset	Register Name	Description
0x0000	I2SRX_CTL	I2SRX Control Register
0x0004	I2SRX_FIFOCTL	I2SRX FIFO Control Register
0x0008	I2SRX_FIFOSTA	I2SRX FIFO State Register
0x000C	I2SRX_DAT	I2SRX FIFO Data Register
0x0010	I2SRX_SRDCTL	I2SRX Slave Mode Sample Rate Detect Control Register
0x0014	I2SRX_SRDSTA	I2SRX Slave Mode Sample Rate Detect State Register

13.3.6 I2SRX Register Description

13.3.6.1 I2SRX_CTL

I2SRX Control Register
Offset = 0x00

Bit(s)	Name	Description	R/W	Reset
31	RX_ER_PD	The RX0 Channel alignment error pending: 0: no irq pending 1: irq pending Write '1' to clean this bit.	R/W	0x0
30	RX_ER_IE	Receive Error interrupt enable: 0：disable 1：enable	R/W	0x0
29:27	-	Reserved	R	0x0
26	PCM_SYNC	PCM Frame Mode Select 0: Short Frame Sync Mode 1: Long Frame Sync Mode	R/W	0x0

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25:24	PCM_CHAN	How many Channels be transmitted per frame 0x0: l slot. 0x1: 2 slot. 0x2: 4 slot. 0x3: 8 slot.	R/W	0x0
23:17	-	Reserved	R	0x0
16	TDMRX_CHAN	0:4 channel 1:8 channel	R/W	0x0
15	TDMRX_MODE	0:TDM-A (I2S format) 1:TDM-B (left-justified format)	R/W	0x0
14:13	TDMRX_SYNC	at the beginning of the data frame, the type of LRCLK set: 00: at the rising edge of LRCLK with a pulse(duration of one BCLK) 01: at the rising edge of LRCLK with a 50% duty cycle; 10: at the falling edge of LRCLK with a 50% duty cycle; 11: Reserved	R/W	0x0
12:7	-	Reserved	R	0x0
6	PCMRX_EN	PCM RX Enable: 0: Disable 1: Enable	R/W	0x0
5:4	RXWIDTH	Effective width 00: data are 16 bit effective 01: data are 20 bit effective 10: data are 24 bit effective 11: reserved	R/W	0x2
3	RXBCLKSET	0: 32bit 1: 16bit	R/W	0x0
2:1	RXMODELSEL	I2S transfer format select: 00 : S format 01: left-justified format 10: right-justified format 11: TDM	R/W	0x0
0	RXEN	I2S RX Enable. 0: Disable 1: Enable	R/W	0x0

13.3.6.2 I2SRX_FIFOCTL

I2SRX FIFO Control Register
Offset = 0x04

Bit(s)	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7	RXFIFO_ DMAWIDTH	I2SRXFIFO DMA transfer width configured 0 : 32bit 1 : 16bit Set this configure match width DMA width, I2SRXFIFO not need to support 8bit and 64bit DMA data width.	RW	0x0
6:4	RXFOS	I2SRX FIFO Output Select: 0x00: CPU 0x01: DMA	RW	0x0

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		0x03: DSP 0x04: DSPDMA Others: Reserved		
3	RXFF_DSPDMA_DEN	I2SRX0_FIFO DSPDMA DRQ Enable. 0x0: Disable 0x1: Enable	R/W	0x0
2	RXFFIE	I2SRX FIFO Half-Filled IRQ Enable. 0: Disable 1: Enable	RW	0x0
1	RXFFDE	I2SRX FIFO Half-Filled DRQ Enable. 0x0: Disable 0x1: Enable	RW	0x0
0	RXFRT	I2SRX FIFO Reset. 0x0: Reset FIFO 0x1: Enable FIFO	RW	0x0

13.3.6.3 I2SRX_FIFOSTA

I2SRX FIFO State Register
Offset = 0x08

Bit(s)	Name	Description	R/W	Reset
31:9	-	Reserved	R	0x0
8	RFEP	RX FIFO Error Pending Bit. 0: No Error 1: Error Writing 1 to the bit will clear it	RW	0x0
7	RXFEF	I2SRX FIFO Empty Flag. 0x0: Not Empty 0x1: Empty Can be accessed by DSP.	R	0x1
6	RXFIP	I2SRX FIFO Half-Filled IRQ Pending Bit. 0x0: No IRQ 0x1: IRQ Writing 1 to the bit is clear it. Can be accessed by DSP.	RW	0x0
5:0	RXFS	I2SRX FIFO Status. This 6 bits shows how many samples fifo filled. when read as 6, means can read 6 samples from fifo. If no I2SRX_CLK register would display zero.	R	0x0

13.3.6.4 I2SRX_DAT

I2SRX FIFO Data Register
Offset = 0x0C

Bit(s)	Name	Description	R/W	Reset
31:8	RXDAT	RX Data FIFO is 24bit x 32 levels.	R	x
7:0	-	Reserved	R	0x0

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13.3.6.5 I2SRX_SRDCTL

I2SRX Slave Mode Sample Rate Detect Control Register
Offset = 0x10

Bit(s)	Name	Description	R/W	Reset
31:13	-	Reserved	R	0x0
12	MUTE_EN	If detect sample rate or channel width changing, mute the TX output as 0. 0:not mute 1:mute **Mute would continue until SRC_PD was clear.**	R/W	0x0
11	TO_IE	sample rate detection **timeout** interrupt enable 0:disable 1:enable	R/W	0x0
10	SRD_IE	**sample rate change** detection interrupt enable 0:disable 1:enable	R/W	0x0
9	-	Reserved	R/W	0x0
8	CHW_IE	**channel width change** interrupt enable 0:disable 1:enable	R/W	0x0
7:6	-	Reserved	R/W	0x0
5:4	CNT_TIM	Slave mode sample rate detect counter period select 0:2 LRCLK cycle 1:4 LRCLK cycle Other: reserved	R/W	0x0
3:1	SRD_TH	The threshold of sample rate detection. 0: 8 1: 16 … 6: 56 7: 64 Value=(SRD_TH+1)*8	R/W	0x0
0	SRD_EN	Slave mode sample rate detect enable: 0:disable 1:enable	R/W	0x0

13.3.6.6 I2SRX_SRDSTA

I2SRX Slave Mode Sample Rate Detect State Register
Offset = 0x14

Bit(s)	Name	Description	R/W	Reset
31:25	-	Reserved	R	0x0
24:12	CNT	CNT of LRCLK which sampling by SRDCLK. CNT= SRDCLK / LRCLK. It would clear when module was enabling at first time.	R	0x0

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11	TO_PD	sample rate changing detection timeout interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit. If CNT overflow (0x9c00), if would cause TIMEOUT irq.	RW	0x0
10	SRC_PD	sample rate changing detection interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit.	RW	0x0
9	-	Reserved for future use	R	0x0
8	CHW_PD	channel width change interrupt pending 0:no irq pending 1:irq pending Write '1' to clean this bit.	RW	0x0
7:3	-	Reserved	R	0x0
2:0	WL	Channel width length (the rate of BCLK to LRCLK)： 000： 32 rate(I2S: 16*2bit) 001： 64 rate(I2S: 32*2bit; TDM4: 16*4bit) 010：128 rate(TDM8: 16*8bit; TDM4: 32*4bit) 011：256 rate(TDM8: 32*8bit) 1xx：Reserved	R	0x0

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14 Display

14.1 LCD Controller (LCDC)

14.1.1 Features

- Support RGB888、RGB666、RGB565、ARGB Source Data Format
- 64*32 level FIFO with DRQ and IRQ
- Support MCU 8bit 8080/6800 interface
- Support (TFT) LCD panels with 3-line 1/2 data lane serial interface optionI/II interface
- Support (TFT) LCD panels with 4-line 1/2 data lane serial interface optionI/II interface
- Support (TFT) LCD panels with quad serial interface(SDR/DDR/SYNC)
- Support TR-J/TR-S parallel HSYNC/VSYNC LCD interface
- Support DMA for Display

14.1.2 MCU Interface

- MCU IF support Moto 6800 and Intel 8080

14.1.2.1 MCU Interface signal

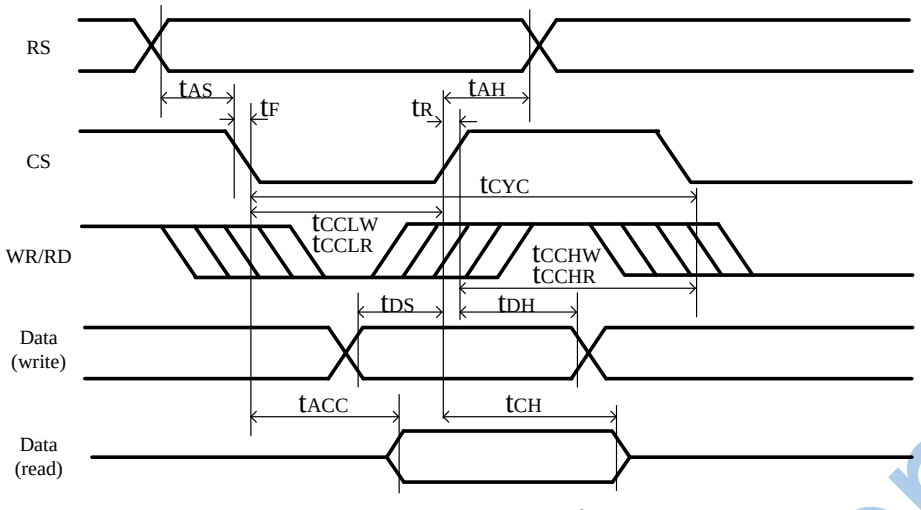
Table 14-1 MCU Interface signal

Signal	Type	Description
D7~D0	I/O	Data bus
E/RD#	O	M6800: Strobe signal I8080：Reads strobe signal
WR#	O	M6800: 1. Read，0. Write. I8080：Writes strobe signal
RS	O	Command/Data selection，0：Command，1：Data
RST#	O	Reset
CS1~0#	O	Chip select
TE	I	Tearing effect output pin to synchronize MCU to frame writing

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14.1.2.2 MCU Interface timing



RS	RD	WR	Function
0	0	1	Read Status
0	1	0	Set Index Register
1	0	1	Read Display Data
1	1	0	Write Display Data

Figure 14-1 I8080 interface control signal Timing

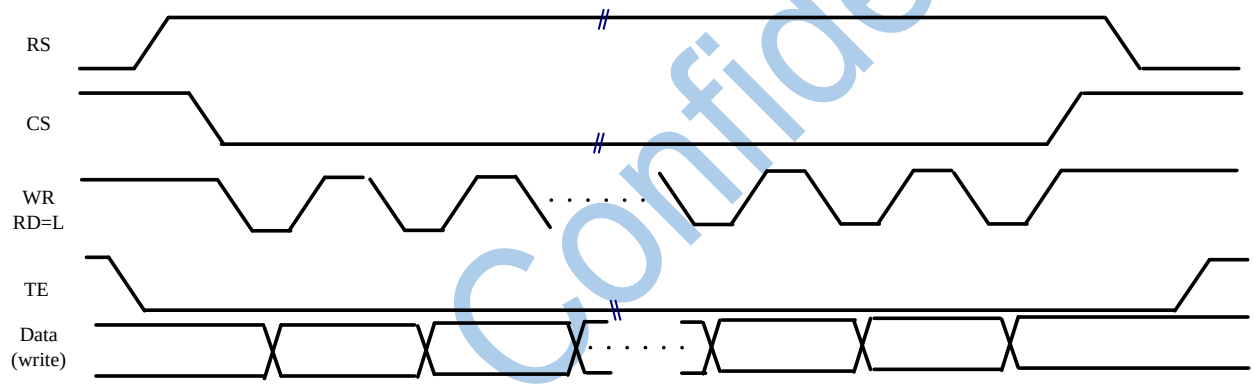
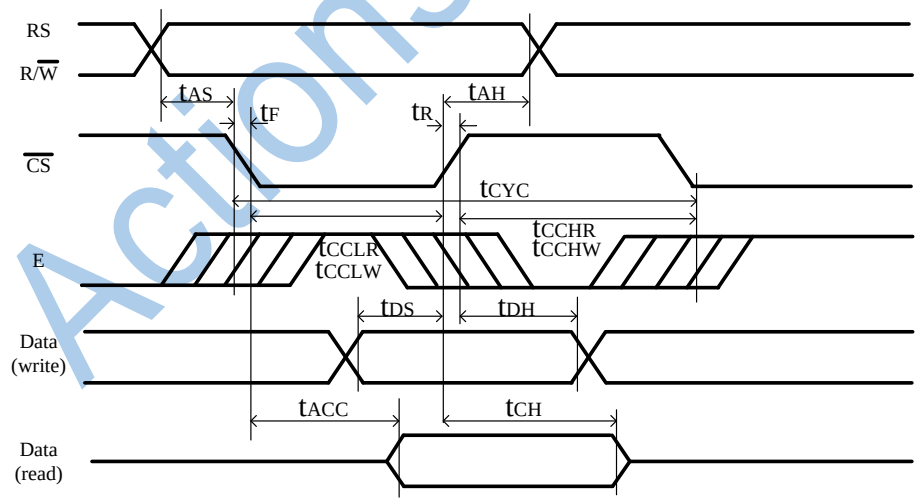


Figure 14-2 I8080 interface image write Timing



RS	R/$\overline{W}$	Function
0	1	Read Status
0	0	Set Index Register
1	0	Write Display Data
1	1	Read Display Data

Figure 14-3 M6800 interface control signal Timing

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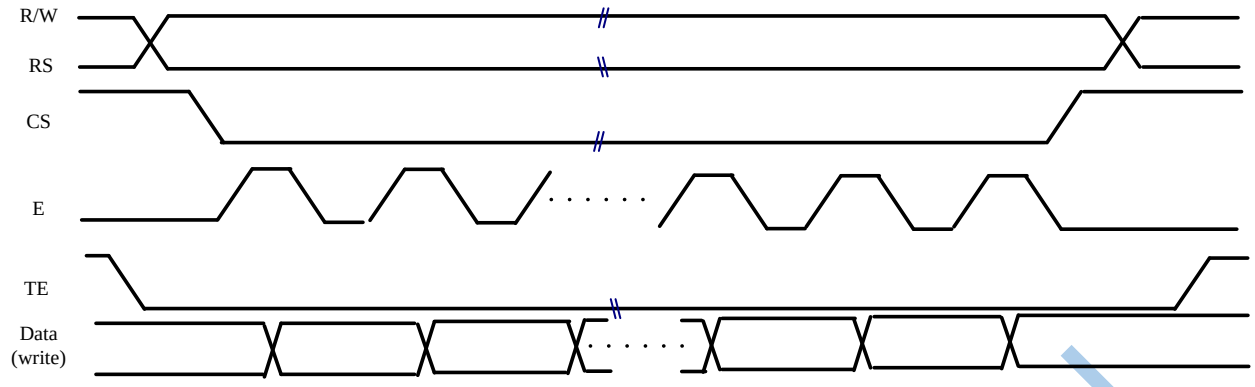


Figure 14-4 M6800 interface image write Timing

14.1.3 SPI Interface

3-lines and 4-line and Quad serial interfaces for communication between host and Lcd Driver are supported.
The Quad serial mode consists of the chip enable input (CSX), the serial clock input (SCL) and serial data Input/Output (SDI_RDX), input(DCX,DB[0],DB[1]).
The 3-line serial mode consists of the chip enable input (CSX), the serial clock input (SCL) and serial data Input (SDI_RDX) and serial data output (SDO).
The 4-line serial mode consists of the Data/ Command selection input (D/CX), chip enable input (CSX), the serial clock input (SCL) and serial data Input(SDI_RDX) and Output (SDO) for data transmission.

14.1.3.1 SPI Interface signal

Table 14-2 3-line serial interface option I

Signal	Type	Description
CSX	O	Chip Select Signal
SCL	O	Clock Signal
SDA	I/O	Data in/out
TE	I	Tearing effort

Table 14-3 3-line serial interface option II

Signal	Type	Description
CSX	O	Chip Select Signal
SCL	O	Clock Signal
SDA	O	Data Output
SDO	I	Data Input
TE	I	Tearing effort

Table 14-4 4-line serial interface option I

Signal	Type	Description
CSX	O	Chip Select Signal
SCL	O	Clock Signal
DCX/CDX	O	Data or Command
SDA	I/O	Data In/Out
TE	I	Tearing effort

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Table 14-5 4-line serial interface option II

Signal	Type	Description
CSX	O	Chip Select Signal
SCL	O	Clock Signal
DCX/CDX	O	Data or Command
SDA	O	Data Output
SDO	I	Data Input

Table 14-6 Quad serial interface

Signal	Type	Description
CSX	O	Chip Select Signal
WRX_SCL	O	CLK
SDA/SDI	I/O	Data Input/Output
SD[3:1]	O	Data Output

14.1.3.2 SPI Interface Command Write timing

3-Line Serial Interface

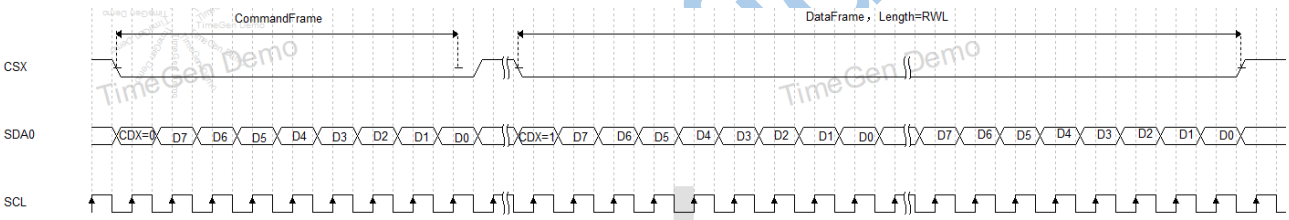


Figure 14-5 3-line serial interface command write

4-Line Serial Interface

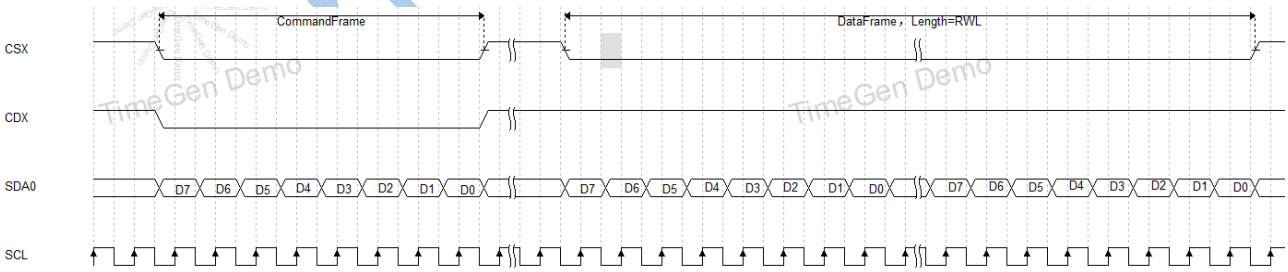


Figure 14-6 4-line serial interface command write

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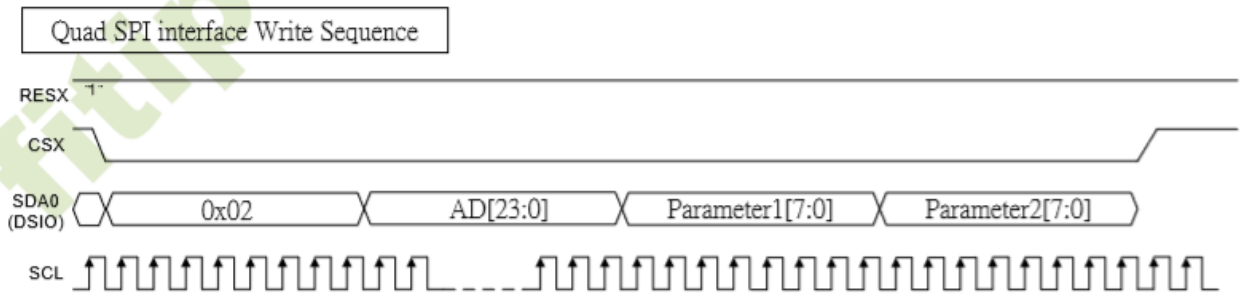


Figure 14-7 Quad serial interface command write

14.1.3.1 SPI Interface Read Timing

3-Line Serial Interface I/II

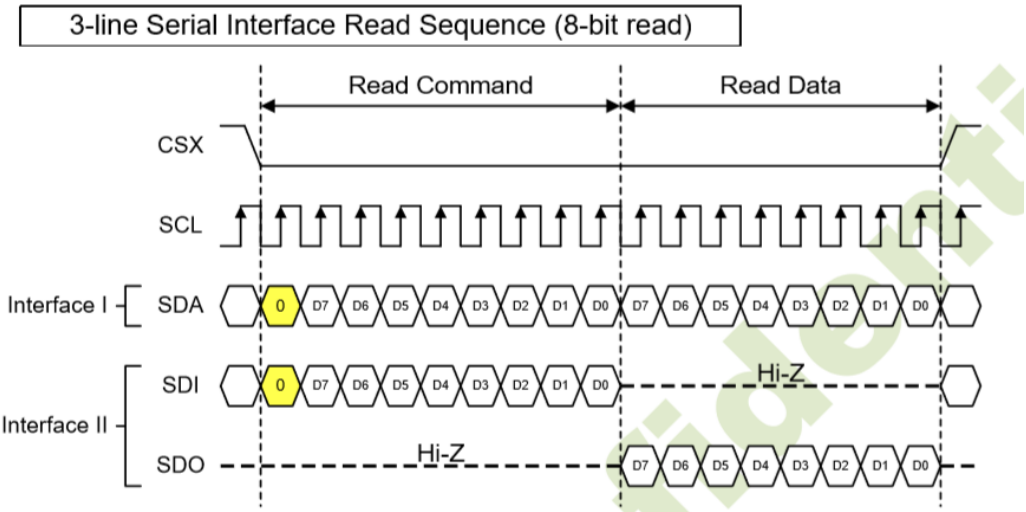


Figure 14-8 3-line serial read timing (8-bit read)

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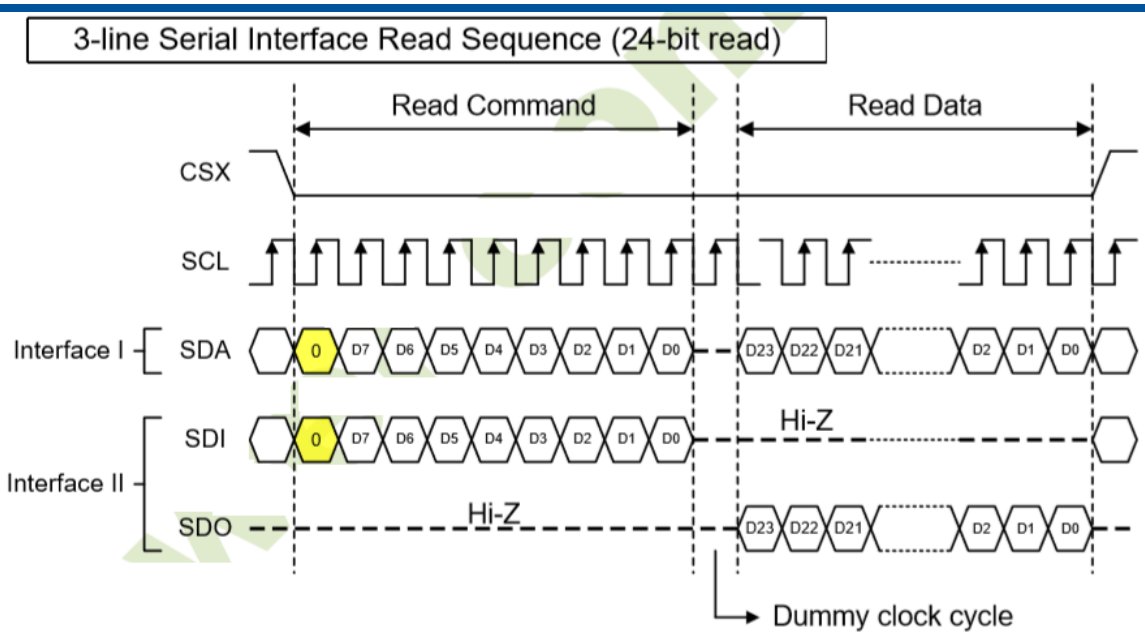


Figure 14-9 3-line serial read timing (24-bit read)

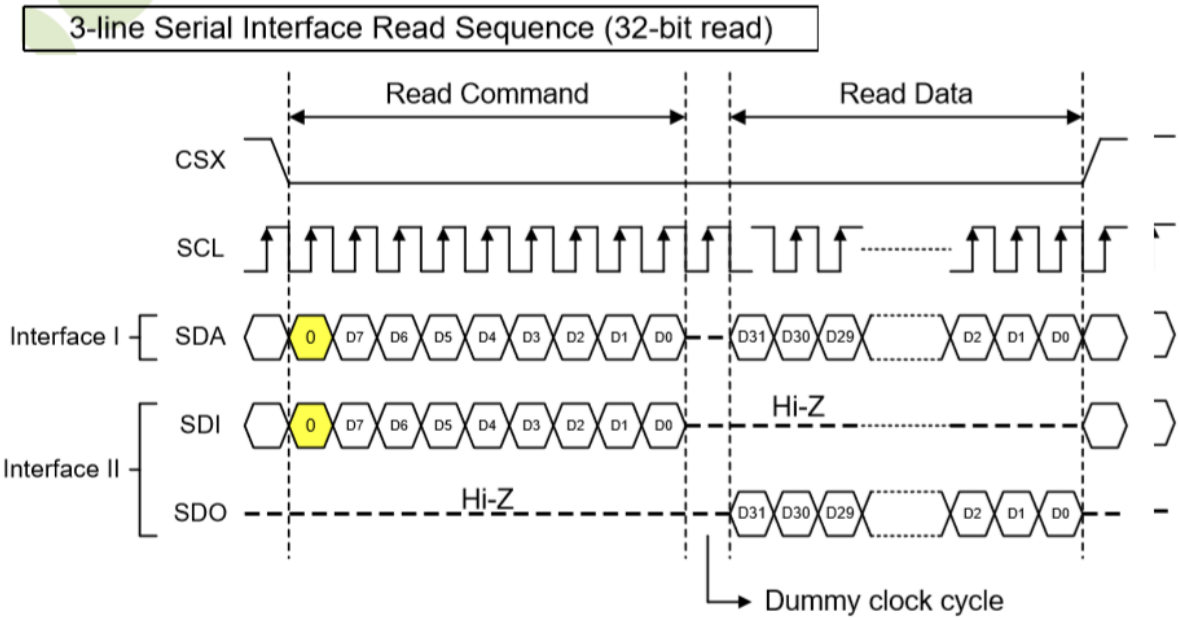


Figure 14-10 3-line serial read timing (32-bit read)

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4-Line Serial Interface I/II

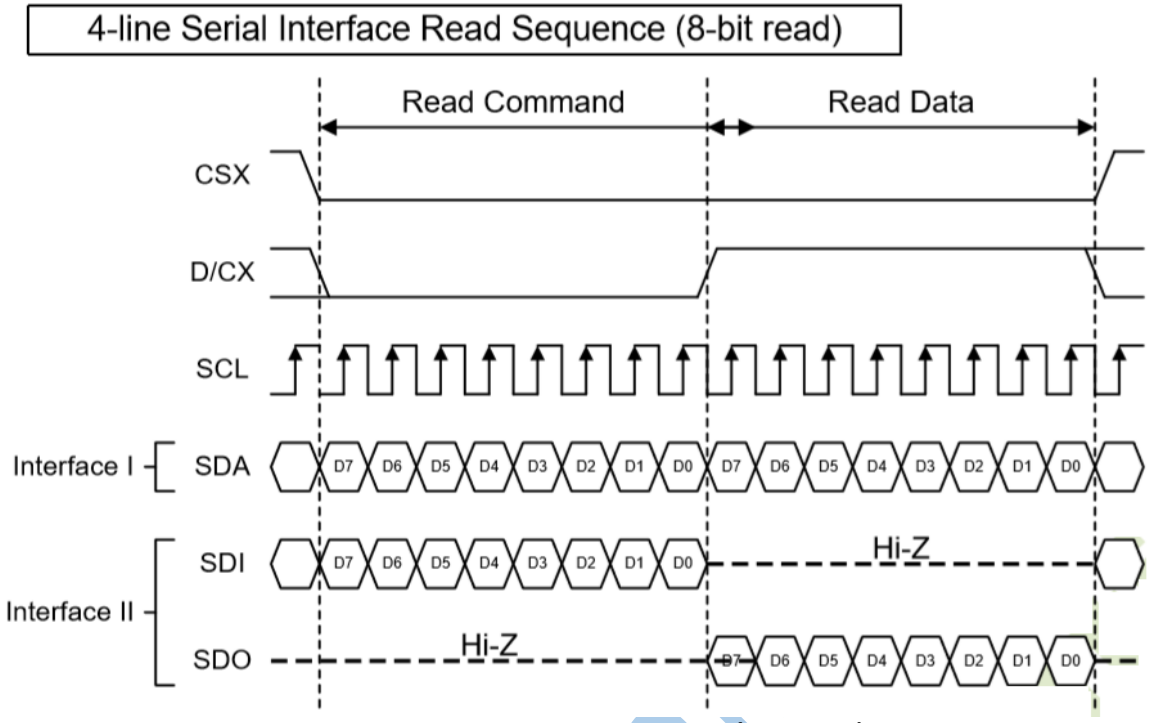


Figure 14-11 4-line serial read timing (8-bit read)

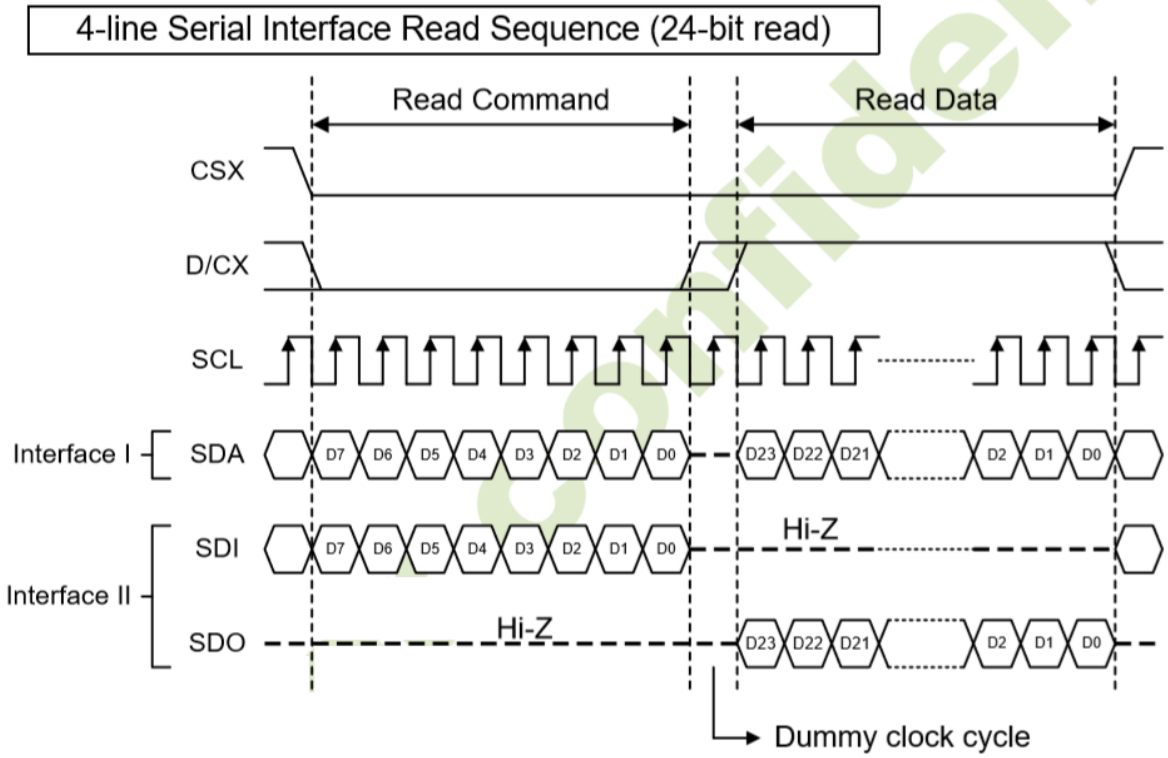


Figure 14-12 4-line serial read timing (24-bit read)

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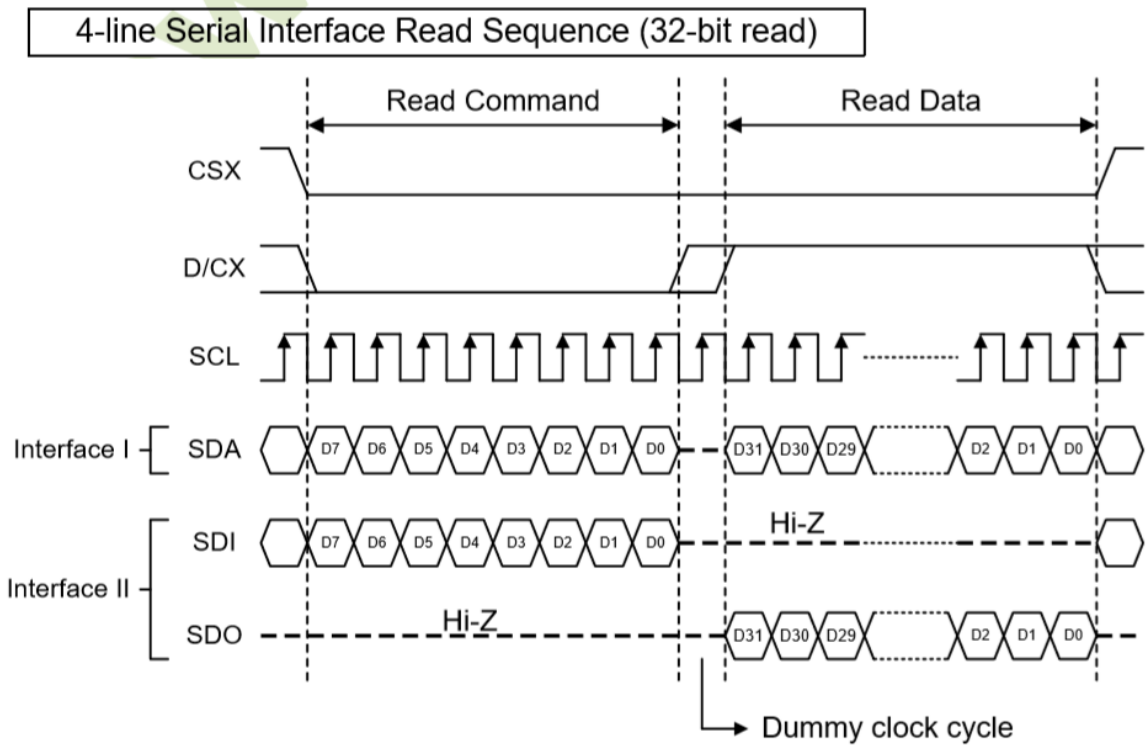


Figure 14-13 4-line serial read timing (for RDDST command: 32-bit read)

Quad Serial Interface

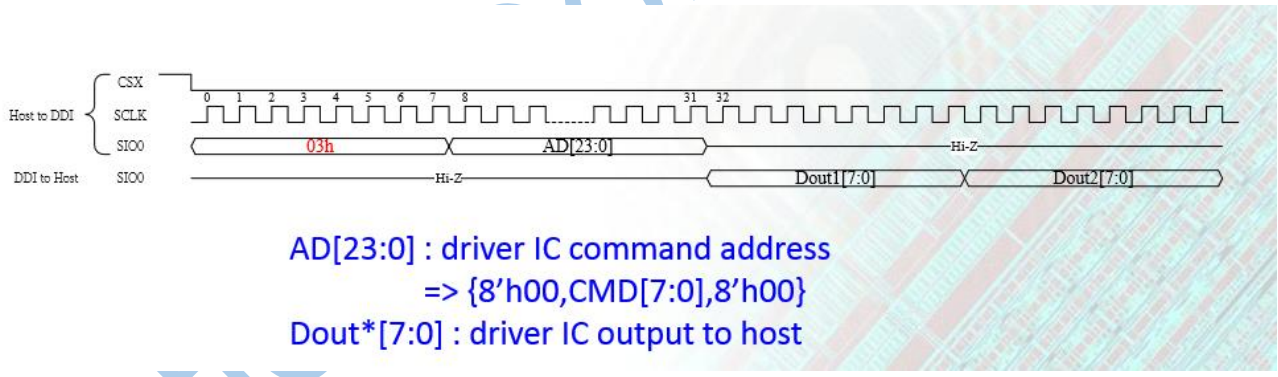


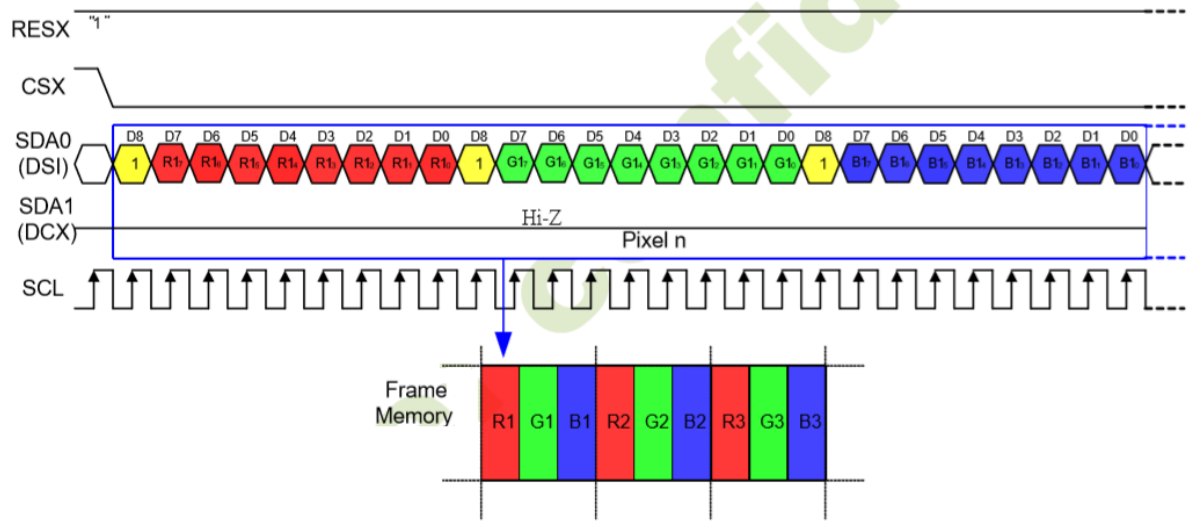
Figure 14-14 QUAD Serial Interface read timing

14.1.3.2 SPI Interface Image Write timing

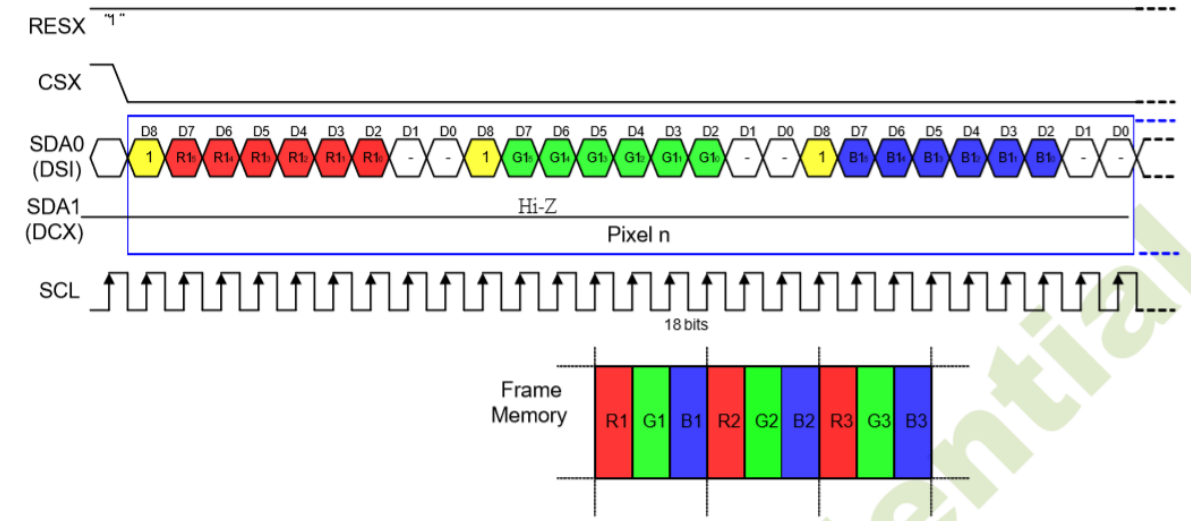
3-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

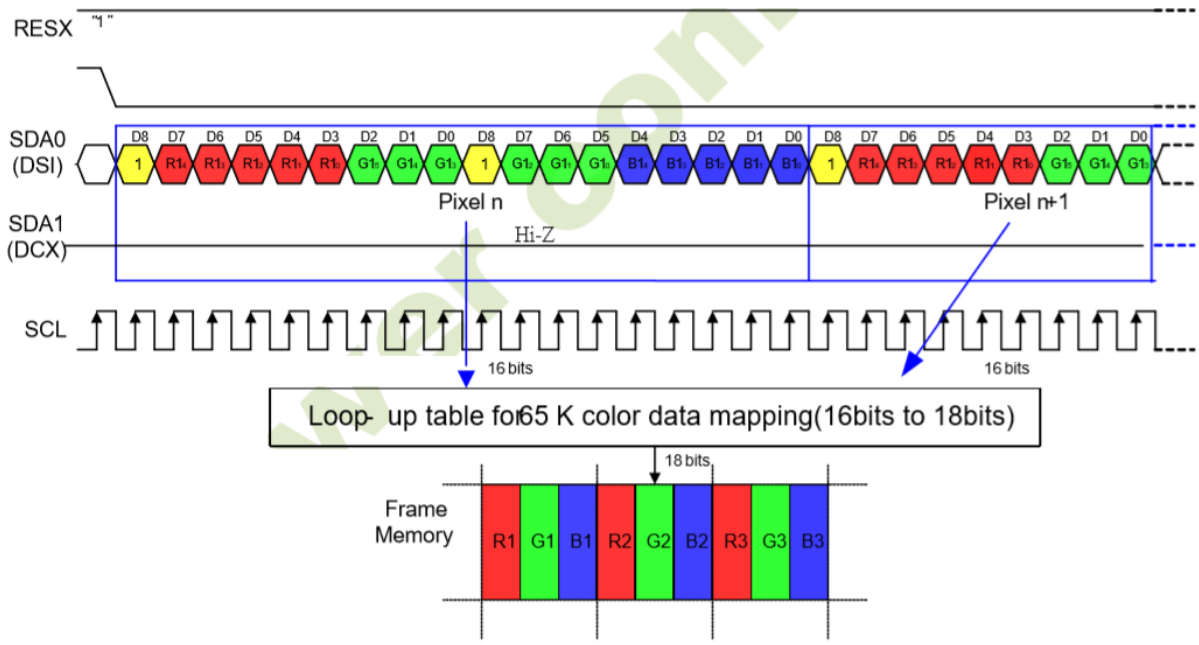
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(2)RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

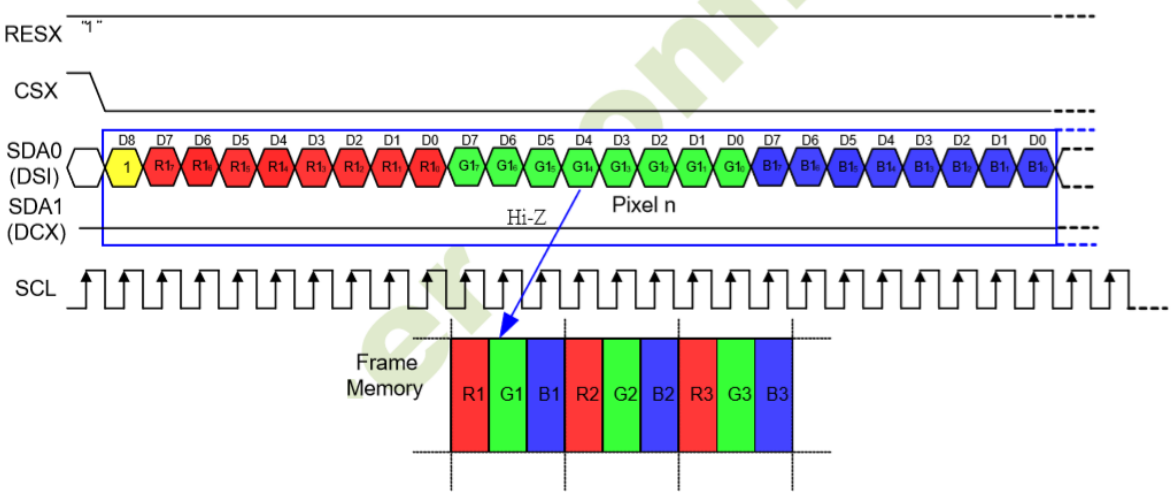


(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

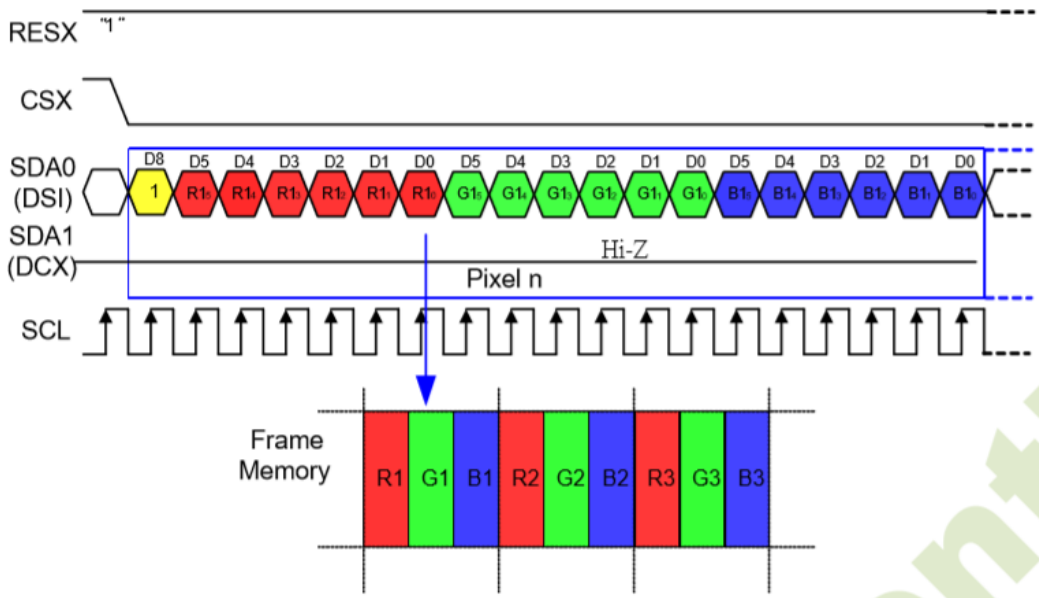


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(4) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

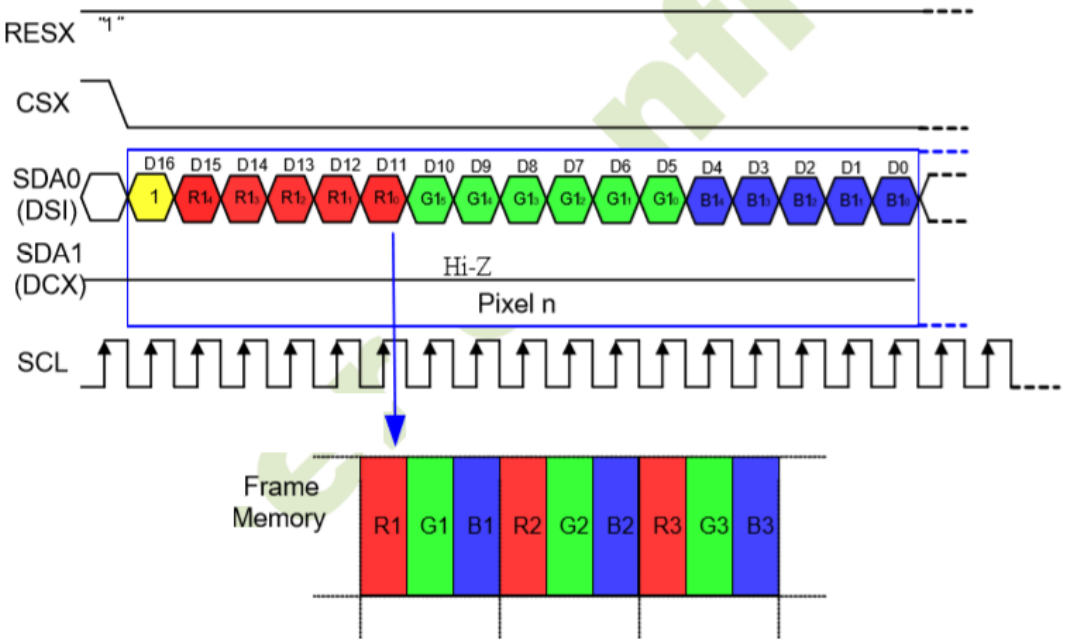


(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1



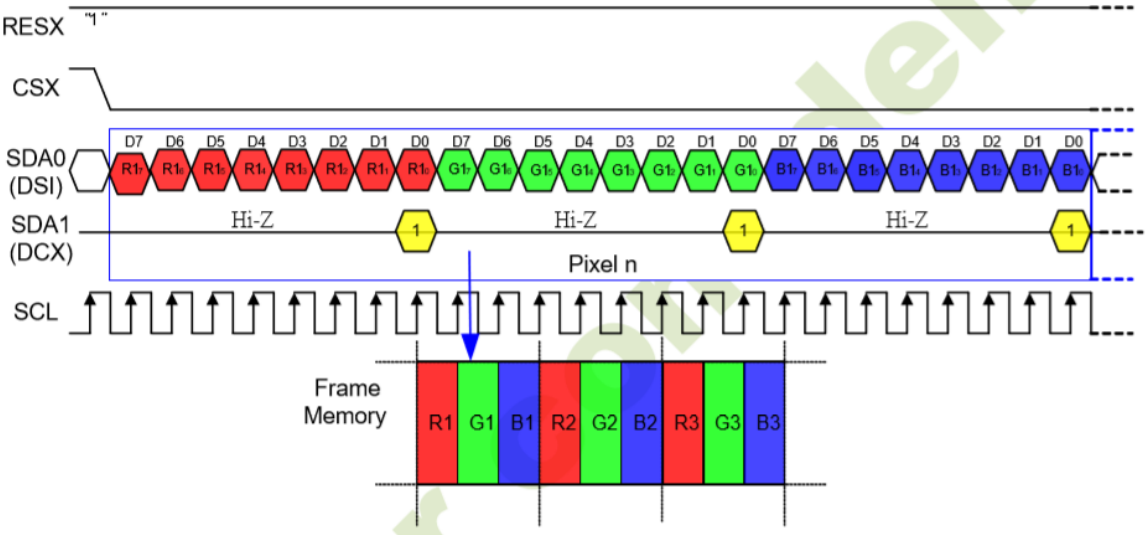
(6) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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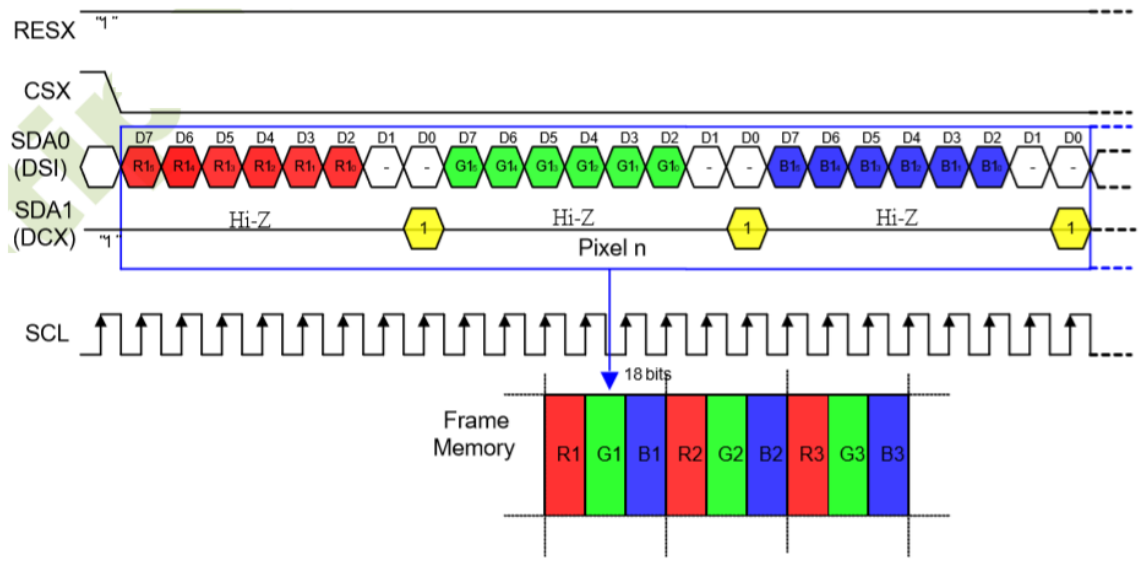
4-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1)RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

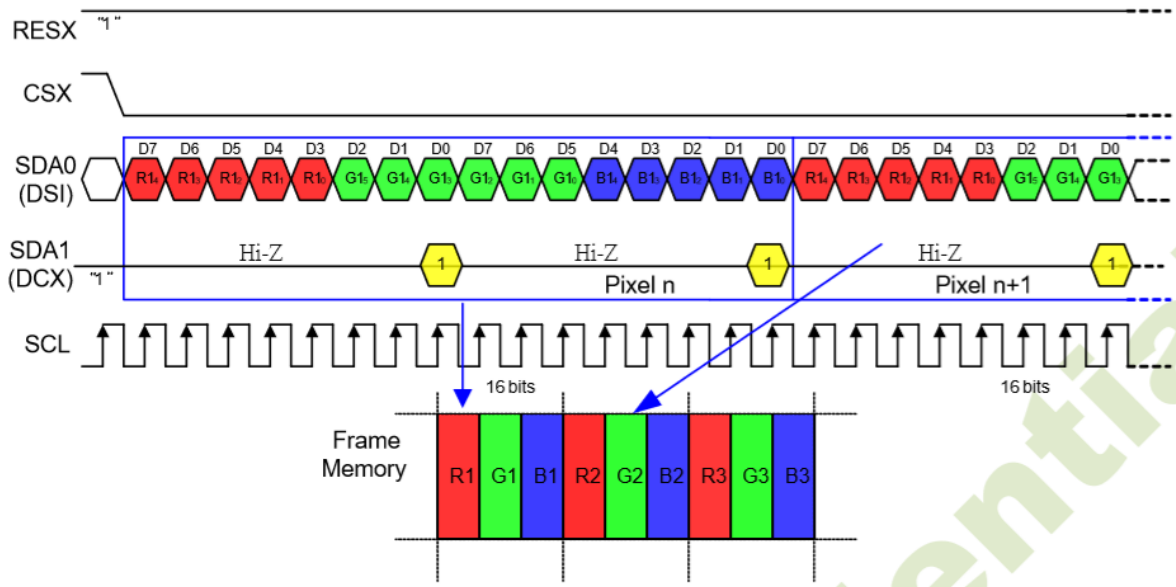


(2) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0

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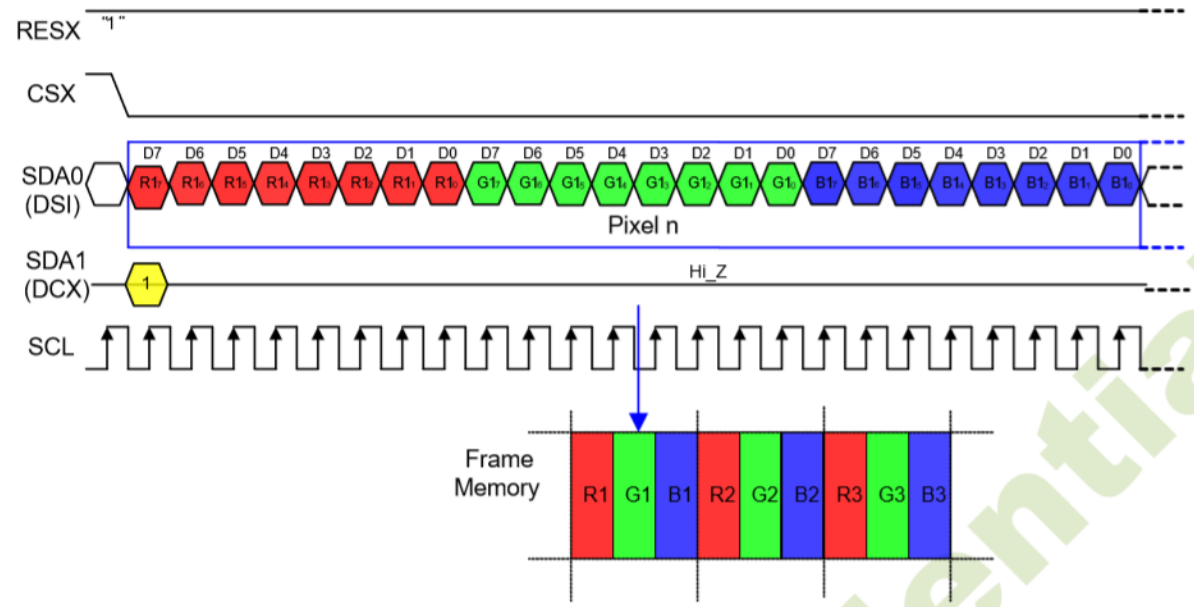
(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=0



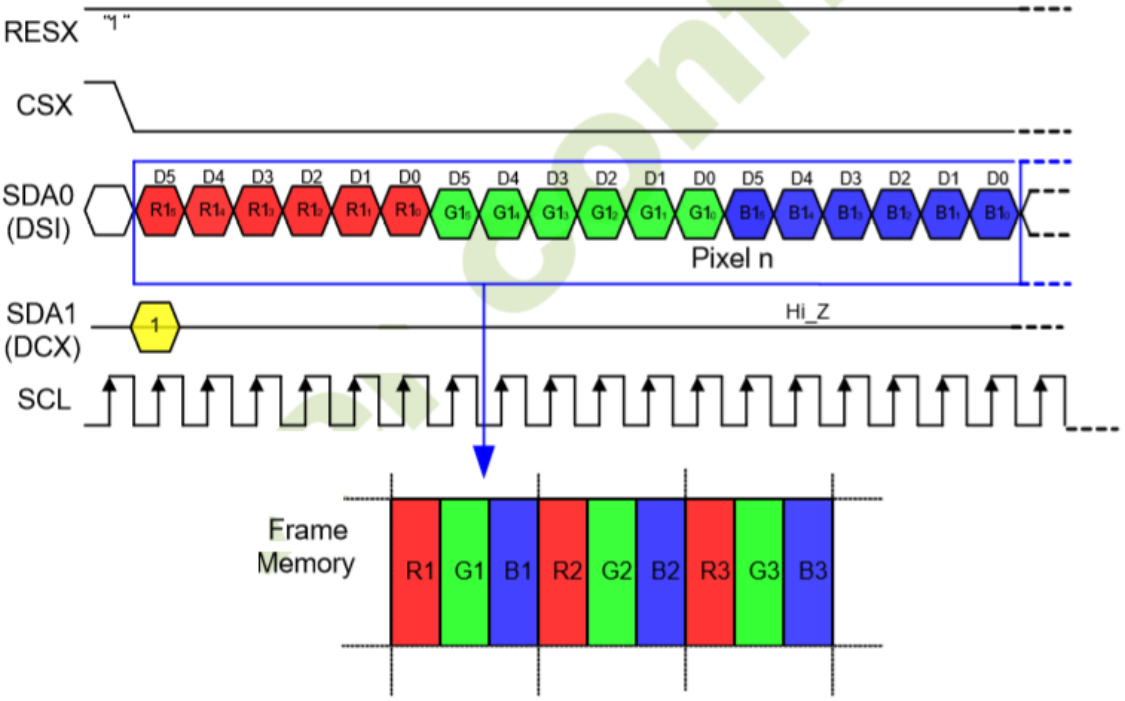
(4) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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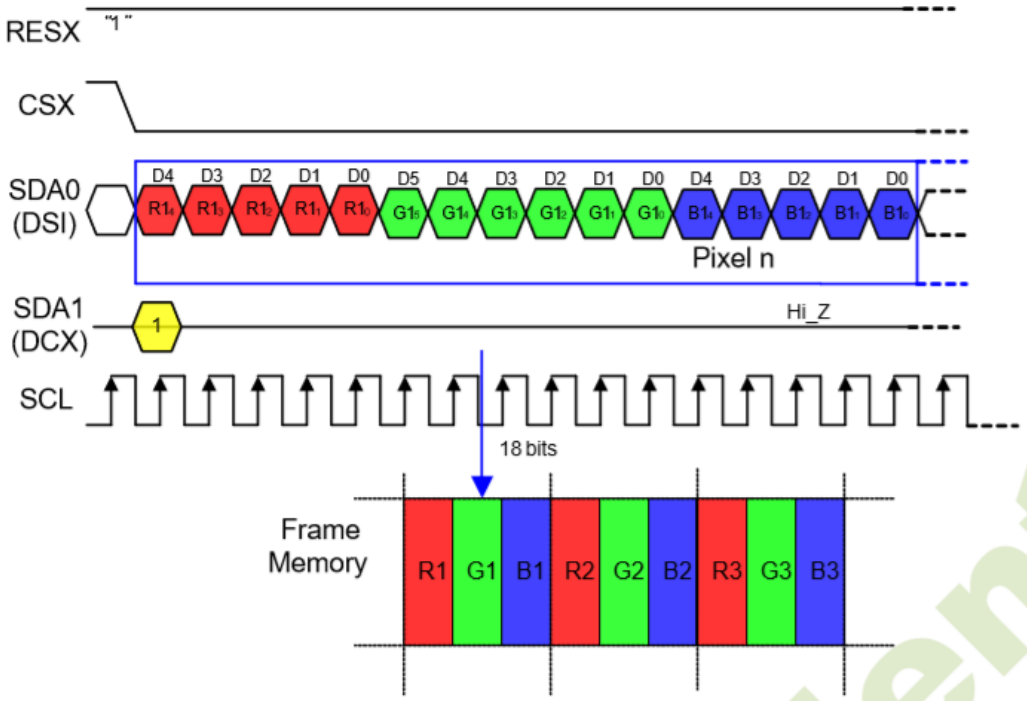


(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1



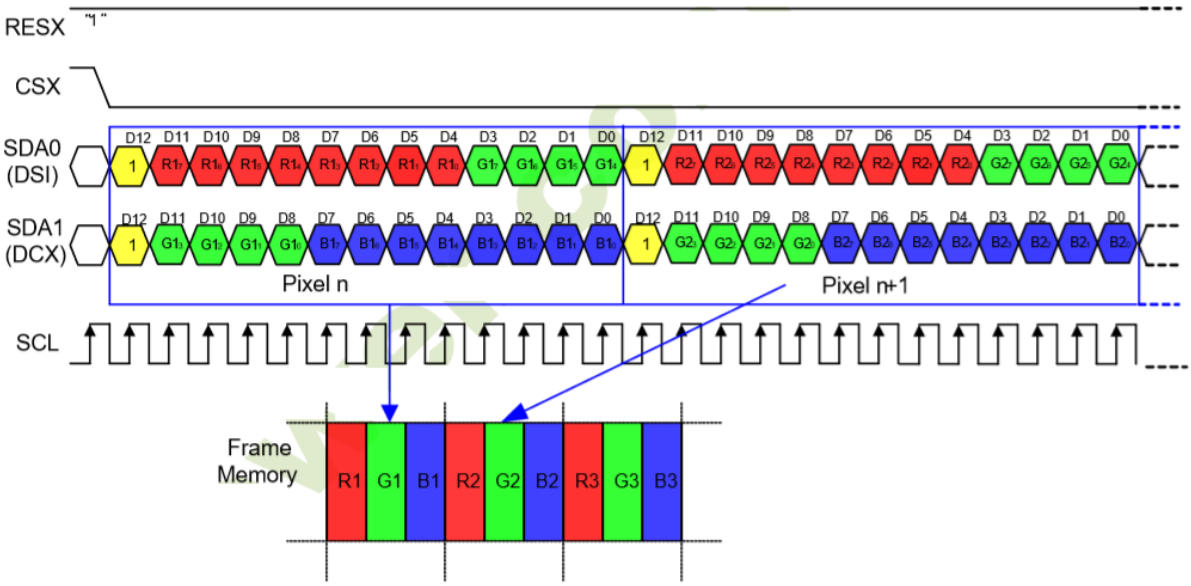
(6) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=1

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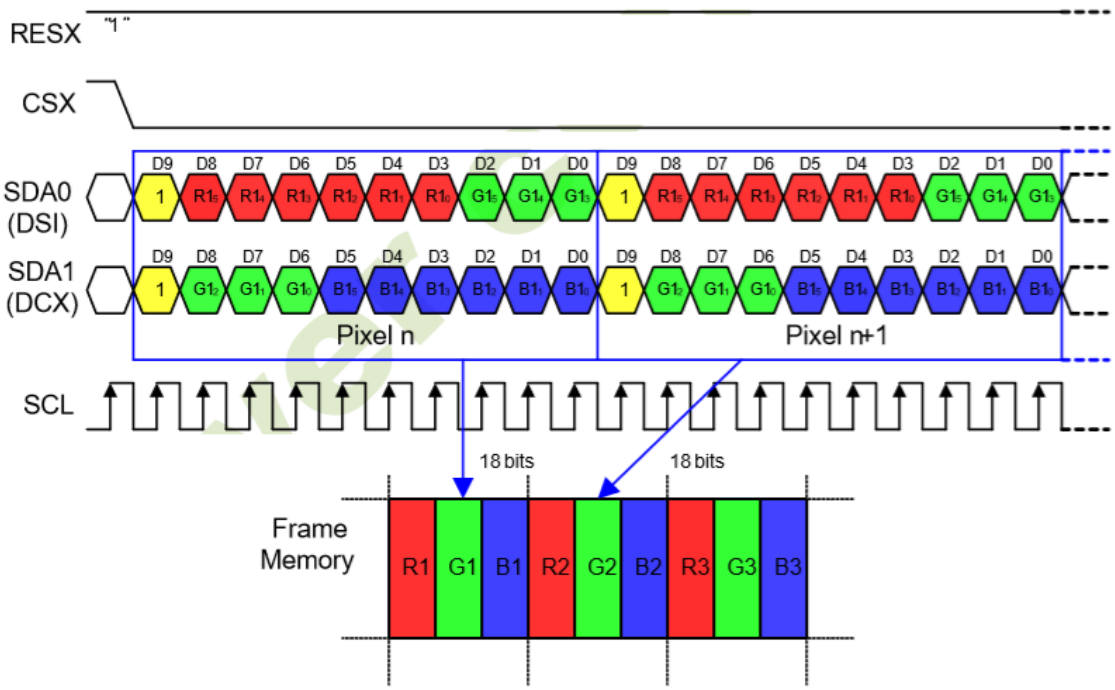
3/4-line Dual Lane Serial Interface

(1)RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE= don’t care

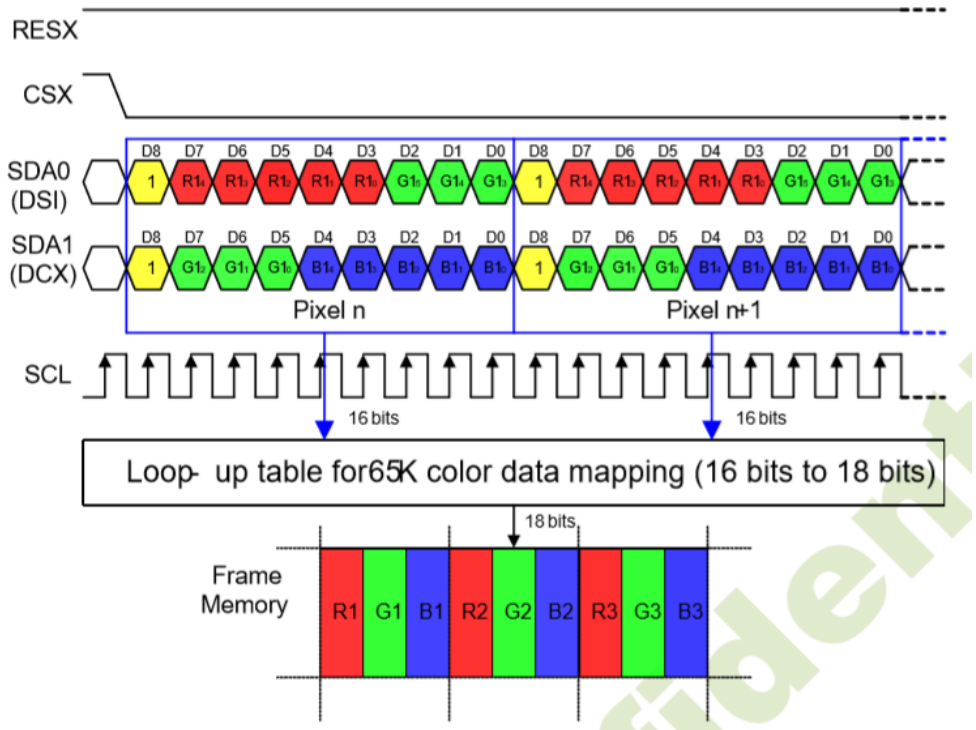


(2) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=don’t care

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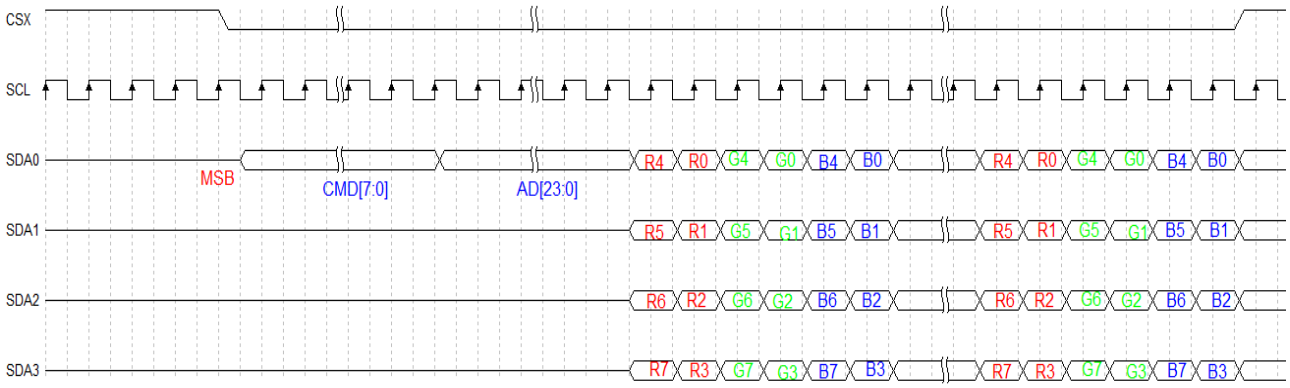
(3) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=don't care



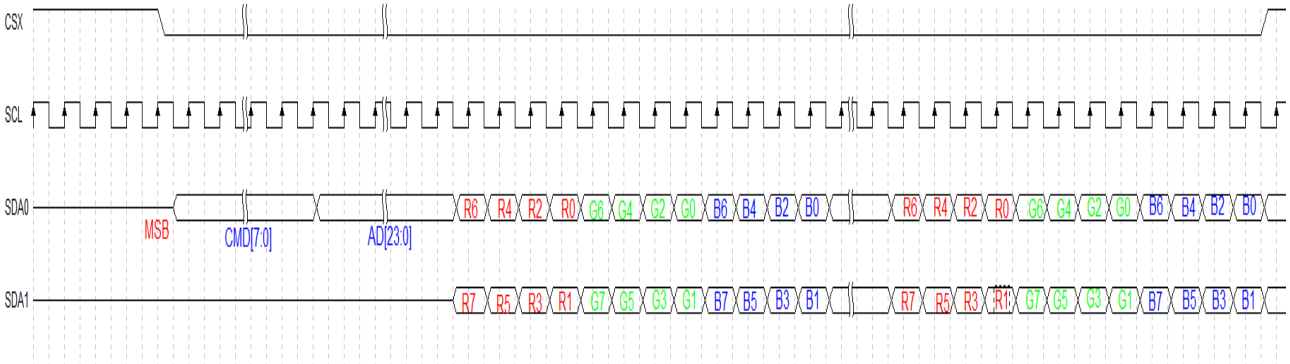
Quad-line Serial Interface

Different display data formats are available for three colors depth supported by the LCM listed below:
(1) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0

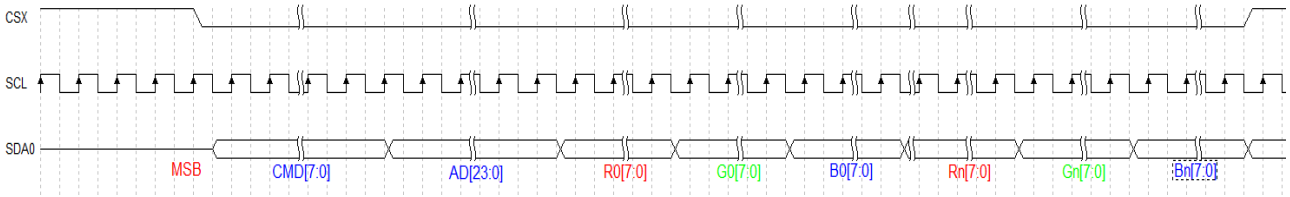
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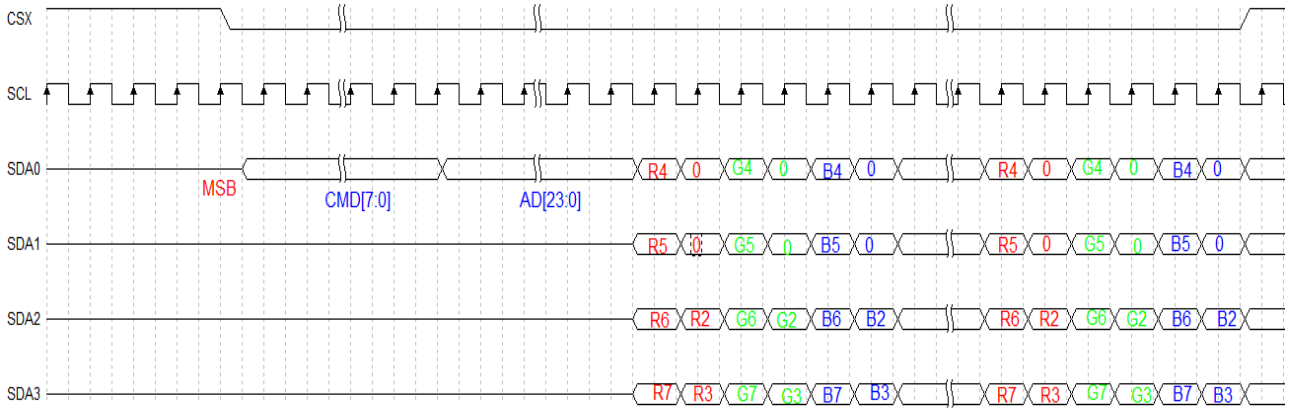
(2) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0



(3) RGB888；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don’t care；DDR_EN=0

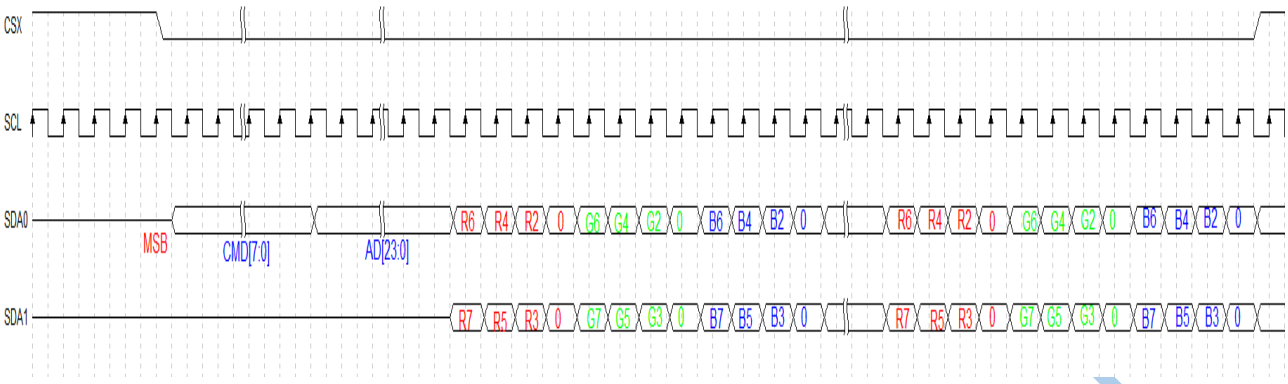


(4) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0

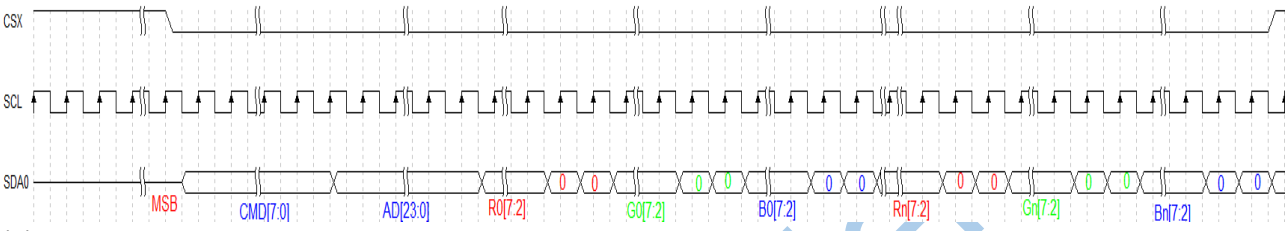


(5) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0

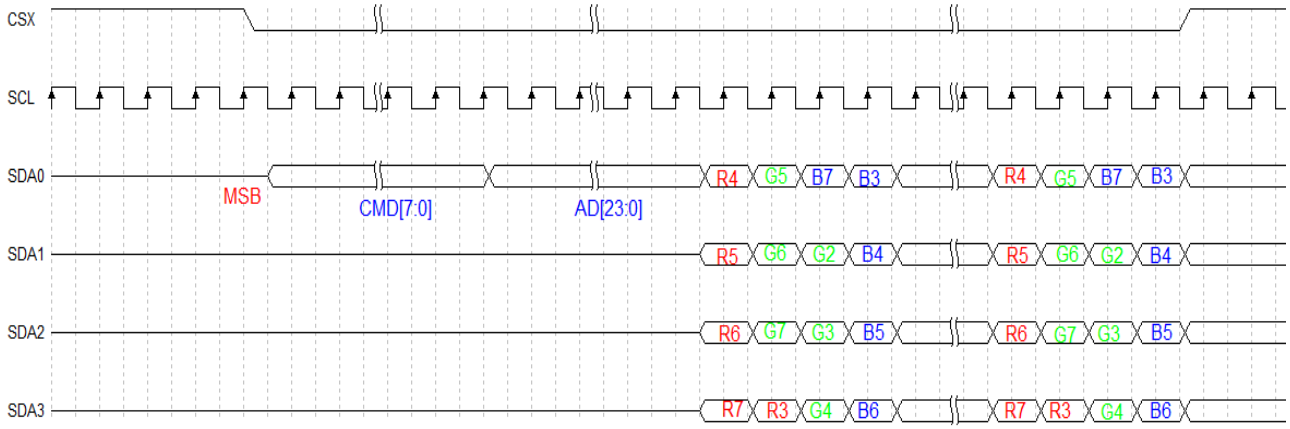
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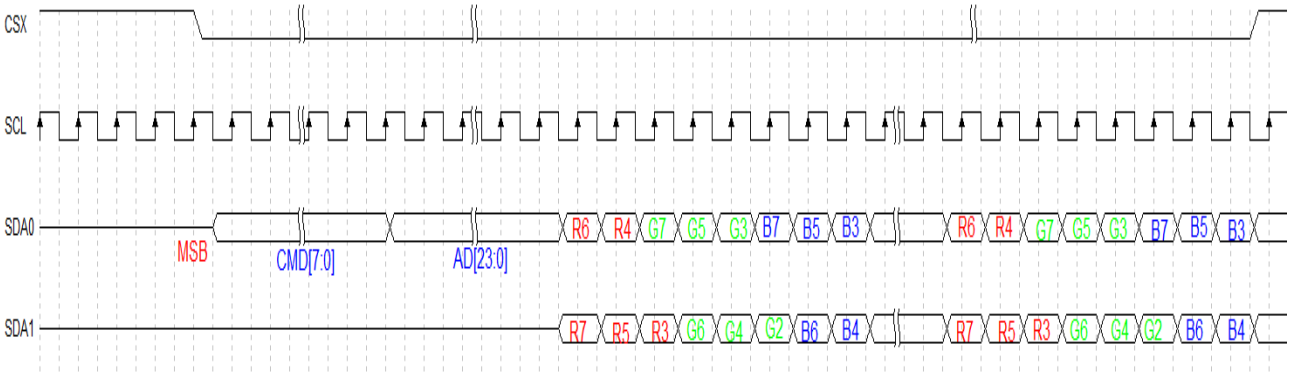
(6) RGB666；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don't care；DDR_EN=0



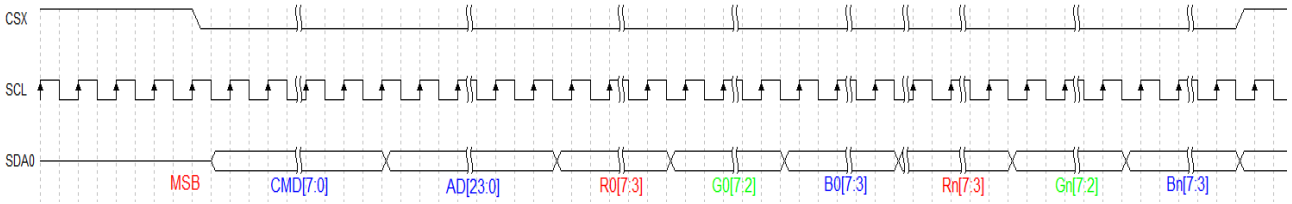
(7) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=0；DDR_EN=0



(8) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=1；DCP_MODE=1；DDR_EN=0



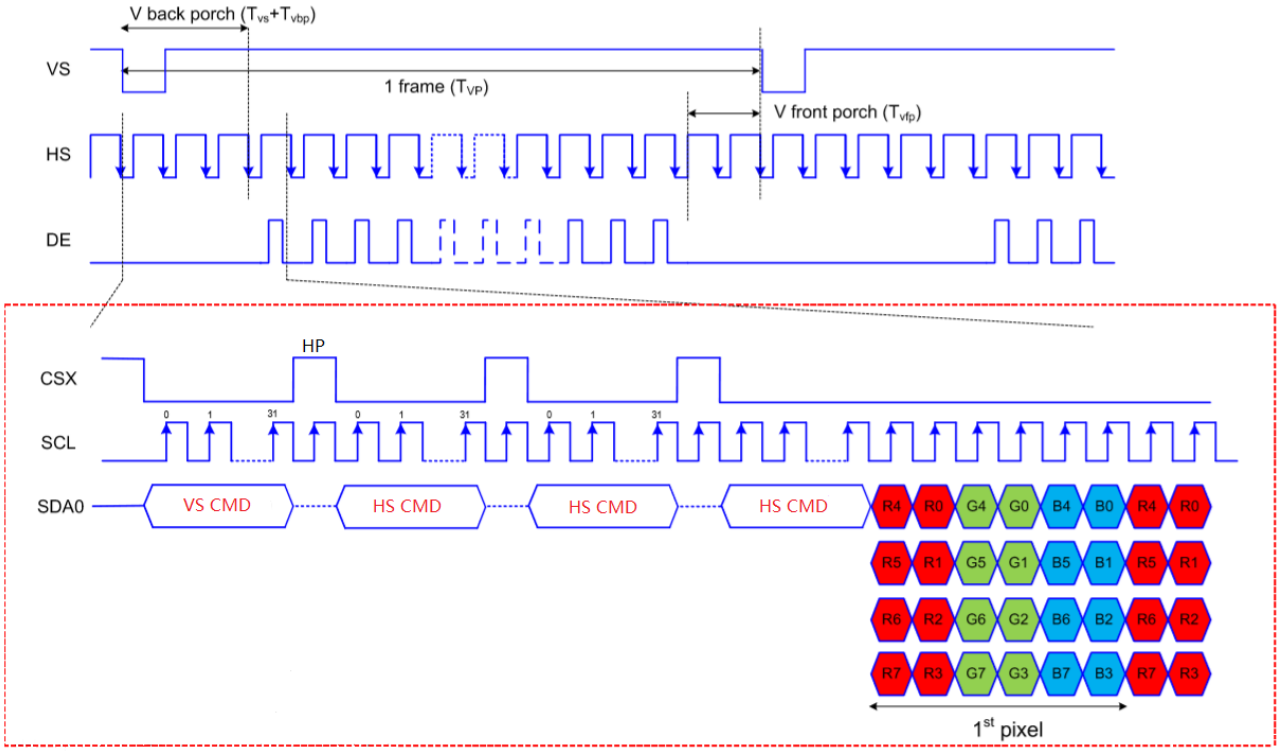
(9) RGB565；CDX=1；AHB_CFG=0；DUAL_LANE=0；DCP_MODE=don't care；DDR_EN=0



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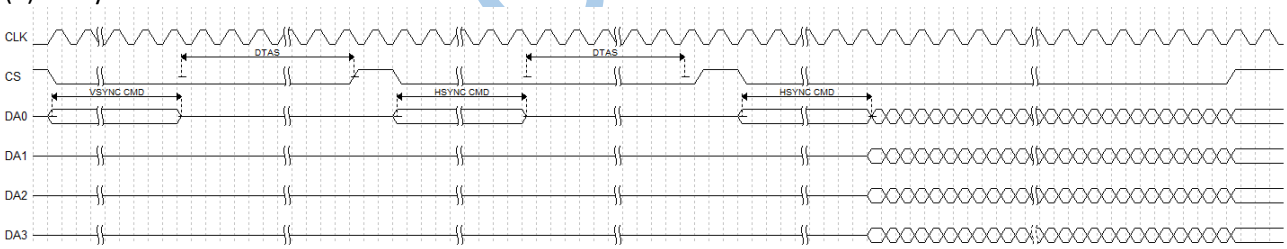
Sync Mode

(1)Sync Timing



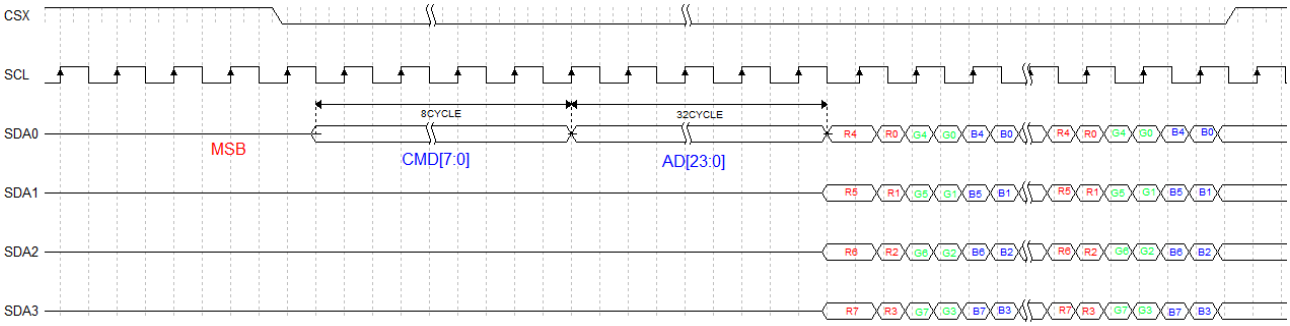
NOTE: Sample for the RGB888,and the same timing with 565/666 format. And Sync Mode support all the quad-line data mode.

(2) Delay Time After SYNC



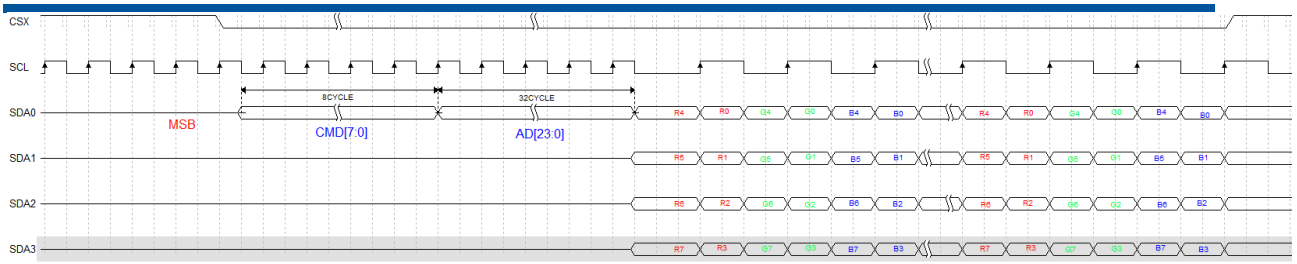
QSPI DDR Mode

(1) RGB888； CDX=1； AHB_CFG=0； DDR_EN=1； DDR_TYPE=0

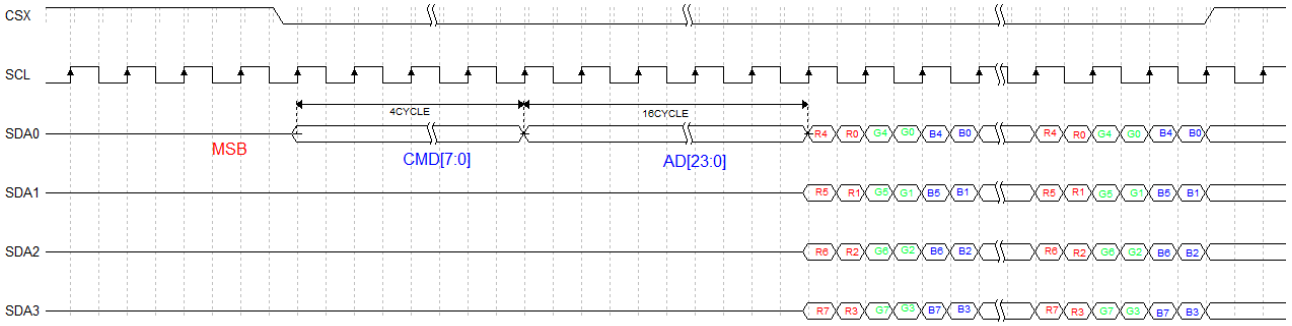


(2)RGB888； CDX=1； AHB_CFG=0； DDR_EN=1； DDR_ TYPE =1 (not support)

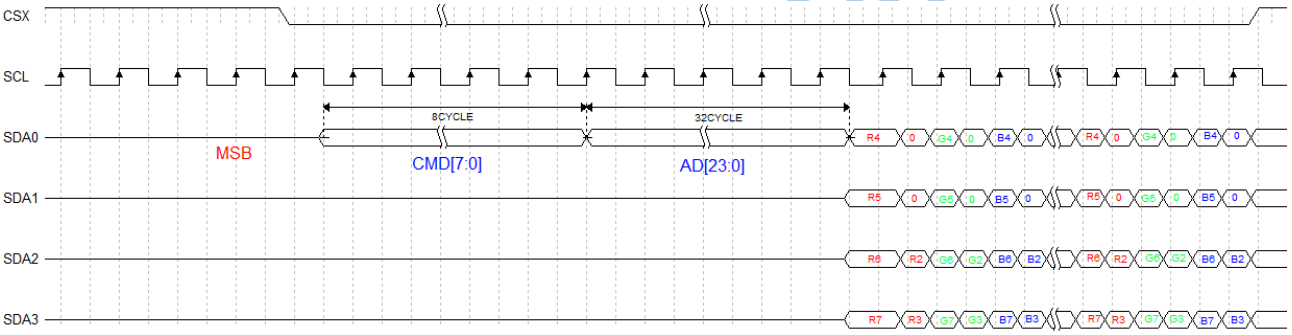
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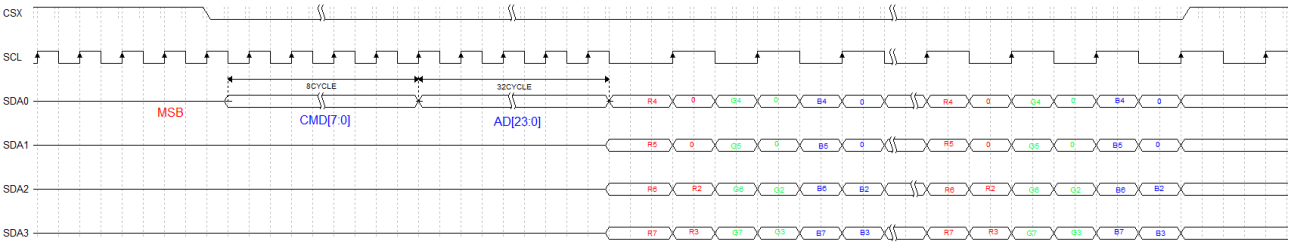
(3)RGB888；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2



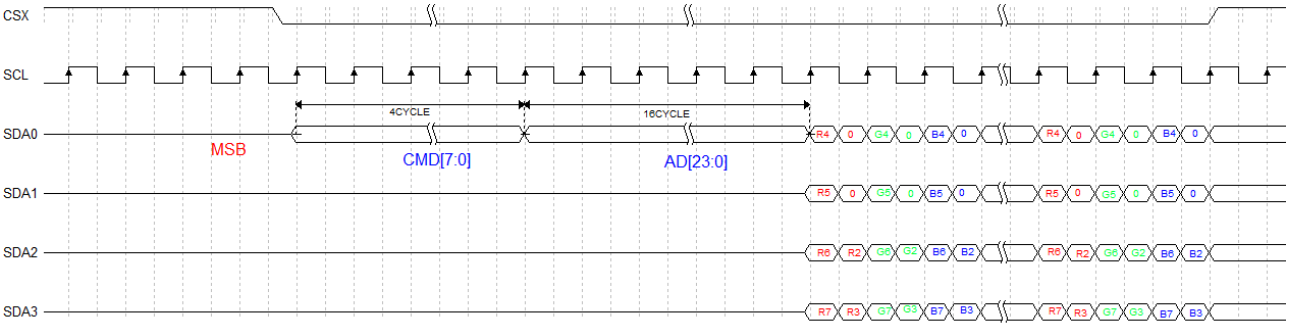
(4)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =0



(5)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =1

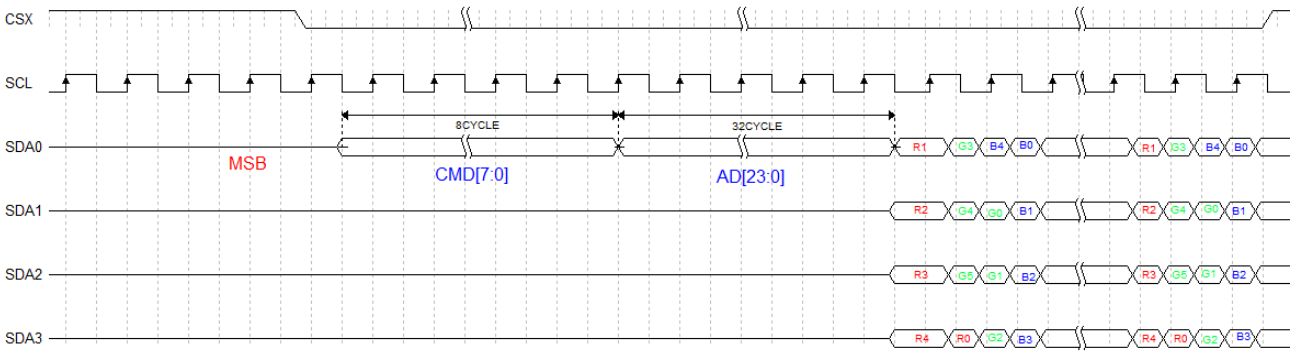


(6)RGB666；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2

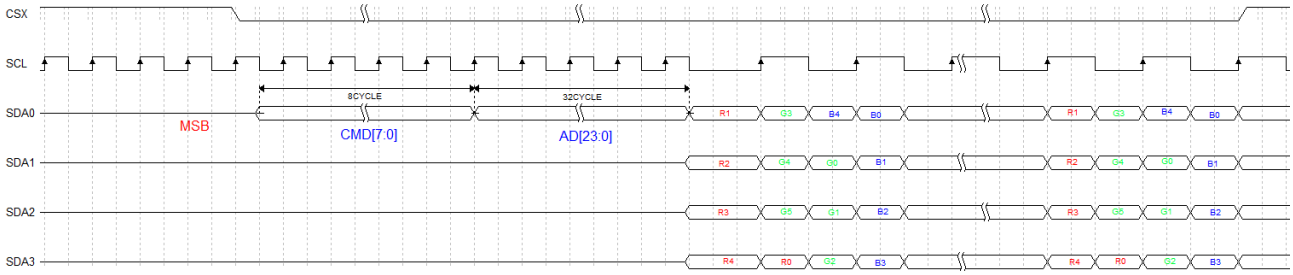


(7)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =0

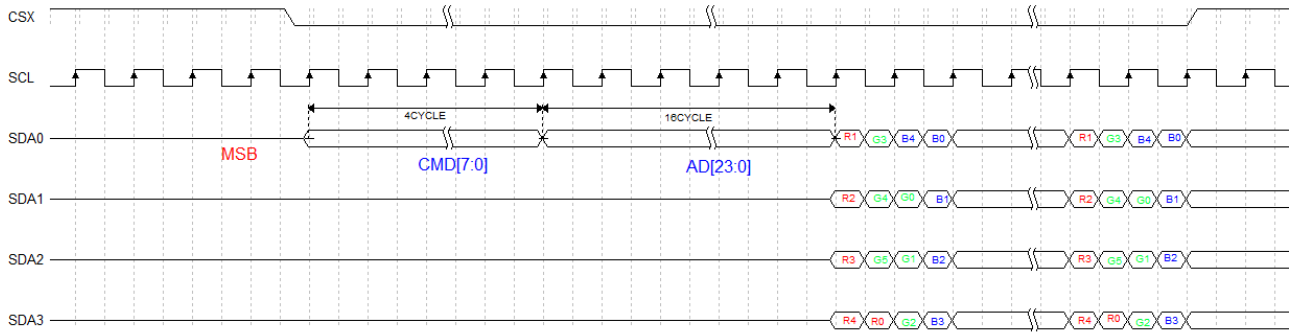
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(8)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =1



(9)RGB565；CDX=1；AHB_CFG=0；DDR_EN=1；DDR_ TYPE =2



14.1.4 TR-LCD Parallel Interface

Support TR-J/TR-S Parallel HSYNC/VSYNC LCD interface.

14.1.4.1 TR-LCD Interface signal

Table 14-7 TR-J interface signal

Signal	Type	Description
B0	O	Blue signal for odd Pixels
B1	O	Blue signal for even Pixels
G0	O	Green signal for odd Pixels
G1	O	Green signal for even Pixels
R0	O	Red signal for odd Pixels
R1	O	Red signal for even Pixels
HST	O	Horizontal Driver Start Pulse
XFRP	O	White Signal
VCOM/ FRP	O	VCOM AC Input/ Black signal
VCK	O	Vertical Driver Clock Pulse

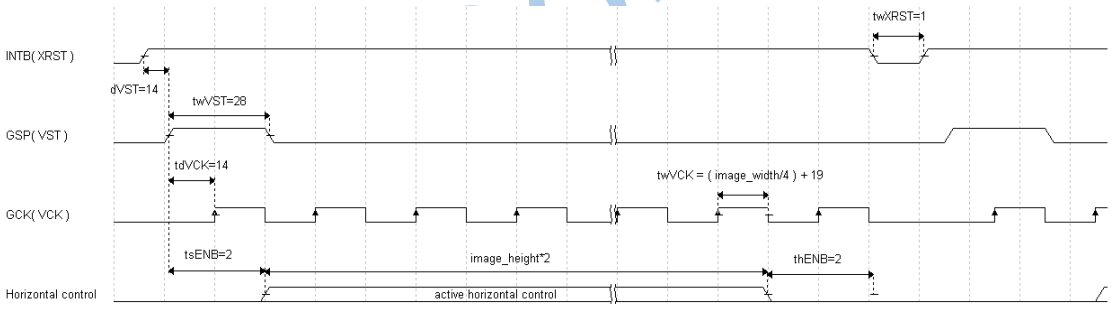
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XRST	O	Reset Pulse
VST	O	Vertical Driver Start Pulse
ENB	O	Gate Enable Pulse
HCK	O	Horizontal Driver Clock Pulse

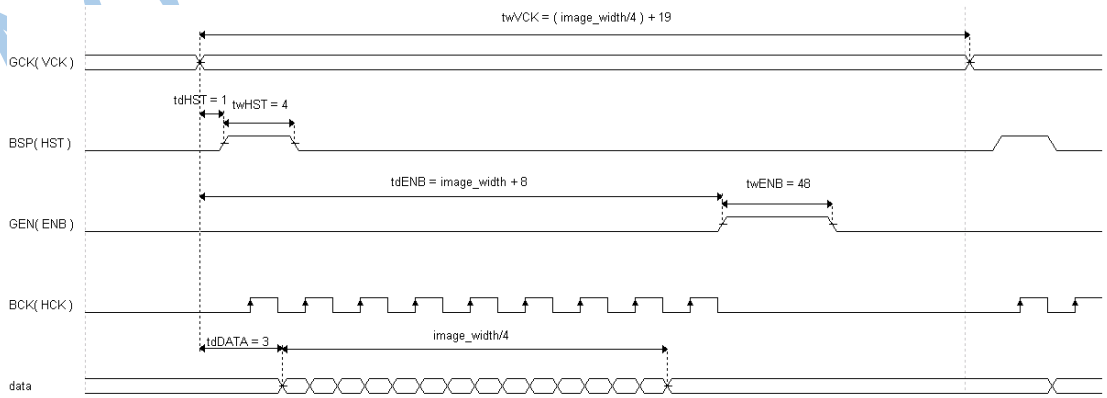
Table 14-8 TR-S interface signal

Signal	Type	Description
B0	O	Blue signal for odd Pixels
B1	O	Blue signal for even Pixels
G0	O	Green signal for odd Pixels
G1	O	Green signal for even Pixels
R0	O	Red signal for odd Pixels
R1	O	Red signal for even Pixels
BSP	O	Start signal for Binary-Driver
VA	O	White signal voltage of LCD
VCOM/VB	O	Common terminal voltage for LCD/ Black signal voltage of LCD
GCK	O	Clock for signal for the Gate-Driver
INTB	O	Initial signal for Binary/Gate-Driver
GSP	O	Start signal for the Gate-Driver
GEN	O	Gate enable signal
BCK	O	Clock for driving of Binary-Driver

14.1.4.2 Veritical



14.1.4.3 Horizontal



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14.1.5 LCD Pin Mapping

The LCD Pin Mapping of different interface is as follows.

Table 14-9 LCD Pin Mapping

GPIO	CPU8080 /6800	3-LineI	3-LineII	4-LineI	4-LineII	QSPI	TR-S	TR-J
GPIO33	E/RD#		SDO		SDO		GEN	ENB
GPIO34	WR#	SCLK	SCLK	SCLK	SCLK	SCLK	BCK	HCK
GPIO32	RS						GCK	VCK
GPIO30	CS0	CS0	CS0	CS0	CS0	CS0	BSP	HST
GPIO35	TE	TE	TE	TE	TE	TE	GSP	VST
GPIO14	D0	SDA0/SDO	SDA0	SDA0/SDO	SDA0	SDA0	R0	R0
GPIO15	D1			SDA1/DCX	SDA1/DCX	SDA1	R1	R1
GPIO16	D2					SDA2	G0	G0
GPIO17	D3					SDA3	G1	G1
GPIO18 /GPIO24	D4						B0	B0
GPIO19 /GPIO25	D5						B1	B1
GPIO20 /GPIO26	D6						VCOM_VB	VCOM_FRP
GPIO21 /GPIO27	D7						VA	XFRP
							INTB	XRST

14.1.6 LCD Register List

Table 14-10 LCD Controller Registers base address

Name	Physical Base Address	KSEG1 Base Address
LCDC_REGISTER	0x40064000	0x40064000

Table 14-11 LCD Controller Registers

Offset	Register Name	Description
0x0000	LCD_CTL	LCD Control Register
0x0008	LCD_DISP_SIZE	LCD DMA Source Display Size Register
0x001C	LCD_CPU_CTL	LCD CPU Mode Control Register
0x0020	LCD_DATA	LCD Data Register
0x0024	LCD_CPU_CLK	LCD CPU Mode Clock Control Register
0x0028	LCD_TPL	LCD Transfer Pixel Length Register
0x002C	LCD_SPI_CTL	LCD Serial Interface Control Register
0x0034	LCD_SPI_CMD	LCD Quad Serial Interface Command and Address register
0x0038	LCD_QSPI_CMD1	LCD Quad Serial Interface Command and Address register
0x003C	LCD_QSPI_SYNC_TIM	LCD QSPI SYNC Mode Time Register
0x0040	LCD_QSPI_IMG_SIZE	LCD QSPI SYNC Mode Display image size Register
0x0044	2DGPU_INTERFACE_CTL	2DGPU Interface Control Register

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0x0048	LCD_PENDING	LCD Quad Serial Interface Command and Address register
0x004c	LCD_QSPI_DTAS	LCD QUAD Serial Interface Delay Time after sync
0x0050	LCD_DATA1	LCD Data1 Register
0x0054	LCD_DATA2	LCD Data2 Register
0x0058	LCD_DATA3	LCD Data3 Register
0x005C	LCD_DATA4	LCD Data4 Register
0x0060	LCD_DATA5	LCD Data5 Register
0x0064	LCD_DATA6	LCD Data6 Register
0x0068	LCD_DATA7	LCD Data7 Register
0x0070	LCD_SPI_CTL1	LCD SPI Control 1 Register
0x0074	LCD_SPI_CTL2	LCD SPI Control 2 Register
0x0078	LCD_QSPI_DDR_CTL	LCD QSPI DDR Control register
0x007c	LCD_SD_CNT	LCD Send Data Counter Register
0x0080	LCD_SD_CHECKSUM	LCD Send Data CheckSum Register
0x0090	LCD_TR_CTL	LCD TR-LCD Display Control Register
0x0094	LCD_TR_IMG_SIZE	LCD TR-LCD Display Image Size Register
0x0098	LCD_TR_TIM0	LCD TR-LCD Display Timing0 Register
0x009C	LCD_TR_TIM1	LCD TR-LCD Display Timing1 Register
0x00A0	LCD_TR_TIM2	LCD TR-LCD Display Timing2 Register
0x00A4	LCD_TR_TIM3	LCD TR-LCD Display Timing3 Register
0x00A8	LCD_TR_TIM4	LCD TR-LCD Display Timing4 Register
0x00AC	LCD_TR_TIM5	LCD TR-LCD Display Timing5 Register

14.1.7 LCD Register Description

14.1.7.1 LCD_CTL

LCD controller control register
Offset=0x0000

Bit(s)	Name	Description	R/W	Reset
31	CLK_GATING_EN	CLK Gating Enable 0:module clock gating 1: mode clock enable	R/W	0x0
30	CHECK_CLK_EN	Pixel counter, checksum clock enable 0: disable clock 1: enable clock	R/W	0x0
29	-	Reserved	R	0x0
28	RGB565_OUT_OPT	RGB565 Out Options 0:RGB565 1:RGB555 When RGB565_OUT_OPT =1 the output Green color G[0] force to 0.	R/W	0x0
27:26	RGB666_IN_OPT	RGB666 source options 0:RGB666_C0 1:RGB666_CLB 2:RGB666_CHB 3:RGB666_C0	R/W	0x0
25:24	RGB565_IN_OPT	RGB565 Source Options	R/W	0x0

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		0:RGB565_C0 1:RGB565_CLB 2:RGB565_CHB 3:RGB565_F555 look at data source and converting for details		
23:17	-	Reserved	R	0x0
16	LCD_HOLD_OPT	LCD Hold Opt 0:not hold when FIFO empty 1:hold when FIFO empty invalid in QUAD-SYNC or RGB mode	R/W	0x0
15	-	Reserved	R	0x0
14	ARST_FIFO_EN	Auto Reset FIFO per frame start 0:disable 1:enable	R/W	0x0
13	FIFO_EN	LCDC FIFO Enable 0:disable(reset FIFO) 1:enable	R/W	0x0
12	TE_EN	Tearing Effort Enable 0 : disable 1 : enable Invalid in RGB/QUAD-SYNC mode	R/W	0x0
11:10	TE_MOD_SEL	Tearing Effort Mode Select 0 : High level 1 : Low level 2 : Rising edge 3: Falling edge Invalid in RGB/QUAD-SYNC mode	R/W	0x0
9	CESEL	Chip Select 0: CE0 1: CE1 for CPU mode and SPI mode	R/W	0x0
8	MLS	Output MSB/LSB Swap 0:MSB Frist 1:LSB Frist Only for CPU mode and IF_OFS=RGB565	R/W	0x0
7:6	-	Reserved	R	0x0
5:4	IF_OFS	IF RGB Out Format Select 0: RGB 565 1: RGB 666 2: RGB 888 Others: Reserved	R/W	0x0
3:1	IFSEL	Mode select 0: I8080 Interface 1: M6800 Interface 2: Reserved 3: SPI Interface 4: TR Interface Others: Reserved	R/W	0x0
0	LCD_EN	LCD Module Enable 0:Disable 1:Enable	R/W	0x0

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14.1.7.2 LCD_DISP_SIZE

LCD Pixel Size Register
Offset=0x0008

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25:16	HEIGHT	Screen Height (in lines) for Screen	R/W	0x0
15:10	-	Reserved	R	0x0
9:0	WIDTH	Screen Width (in pixels) for Screen NOTE: include the half word pixel	R/W	0x0

NOTE: Only for DMA Source.

14.1.7.3 LCD_CPU_CTL

LCD CPU Control Register
Offset=0x001C

Bit(s)	Name	Description	R/W	Reset
31	START	Start CPU Mode Transfer 0: normal 1: start transfer, clean when complete	R/W	0x0
30	FEND_IRQ_EN	Frame End IRQ Enable 0 : disable 1 : enable	R/W	0x0
29:18	-	Reserved	R	0x0
17:15	SDT	Source Data Type Select: 0: RGB565 1: RGB666/BGR666 2: RGB888/BGR888 3: ARGB888 4: BGRA888 Others: Reserved For AHB & DMA mode	R/W	0x0
14	-	Reserved	R	0x0
13	RB_SWAP	Input color R/B Swap 0:color R/B not swap 1:color R/B swap For CPU AHB & DMA Mode	R/W	0x0
12:9	-	Reserved	R	0x0
8	RD_OPT	Read option 0:read data when RD rising 1:read data when RD rising delay 1 ahb_clk	R/W	0x0
7:6	-	Reserved	R	0x0
5	F565_SEL	RGB565 Format Select 0: R[4:0]->LCD_CPU_DATA[23:19] G[5:0]->LCD_CPU_DATA[15:10] B[4:0]->LCD_CPU_DATA[7:3] 1: R[4:0]->LCD_CPU_DATA[15:11] G[5:0]->LCD_CPU_DATA[10:5] B[4:0]->LCD_CPU_DATA[4:0] Note: Only CPU AHB MODE	R/W	0x0

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4	RS	RS select 0: RS output low voltage level 1: RS output high voltage level	R/W	0x0
3:2	SRC_SEL	Input Source Select 0: AHB 1: 2DGPU 2: DMA Others: Reserved	R/W	0x0
1	AHB_CFG_DATA	AHB CFG/DATA Select 0: Image Data 1: Config Data	R/W	0x0
0	CSX	Chip Select pin Level 0: CSX=Low Level 1: CSX=High Level Only for AHB mode	R/W	0x1

14.1.7.4 LCD_DATA

LCD Data Register
Offset=0x0020

Data for AHB mode：

ARGB888:

Bit(s)	Name	Description	R/W	Reset
31:24	A	panel's color A	R/W	0x0
23:16	R	panel's color R	R/W	0x0
15:8	G	panel's color G	R/W	0x0
7:0	B	panel's color B	R/W	0x0

BGRA888:

Bit(s)	Name	Description	R/W	Reset
31:24	B	panel's color B	R/W	0x0
23:16	G	panel's color G	R/W	0x0
15:8	R	panel's color R	R/W	0x0
7:0	A	panel's color A	R/W	0x0

RGB888:

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0x0
23:16	R	panel's color R	R/W	0x0
15:8	G	panel's color G	R/W	0x0
7:0	B	panel's color B	R/W	0x0

RGB666:

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0x0
23:18	R	panel's color R	R/W	0x0
17:16	-	Reserved	R/W	0x0
15:10	G	panel's color G	R/W	0x0

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9:8	-	Reserved	R/W	0x0
7:2	B	panel's color B	R/W	0x0
1:0	-	Reserved	R/W	0x0

RGB565, F565_SEL=0

Bit(s)	Name	Description	R/W	Reset
31:24	-	Reserved	R/W	0x0
23:19	R	panel's color R	R/W	0x0
18:16	-	Reserved	R/W	0x0
15:10	G	panel's color G	R/W	0x0
9:8	-	Reserved	R/W	0x0
7:3	B	panel's color B	R/W	0x0
2:0	-	Reserved	R/W	0x0

RGB565, F565_SEL=1

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R/W	0x0
15:11	R	panel's color R	R/W	0x0
10:5	G	panel's color G	R/W	0x0
4:0	B	panel's color B	R/W	0x0

Command for AHB mode:

Bit(s)	Name	Description	R/W	Reset
31:8	-	Reserved	R/W	0x0
7:0	CMD	Transmitted as a command	R/W	0x0

CMD[7:0] transmitted as CMD when 3/4-Line Serial Interface and CPU mode

Data for AHB SPI mode

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Transmitted Data when 3/4-Line Serial and QSPI Interface

Note: set LCD_DATA1 ~ LCD_DATA7 and RWL bits before write data to LCD_DATA.

Data for DMA mode:

RGB565 F565_SEL=0

G1[2:0]	B1[4:0]	R1[4:0]	G1[5:3]	G0[2:0]	B0[4:0]	R0[4:0]	G0[5:3]

31bit 24 23 16 15 8 7 0bit

RGB565 F565_SEL=1

R1[4:0]	G1[5:3]	G1[2:0]	B1[4:0]	R0[4:0]	G0[5:3]	G0[2:0]	B0[4:0]

31bit 24 23 16 15 8 7 0bit

RGB666

R1[5:0]	2b00	B0[5:0]	2b00	G0[5:0]	2b00	R0[5:0]	2b00
G2[5:0]	2b00	R2[5:0]	2b00	B1[5:0]	2b00	G1[5:0]	2b00
B3[5:0]	2b00	G3[5:0]	2b00	R3[5:0]	2b00	B2[5:0]	2b00

31bit 24 23 16 15 8 7 0bit

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RGB888

R1[7:0]	B0[7:0]	G0[7:0]	R0[7:0]
G2[7:0]	R2[7:0]	B1[7:0]	G1[7:0]
B3[7:0]	G3[7:0]	R3[7:0]	B2[7:0]

31bit 24 23 16 15 8 7 0bit

ARGB888

A0[7:0]	R0[7:0]	G0[7:0]	B0[7:0]
A1[7:0]	R1[7:0]	G1[7:0]	B1[7:0]
A2[7:0]	R2[7:0]	G2[7:0]	B2[7:0]

31bit 24 23 16 15 8 7 0bit

BGRA888:

B0[7:0]	G0[7:0]	R0[7:0]	A0[7:0]
B1[7:0]	G1[7:0]	R1[7:0]	A1[7:0]
B2[7:0]	G2[7:0]	R2[7:0]	A2[7:0]

31bit 24 23 16 15 8 7 0bit

14.1.7.5 LCD_CPU_CLK

LCD CPU Clock Register
Offset=0x0024

Bit(s)	Name	Description	R/W	Reset
31:22	-	Reserved	R	0x0
21:16	CLKLDU2	Clock Low Level Duration (from AHB_CLK) from 1 to 64 (EXCL2KL +1)	R/W	0x0
15:14	-	Reserved	R	0x0
13:8	CLKLDU	Clock Low Level Duration (from AHB_CLK) from 1 to 64 (EXCLKL +1)	R/W	0xF
7:6	-	Reserved	R	0x0
5:0	CLKHDU	Clock High Level Duration (from AHB_CLK). from 1 to 64 (EXCLKH +1)	R/W	0xF

NOTE:
For 2DGPU&DMA source write freq = CMU_LCDCLK/((clkldu+1)+(Clkhdu+1))
For AHB source write freq = AHB_CLK/((clkldu+1)+(Clkhdu+1))

14.1.7.6 LCD_TPL

LCD Transfer Pixel Length
Offset=0x0028

Bit(s)	Name	Description	R/W	Reset
31:0	TPL	Transfer Pixel Length(only for SI/CPU in 2DGPU Transfer mode)	R/W	0x0

NOTE: TPL=WIDTH*HEIGHT-1

14.1.7.7 LCD_SPI_CTL

LCD Serial Interface Control Register
Offset=0x002C

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Bit(s)	Name	Description	R/W	Reset
31	START	START Transfer 0:normal 1:start transfer, auto clean when complete	R/W	0x0
30	FTC_IRQ_EN	Frame Transfer Complete IRQ Enable 1'b0 : disable 1'b1 : enable Not for sync mode	R/W	0x0
29	FEND_PD_EN	Frame End Pending Enable 0:disable 1:enable Only for sync mode	R/W	0x0
28	AHB_CFG_DATA	AHB CFG/DATA Select 0：Image Data 1：Config Data Quad mode	R/W	0x0
27	-	Reserved	R/W	0x0
26:24	SDT	Source Data Type Select: 0:RGB565 1:RGB666/BGR666 2:RGB888/BGR888 3:ARGB888 4: BGRA888 Others: Reserved for AHB&DMA Mode	R/W	0x0
23:18	RWL	SPI Read/Write Data Length 0~32Byte(only in quad mode) 1~4Byte(in SPI 3/4 wire mode) For read:0~4Byte,others 1Byte Write LCD_DAT when send CMD and RWL=0	R/W	0x0
17	F565_SEL	RGB565 Format Select 0: R[4:0]->LCD_CPU_DATA[23:19] G[5:0]->LCD_CPU_DATA[15:10] B[4:0]->LCD_CPU_DATA[7:3] 1: R[4:0]->LCD_CPU_DATA[15:11] G[5:0]->LCD_CPU_DATA[10:5] B[4:0]->LCD_CPU_DATA[4:0] Note: for AHB & DMA Mode	R/W	0x0
16	DUAL_LANE	Dual Data Lane Sel 0:signle data lane 1:dual/quad data lane	R/W	0x0
15:12	DELAY_CHAIN	Delay chain 0~15ns Delay Valid in read operation	R/W	0x0
11:10	SRC_SEL	Input Source Select 0: AHB 1: 2DGPU 2: DMA Others: Reserved	R/W	0x0
9:8	QSPI_RDP_SEL	QUAD SPI Read Data Port Select 0:D0 1:D1 2:D2	R/W	0x0

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		3:D3 The SPI Command and Dummy will go out by D0, and Read Data Port Decision by QSPI_RDP_SEL. Only in Quad Mode		
7	DCP_MODE	Data Compact Mode SEL 0:normal mode 1:compact mode	R/W	0x0
6	-	Reserved	R/W	0x0
5	CSX	ChipSel 0:CSX=0 1:CSX=1 Only for AHB mode	R/W	0x1
4	RB_SWAP	Input color R/B Swap 0:color R/B not swap 1:color R/B swap For SI AHB or DMA Mode	R/W	0x0
3	CDX	Command/Data Control 0: Command 1: Display Data or Parameter	R/W	0x0
2:0	TYPE_SEL	SPI type select 0: 3-line serial interface type I 1: 3-line serial interface type II 2: 4-line serial interface type I 3 : 4-line serial interface type II 4: quad serial interface 5: quad serial interface sync mode Others: Reserved	R/W	0x0

14.1.7.8 LCD_SPI_CMD

LCD Quad Serial Interface Command and Address Register
Offset=0x0034
For 3/4 line Read

Bit(s)	Name	Description	R/W	Reset
31:24	CMD	Quad Serial Interface Command	R/W	0x0
23:0	-	Reserved	R	0x0

For Quad Mode

Bit(s)	Name	Description	R/W	Reset
31:24	CMD	Quad Serial Interface Command	R/W	0x0
23:0	AD	Quad Serial Interface Address	R/W	0x0

NOTE: QSI Transfer start CMD&AD or VS start command for sync mode

14.1.7.9 LCD_QSPI_CMD1

LCD Quad Serial Interface Command and Address Register
Offset=0x0038

Bit(s)	Name	Description	R/W	Reset

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31:24	CMD	Quad Serial Interface Command	R/W	0x0
23:0	AD	Quad Serial Interface Address	R/W	0x0

Note: HS Start Command for sync mode

14.1.7.10 LCD_QSPI_SYNC_TIM

LCD Quad Serial Interface sync timing control Register
Offset=0x003C

Bit(s)	Name	Description	R/W	Reset
31:16	HP	Horizontal Porch Delay(in SCL, Min 200ns)	R/W	0x0
15:8	VFP	Vertical Front Porch Delay(in lines)=VFP+1	R/W	0x0
7:0	VBP	Vertical Back Porch Delay(in lines)=VBP+1	R/W	0x0

NOTE: only for sync mode

14.1.7.11 LCD_QSPI_IMG_SIZE

LCD Quad Serial Interface Display Image Size Register
Offset=0x0040

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25:16	Y	Screen Height (in lines) for QSI IF	R/W	0x0
15:10	-	Reserved	R	0x0
9:0	X	Screen Width (in pixels) for QSI IF	R/W	0x0

NOTE: only for sync mode

14.1.7.12 2DGPU_INTERFACE_CTL

2DGPU Interface Control Register
Offset=0x0044

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29:20	UNIT	Unit Note: Unit can't < 4	R/W	0x0F
19:10	VSFP	Vertical Front Porch Delay	R/W	0x0
9:0	VSPW	Vertical Sync Pulse Width	R/W	0x0

14.1.7.13 LCD_PENDING

LCD Pending Register
Offset=0x0048

Bit(s)	Name	Description	R/W	Reset
31:8	-	Reserved	R	0x0
7	TR_TF_PD	2DGPU/DMA Mode Transfer Finish Pending 0 : busy	R/W	0x0

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		1 : finish Write 1auto clean Only for TR mode		
6	FIFO_UDF	FIFO UnderFlow 0:normal 1:flow	R/W	0x0
5	FEND_PD	QSPI Sync Frame End Pending 0:normal 1:Frame End Write 1auto clean, Only for QSPI sync mode	R/W	0x0
4	TES	TE signal status	R	0x0
3	SPI_TF_PD	2DGPU/DMA Mode Transfer Finish Pending 0 : busy 1 : finish Write 1auto clean For normal SPI mode not include QSPI Sync mode	R/W	0x0
2	CPU_TF_PD	2DGPU/DMA Mode Transfer Finish Pending 0 : busy 1 : finish Write 1 auto clean For CPU mode	R/W	0x0
1:0	-	Reserved	R	0x0

14.1.7.14 LCD_QSPI_DTAS

QSPI QUAD Serial Interface Delay Time after sync Register
Offset=0x004C

Bit(s)	Name	Description	R/W	Reset
31:17	-	Reserved	R	0x0
16	DCKEN	SCK output enable when delay time after sync cmd 0:disable, the sck output low level 1:enable	R/W	0x0
15:0	DTAS	Delay time after sync (in sclk)	R/W	0x0

14.1.7.15 LCD_DATA1

LCD Data1 register for AHB QSPI mode
Offset=0x0050

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.16 LCD_DATA2

LCD Data2 register for AHB QSPI mode
Offset=0x0054

Bit(s)	Name	Description	R/W	Reset

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31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.17 LCD_DATA3

LCD Data3 register for AHB QSPI mode
Offset=0x0058

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.18 LCD_DATA4

LCD Data4 register for AHB QSPI mode
Offset=0x005C

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.19 LCD_DATA5

LCD Data5 register for AHB QSPI mode
Offset=0x0060

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.20 LCD_DATA6

LCD Data6 register for AHB QSPI mode
Offset=0x0064

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.21 LCD_DATA7

LCD Data7 register for AHB QSPI mode
Offset=0x0068

Bit(s)	Name	Description	R/W	Reset
31:0	DATA	DATA	R/W	0x0

Note: transmitted Data when QSPI Interface

14.1.7.22 LCD_SPI_CTL1

LCD SPI Control 1 register
Offset=0x0070

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Bit(s)	Name	Description	R/W	Reset
31:18	-	reserved	R	0x0
17:16	AHB_CLK_DIV	AHB clk division 0:1 1:1/2 2:1/4 3:1/8 Note: Valid only in AHB mode	R/W	0x0
15:9	-	Reserved	R	0x0
8:4	RDLC	Read Dummy Delay Cycle Valid in 3/4wire & quad spi mode Note:0~31	R/W	0x0
3:0	-	Reserved	R	0x0

14.1.7.23 LCD_SPI_CTL2

LCD SPI Control 2 register
Offset=0x0074

Bit(s)	Name	Description	R/W	Reset
31:30	-	Reserved	R	0x0
29:25	DELAY_CSX	Delay control for CSX (0.5ns/step) Valid for input or output operation	R/W	0x0
24:20	DELAY_SCL	Delay control for SCL (0.5ns/step) Valid for input or output operation	R/W	0x0
19:15	DELAY_D3	Out Delay control for D3 (0.5ns/step) Valid for output operation	R/W	0x0
14:10	DELAY_D2	Out Delay control for D2 (0.5ns/step) Valid for output operation	R/W	0x0
9:5	DELAY_D1	Out Delay control for D1 (0.5ns/step) Valid for output operation	R/W	0x0
4:0	DELAY_D0	Out Delay control for D0 (0.5ns/step) Valid for output operation	R/W	0x0

14.1.7.24 LCD_QSPI_DDR_CTL

LCD QSPI DDR Control register
Offset=0x0078

Bit(s)	Name	Description	R/W	Reset
31:30	SPI_MODE	SPI protocol mode({ cpol, cpha }) 0: mode 0 1: mode 1 2: mode 2 3: mode 3 Valid when DDR_EN=1 & DDR_TYPE=0.	R/W	0x3
29	SPI_CLK_OPT	SPI CLK Option 0: Keep 1: Recovery	R/W	0x0
28:3	-	reserved	R	0x0

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2:1	DDR_TYPE	DDR Type 0:sdr mode 1:ddr type0 2:ddr type1 3:ddr type2 Invalid QUAD-SYNC mode	R/W	0x0
0	DDR_EN	DDR mode enable 0:disable 1:enable Invalid QUAD-SYNC mode	R/W	0x0

14.1.7.25 LCD_SD_CNT

LCD Send Counter register
Offset=0x007C

Bit(s)	Name	Description	R/W	Reset
31	EN	Check Sum enable 0:disable 1:enable	R/W	0x0
30:0	CNT	LCDC send data counter(units: pixel) invalid in sync mode and valid in DMA or 2DGPU souce	R	0x0

14.1.7.26 LCD_SD_CHECKSUM

LCD Send Data Checksum register
Offset=0x0080

Bit(s)	Name	Description	R/W	Reset
31	EN	Check Sum enable 0:disable 1:enable	R/W	0x0
30:0	SUM	LCDC send data CheckSum(Checksum for data in FIFO) valid in DMA or 2DGPU mode	R	0x0

Note:
If RGB565 out, SUM+=color_h<<8|color;
Else SUM+=(color_r<<16)| (color_g<<8)|(color_b<<0);

14.1.7.27 LCD_TR_CTL

LCD TR-LCD Control register
Offset=0x0090

Bit(s)	Name	Description	R/W	Reset
31	START	Start Transfer 0:normal 1:start Transfer Note: Auto clean when Transfer complete.	R/W	0x0
30	FTC_IRQ_EN	Frame Transfer Complete IRQ Enable 0 : IRQ disable 1 : IRQ enable	R/W	0x0
29	FRP_END_MOD	FRP End mode	R	0x0

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		0:Frp end immediately 1:Frp end when current cycle		
28:27	-	Reserved	R	0x0
26:24	SDT	Source Data Type Select: 0: RGB565 1: RGB666/BGR666 2: RGB888/BGR888 3: ARGB888 4: BGRA888 Others: Reserved for DMA Mode	R/W	0x0
23:22	LB_SEL	Low Bit Select 0: bit3 1: bit4 2: bit5 3: bit6 LSB select.	R/W	0x0
21:19	-	Reserved	R	0x0
18	HCK_TAIL_OPT	HCK Tail Option 0: Tail disable 1: Tail enable	R/W	0x1
17	F565_SEL	RGB565 Format Select 0: R[4:0]->LCD_CPU_DATA[23:19] G[5:0]->LCD_CPU_DATA[15:10] B[4:0]->LCD_CPU_DATA[7:3] 1: R[4:0]->LCD_CPU_DATA[15:11] G[5:0]->LCD_CPU_DATA[10:5] B[4:0]->LCD_CPU_DATA[4:0] Note: for DMA Mode	R/W	0x0
16	RB_SWAP	Input color R/B Swap 0:color R/B not swap 1:color R/B swap For DMA Mode	R/W	0x0
15	VCK_CONT	VCK continue flips during XRST low.	R/W	0x0
14	VCK_MODE	VCK flips during idle period.	R/W	0x0
13	HCK_MODE	VCK continue flips outside active date period.	R/W	0x0
12	PU_MODE	Partition update mode	R/W	0x0
11:10	SRC_SEL	Input Source Select 0: Reserved 1: 2DGPU 2: DMA Others: Reserved	R/W	0x0
9	FRP_EN	Frame refresh pulse enable	R/W	0x0
8	VCOM_INV	0: normal 1: inverse	R/W	0x0
7	FRP_INV	0: normal 1: inverse	R/W	0x0
6	XFRP_INV	0: normal 1: inverse	R/W	0x0
5	XRST_INV	0: normal 1: inverse	R/W	0x0

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4	VST_INV	0: normal 1: inverse	R/W	0x0
3	HST_INV	0: normal 1: inverse	R/W	0x0
2	VCK_INV	0: normal 1: inverse	R/W	0x0
1	HCK_INV	0: normal 1: inverse	R/W	0x0
0	ENB_INV	0: normal 1: inverse	R/W	0x0

14.1.7.28 LCD_TR_IMG_SIZE

LCD TR-LCD Display Size register0
Offset=0x0094

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25:16	IMG_HEIGHT	Screen Height (in lines)	R/W	0x0
15:10	-	Reserved	R	0x0
9:0	IMG_WIDTH	Screen Width (in pixels)	R/W	0x0

Note: Valid in 2DGPU&DMA mode

14.1.7.29 LCD_TR_TIM0

LCD TR-LCD Display Timing register0
Offset=0x0098

Bit(s)	Name	Description	R/W	Reset
31:24	TWXRST	Low level width of XRST	R/W	0x0
23	-	Reserved	R	0x0
22:0	TWVCOM	High/Low level width of VCOM	R/W	0x0

14.1.7.30 LCD_TR_TIM1

LCD TR-LCD Display Timing register1
Offset=0x009C

Bit(s)	Name	Description	R/W	Reset
31:24	TDVST	Time interval from the rising edge of XRST to the rising edge of VST	R/W	0x0
23:16	TWVST	High level width of VST	R/W	0x0
15:8	TDHST	Time interval from the rising/falling edge of VCK to the rising edge of HST	R/W	0x0
7:0	TWHST	High level width of HST	R/W	0x0

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14.1.7.31 LCD_TR_TIM2

LCD TR-LCD Display Timing register2
Offset=0x00A0

Bit(s)	Name	Description	R/W	Reset
31:24	TDVCK	Time interval from the rising edge of VST to the rising edge of VCK	R/W	0x0
23:13	TWVCK	High/Low level width of VCK	R/W	0x0
12:11	TPHCK	HCK phase selection	R/W	0x2
10:8	-	Reserved	R	0x0
7:0	TDHCK	Div coefficient of CMU_LCD_CLK, Must be a multiple of 4	R/W	0x4

14.1.7.32 LCD_TR_TIM3

LCD TR-LCD Display Timing register3
Offset=0x00A4

Bit(s)	Name	Description	R/W	Reset
31:21	TSENB	Time interval from the rising edge of VST to the first line valid data	R/W	0x0
20:10	THENB	Time interval from the falling edge of XRST to the last valid data	R/W	0x0
9:8	-	Reserved	R	0x0
7:0	TDDATA	Time interval from the rising/falling edge of VCK to the first valid data	R/W	0x0

14.1.7.33 LCD_TR_TIM4

LCD TR-LCD Display Timing register4
Offset=0x00A8

Bit(s)	Name	Description	R/W	Reset
31:29	-	Reserved	R	0x0
28:16	TDENB	Time interval from the rising/falling edge of VCK to the rising edge of ENB	R/W	0x0
15:13	-	Reserved	R	0x0
12:0	TWENB	High level width of ENB	R/W	0x0

14.1.7.34 LCD_TR_TIM5

LCD TR-LCD Display Timing register5
Offset=0x00AC

Bit(s)	Name	Description	R/W	Reset
31:21	TSMENB	Time interval from the rising edge of VST to the first minimum VCK(units: VCK/4)	R/W	0x3
20:10	THMENB	Time interval from the last minimum VCK to the falling edge	R/W	0x0

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		of XRST(units: VCK/4)		
9:0	TWMVCK	Minimum VCK width(units: HCK)	R/W	0x1

Note: Active only when pu_mode =1.

14.2 2.5D GPU

14.2.1 Features

- 192MHz 2.5D GPU
- Support any curves and any font rendering by HW tessellation, NO CPU overhead.
- 2.5D perspective transformation.
- Any angle primitive rotation, flip and mirror.
- Anti-aliasing: 16x Anti-aliasing without extra performance drop and memory overhead.
- Solid, linear gradient paint and pattern paint.
- Texture mapping; Point sampling, linear and bilinear filtering.
- Scissoring and viewport clipping.
- Porter Duff Alpha Blending plus additional blend modes.
- 8/16/24/32 bits source image and frame color formats.
- Index 1/2/4/8 format source image supported.
- Command buffer based DMA to minimize CPU overhead.
- Lossy compression to save system memory.

14.3 2D GPU

14.3.1 Features

- RGB565\666\888\ARGB\ARGB8565\ARGB6666\ARGB1555\Ax\ABMP\XRGB\CLUT source data format
- 256*32 deep FIFO with DRQ and IRQ pre Layer
- 512*32 deep FIFO for Scaling function
- Support 256 Level Alpha
- Support three Image Layer
- Support Master DMA Transfer
- Support any angle rotate with RGB565/ARGB/BGRA source
- The Layer1/2 Support Round Rotation Function
- The Layer1/2 Support Rotation Function
- The Layer1/2 Support 8Phase 2-TAP (8x~1/15x)Up/down scaling
- All layers support Horizontal Anti-aliasing
- The Layer1/2 Support Super-Sampling Anti-Aliasing(SSAA)

14.4 JPEG decoder

14.4.1 Features

- Input mode: JPEG Baseline
- Resolution: 48x48 ~ 480x480

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- Input data format: YUV 420
- Input data format: RGB888/RGB565

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15 GPIO and I/O Multiplexer

15.1 Features

GPIO (General Purpose Input /Output) MFP:

GPIO can output 0 or 1 and detect the signal level of the external circuit. Each GPIO has its own enable control bit and data registers. But the PADs are limited, so MFP module is designed for multiplexing these PADs. Features of GPIO are listed below:

- 55 Programmable GPIOs, and 6 analog IOs can also config as GPIOs.
- All PADs have internal pull down resistors (100KOhm) or pull up resistors (50 KOhm/10KOhm)
- Driving strength adjustable
- Automatically switching PAD function
- All PADs have external Interrupts function

WIO (Wake up I/O) MFP:

- 5 wake up I/O can wake up SOC externally, and these I/O can be config as Special I/O. The Maximum work frequency should no more than 10KHz.
- WIO2/3 also can be config as LOSCI/LOSCO for connect the low frequency crystal.

Settings in actual practice please consult our engineers. The multiplexing relationship can be found in *Pin Description list*.

15.2 Operation Manual

15.2.1 Multi-function Switch Operation

1. Some pin can be multiplexed as three kinds of functions: module function (MFP), GPIO function, Analog function, which can be configured by setting AD_Select bit and MFP bit of GPIOx_CTL registers.
2. GPIO and MFP are digital functions. Once the Analog function is selected, the digital functions will fail. The function priority of some multiplexed pin are Analog function > GPIO function > MFP function.
3. Some pin can be multiplexed as analog function and digital function. If the pin is used as digital function, it must be disabled analog function firstly by setting AD_Select register.
4. Some MFP modules have itself pull-up and pull-down resistors, referring the chapter Pad PU control register and Pad PD control register; when the pin is multiplexed as MFP module function, the modules pull-up/pull-down resistors will be enabled automatically and the pull-up/pull-down resistors of GPIOs must be DISABLED, or the voltage level and functions will be abnormal.

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15.2.2 GPIO Output

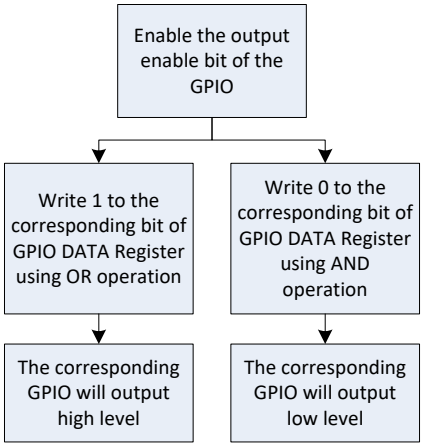


Figure 15-1 GPIO Output Configuration

15.2.3 GPIO Input

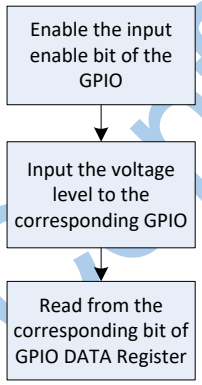


Figure 15-2 GPIO Input Configuration

15.2.4 GPIO Output/Input Loop Test

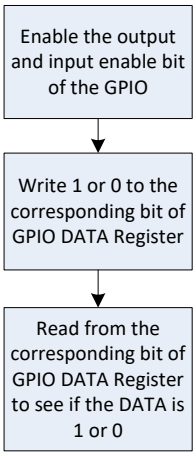
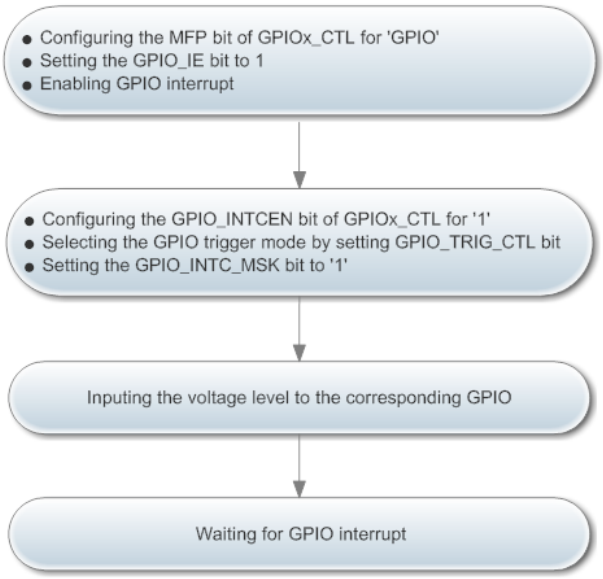


Figure 15-3 GPIO In/Out Loop Test

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15.2.5 GPIO INTC



15.3 GPIO Register List

Table 15-1 GPIO_MFP Controller Registers Address

Name	Physical Base Address	KSEG1 Base Address
GPIO_MFP_REGISTER	0x40068000	0x40068000

Table 15-2 GPIO&MFP Controller Registers

Offset	Register Name	Description	Register Voltage	IO power domain
0x0010	GPIO4_CTL	GPIO4 control Register	VDD	IOVCC1
0x0014	GPIO5_CTL	GPIO5 control Register	VDD	IOVCC1
0x0020	GPIO8_CTL	GPIO8 control Register	VDD	IOVCC1
0x0024	GPIO9_CTL	GPIO9 control Register	VDD	IOVCC1
0x0028	GPIO10_CTL	GPIO10 control Register	VDD	IOVCC1
0x002C	GPIO11_CTL	GPIO11 control Register	VDD	IOVCC1
0x0030	GPIO12_CTL	GPIO12 control Register	VDD	IOVCC1
0x0034	GPIO13_CTL	GPIO13 control Register	VDD	IOVCC1
0x0038	GPIO14_CTL	GPIO14 control Register	VDD	IOVCC2
0x003C	GPIO15_CTL	GPIO15 control Register	VDD	IOVCC2
0x0040	GPIO16_CTL	GPIO16 control Register	VDD	IOVCC2
0x0044	GPIO17_CTL	GPIO17 control Register	VDD	IOVCC2
0x0048	GPIO18_CTL	GPIO18 control Register	VDD	VCC
0x004C	GPIO19_CTL	GPIO19 control Register	VDD	VCC
0x0050	GPIO20_CTL	GPIO20 control Register	VDD	VCC
0x0054	GPIO21_CTL	GPIO21 control Register	VDD	VCC
0x0058	GPIO22_CTL	GPIO22 control Register	VDD	VCC
0x005C	GPIO23_CTL	GPIO23 control Register	VDD	VCC
0x0060	GPIO24_CTL	GPIO24 control Register	VDD	VCC
0x0064	GPIO25_CTL	GPIO25 control Register	VDD	VCC
0x0068	GPIO26_CTL	GPIO26 control Register	VDD	VCC
0x006C	GPIO27_CTL	GPIO27 control Register	VDD	VCC
0x0070	GPIO28_CTL	GPIO28 control Register	VDD	VCC

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0x0074	GPIO29_CTL	GPIO29 control Register	VDD	VCC
0x0078	GPIO30_CTL	GPIO30 control Register	VDD	IOVCC2
0x007C	GPIO31_CTL	GPIO31 control Register	VDD	VCC
0x0080	GPIO32_CTL	GPIO32 control Register	VDD	IOVCC2
0x0084	GPIO33_CTL	GPIO33 control Register	VDD	IOVCC2
0x0088	GPIO34_CTL	GPIO34 control Register	VDD	IOVCC2
0x008C	GPIO35_CTL	GPIO35 control Register	VDD	IOVCC2
0x00A8	GPIO42_CTL	GPIO42 control Register	VDD	IOVCC3
0x00BC	GPIO47_CTL	GPIO47 control Register	VDD	IOVCC3
0x00C4	GPIO49_CTL	GPIO49 control Register	VDD	IOVCC4
0x00C8	GPIO50_CTL	GPIO50 control Register	VDD	IOVCC4
0x00CC	GPIO51_CTL	GPIO51 control Register	VDD	IOVCC4
0x00D0	GPIO52_CTL	GPIO52 control Register	VDD	IOVCC4
0x00D4	GPIO53_CTL	GPIO53 control Register	VDD	IOVCC4
0x00D8	GPIO54_CTL	GPIO54 control Register	VDD	IOVCC4
0x00DC	GPIO55_CTL	GPIO55 control Register	VDD	IOVCC4
0x00E0	GPIO56_CTL	GPIO56 control Register	VDD	IOVCC4
0x00E4	GPIO57_CTL	GPIO57control Register	VDD	IOVCC4
0x00E8	GPIO58_CTL	GPIO58 control Register	VDD	IOVCC4
0x00EC	GPIO59_CTL	GPIO59 control Register	VDD	IOVCC4
0x00F0	GPIO60_CTL	GPIO60 control Register	VDD	IOVCC4
0x00F4	GPIO61_CTL	GPIO61 control Register	VDD	IOVCC4
0x00F8	GPIO62_CTL	GPIO62 control Register	VDD	IOVCC4
0x00FC	GPIO63_CTL	GPIO63 control Register	VDD	IOVCC4
0x0100	GPIO64_CTL	GPIO64 control Register	VDD	IOVCC1
0x0128	GPIO74_CTL	GPIO74 control Register	VDD	IOVCC1
0x012C	GPIO75_CTL	GPIO75 control Register	VDD	VCC
0x0130	GPIO76_CTL	GPIO76 control Register	VDD	VCC
0x0134	GPIO77_CTL	GPIO77 control Register	VDD	IOVCC2
0x0138	GPIO78_CTL	GPIO78 control Register	VDD	IOVCC4
0x013C	GPIO79_CTL	GPIO79 control Register	VDD	IOVCC4
0x0140	GPIO80_CTL	GPIO80 control Register	VDD	IOVCC4
0x0168	GPIO90_CTL	GPIO90 control Register	VDD	AVCC
0x016C	GPIO91_CTL	GPIO91 control Register	VDD	AVCC
0x0170	GPIO92_CTL	GPIO92 control Register	VDD	AVCC
0x0174	GPIO93_CTL	GPIO93 control Register	VDD	AVCC
0x0178	GPIO94_CTL	GPIO94 control Register	VDD	AVCC
0x017C	GPIO95_CTL	GPIO95 control Register	VDD	AVCC
0x0200	GPIO_ODAT0	GPIO Output Data register0	VDD	
0x0204	GPIO_ODAT1	GPIO Output Data register1	VDD	
0x0208	GPIO_ODAT2	GPIO Output Data register2	VDD	
0x0210	GPIO_BSR0	GPIO Output Data bit set register0	VDD	
0x0214	GPIO_BSR1	GPIO Output Data bit set register1	VDD	
0x0218	GPIO_BSR2	GPIO Output Data bit set register2	VDD	
0x0220	GPIO_BRR0	GPIO Output Data bit set register0	VDD	
0x0224	GPIO_BRR1	GPIO Output Data bit set register1	VDD	
0x0228	GPIO_BRR2	GPIO Output Data bit set register2	VDD	
0x0230	GPIO_IDAT0	GPIO Output Data register0	VDD	
0x0234	GPIO_IDAT1	GPIO Output Data register1	VDD	
0x0238	GPIO_IDAT2	GPIO Output Data register2	VDD	
0x0240	GPIO_PD0	GPIO IRQ Pending register0	VDD	

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0x0244	GPIO_PD1	GPIO IRQ Pending register1	VDD	
0x0248	GPIO_PD2	GPIO IRQ Pending register2	VDD	
0x0300	WIO0_CTL	Wake up IO0 control Register	SVCC	SVCC
0x0304	WIO1_CTL	Wake up IO1 control Register	SVCC	SVCC
0x0308	WIO2_CTL	Wake up IO2 control Register	SVCC	SVCC
0x030C	WIO3_CTL	Wake up IO3 control Register	SVCC	SVCC
0x0310	WIO4_CTL	Wake up IO4 control Register	SVCC	SVCC
0x0320	DC5V_CTL	DC5V control Register	SVCC	SVCC

15.4 GPIO Register Description

15.4.1 GPIO4_CTL

GPIO4 control Register
Offset=0x0010

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable	R/W	0x0

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		0x1: Enable		
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO4 0x03: SPI2_SS 0x06: UART1_TX 0x08: I2C0_CLK 0x0A: SD0_DAT4 0x0B: PPI_TRIGGER0 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x0E: I2C2_DAT 0x12: PWM0 0x14: SD1_DAT2 0x19: SPIMT1_SS0 0x1C: LRADC4 Others: Reserved	R/W	0x0

15.4.2 GPIO5_CTL

GPIO5 control Register
Offset=0x0014

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable.	R/W	0x0

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		0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.		
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO5 0x03: SPI2_CLK 0x06: UART1_RX 0x08: I2C0_DAT 0x0A: SD0_DAT5 0x0B: PPI_TRIGGER1 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0E: I2C2_CLK 0x12: PWM1 0x14: SD1_DAT3 0x19: SPIMT1_CLK 0x1C: LRADC5 Others: Reserved	R/W	0x0

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15.4.3 GPIO8_CTL

GPIO8 control Register
Offset=0x0020

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO8 0x03: SPI2_MISO 0x04: SPI3_IO2 0x07: UART2_TX 0x0A: SD0_DAT2 0x0B: PPI_TRIGGER2 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0E: I2C2_DAT 0x0F: I2C3_CLK 0x12: PWM0 0x14: SD1_CMD 0x19: SPIMT1_MISO 0x1C: LRADC2 Others: Reserved	R/W	0x0

15.4.4 GPIO9_CTL

GPIO9 control Register
Offset=0x0024

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1	R/W	0x1

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		0x1: Level 2 0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO9 0x03: SPI2_MOSI 0x04: SPI3_IO3 0x07: UART2_RX 0x0A: SD0_DAT3 0x0B: PPI_TRIGGER3 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0F: I2C3_DAT 0x12: PWM1 0x14: SD1_CLK 0x19: SPIMT1_MOSI 0x1C: LRADC3 Others: Reserved	R/W	0x0

15.4.5 GPIO10_CTL

GPIO10 control Register
Offset=0x0028

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0

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23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO10 0x04: SPI3_SS 0x05: UART0_TX 0x07: UART2_CTS 0x08: I2C0_CLK 0x0A: SD0_CMD 0x12: PWM2 0x14: SD1_DAT2	R/W	0x0

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		0x15: IISTX1_DOUT 0x16: SPI0_SS 0x19: SPIMT1_SS1 Others: Reserved		

15.4.6 GPIO11_CTL

GPIO11 control Register
Offset=0x002C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable	R/W	0x0

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		0x1: Enable		
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO11 0x04: SPI3_CLK 0x05: UART0_RX 0x07: UART2_RTS 0x08: I2C0_DAT 0x0A: SD0_DAT0 0x0B: PPI_TRIGGER5 0x0E: I2C2_DAT 0x12: PWM3 0x14: SD1_CLK 0x16: SPI0_CLK 0x18: SPIMT0_SS3 0x19: SPIMT1_SS2 Others: Reserved	R/W	0x0

15.4.7 GPIO12_CTL

GPIO12 control Register
Offset=0x0030

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0

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15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO12 0x04: SPI3_MISO 0x05: UART0_CTS 0x06: UART1_TX 0x09: I2C1_CLK 0x0A: SD0_CLK0 0x0B: PPI_TRIGGER6 0x0E: I2C2_CLK 0x11: DMIC01_CLK 0x12: PWM1 0x14: SD1_DAT3 0x15: IISTX1_DOUT 0x16: SPI0_MISO 0x18: SPIMT0_SS2 0x19: SPIMT1_SS3 Others: Reserved	R/W	0x0

15.4.8 GPIO13_CTL

GPIO13 control Register
Offset=0x0034

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Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO	R/W	0x0

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		0x00: GPIO13 0x04: SPI3_MOSI 0x05: UART0_RTS 0x06: UART1_RX 0x09: I2C1_DAT 0x0A: SD0_DAT1 0x0B: PPI_TRIGGER7 0x0D: IISRX0_DIN 0x0F: I2C3_CLK 0x11: DMIC01_DAT 0x14: SD1_DAT0 0x16: SPI0_MOSI 0x18: SPIMT0_SS1 0x1F: WCI_TX Others: Reserved		

15.4.9 GPIO14_CTL

GPIO14 control Register
Offset=0x0038

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO14 0x05: UART0_TX 0x08: I2C0_CLK 0x0B: PPI_TRIGGER0 0x0C: IISTX0_DOUT 0x12: PWM0 0x13: LCD_D0 0x17: BT_UART_TX 0x18: SPIMT0_SS3 0x19: SPIMT1_MISO 0x1A: I2CMT0_CLK Others: Reserved	R/W	0x0

15.4.10 GPIO15_CTL

GPIO15 control Register
Offset=0x003C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level	R/W	0x0

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		Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO15 0x05: UART0_RX 0x08: I2C0_DAT 0x0B: PPI_TRIGGER1 0x0D: IISRX0_DIN 0x12: PWM1 0x13: LCD_D1 0x17: BT_UART_RX 0x18: SPIMT0_SS2 0x19: SPIMT1_MOSI 0x1A: I2CMT0_DAT Others: Reserved	R/W	0x0

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15.4.11 GPIO16_CTL

GPIO16 control Register
Offset=0x0040

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO16 0x03: SPI2_SS 0x06: UART1_TX 0x09: I2C1_DAT 0x0B: PPI_TRIGGER2 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x12: PWM2 0x13: LCD_D2 0x17: BT_REQ 0x19: SPIMT1_SS1 0x1B: I2CMT1_DAT Others: Reserved	R/W	0x0

15.4.12 GPIO17_CTL

GPIO17 control Register
Offset=0x0044

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3	R/W	0x1

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		0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO17 0x03: SPI2_CLK 0x06: UART1_RX 0x09: I2C1_CLK 0x0B: PPI_TRIGGER3 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0F: I2C3_CLK 0x12: PWM3 0x13: LCD_D3 0x17: BT_ACCESS 0x19: SPIMT1_SS0 0x1B: I2CMT1_CLK OTHER: Reserved	R/W	0x0

15.4.13 GPIO18_CTL

GPIO18 control Register
Offset=0x0048

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge	R/W	0x0

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		0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO18 0x03: SPI2_MISO 0x04: SPI3_MISO 0x06: UART1_CTS 0x08: I2C0_CLK 0x0A: SD0_DAT0 0x0B: PPI_TRIGGER4 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x12: PWM1	R/W	0x0

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		0x13: LCD_D4 0x14: SD1_DAT2 0x15: IISTX1_DOUT 0x17: PTA_GRANT 0x18: SPIMT0_MISO 0x19: SPIMT1_MISO 0x1A: I2CMT0_CLK 0x1B: I2CMT1_CLK OTHER: Reserved		

15.4.14 GPIO19_CTL

GPIO19 control Register
Offset=0x004C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO19 0x03: SPI2_MOSI 0x04: SPI3_MOSI 0x06: UART1_RTS 0x08: I2C0_DAT 0x0A: SD0_DAT1 0x0B: PPI_TRIGGER5 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x12: PWM2 0x13: LCD_D5 0x14: SD1_DAT3 0x18: SPIMT0_MOSI 0x19: SPIMT1_MOSI 0x1A: I2CMT0_DAT 0x1B: I2CMT1_DAT 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.15 GPIO20_CTL

GPIO20 control Register
Offset=0x0050

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level	R/W	0x0

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		0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO20 0x03: SPI2_SS 0x04: SPI3_SS 0x06: UART1_TX 0x07: UART2_TX 0x09: I2C1_CLK 0x0A: SD0_DAT2 0x0B: PPI_TRIGGER6 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x12: PWM3 0x13: LCD_D6 0x18: SPIMT0_SS0	R/W	0x0

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		0x19: SPIMT1_SS0 0x1B: I2CMT1_CLK 0x1C: LRADC2 OTHER: Reserved		

15.4.16 GPIO21_CTL

GPIO21 control Register
Offset=0x0054

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable	R/W	0x0

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		0x1: Enable		
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO21 0x03: SPI2_CLK 0x04: SPI3_CLK 0x06: UART1_RX 0x07: UART2_RX 0x08: I2C0_CLK 0x09: I2C1_DAT 0x0A: SD0_DAT3 0x0B: PPI_TRIGGER7 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x12: PWM0 0x13: LCD_D7 0x18: SPIMT0_CLK 0x19: SPIMT1_CLK 0x1B: I2CMT1_DAT 0x1C: LRADC3 OTHER: Reserved	R/W	0x0

15.4.17 GPIO22_CTL

GPIO22 control Register
Offset=0x0058

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC.	R/W	0x0

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		0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.		
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO22 0x04: SPI3_MISO 0x07: UART2_CTS 0x08: I2C0_DAT 0x0A: SD0_DAT6 0x0B: PPI_TRIGGER8 0x0E: I2C2_CLK 0x12: PWM1 0x14: SD1_DAT0 0x15: IISTX1_DOUT 0x18: SPIMT0_SS3 0x19: SPIMT1_MISO 0x1A: I2CMT0_CLK 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

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15.4.18 GPIO23_CTL

GPIO23 control Register
Offset=0x005C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO23 0x04: SPI3_MOSI 0x07: UART2_RTS 0x09: I2C1_CLK 0x0A: SD0_DAT7 0x0B: PPI_TRIGGER9 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0E: I2C2_DAT 0x14: SD1_DAT1 0x18: SPIMT0_SS2 0x19: SPIMT1_MOSI 0x1A: I2CMT0_DAT 0x1C: LRADC2 0x1F: WCI_TX OTHER: Reserved	R/W	0x0

15.4.19 GPIO24_CTL

GPIO24 control Register
Offset=0x0060

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control	R/W	0x1

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		0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO24 0x04: SPI3_CLK 0x06: UART1_TX 0x08: I2C0_CLK 0x09: I2C1_DAT 0x0A: SD0_CLK0 0x0B: PPI_TRIGGER10 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0F: I2C3_CLK 0x13: LCD_D4 0x18: SPIMT0_SS1 0x19: SPIMT1_SS0 0x1A: I2CMT0_CLK 0x1F: WCI_RX OTHER: Reserved	R/W	0x0

15.4.20 GPIO25_CTL

GPIO25 control Register
Offset=0x0064

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module;	R/W	0x0

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		0x1: Send interrupt to the INTC, when the GPIO trigger event is detect		
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO25 0x04: SPI3_IO2 0x06: UART1_RX 0x08: I2C0_DAT 0x0A: SD0_CMD	R/W	0x0

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		0x0B: PPI_TRIGGER11 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0E: I2C2_CLK 0x0F: I2C3_DAT 0x13: LCD_D5 0x19: SPIMT1_CLK 0x1A: I2CMT0_DAT 0x1C: LRADC5 0x1D: HOSCOUT 0x1E : LOSCOUT 0x1F: WCI_TX OTHER: Reserved		

15.4.21 GPIO26_CTL

GPIO26 control Register
Offset=0x0068

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO26 0x04: SPI3_SS 0x07: UART2_TX 0x09: I2C1_CLK 0x0A: SD0_DAT4 0x0B: PPI_TRIGGER0 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x0E: I2C2_DAT 0x13: LCD_D6 0x14: SD1_CMD 0x19: SPIMT1_MISO 0x1B: I2CMT1_CLK 0x1F: VCC_OUT(3.3v) OTHER: Reserved	R/W	0x0

15.4.22 GPIO27_CTL

GPIO27 control Register
Offset=0x006C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level	R/W	0x0

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		0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO27 0x04: SPI3_IO3 0x07: UART2_RX 0x09: I2C1_DAT 0x0A: SD0_DAT5 0x0B: PPI_TRIGGER1 0x0C: IISTX0_DOUT 0x0F: I2C3_CLK 0x13: LCD_D7 0x14: SD1_CLK 0x19: SPIMT1_MOSI 0x1B: I2CMT1_DAT 0x1C: LRADC6	R/W	0x0

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		0x1F: VCC_OUT(3.3v) OTHER: Reserved		

15.4.23 GPIO28_CTL

GPIO28 control Register
Offset=0x0070

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO28 0x04: SPI3_IO2 0x05: UART0_TX 0x08: I2C0_CLK 0x0A: SD0_DAT6 0x0B: PPI_TRIGGER2 0x0C: IISTX0_DOUT 0x0E: I2C2_CLK 0x14: SD1_DAT1 0x17: BT_UART_TX 0x19: SPIMT1_SS0 0x1A: I2CMT0_CLK OTHER: Reserved	R/W	0x0

15.4.24 GPIO29_CTL

GPIO29 control Register
Offset=0x0074

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0

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14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO29 0x03: SPI2_SS 0x04: SPI3_IO3 0x05: UART0_RX 0x08: I2C0_DAT 0x0A: SD0_DAT7 0x0B: PPI_TRIGGER3 0x0D: IISRX0_DIN 0x0E: I2C2_DAT 0x14: SD1_DAT0 0x15: IISTX1_DOUT 0x17: BT_UART_RX 0x19: SPIMT1_SS1 0x1A: I2CMT0_DAT OTHER: Reserved	R/W	0x0

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15.4.25 GPIO30_CTL

GPIO30 control Register
Offset=0x0078

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the	R/W	0x0

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		INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect		
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO30 0x03: SPI2_SS 0x0B: PPI_TRIGGER0 0x0C: IISG0_LRCLK	R/W	0x0

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		0x0D: IISG1_LRCLK 0x0E: I2C2_DAT 0x0F: I2C3_DAT 0x11: DMIC01_CLK 0x13: LCD_CS0 0x19: SPIMT1_CLK OTHER: Reserved		

15.4.26 GPIO31_CTL

GPIO31 control Register
Offset=0x007C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO31 0x03: SPI2_CLK 0x04: SPI3_CLK 0x0B: PPI_TRIGGER4 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0F: I2C3_DAT 0x11: DMIC01_DAT 0x14: SD1_DAT2 0x19: SPIMT1_CLK 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.27 GPIO32_CTL

GPIO32 control Register
Offset=0x0080

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0

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19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO32 0x03: SPI2_MISO 0x08: I2C0_CLK 0x0B: PPI_TRIGGER1 0x0E: I2C2_CLK 0x13: LCD_RS_CD 0x15: IISTX1_DOUT 0x18: SPIMT0_MISO 0x1A: I2CMT0_CLK 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.28 GPIO33_CTL

GPIO33 control Register
Offset=0x0084

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0

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25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO33 0x03: SPI2_MOSI	R/W	0x0

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		0x08: I2C0_DAT 0x0B: PPI_TRIGGER2 0x13: LCD_RD# 0x18: SPIMT0_MOSI 0x1A: I2CMT0_DAT 0x1F: IOVCC2_OUT(1.8/3.3v) OTHER: Reserved		

15.4.29 GPIO34_CTL

GPIO34 control Register
Offset=0x0088

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO34 0x03: SPI2_CLK 0x09: I2C1_CLK 0x0B: PPI_TRIGGER3 0x11: DMIC01_CLK 0x13: LCD_WR# 0x17: BT_UART_TX 0x18: SPIMT0_CLK 0x1B: I2CMT1_CLK OTHER: Reserved	R/W	0x0

15.4.30 GPIO35_CTL

GPIO35 control Register
Offset=0x008C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate	R/W	0x0

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		0x1: Slow slew rate		
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO35 0x03: SPI2_MISO 0x09: I2C1_DAT 0x0B: PPI_TRIGGER4 0x0F: I2C3_CLK 0x11: DMIC01_DAT 0x13: LCD_TE 0x17: BT_UART_RX 0x18: SPIMT0_SS1 0x1A: I2CMT0_CLK 0x1B: I2CMT1_DAT 0x1C: LRADC4 OTHER: Reserved	R/W	0x0

15.4.31 GPIO42_CTL

GPIO42 control Register
Offset=0x00A8

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module;	R/W	0x0

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		0x1: Send interrupt to the INTC, when the GPIO trigger event is detect		
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO42 0x03: SPI2_CLK 0x07: UART2_RX 0x09: I2C1_DAT 0x0B: PPI_TRIGGER6	R/W	0x0

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		0x0D: IISG1_LRCLK 0x12: PWM2 0x14: SD1_DAT3 0x19: SPIMT1_CLK OTHER: Reserved		

15.4.32 GPIO47_CTL

GPIO47 control Register
Offset=0x00BC

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO47 0x0F: I2C3_CLK 0x15: IISTX1_DOUT 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.33 GPIO49_CTL

GPIO49 control Register
Offset=0x00C4

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1	R/W	0x1

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		0x1: Level 2 0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO49 0x03: SPI2_SS 0x06: UART1_TX 0x08: I2C0_CLK 0x0B: PPI_TRIGGER0 0x0E: I2C2_CLK 0x12: PWM0 0x14: SD1_CMD 0x17: BT_REQ 0x18: SPIMT0_SS0 0x1A: I2CMT0_CLK OTHER: Reserved	R/W	0x0

15.4.34 GPIO50_CTL

GPIO50 control Register
Offset=0x00C8

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge	R/W	0x0

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		0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO50 0x03: SPI2_CLK 0x06: UART1_RX 0x08: I2C0_DAT 0x0B: PPI_TRIGGER1 0x0E: I2C2_DAT 0x12: PWM1 0x14: SD1_CLK 0x17: BT_ACCESS 0x18: SPIMT0_CLK	R/W	0x0

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		0x1A: I2CMT0_DAT OTHER: Reserved		

15.4.35 GPIO51_CTL

GPIO51 control Register
Offset=0x00CC

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO51 0x03: SPI2_MISO 0x06: UART1_CTS 0x09: I2C1_CLK 0x0B: PPI_TRIGGER2 0x0F: I2C3_CLK 0x12: PWM2 0x14: SD1_DAT1 0x17: PTA_GRANT 0x18: SPIMT0_MISO 0x1B: I2CMT1_CLK OTHER: Reserved	R/W	0x0

15.4.36 GPIO52_CTL

GPIO52 control Register
Offset=0x00D0

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0

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13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO52 0x03: SPI2_MOSI 0x06: UART1_RTS 0x08: I2C0_CLK 0x09: I2C1_DAT 0x0B: PPI_TRIGGER3 0x0C: IISTX0_DOUT 0x0F: I2C3_DAT 0x12: PWM3 0x14: SD1_DAT0 0x15: IISTX1_DOUT 0x18: SPIMT0_MOSI 0x1B: I2CMT1_DAT OTHER: Reserved	R/W	0x0

15.4.37 GPIO53_CTL

GPIO53 control Register
Offset=0x00D4

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger	R/W	0x0

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		event is detect		
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO53 0x03: SPI2_SS 0x04: SPI3_SS 0x07: UART2_TX 0x08: I2C0_DAT 0x0A: SD0_DAT4	R/W	0x0

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		0x0B: PPI_TRIGGER4 0x0D: IISG1_MCLK 0x0E: I2C2_CLK 0x12: PWM0 0x14: SD1_DAT2 0x19: SPIMT1_SS0 0x1A: I2CMT0_CLK 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved		

15.4.38 GPIO54_CTL

GPIO54 control Register
Offset=0x00D8

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO54 0x03: SPI2_CLK 0x04: SPI3_CLK 0x07: UART2_RX 0x09: I2C1_CLK 0x0A: SD0_DAT5 0x0B: PPI_TRIGGER5 0x0D: IISG1_BCLK 0x0E: I2C2_DAT 0x12: PWM1 0x14: SD1_DAT3 0x19: SPIMT1_CLK 0x1A: I2CMT0_DAT OTHER: Reserved	R/W	0x0

15.4.39 GPIO55_CTL

GPIO55 control Register
Offset=0x00DC

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable;	R/W	0x0

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		Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.		
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO55 0x03: SPI2_MISO 0x04: SPI3_MISO 0x07: UART2_CTS 0x09: I2C1_DAT 0x0A: SD0_DAT6 0x0B: PPI_TRIGGER6 0x0D: IISG1_LRCLK 0x0F: I2C3_CLK 0x12: PWM2 0x14: SD1_CLK 0x19: SPIMT1_MISO 0x1B: I2CMT1_CLK OTHER: Reserved	R/W	0x0

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15.4.40 GPIO56_CTL

GPIO56 control Register
Offset=0x00E0

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO56 0x03: SPI2_MOSI 0x04: SPI3_MOSI 0x07: UART2_RTS 0x0A: SD0_DAT7 0x0B: PPI_TRIGGER7 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0E: I2C2_CLK 0x0F: I2C3_DAT 0x12: PWM3 0x19: SPIMT1_MOSI 0x1B: I2CMT1_DAT OTHER: Reserved	R/W	0x0

15.4.41 GPIO57_CTL

GPIO57 control Register
Offset=0x00E4

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2	R/W	0x1

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		0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO57 0x03: SPI2_SS 0x08: I2C0_CLK 0x0A: SD0_DAT2 0x0B: PPI_TRIGGER8 0x0E: I2C2_DAT 0x12: PWM1 0x15: IISTX1_DOUT 0x18: SPIMT0_SS2 0x1A: I2CMT0_CLK 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.42 GPIO58_CTL

GPIO58 control Register
Offset=0x00E8

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge	R/W	0x0

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		0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO58 0x03: SPI2_CLK 0x08: I2C0_DAT 0x0A: SD0_DAT3 0x0B: PPI_TRIGGER9 0x0C: IISTX0_DOUT 0x0F: I2C3_CLK 0x18: SPIMT0_SS3 0x1A: I2CMT0_DAT 0x1F: WCI_TX	R/W	0x0

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		OTHER: Reserved		

15.4.43 GPIO59_CTL

GPIO59 control Register
Offset=0x00EC

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable	R/W	0x0

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		0x1: Enable		
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO59 0x03: SPI2_MISO 0x09: I2C1_CLK 0x0A: SD0_CLK1 0x0B: PPI_TRIGGER10 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0F: I2C3_DAT 0x18: SPIMT0_SS1 0x1B: I2CMT1_CLK 0x1F: WCI_RX OTHER: Reserved	R/W	0x0

15.4.44 GPIO60_CTL

GPIO60 control Register
Offset=0x00F0

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control	R/W	0x1

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		0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO60 0x03: SPI2_MOSI 0x09: I2C1_DAT 0x0A: SD0_CLK0 0x0B: PPI_TRIGGER11 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0F: I2C3_CLK 0x1B: I2CMT1_DAT 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.45 GPIO61_CTL

GPIO61 control Register
Offset=0x00F4

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode	R/W	0x0

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		0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved		
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO61 0x04: SPI3_IO2 0x0A: SD0_DAT0 0x0B: PPI_TRIGGER4 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x0E: I2C2_CLK 0x19: SPIMT1_SS3 0x1C: LRADC5	R/W	0x0

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		0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved		

15.4.46 GPIO62_CTL

GPIO62 control Register
Offset=0x00F8

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO62 0x04: SPI3_IO3 0x05: UART0_RX 0x06: UART1_RX 0x07: UART2_RX 0x0A: SD0_DAT1 0x0B: PPI_TRIGGER5 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0E: I2C2_DAT 0x19: SPIMT1_SS1 0x1C: LRADC4 OTHER: Reserved	R/W	0x0

15.4.47 GPIO63_CTL

GPIO63 control Register
Offset=0x00FC

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate	R/W	0x0

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		0x1: Slow slew rate		
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO63 0x03: SPI2_MISO 0x04: SPI3_MISO 0x05: UART0_TX 0x06: UART1_TX 0x07: UART2_TX 0x0A: SD0_CMD 0x0B: PPI_TRIGGER6 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0F: I2C3_DAT 0x15: IISTX1_DOUT 0x19: SPIMT1_SS2 0x1C: LRADC3 0x1D: HOSCOUT 0x1E: LOSCOUT 0x1F: IOVCC4_OUT(1.8v) OTHER: Reserved	R/W	0x0

15.4.48 GPIO64_CTL

GPIO64 control Register
Offset=0x0100

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Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO	R/W	0x0

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		0x00: GPIO64 0x03: SPI2_MISO 0x04: SPI3_MISO 0x05: UART0_TX 0x06: UART1_TX 0x07: UART2_TX 0x0A: SD0_DAT7 0x0B: PPI_TRIGGER11 0x0C: IISG0_MCLK 0x0D: IISG1_MCLK 0x0E: I2C2_CLK 0x0F: I2C3_DAT 0x12: PWM0 0x14: SD1_DAT1 0x15: IISTX1_DOUT 0x18: SPIMT0_SS0 0x1B: I2CMT1_DAT 0x1D: HOSCOUT 0x1E: LOSCOUT 0x1F: IOVCC1_OUT(3.3v) OTHER: Reserved		

15.4.49 GPIO74_CTL

GPIO74 control Register
Offset=0x0128

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0

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13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO74 0x03: SPI2_MOSI 0x04: SPI3_MOSI 0x05: UART0_RX 0x07: UART2_RX 0x08: I2C0_CLK 0x09: I2C1_DAT 0x0A: SD0_DAT6 0x0B: PPI_TRIGGER4 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0E: I2C2_DAT 0x0F: I2C3_CLK 0x11: DMIC01_CLK 0x12: PWM1 0x14: SD1_DAT0 0x18: SPIMT0_CLK 0x1B: I2CMT1_CLK 0x1C: LRADC6 0x1D: HOSCOUT 0x1E: LOSCOUT 0x1F: WCI_RX OTHER: Reserved	R/W	0x0

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15.4.50 GPIO75_CTL

GPIO75 control Register
Offset=0x012C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO75 0x03: SPI2_MISO 0x04: SPI3_MISO 0x05: UART0_TX 0x07: UART2_RX 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0E: I2C2_CLK 0x10: USBDP 0x11: DMIC01_CLK 0x12: PWM2 0x14: SD1_DAT3 0x17: BT_UART_TX 0x19: SPIMT1_SS3 0x1A: I2CMT0_CLK 0x1B: I2CMT1_CLK 0x1F: WCI_TX OTHER: Reserved	R/W	0x0

15.4.51 GPIO76_CTL

GPIO76 control Register
Offset=0x0130

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0

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14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO76 0x03: SPI2_MOSI 0x04: SPI3_MOSI 0x05: UART0_RX 0x07: UART2_TX 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x0E: I2C2_DAT 0x10: USBDM 0x11: DMIC01_DAT 0x12: PWM3 0x14: SD1_CLK 0x17: BT_UART_RX 0x19: SPIMT1_SS2 0x1A: I2CMT0_DAT 0x1B: I2CMT1_DAT 0x1F: WCI_RX OTHER: Reserved	R/W	0x0

15.4.52 GPIO77_CTL

GPIO77 control Register
Offset=0x0134

Bit(s)	Name	Description	R/W	Reset

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31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO77	R/W	0x0

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		0x03: SPI2_MOSI 0x0A: SD0_CLK0 0x0B: PPI_TRIGGER4 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0F: I2C3_DAT 0x11: DMIC01_DAT 0x13: LCD_CS1 0x14: SD1_CLK 0x15: IISTX1_DOUT 0x18: SPIMT0_SS0 0x1A: I2CMT0_DAT 0x1C: LRADC6 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved		

15.4.53 GPIO78_CTL

GPIO78 control Register
Offset=0x0138

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1

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11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO78 0x03: SPI2_SS 0x04: SPI3_SS 0x06: UART1_TX 0x08: I2C0_DAT 0x0A: SD0_DAT4 0x0B: PPI_TRIGGER8 0x0C: IISG0_BCLK 0x0D: IISG1_BCLK 0x0E: I2C2_CLK 0x0F: I2C3_CLK 0x11: DMIC01_DAT 0x12: PWM2 0x14: SD1_CLK 0x18: SPIMT0_MISO 0x1A: I2CMT0_CLK 0x1B: I2CMT1_DAT 0x1F: WCI_TX OTHER: Reserved	R/W	0x0

15.4.54 GPIO79_CTL

GPIO79 control Register
Offset=0x013C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger	R/W	0x0

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		event is detect		
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO79 0x03: SPI2_CLK 0x04: SPI3_CLK 0x06: UART1_RX 0x07: UART2_RX 0x08: I2C0_CLK	R/W	0x0

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		0x09: I2C1_CLK 0x0A: SD0_DAT5 0x0B: PPI_TRIGGER9 0x0C: IISG0_LRCLK 0x0D: IISG1_LRCLK 0x0F: I2C3_CLK 0x11: DMIC01_CLK 0x12: PWM3 0x14: SD1_CMD 0x18: SPIMT0_MOSI 0x1A: I2CMT0_DAT 0x1B: I2CMT1_CLK 0x1F: WCI_RX OTHER: Reserved		

15.4.55 GPIO80_CTL

GPIO80 control Register
Offset=0x0140

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	GPIO_INTC_MSK	GPIO INTC mask. 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the GPIO trigger event is detect	R/W	0x0
24	-	Reserved	R	0x0
23:21	GPIO_TRIG_CTL	GPIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	GPIO_INTCEN	GPIO INTC Enable. 0x0: Disable; Do not generate IRQ pending and do not send IRQ to INTC. 0x1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when GPIO_INTC_MSK is '1'.	R/W	0x0
19:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable	R/W	0x0

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		0x1: Enable		
10	GPIO4K7PUEN	GPIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4:0	MFP	Multi-Function of GPIO 0x00: GPIO80 0x03: SPI2_MOSI 0x04: SPI3_MOSI 0x06: UART1_RX 0x07: UART2_TX 0x09: I2C1_DAT 0x0A: SD0_DAT7 0x0B: PPI_TRIGGER8 0x0C: IISTX0_DOUT 0x0D: IISRX0_DIN 0x0E: I2C2_DAT 0x0F: I2C3_DAT 0x11: DMIC01_DAT 0x14: SD1_DAT0 0x15: IISTX1_DOUT 0x19: SPIMT1_SS0 0x1D: HOSCOUT 0x1E: LOSCOUT OTHER: Reserved	R/W	0x0

15.4.56 GPIO90_CTL

GPIO90 control Register
Offset=0x0168

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1	R/W	0x1

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		0x1: Level 2 0x2: Level 3 0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(INPUT0P)	R/W	0x1
3:0	MFP	0:GPIO90 Other: Reserved	R/W	0x0

Note:
If the user want to configure INPUT0P PIN to be GPIO90, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT0P PIN to be GPIO90 through GPIO90_CTL.

15.4.57 GPIO91_CTL

GPIO91 control Register
Offset=0x016C

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3	R/W	0x1

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		0x3: Level 4		
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(INPUT0N)	R/W	0x1
3:0	MFP	0:GPIO91 Other: Reserved	R/W	0x0

Note:
If the user want to configure INPUT0N PIN to be GPIO91, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT0N PIN to be GPIO91 through GPIO91_CTL.

15.4.58 GPIO92_CTL

GPIO92 control Register
Offset=0x0170

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(INPUT1P)	R/W	0x1
3:0	MFP	0:GPIO92 Other: Reserved	R/W	0x0

Note:
If the user want to configure INPUT1P PIN to be GPIO92, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT1P PIN to be GPIO92 through GPIO92_CTL.

15.4.59 GPIO93_CTL

GPIO93 control Register
Offset=0x0174

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(INPUT1N)	R/W	0x1
3:0	MFP	0:GPIO93 Other: Reserved	R/W	0x0

Note:
If the user want to configure INPUT1N PIN to be GPIO93, should do as flow:
1. Enable the power of ADC digital through set the bit26 of PWRGATE_DIG to 1;
2. Wait until the bit26 of PWRGATE_DIG_ACK become to 1;
3. Enable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 1;
4. Release the reset of ADC digital through set the bit3 of MRCR1 to 1;
5. Release the Isolation of ADC IO through set the bit23&bit22 of ADC_VMIC_CTL to 1;
6. Disable the clock of ADC digital through set the bit3 of CMU_DEVCLKEN1 to 0;
7. Set the INPUT1N PIN to be GPIO93 through GPIO93_CTL.

15.4.60 GPIO94_CTL

GPIO94 control Register
Offset=0x0178

Bit(s)	Name	Description	R/W	Reset
31:15	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable.	R/W	0x0

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		0x0: Disable 0x1: Enable		
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(AOUT/AOUT_P)	R/W	0x1
3:0	MFP	0:GPIO94 Other: Reserved	R/W	0x0

15.4.61 GPIO95_CTL

GPIO95 control Register
Offset=0x017C

Bit(s)	Name	Description	R/W	Reset
31:16	-	Reserved	R	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	GPIO50KPUEN	GPIO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	-	Reserved	R	0x0
9	GPIO100KPDEN	GPIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	GPIO10KPUEN	GPIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	GPIOINEN	GPIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	GPIOOUTEN	GPIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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5	SMIT	PAD Schmitt enable bit of GPIO: 0x0: disable 0x1: enable	R/W	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0: MFP 1: Analog Function(AOUT_N)	R/W	0x1
3:0	MFP	0:GPIO95 Other: Reserved	R/W	0x0

15.4.62 GPIO_ODAT0

GPIO Output Data register0
Offset = 0x200

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_ODAT	GPIO[31:0] Output Data.	R/W	0x0

15.4.63 GPIO_ODAT1

GPIO Output Data register1
Offset = 0x204

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_ODAT	GPIO[63:32] Output Data.	R/W	0x0

15.4.64 GPIO_ODAT2

GPIO Output Data register2
Offset = 0x208

Bit(s)	Name	Description	R/W	Reset
31:26	GPIO_ODAT2_1	GPIO[95:90] Output Data.	R/W	0x0
25:17	-	Reserved	R	0x0
16:0	GPIO_ODAT2_0	GPIO[80:64] Output Data.	R/W	0x0

15.4.65 GPIO_BSR0

GPIO Output Data bit set register0
Offset = 0x210

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_BSR	GPIO[31:0] Output Data bit set register 0: Not care; 1: Set the opposite bit of GPIO_ODAT0 as '1' and Set the GPIO to high level; Write this bit to '1', then will be clear automatically	R/W	0x0

15.4.66 GPIO_BSR1

GPIO Output Data bit set register1
Offset = 0x214

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Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_BSR	GPIO[63:32] Output Data bit set register 0: Not care; 1: Set the opposite bit of GPIO_ODAT1 as '1' and Set the GPIO to high level; Write this bit to '1', then will be clear automatically	R/W	0x0

15.4.67 GPIO_BSR2

GPIO Output Data bit set register2
Offset = 0x218

Bit(s)	Name	Description	R/W	Reset
31:26	GPIO_BSR2_1	GPIO[95:90] Output Data bit set register 0: Not care; 1: Set the opposite bit of GPIO_ODAT2 as '1' and Set the GPIO to high level; Write this bit to '1', then will be clear automatically	R/W	0x0
25:17	-	-	R	0x0
16:0	GPIO_BSR2_0	GPIO[80:64] Output Data bit set register 0: Not care; 1: Set the opposite bit of GPIO_ODAT2 as '1' and Set the GPIO to high level; Write this bit to '1', then will be clear automatically	R/W	0x0

15.4.68 GPIO_BRR0

GPIO Output Data bit reset register0
Offset = 0x220

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_BRR	GPIO[31:0] Output Data bit reset register 0: Not care; 1: Set the opposite bit of GPIO_ODAT0 as '0' and Set the GPIO to low level; Write this bit to '1', then will be clear automatically	R/W	0x0

15.4.69 GPIO_BRR1

GPIO Output Data bit reset register1
Offset = 0x224

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_BRR	GPIO[63:32] Output Data bit reset register 0: Not care; 1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level; Write this bit to '1', then will be clear	R/W	0x0

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		automatically		

15.4.70 GPIO_BRR2

GPIO Output Data bit reset register2
Offset = 0x228

Bit(s)	Name	Description	R/W	Reset
31:26	GPIO_BRR2_1	GPIO[95:90] Output Data bit reset register 0: Not care; 1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level; Write this bit to '1', then will be clear automatically	R/W	0x0
25:17	-	-	R	0x0
16:0	GPIO_BRR2_0	GPIO[80:64] Output Data bit reset register 0: Not care; 1: Set the opposite bit of GPIO_ODAT1 as '0' and Set the GPIO to low level; Write this bit to '1', then will be clear automatically	R/W	0x0

15.4.71 GPIO_IDAT0

GPIO Input Data register0
Offset = 0x230

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_IDAT	GPIO[31:0] Input Data.	R	0x0

15.4.72 GPIO_IDAT1

GPIO Input Data register1
Offset = 0x234

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_IDAT	GPIO[63:32] Input Data.	R	0x0

15.4.73 GPIO_IDAT2

GPIO Input Data register2
Offset = 0x238

Bit(s)	Name	Description	R/W	Reset
31:26	GPIO_IDAT2_1	GPIO[95:90] Input Data	R	0x0
25:17	-	Reserved	R	0x0
16:0	GPIO_IDAT2_0	GPIO[80:64] Input Data.	R	0x0

15.4.74 GPIO_PD0

GPIO IRQ Pending register0
Offset = 0x240

Bit(s)	Name	Description	R/W	Reset

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31:0	GPIO_PD	GPIO[31:0] IRQ Pending register. '0': No IRQ '1': IRQ Writing 1 to the bit is clear it.	R/W	0x0

15.4.75 GPIO_PD1

GPIO IRQ Pending register1
Offset = 0x244

Bit(s)	Name	Description	R/W	Reset
31:0	GPIO_PD	GPIO[63:32] IRQ Pending register. '0': No IRQ '1': IRQ Writing 1 to the bit is clear it.	R/W	0x0

15.4.76 GPIO_PD2

GPIO IRQ Pending register2
Offset = 0x248

Bit(s)	Name	Description	R/W	Reset
31:17	-	-	R	0x0
16:0	GPIO_PD	GPIO[80:64] IRQ Pending register. '0': No IRQ '1': IRQ Writing 1 to the bit is clear it.	R/W	0x0

15.4.77 WIO0_CTL

WIO0 control Register (SVCC)
Offset=0x300

Bit(s)	Name	Description	R/W	Reset
31:26	Reserved	Reserved	R	0x0
25	WIO_INTC_MSK	WIO INTC mask. 0: Mask the interrupt, do not send the interrupt to the INTC module; 1: Send interrupt to the INTC, when the WIO trigger event is detect	R/W	0x0
24	WIO_INTC_PD	WIO INTC Pending. 0: no pending; 1: pending Write 1 to clear	R/W	0x0
23:21	WIO_TRIG_CTL	WIO trigger mode 000: rising edge 001: falling edge 010:dual edge 011: high level 100: low level Others: Reserved	R/W	0x0
20	WIO_INTCEN	WIO INTC Enable. 0: Disable; Do not generate IRQ pending and do not send	R/W	0x0

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		IRQ to INTC. 1: Enable; Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is '1'.		
19:17	Reserved	Reserved	R	0x0
16	WIODAT	WIO Input/Output Data.	R/W	0x0
15	SR	WIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	WIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	WIO100KPUEN	WIO 100K PU Enable. 0: Disable 1: Enable	R/W	0x0
10	WIO4K7PUEN	WIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	WIO100KPDEN	WIO 100K PD Enable. 0: Disable 1: Enable	R/W	0x0
8	10KPUEN	WIO 10K PU Enable. 0: Disable 1: Enable	R/W	0x0
7	WIOINEN	WIO Input Enable. 0: Disable 1: Enable	R/W	0x0
6	WIOOUTEN	WIO Output Enable. 0: Disable 1: Enable	R/W	0x0
5:4	Reserved	Reserved	R	0x0
3:0	MFP	Multi-Function of WIO 0:WIO0 1:ONOFF 2:I2C1_CLK 3:LRADC1 OTHER: Reserved	R/W	0x3

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

15.4.78 WIO1_CTL

Wake up IO1 control Register (SVCC)
Offset=0x0304

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	WIO_INTC_MASK	WIO INTC mask	R/W	0x0

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		0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the WIO trigger event is detect.		
24	WIO_INTC_PD	WIO INTC Pending 0x0: no pending 0x1: pending Write 1 to clear	R/W	0x0
23:21	WIO_TRIG_CTL	WIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	WIO_INTCEN	WIO INTC Enable 0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC) 0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1")	R/W	0x0
19:17	-	Reserved	R	0x0
16	WIODAT	WIO Input/Output Data.	R/W	0x0
15	SR	WIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	WIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	WIO100KPUEN	WIO 100K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	WIO4K7PUEN	WIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	WIO100KPDEN	WIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	WIO10KPUEN	WIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	WIOINEN	WIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	WIOOUTEN	WIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5:4	-	Reserved	R	0x0
3:0	MFP	Multi-Function of WIO 0x0: WIO1	R/W	0x0

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		0x2: I2C1_DAT 0x3: LRADC1 0x6: LOSCOUT OTHER: Reserved		

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

15.4.79 WIO2_CTL

Wake up IO2 control Register (SVCC)
Offset=0x0308

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	WIO_INTC_MASK	WIO INTC mask 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the WIO trigger event is detect.	R/W	0x0
24	WIO_INTC_PD	WIO INTC Pending 0x0: no pending 0x1: pending Write 1 to clear	R/W	0x0
23:21	WIO_TRIG_CTL	WIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	WIO_INTCEN	WIO INTC Enable 0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC) 0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1")	R/W	0x0
19:17	-	Reserved	R	0x0
16	WIODAT	WIO Input/Output Data.	R/W	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	WIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	WIO100KPUEN	WIO 100K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	WIO4K7PUEN	WIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0

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9	WIO100KPDEN	WIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	WIO10KPUEN	WIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	WIOINEN	WIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	WIOOUTEN	WIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	-	Reserved	R	0x0
4	AD_SELECT	WIO Analog/Digital Select Register 0: MFP 1: Analog Function(LOSCI)	R/W	0x1
3:0	MFP	Multi-Function of WIO 0x0: WIO2 0x2: I2C0_CLK OTHER: Reserved	R/W	0x0

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

15.4.80 WIO3_CTL

Wake up IO3 control Register (SVCC)
Offset=0x030C

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	WIO_INTC_MASK	WIO INTC mask 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the WIO trigger event is detect.	R/W	0x0
24	WIO_INTC_PD	WIO INTC Pending 0x0: no pending 0x1: pending Write 1 to clear	R/W	0x0
23:21	WIO_TRIG_CTL	WIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	WIO_INTCEN	WIO INTC Enable 0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC) 0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1")	R/W	0x0
19:17	-	Reserved	R	0x0

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16	WIODAT	WIO Input/Output Data.	R/W	0x0
15	SR	WIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	WIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	WIO100KPUEN	WIO 100K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	WIO4K7PUEN	WIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	WIO100KPDEN	WIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	WIO10KPUEN	WIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	WIOINEN	WIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	WIOOUTEN	WIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	-	Reserved	R	0x0
4	AD_SELECT	WIO Analog/Digital Select Register 0: MFP 1: Analog Function(LOSCO)	R/W	0x1
3:0	MFP	Multi-Function of WIO 0x0: WIO3 0x2: I2C0_DAT 0x6: LOSCOUT OTHER: Reserved	R/W	0x0

Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

15.4.81 WIO4_CTL

Wake up IO4 control Register (SVCC)
Offset=0x0310

Bit(s)	Name	Description	R/W	Reset
31:26	-	Reserved	R	0x0
25	WIO_INTC_MASK	WIO INTC mask 0x0: Mask the interrupt, do not send the interrupt to the INTC module; 0x1: Send interrupt to the INTC, when the WIO trigger event is detect.	R/W	0x0

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24	WIO_INTC_PD	WIO INTC Pending 0x0: no pending 0x1: pending Write 1 to clear	R/W	0x0
23:21	WIO_TRIG_CTL	WIO trigger mode 0x0: rising edge 0x1: falling edge 0x2: dual edge 0x3: high level 0x4: low level Others: Reserved	R/W	0x0
20	WIO_INTCEN	WIO INTC Enable 0x0: Disable (Don't generate IRQ pending and do not send IRQ to INTC) 0x1: Enable (Generate IRQ pending when the GPIO trigger event is detect, and send IRQ to INTC when WIO_INTC_MSK is "1")	R/W	0x0
19:17	-	Reserved	R	0x0
16	WIODAT	WIO Input/Output Data.	R/W	0x0
15	SR	GPIO Slew Rate 0x0: Fast slew rate 0x1: Slow slew rate	R/W	0x0
14	-	Reserved	R	0x0
13:12	PADDRV	GPIO PAD Drive Control 0x0: Level 1 0x1: Level 2 0x2: Level 3 0x3: Level 4	R/W	0x1
11	WIO100KPUEN	WIO 100K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	WIO4K7PUEN	WIO 4.7K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
9	WIO100KPDEN	WIO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	WIO10KPUEN	WIO 10K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	WIOINEN	WIO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	WIOOUTEN	WIO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5:4	-	Reserved	R	0x0
3:0	MFP	Multi-Function of WIO 0x0: WIO4 0x2: I2C1_CLK 0x5: Reserved 0x6: LOSCOUT OTHER: Reserved	R/W	0x0

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Note:
WIO and its' control registers are work in SVCC domain, which will not be power off in all PMU state.

15.4.82 DC5V_CTL

DC5V control Register (VDD)
Offset=0x320

Bit(s)	Name	Description	R/W	Reset
31:17	-	Reserved	R	0x0
16	IODAT	Input/Output Data.	R/W	0x0
15:12	-	Reserved	R	0x0
11	IO50KPUEN	IO 50K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
10	-	Reserved	R	0x0
9	IO100KPDEN	IO 100K PD Enable. 0x0: Disable 0x1: Enable	R/W	0x0
8	IO5K6PUEN	IO 5.6K PU Enable. 0x0: Disable 0x1: Enable	R/W	0x0
7	IOINEN	IO Input Enable. 0x0: Disable 0x1: Enable	R/W	0x0
6	IOOUTEN	IO Output Enable. 0x0: Disable 0x1: Enable	R/W	0x0
5	-	Reserved	R	0x0
4	AD_SELECT	GPIO Analog/Digital Select Register 0x0: Auto 0x1: Analog Function(DC5V)	R/W	0x1
3:0	MFP	Multi-Function of GPIO 0x0: GPIO 0x1: UART1_TX 0x2: UART1_RX 0x3: UART2_TX 0x4: UART2_RX 0x5: BT_UART_TX 0x6: BT_UART_RX OTHER: Reserved	R/W	0x0

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16 Electrical Characteristics

16.1 Absolute Maximum Ratings

Table 16-1 Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Unit
Ambient Temperature	Tamb	-40	85	°C
Storage temperature	Tstg	-55	+150	°C
ESD Stress voltage	Vesd (Human body model)	3000	-	V
Supply Voltage	DC5V	-0.3	7	V
	BAT	-0.3	5	V
Input Voltage	3.3V IO	-0.3	IOVCC+0.3	V
	1.8V IO	-0.3	IOVCC+0.3	V
Thermal Resistance	ΘJA, junction to ambient thermal resistance	54		℃/W

Note:
Even if one of the above parameters exceeds the absolute maximum ratings momentarily, the quality of the product may be degraded. The absolute maximum ratings, therefore, specify the value exceeding, which the product may be physically damaged. Use the product well within these ratings.

16.2 Recommended PWR Supply

Table 16-2 Recommended PWR Supply

Supply Voltage	Min	Typ	Max	Unit
BAT (Li)	3.2	3.8	4.5	V
DC5V	4.5	5.0	7.0	V
VCC	2.9	3.3	3.5	V
VCC2	2.9	3.1	3.5	V
SVCC	--	2.5	--	V
VDD	0.6	1.1	1.3	V
VD12	0.7	1.2	1.45	V
VC18	1.74	1.8	2.1	V
VREF	--	1.5	--	V

16.3 DC Characteristics

Table 16-3 DC Parameters for +3.3V IO Pin with Schmitt Trigger Off

Parameter	Symbol	Min.	Max.	Unit	Condition
Low-level input voltage	VIL	-	0.8	V	VCC = 3.1V Tamb = -10 to 70 °C
High-level input voltage	VIH	2.0	-	V	
Low-level output voltage	VOL	-	0.4	V	
High-level output voltage	VOH	2.4	-	V	

Table 16-4 DC Parameter for +3.3V IO Pin with Schmitt Trigger On

Parameter	Symbol	Min.	Max.	Unit	Condition
Schmitt trigger positive-going threshold	VT+	-	1.9	V	VCC=3.1V Tamb = -10 to 70 °C
Schmitt trigger negative-going threshold	VT-	1.2	-	V	

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Table 16-5 DC Parameters for +1.8V IO Pin with Schmitt Trigger Off

Parameter	Symbol	Min.	Max.	Unit	Condition
Low-level input voltage	VIL	-	0.4	V	VCC = 1.8V Tamb = -10 to 70 °C
High-level input voltage	VIH	1.2	-	V	
Low-level output voltage	VOL	-	0.36	V	
High-level output voltage	VOH	1.4	-	V	

Table 16-6 DC Parameter for +1.8V IO Pin with Schmitt Trigger On

Parameter	Symbol	Min.	Max.	Unit	Condition
Schmitt trigger positive-going threshold	VT+	-	1.1	V	VCC=1.8V Tamb = -10 to 70 °C
Schmitt trigger negative-going threshold	VT-	0.7	-	V	

16.4 PWR Consumption

Table 16-7 PWR Consumption Table

VDD = 1.2V @ 25°C unless otherwise specified

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
BR Sniff Mode	Cs	500ms		90		µA
BLE Sniff Mode	Cs	500ms		65		µA
Standby	Cd	Vbat = 3.8V		20		µA

16.5 Bluetooth Characteristics

16.5.1 Transmitter

Table 16-8 Basic Data Rate of Transmitter

Parameter	Condition	Min.	Typ.	Max.	Unit
Maximum RF Transmit PWR	-	-		13	dBm
RF PWR Control Step	-	2	4	8	dB
20dB Bandwidth for Modulated Carrier	-	-	914	1500	KHz
Adjacent Channel Transmit	+2 MHz	-		-20	dBm
	-2 MHz	-		-20	dBm
	+3 MHz	-		-40	dBm
	-3 MHz	-		-40	dBm
Frequency Deviation	Δf1avg Maximum Modulation	140	166	175	KHz
	Δf2max Maximum Modulation	115	162		KHz
	Δf1avg/Δf2avg	0.8	1		
Initial Carrier Frequency Tolerance	-	-75	<+/-8	75	KHz
Frequency Drift	DH1 Packet	-25	<+/-10	25	KHz
	DH3 Packet	-40	<+/-10	40	KHz
	DH5 Packet	-40	<+/-10	40	KHz

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Frequency Drift Rate	-	-20	3	20	KHz/50us

Table 16-9 Enhanced Data Rate of Transmitter

Description		Min	Typ.	Max.	Unit
Relative Transmit PWR(EDR)		-4	-2.5	1	dB
π/4 DQPSK max carrier frequency stability|ω_0|		-10	+/-3	10	KHz
π/4 DQPSK max carrier frequency stability|ω_i|		-75	+/-5	75	KHz
π/4 DQPSK max carrier frequency stability|$\omega_0+\omega_i$|		-75	+/-4	75	KHz
8DPSK max carrier frequency stability |ω_0|		-10	+/-3	10	KHz
8DPSK max carrier frequency stability |ω_i|		-75	+/-5	75	KHz
8DPSK max carrier frequency stability|$\omega_0+\omega_i$|		-75	+/-5	75	KHz
π/4 DQPSK Modulation Accuracy	RMS DEVM		<7	20	%
	99% DEVM	99	100		%
	Peak DEVM		<16	35	%
In-band spurious emissions	F >F0 + 3MHz			-40	dBm
	F < F0 - 3MHz			-40	dBm
	F = F0 + 3MHz			-40	dBm
	F = F0 - 3MHz			-40	dBm
	F = F0 + 2MHz			-20	dBm
	F = F0 - 2MHz			-20	dBm
	F = F0 + 1MHz			-26	dBm
	F = F0 - 1MHz			-26	dBm
EDR Differential Phase Encoding		99	100		%

Table 16-10 BLE 1Mbps of Transmitter

Description		Min.	Typ.	Max.	Unit
Maximum RF Transmit PWR				14	dBm
In-band emissions	+2 MHz		-44	-20	dBm
	-2 MHz		-44	-20	dBm
	+3 MHz		-41	-30	dBm
	-3 MHz		-41	-30	dBm
Modulation Characteristics	Δf1avg Maximum Modulation	225	255	275	KHz
	Δf2max Maximum Modulation	185	235		KHz
	Δf1avg/Δf2avg	0.8	0.96		
Carrier Frequency Offset		-150	<+/-10	150	KHz
Frequency Drift		-50	<+/-5	50	KHz

Table 16-11 BLE 2Mbps of Transmitter

Description		Min.	Typ.	Max.	Unit
Maximum RF Transmit PWR				14	dBm
Adjacent Channel Transmit	+2 MHz		-28	-20	dBm
	-2 MHz		-28	-20	dBm
	+3 MHz		-42	-30	dBm
	-3 MHz		-42	-30	dBm
Frequency Deviation	Δf1avg Maximum Modulation	450	495	550	KHz
	Δf2max Maximum Modulation	370	425		KHz
	Δf1avg/Δf2avg	0.8	0.89		
Carrier Frequency Offset		-150	<+/-10	150	KHz
Frequency Drift		-50	<+/-5	50	KHz

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Frequency Drift Rate	-20	<+/-5	20	KHz/50us

16.5.2 Receiver

Table 16-12 Basic Data Rate of Receiver

RF Characteristics		Min.	Typ.	Max.	Specification
DH1/DH5 Rx sensitivity with dirty transmit on@0.1% BER			-96.5		<-70dBm
Maximum Input PWR at 0.1% BER		-20	0		<-20dBm
C/I (dB)	Co-channel		7		<11dB
	Adjacent 1Mhz		-9		<0dB
	Adjacent 2MHz		-43		<-30dB
	Adjacent >=3Mhz		-50		<-40dB
	Image interference		-44		<-9dB
	Image+/-1Mhz		-44/-54		<-20dB
Blocking(dBm)	30Mhz~2000Mhz	-30			-30
	2000Mhz~2400MHz	-35			-35
	2500MHz~3000MHz	-35			-35
	3000MHz~12.75Ghz	-30			-30
Inter-Modulation	Payload length: 37, n: 3,4,5 Packets: 1500 RF Level: -64 dBm		-30		PER≦30.8% ≧-50

Table 16-13 Enhanced Data Rate of Receiver(2M)

RF Characteristics		Min.	Typ.	Max.	Specification
2DH5 Rx sensitivity with dirty transmit on@0.1% BER			-95		<-70dBm
Maximum Input PWR at 0.1% BER			0		>-20dBm
C/I (dB)	Co-channel		11		<13dB
	Adjacent 1Mhz		-9		<0dB
	Adjacent 2MHz		-41		<-30dB
	Adjacent >=3Mhz		-50		<-40dB
	Image interference		-43		<-7dB
	Image+/-1Mhz		-42/-52		<-20dB
Blocking(dBm)	30Mhz~2000Mhz	-30			-30
	2000Mhz~2400MHz	-35			-35
	2500MHz~3000MHz	-35			-35
	3000MHz~12.75Ghz	-30			-30
Inter-Modulation	Payload length: 37, n: 3,4,5 Packets: 1500 RF Level: -64 dBm		-30		PER≦30.8% ≧-50

Table 16-14 Enhanced Data Rate of Receiver(3M)

RF Characteristics		Min.	Typ.	Max.	Specification
3DH5 Rx sensitivity (8DPSK)			-89.5		<-65dBm
Maximum Input PWR at 0.1% BER (8DPSK)			0		<-20dBm
C/I (dB)	Co-channel		16		<21dB
	Adjacent 1Mhz		-3		<5dB
	Adjacent 2MHz		-38		<-25dB
	Adjacent >=3Mhz		-44		<-33dB
	Image interference		-37		<0dB

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	Image+/-1Mhz		-37/-47		<-13dB
Blocking(dBm)	30Mhz~2000Mhz	-30			-30
	2000Mhz~2400MHz	-35			-35
	2500MHz~3000MHz	-35			-35
	3000MHz~12.75Ghz	-30			-30
Inter-Modulation	Payload length: 37, n: 3,4,5 Packets: 1500 RF Level: -64 dBm		-30		PER≦30.8% ≧-50

Table 16-15 BLE 1Mbps of Receiver

RF Characteristics		Min.	Typ.	Max.	Specification
Sensitivity LE 1M (dBm)			-99.5		<-70
Maximum Input PWR at 0.1% BER		-10	-2		-10
C/I (dB)	Co-channel		7		<21dB
	Adjacent 1Mhz		-7		<15dB
	Adjacent 2MHz		-41		<-17dB
	Adjacent >=3Mhz		-47		<-27dB
	Image interference		-45		<-9dB
	Image+/-1Mhz		-44/-51		<-15dB
Blocking(dBm)	30Mhz~2000Mhz	-30			-30
	2000Mhz~2400MHz	-35			-35
	2500MHz~3000MHz	-35			-35
	3000MHz~12.75Ghz	-30			-30
Inter-Modulation	Payload length: 37, n: 3, 4, 5 Packets: 1500 RF Level: -64 dBm		-30		PER≦30.8% ≧-50

Table 16-16 BLE 2Mbps of Receiver

RF Characteristics		Min.	Typ.	Max.	Specification
Sensitivity LE 2M(dBm)			-96		<-70
Maximum Input PWR at 0.1% BER		-10	0		-10
C/I (dB)	Co-channel		8		<21dB
	Adjacent 2Mhz		-7		<15dB
	Adjacent 4MHz		-30		<-17dB
	Adjacent >=6Mhz		-38		<-27dB
	Image interference		-25		<-9dB
	Image +2Mhz		-7		<15dB
	Image -2Mhz		-38		<-15dB
Blocking(dBm)	30Mhz~2000Mhz	-30			-30
	2000Mhz~2400MHz	-35			-35
	2500MHz~3000MHz	-35			-35
	3000MHz~12.75Ghz	-30			-30
Inter-Modulation	Payload length: 37, n: 3,4,5 Packets: 1500 RF Level: -64 dBm		-30		PER≦30.8% ≧-50

16.6 Audio ADC

Table 16-17 Audio ADC Parameters

Analogue to Digital Converter					
Parameter	**Conditions**	**Min**	**Typ**	**Max**	**Unit**

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Full Scale Input Voltage	THD+N < 1% (single-ended input aux input)		-	-	550m	Vrms
	THD+N < 1% (diff mode input aux input)		-	-	1.10	Vrms
Analogue gain	OP	Single Ended	-18	-	25.5	dB
		Full Differential	-12	-	31.5	
Analogue to Digital Converter						
Resolution	-		-	-	24	Bits
Input Sample Rate	-		8	-	96	kHz
SNR	**Single end input:** fin = 1kHz@Full Scale Input Voltage B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	97	-	dB
	Differential input: fin = 1kHz@Full Scale Input Voltage B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	98	-	dB
Dynamic Range	**Single end input:** fin = 1kHz@-40dBFS Input Voltage B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	97	-	dB
	Differential input: fin = 1kHz@-40dBFS Input Voltage B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	98	-	dB
THD+N	**Single end input:** fin = 1kHz B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	-	-81	dB
	Differential input: fin = 1kHz B/W = 22Hz~1/2FS Fs=16kHz(a-weighting)		-	-	-88	dB
Digital gain	-		0	-	52.5	dB

16.7 Audio DAC

Table 16-18 Audio DAC Parameters

Digital to Analogue Converter(Differential Output)					
Parameter	**Conditions**	**Min**	**Typ**	**Max**	**Unit**
Resolution	-	-	-	24	Bits
Output Sample Rate	-	8	-	96	kHz
SNR	fin = 1kHz@0dBFS input B/W = 22Hz~22kHz Fs=48kHz,Load=32Ω (A-Weighting)	-	113	-	dB
Dynamic Range	fin = 1kHz@-48dBFS input B/W = 22Hz~22kHz Fs=48kHz,Load=32 Ω (A-Weighting)	-	111	-	dB

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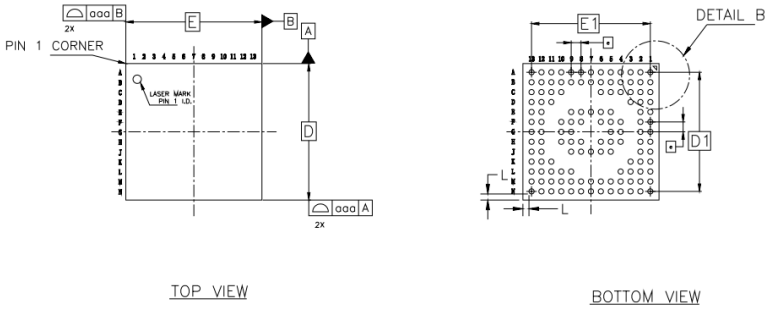
THD+N	fin = 1kHz@0dBFS input B/W = 22Hz~22kHz Fs=48kHz,Load=10KΩ(A-Weighting)				-98	dB
	fin = 1kHz@0dBFS input B/W = 22Hz~22kHz Fs=48kHz,Load=32Ω (A-Weighting)		-	-	-88	dB
Digital gain	0.375db/step Can set to mute		-98	-	24	dB
Max Amplitude/PWR	fin = 1kHz@0dBFS input Fs=48kHz,Load=10KΩ	Differential Output	-	-	0.95	Vrms

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17 Package information

17.1 Package Dimensions



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.85	0.93	1.01
A1	0.10	0.15	0.20
A2	0.73	0.78	0.83
A3		0.60 BASIC	
c	0.15	0.18	0.21
D	5.40	5.50	5.60
D1		4.80 BASIC	
E	5.40	5.50	5.60
E1		4.80 BASIC	
e		0.40 BASIC	
L		0.245 REF	
b	0.16	0.21	0.26
aaa		0.10	
ccc		0.08	
ddd		0.08	
eee		0.15	
fff		0.05	

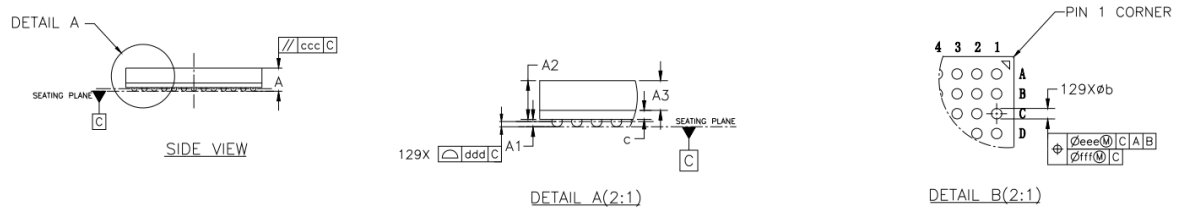
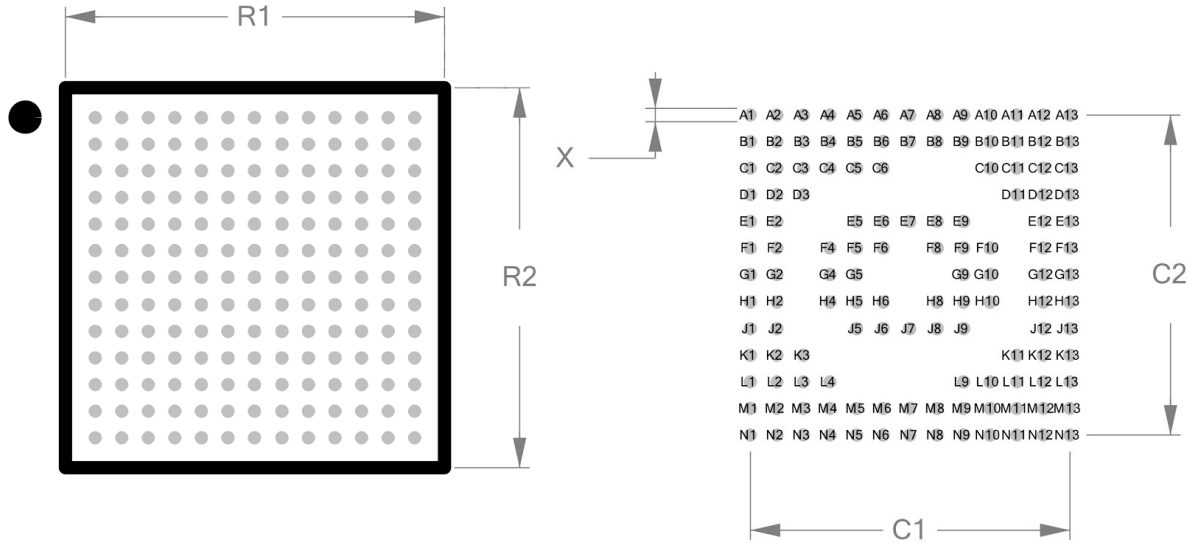


*Figure 17-1 TFBGA129(5.5*5.5*0.93mm, Pitch 0.4mm) Dimensions*

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17.2 Recommended Footprint



SYMBOL	MILLIMETER
R1	5.50
R2	5.50
C1	4.80
C2	4.80
X	0.20

Figure 17-2 TFBGA129(5.5*5.5*0.93mm, Pitch 0.4mm) recommended footprint

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18 Reflow Solder Information

The ATS3089x is constructed with all RoHS compliant material and should be reflowed accordingly. This chip is Moisture Sensitivity Level MSL3 and must be stored and handled accordingly.

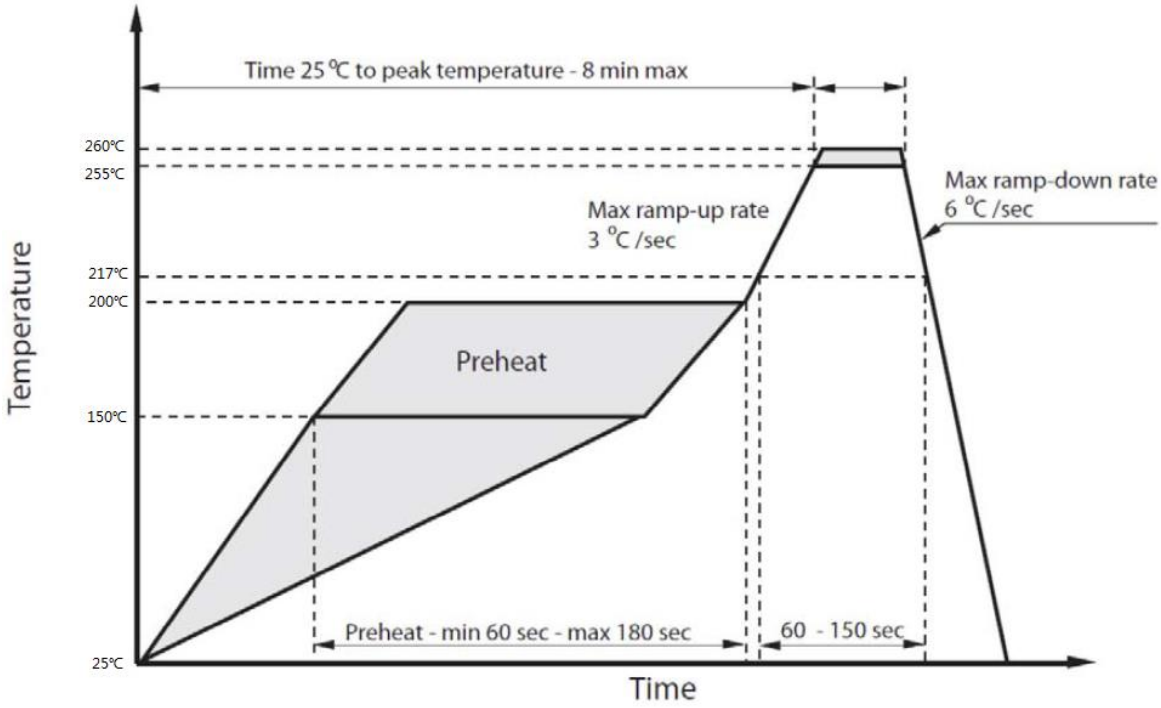
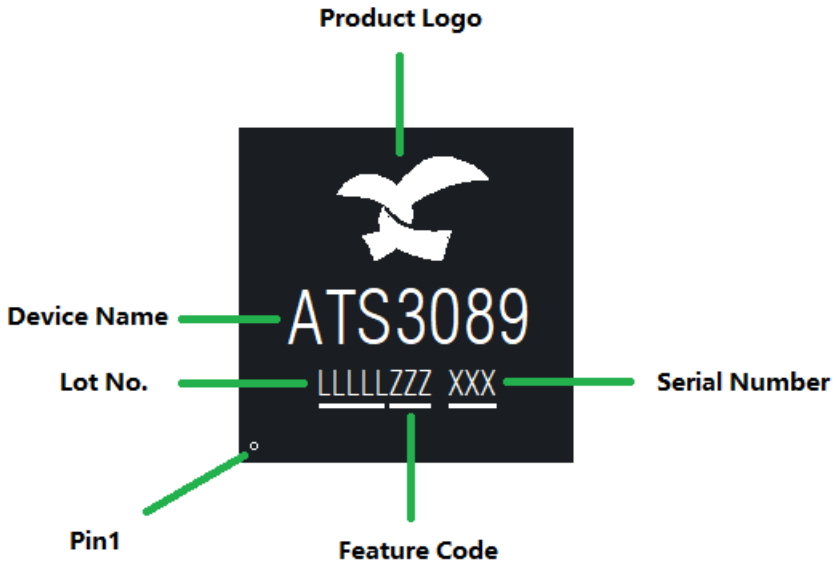


Figure 18-1 Typical Reflow Solder Profile

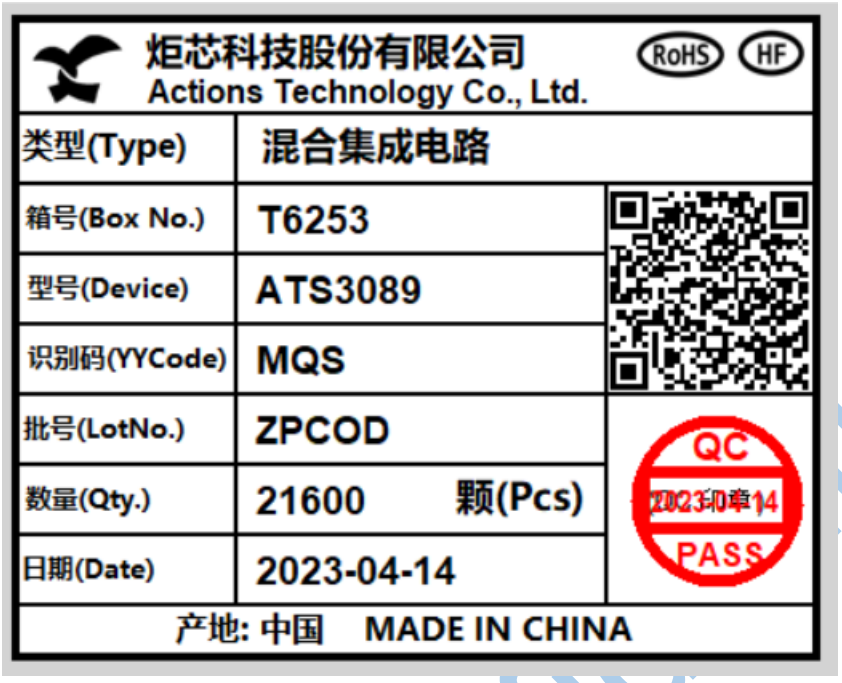
19 Device Marking and Label Information

19.1 Device Marking



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19.2 Label of Outer Packing Box



19.3 Label of Inner Box and Packing Bag



20 Ordering Information

Order Number				Package	Package Size	MOQ (Tray)	MOQ (Tape & Reel Packaging)
Device Name	PSRAM Density	Nor Flash Density	Feature Code				
ATS3089P	256Mbit	64Mbit	MQE (1)	TFBGA129	5.5mm x 5.5mm x 0.93mm	3600	NA
ATS3089	128Mbit	64Mbit	MQS (1)	TFBGA129	5.5mm x 5.5mm x 0.93mm	3600	NA

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ATS3089C	64Mbit	32Mbit	MQD (1)	TFBGA129	5.5mm x 5.5mm x 0.93mm	3600	NA

(1) The difference between feature codes lies in the density of built-in PSRAM/Nor Flash devices. For further information, please contact sales team.

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Acronyms and Abbreviations

Abbreviations	Descriptions
AEC	acoustic echo cancellers
ADC	Analog-to-Digital-Converter
ALU	Arithmetic Logic Unit
CC	Constant Current
DAC	Digital-to-Analog-Converter
DMA	Direct Memory Access
GPIO	General Purpose Input Output
HOSC	High Frequency OSC
HCL	High Frequency Calibration Low Frequency
INTC	Interrupt Controller
IRQ	Interrupt Request
LED	Light Emitting Diode
LCD	Liquid Crystal Display
LOSC	Internal Low Frequency OSC
MIC	Microphone
MMU	Memory Management Unit
MFP	Multiple Function PAD
NMI	Nonmaskable Interrupt
OSC	Oscillator
PA	Power Amplifier
UTMI	USB Transceiver Macro Interface

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Actions Technology Co., Ltd.
Address:No. 1 / C, Ke Ji Si Road, Hi-Tech Zone, Tangjia, Zhuhai
Tel:+86-756-3392353
Fax:+86-756-3392251
Post Code:519085
http://www.actions-semi.com
Business Email:mp-sales@actions-semi.com
Technical Service Email:mp-cs@actions-semi.com

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